

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-01477	1	SCH,MLB,X647	SCH	CRITICAL	
820-00597	1	PCBF,MLB,X647	PCB	CRITICAL	
685-00101	1	PCBA,MLB,COMMON PARTS,X647	CMNPTS		MLB_CMNPTS

BOM Groups

BOM GROUP	BOM OPTIONS
X647_COMMON	COMMON, ALTERNATE, ALT_CMN, X647_PROGPARTS, XDP, RTCRST:Y, GPULCK:OSC, 1V8_AUDIO_DLY:Y
X647_PROGPARTS	SMC:PROG, BOOTROM:PROG, CAMROM:PROG, TBTRON:PROG, ENETROM:PROG
X647_DEVEL	XDP_CONN, SAMCONN

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S90659	1	IC, EFI, V0855, Proto1, X647	U5210	CRITICAL	BOOTROM: PROG
335S90806	1	IC, SERIAL FLASH, 64MBIT, 3V, SP, NS08, QE=1	U5210	CRITICAL	BOOTROM: BLANK
341S90567	1	IC, SMC-B1, EXTERNAL, VSMC2, 36A6, X647	U5000	CRITICAL	SMC: PROG
338S1214	1	IC, SMC12-B1, 40MHZ/500MIPS, MCU, 157B6A	U5000	CRITICAL	SMC: BLANK
341S90660	1	IC, CAMERA, FLASH, VT80, Proto1, x647	U4202	CRITICAL	CAMROM: PROG
335S90852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM: BLANK
341S90796	1	IC, AR, V10 .2. EVT1, X647	U2890	CRITICAL	TBTROM: PROG
335S90133	1	IC, SPI FLASH, 8MBIT, 3.0V, US0N8	U2890	CRITICAL	TBTROM: BLANK
341S3912	1	IC, ENET ROM, NYMONYX, V1. 15	U3990	CRITICAL	ENETROM: PROG
335S1025	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REV F	U3990	CRITICAL	ENETROM: BLANK

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S08341	1	CPU,KBL,QKQ,80,(0,3,0,65W,1,0,1,0A1151	CPU	CRITICAL	CPU:KBL_3.0
337S08340	1	CPU,KBL,QHK,80,(0,3,4,65W,1,3,1,0A1151	CPU	CRITICAL	CPU:KBL_3.4
337S08342	1	CPU,KBL,QHA,80,(0,3,6,65W,1,15,1,0A1151	CPU	CRITICAL	CPU:KBL_3.6

GPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337598387	1	IC, GPU, AND, BUFF IN, LEA, A1, 1F, Q5, BGA769	U8800	CRITICAL	GPU:BF_LEA
337598386	1	IC, GPU, AND, BUFF IN, PROA, A1, 1F, Q5, BGA769	U8800	CRITICAL	GPU:BF_PROA
337598321	1	IC, GPU, AND, BUFF IN, XLA, A1, 1F, Q5, BGA769	U8800	CRITICAL	GPU:BF_XLA

Sockets & Connectors

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
5119-00802	1	CONN, CPU, SOCKET, UGA1155, SKYLAKE, HF	U0500	CRITICAL	
514-00853	1	CONN, RPT, USB -C, DUAL, XE40	J3300	CRITICAL	
998-04867	1	INTERPOSER_AMD, C889, BSA769, VDDC	U8800	CRITICAL	GPU:SKT_VCORE
998-04866	1	INTERPOSER_AMD, C889, BSA769, VDDCI/VDDC	U8800	CRITICAL	GPU:SKT_VDDCI

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33790298	1	IC,PCH,SM,SSKU,Q36D,D1,ES,B0687	U1100	CRITICAL	
33850254	1	IC,TBT,ALFANE,R100E,SLLOR,PQO,CJ,CP337	U2800	CRITICAL	
353500961	2	IC,C03215,ACE,C89,USB,PAR,SW,BLW,86A06	U3100, U3200	CRITICAL	
343S9616	1	IC,BCM57166A,CV+,AB,Bx8	U3900	CRITICAL	

FB VDRAM Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S00044	4	VRAM, MICRON, 4GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:2GB_MICRON
333S00078	4	VRAM, SAMSUNG, 4GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:2GB_SAMSUNG
333S00043	4	VRAM, HYNIX, 4GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:2GB_HYNIX
333S00075	4	VRAM, MICRON, 8GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:4GB_MICRON
333S00074	4	VRAM, SAMSUNG, 8GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:4GB_SAMSUNG
333S00100	4	VRAM, HYNIX, 8GBIT, 7GBPS, 8GA170	U9200, U9250, U9300, U9350	CRITICAL	FB:4GB_HYNIX

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0124	377S0057	ALT_CMN	ALL	TVS
155S0578	155S0367	ALT_CMN	ALL	1200HM EMI BEAD
138S0681	138S0638	ALT_CMN	ALL	Taiyo 10uF 80S alt
197S0479	197S0478	ALT_CMN	ALL	12 MHz Cam. Xtal
197S0486	197S0478	ALT_CMN	ALL	12 MHz Cam. Xtal
107S0251	107S0249	ALT_CMN	ALL	2mOhm Sense resistor
107S0255	107S0240	ALT_CMN	ALL	1mOhm Sense resistor
197S0481	197S0480	ALT_CMN	ALL	25MHZ Xtal
197S0343	197S0480	ALT_CMN	ALL	25MHZ Xtal
197S0369	197S0392	ALT_CMN	ALL	32 KHz PCH Xtal
197S0399	197S0392	ALT_CMN	ALL	32 KHz PCH Xtal
138S0860	138S0775	ALT_CMN	ALL	Single-source 1uF 402
378S0391	378S0390	ALT_CMN	ALL	Debug LEDs
341S00016	341S3912	ALT_CMN	ALL	ENET ROM, ADESTO, V1.15
376S0572	376S0659	ALT_CMN	ALL	Single P-ch FET
128S00042	128S0329	ALT_CMN	ALL	CAP, 220uF, 6.3V, 35mOhm
128S0311	128S0329	ALT_CMN	ALL	CAP, 220uF, 6.3V, 35mOhm
371S00095	371S0567	ALT_CMN	ALL	Diode, 1A0V, 0.25A
138S00084	138S00060	ALT_CMN	ALL	Cap, 47uF, 6.3V, 8603
152S00403	152S00322	ALT_CMN	ALL	Ind, 0.68uH, 20%, 6.1A, 20mOhm
138S0676	138S0691	ALT_CMN	ALL	Cap, Cer, 22uF, 20%, 6.3V
155S00007	155S0667	ALT_CMN	ALL	F1Tr, Common Mode, 900Hz, 100mA
155S0935	155S0667	ALT_CMN	ALL	F1Tr, Common Mode, 900Hz, 100mA
353S00769	353S4398	ALT_CMN	ALL	IC, TPS22966, LOADSW, 6A
353S00993	353S2541	ALT_CMN	ALL	IC, NC520001, OP-AMP, R-R
132S0401	132S00012	ALT_CMN	ALL	CAP, 0.22uF, 10V, 25V, X7R, 0402
371S0684	371S0495	ALT_CMN	ALL	DIODE, DUAL, SCHOTTKY, 30V
376S00015	376S0610	ALT_CMN	ALL	XSTR, FET, DL, NCH, 60V
138S0933	138S0931	ALT_CMN	ALL	CAP, 22UF, 20%, 4V, X5S, 8063
128S00008	128S0380	ALT_CMN	ALL	CAP, TANT, 210uF, 20%, 2.5V
152S00368	152S00269	ALT_CMN	ALL	Ind, 1.0uH, 20%, 14A, 18.7mOhm
128S0412	128S0237	ALT_CMN	ALL	CAP, TANT, 330uF, 20%, 6.3V

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
3375900189	08	CPU, SKL, 4C+2, 65W, 3.2GHZ
3375900187	08	PCH, SSDKUZ170, SR2C9
3385900229	08	TBT, Alpine Ridge -4c
3375900225	09	GPU, BAFIN_M, BGA769
3335900044	09	VRAM, MICRON, 4GBIT, 7GBPS
3335900078	09	VRAM, SAMSUNG, 4GBIT, 7GBPS
3335900075	09	VRAM, MICRON, 8GBIT, 7GBPS
3335900074	09	VRAM, SAMSUNG, 8GBIT, 7GBPS
3535900422	09	ACE, USB-C port power ctrl
353A4442	09	TPS2559, USB load switch
3535900875	09	TAS5764L, Audio amp

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S0030	155S0316	ALT_CMN	ALL	FER BD,600 OHM,0.5A,9663
155S00076	155S0546	ALT_CMN	ALL	FER BD,600 OHM,308mA,0402
152S1876	152S1373	ALT_CMN	ALL	IND,PWZ,7.8 mOhm,13A,7.1x5.53mm
376S1089	376S1128	ALT_CMN	ALL	Single N-Ch FET
155S0600	155S0513	ALT_CMN	ALL	220HM EMI BEAD
155S0694	155S0387	ALT_CMN	ALL	4700HM EMI BEAD
376S1129	376S00074	ALT_CMN	ALL	Dual N-Ch FET
376S0855	376S00074	ALT_CMN	ALL	Dual N-Ch FET
311S0649	311S0541	ALT_CMN	ALL	Single AND Gate
353S4376	353S3384	ALT_CMN	ALL	HDD 00Bv1 comparator
371S00019	371S0463	ALT_CMN	ALL	Rohm Schottky Barrier Diode
138S0746	138S0705	ALT_CMN	ALL	CAP,CER,X5R,10uF,20V,10V,402
311S00004	311S0370	ALT_CMN	ALL	SPI Quad I/O Mux
376S00001	376S0659	ALT_CMN	ALL	Single P-Ch FET
376S0972	376S00075	ALT_CMN	ALL	Single N-Ch FET
376S00124	376S00200	ALT_CMN	ALL	NFET,30V,22A,8P DFN
371S00043	371S0463	ALT_CMN	ALL	Diode,Schottky,30V,100mA
152S00358	152S00208	ALT_CMN	ALL	Ind,1uH,2.1A,120mOhm
138S0714	138S0713	ALT_CMN	ALL	Cap,10uF,6.3V,0402
116S00007	116S00006	ALT_CMN	ALL	Res,00hm,3A,0402
152S1499	152S0864	ALT_CMN	ALL	Ind,47uH,0.580Ohm,0402
353S00054	353S4342	ALT_CMN	ALL	1KCRF3330,V-Ref,3V,5mA
155S0659	155S0382	ALT_CMN	ALL	Filtr,CER,Ferr,BD,300hm,1.7A
179S00047	179S00036	ALT_CMN	ALL	XTAL,20MHz,5ppm,20pf,500hm
107S00030	107S00029	ALT_CMN	ALL	RES,DCDC SENSE,0.0050HM,1%
107S00087	107S00029	ALT_CMN	ALL	RES,DCDC SENSE,0.0050HM,1%
107S00015	107S00011	ALT_CMN	ALL	RES,DCDC SENSE,0.750HM,1%
152S00388	152S00182	ALT_CMN	ALL	IND,2.2uH,20%,43MOHM,4.5A
155S00155	155S0441	ALT_CMN	ALL	FLTR,EMI,FER BD,2200HM,1.4A
371S0718	371S0531	ALT_CMN	ALL	DIODE,BAT54,SHOITKY,30V
155S00115	155S00114	ALT_CMN	ALL	FERR BD,10 OHM,75%,8A
197S0588	197S0591	ALT_CMN	ALL	XTAL,24.0000MHZ,20PPM,9.5PF
197S0590	197S0591	ALT_CMN	ALL	XTAL,24.0000MHZ,20PPM,9.5PF
353S00711	353S2073	ALT_CMN	ALL	IC,1NA219,CURRENT SENSE
376S0977	376S00078	ALT_CMN	ALL	XSTR,FET,N-Ch,DUAL,20V
155S00116	311S0324	ALT_CMN	ALL	74LVCT26346F/5590
106S00017	106S4171	ALT_CMN	ALL	RES,MF,30.10HM,0.125W,1.805
132S00086	132S00045	ALT_CMN	ALL	CAP,X5S,0.822UF,6.3V,0201
132S00057	132S0668	ALT_CMN	ALL	CAP,2208PF,18%,25V,X7R,0201
138S0758	138S0742	ALT_CMN	ALL	CAP,X5R,22uF,MURATA
138S0759	138S0762	ALT_CMN	ALL	CAP,1uF,20V,10V,X5S,0603
138S00015	138S0777	ALT_CMN	ALL	CAP,18uF,20V,16V,X5S,0603
128S0444	128S0323	ALT_CMN	ALL	CAP,330uF,6.3V,D2E
311S00008	311S0271	ALT_CMN	ALL	AND GATE
353S00519	353S00831	ALT_CMN	ALL	IC,FDMF5504,DR MOS
376S00016	376S0644	ALT_CMN	ALL	MOSFET,P-Ch,20V,1A
138S0641	138S0700	ALT_CMN	ALL	CAP,2.2uF,10V,10V,X5R,0402
138S0739	138S0706	ALT_CMN	ALL	CAP,1uF,20V,10V,X5R,0201
155S00204	155S0731	ALT_CMN	ALL	FERR BD,120 OHM,25%,0402
152S00543	152S00484	ALT_CMN	ALL	IND,0.24uH,28%,10A,120MH

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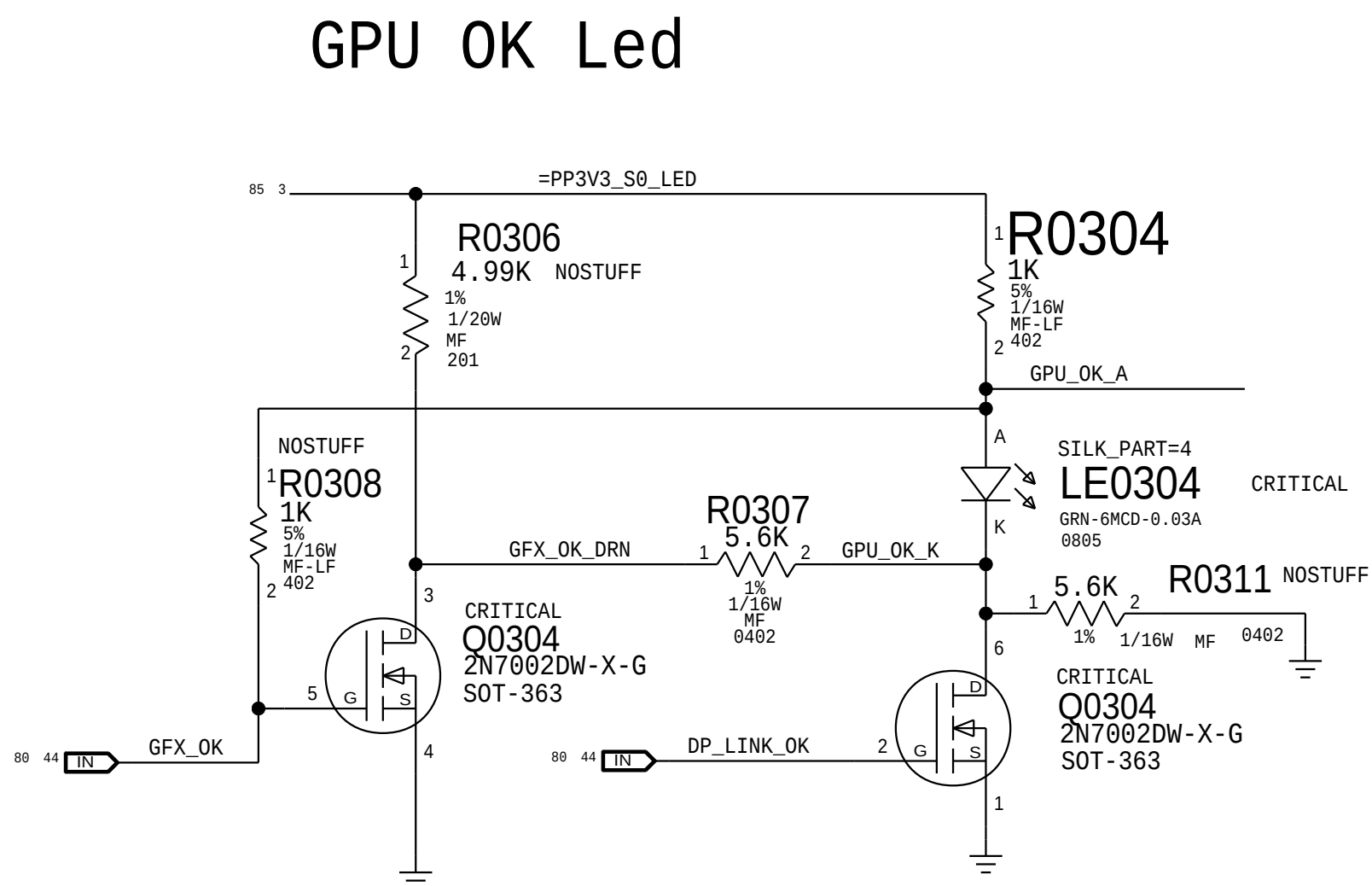
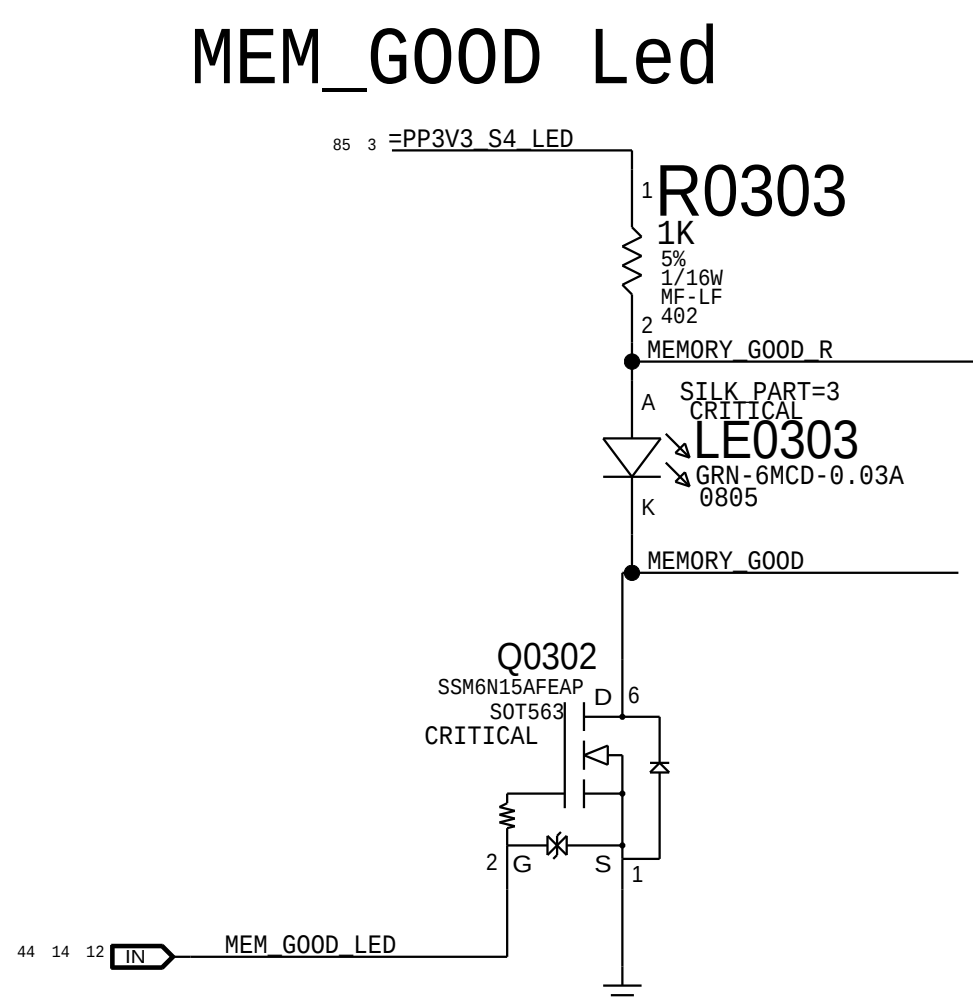
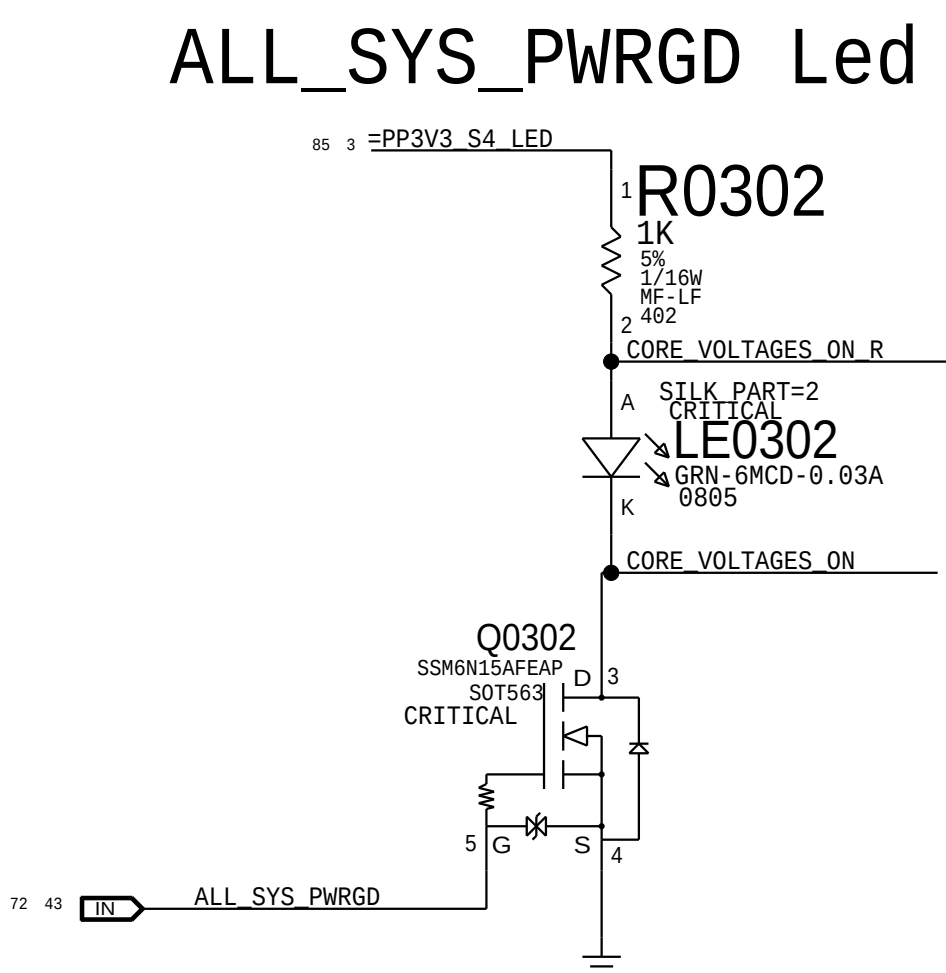
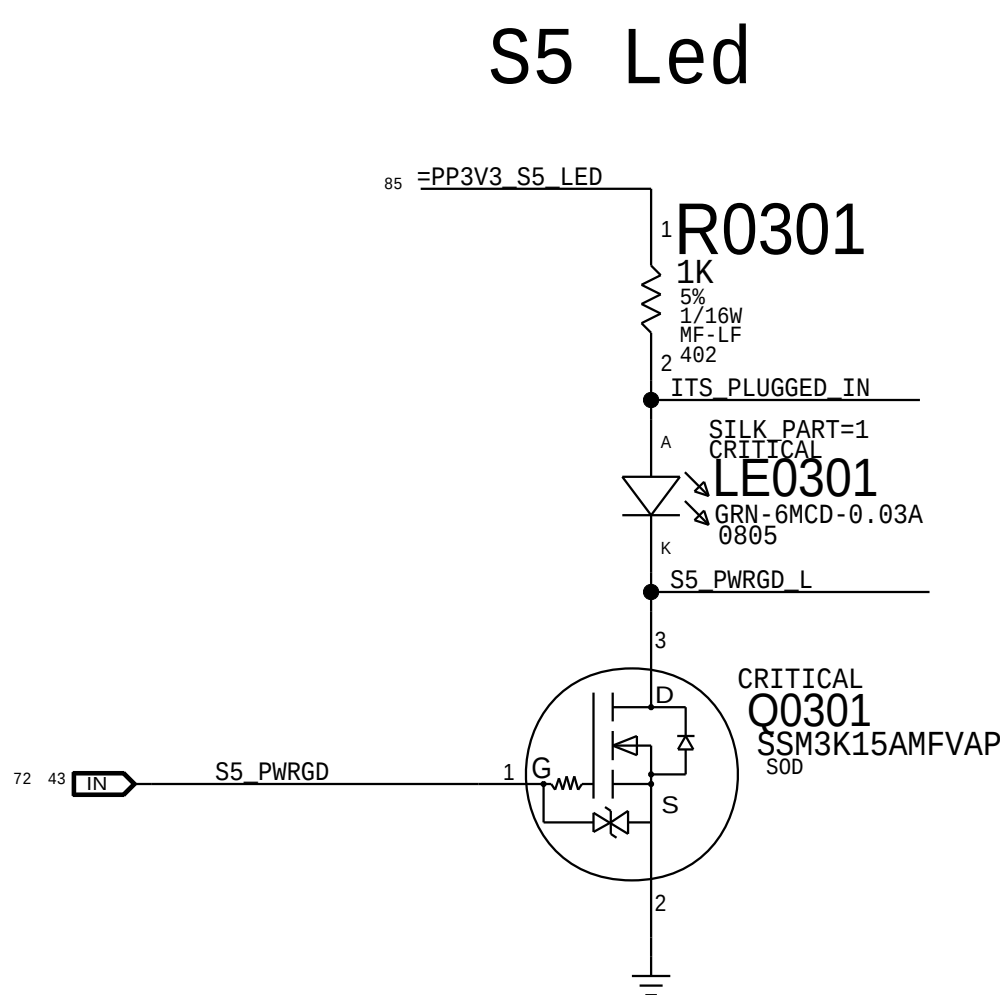
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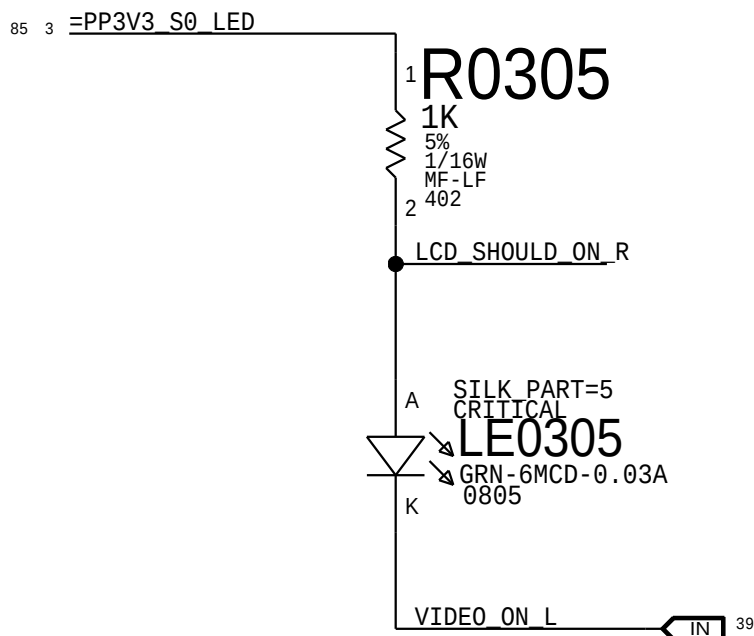
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THIS LED IS DRIVEN BY GPU:
GFX_OK HIGH LIGHTS THE LED DIMLY
DP_LINK_OK HIGH LIGHTS THE LED FULLY

VIDEO ON Led



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Debug LEDS			
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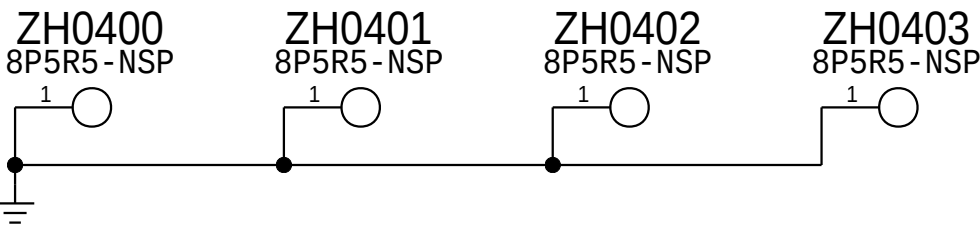
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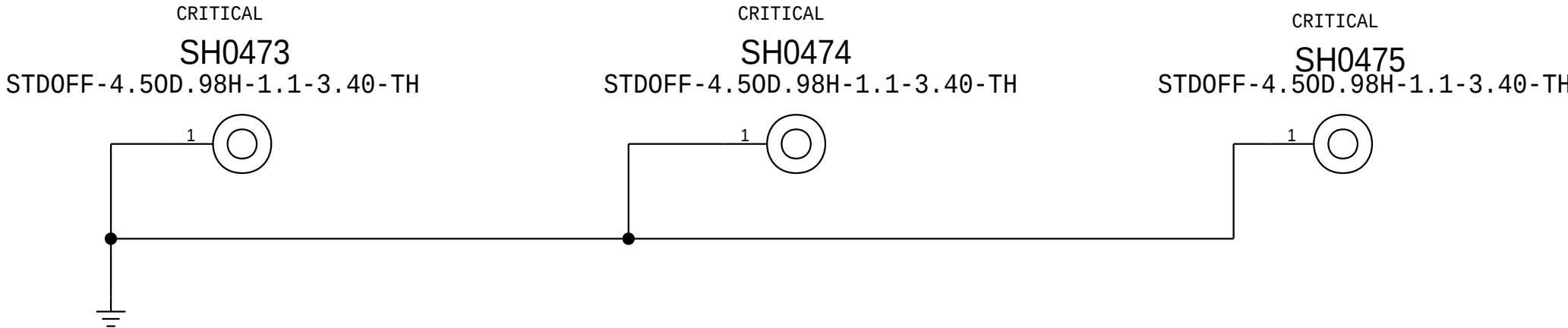
CPU HEATSINK

4mm Plated Holes (APN 998-4158)



GPU HEATSINK MOUNTING FEATURES

APN:860-1532



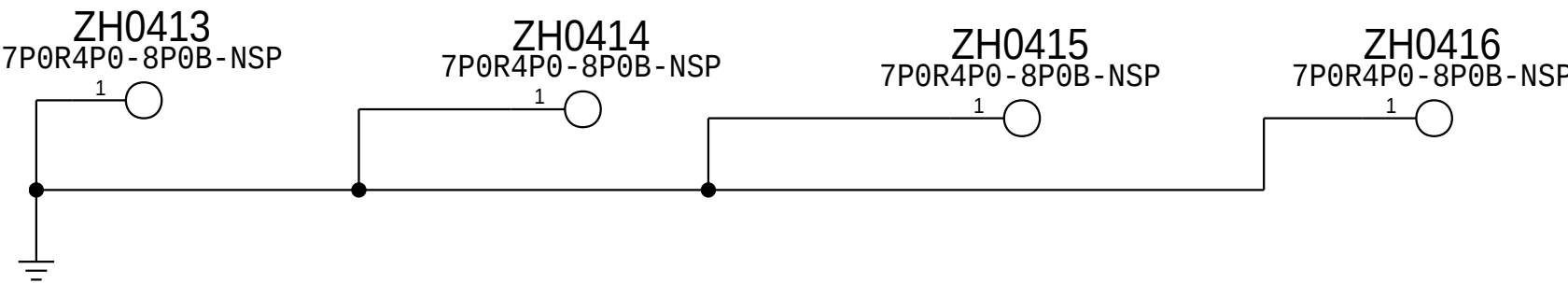
HEATSINK STABILITY MOUNTING FEATURES

APN:860-1532



Rear Cover

APN:998-4559 (Plated holes, 4mm inner diameter, 8mm pad)

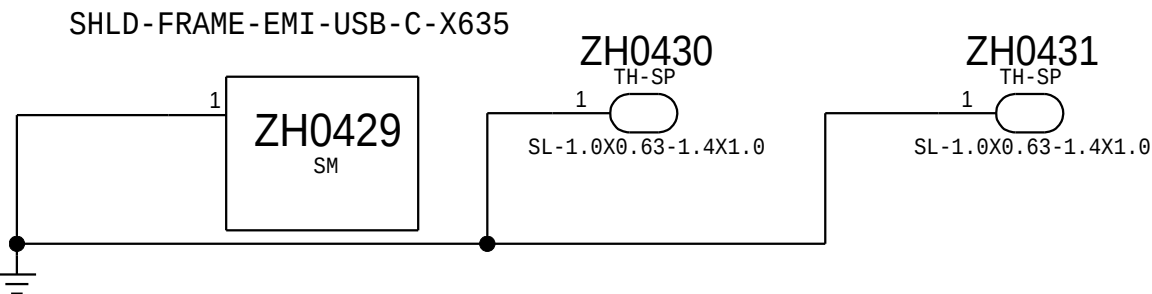


WIRELESS CARD MTG HOLES

998-06186 (PLATED HOLES, 3.3mm INNER DIAMETER, 5.5mm pad)



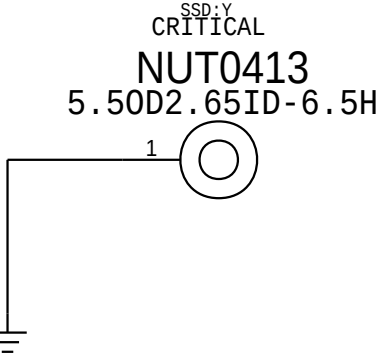
USB-C EMI Can



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
806-07339	1	COVER,EMI SHIELD,USB-C,X635	EMI_SHIELD	CRITICAL	

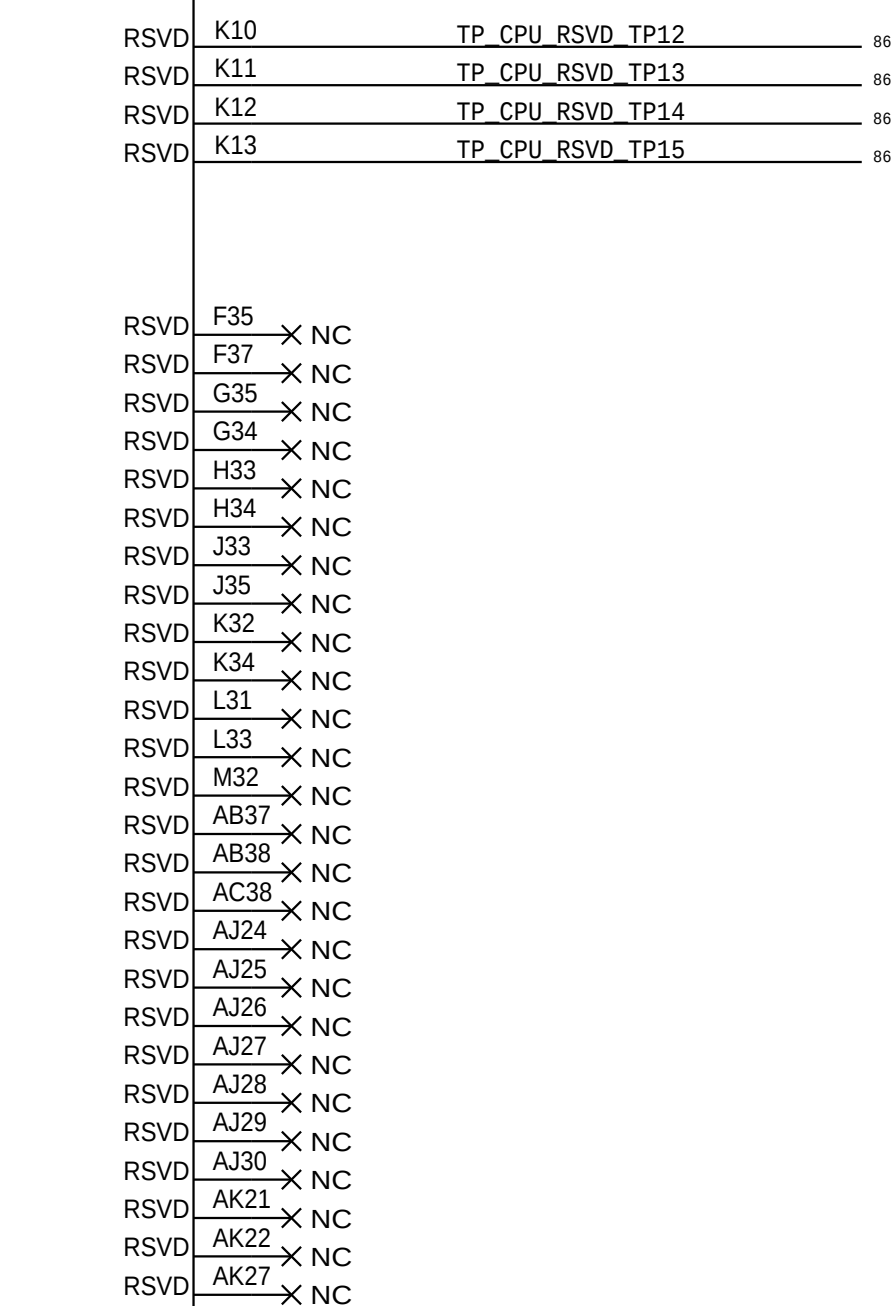
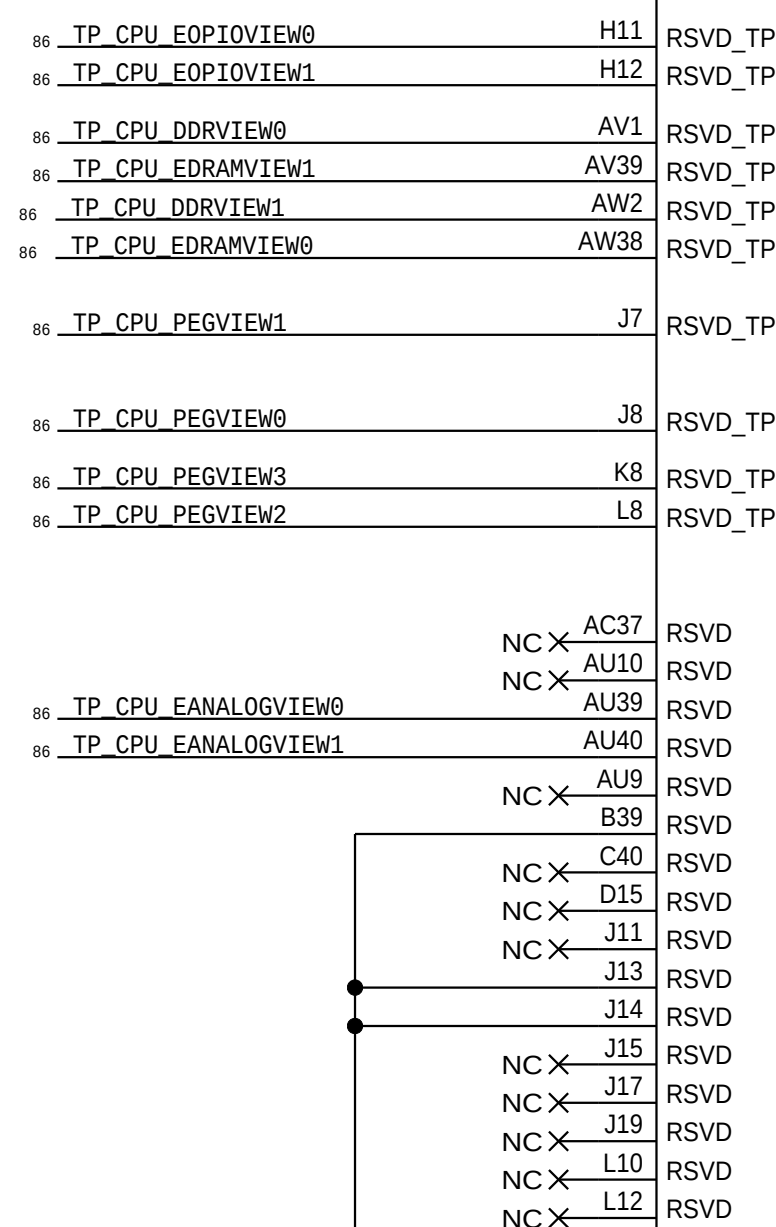
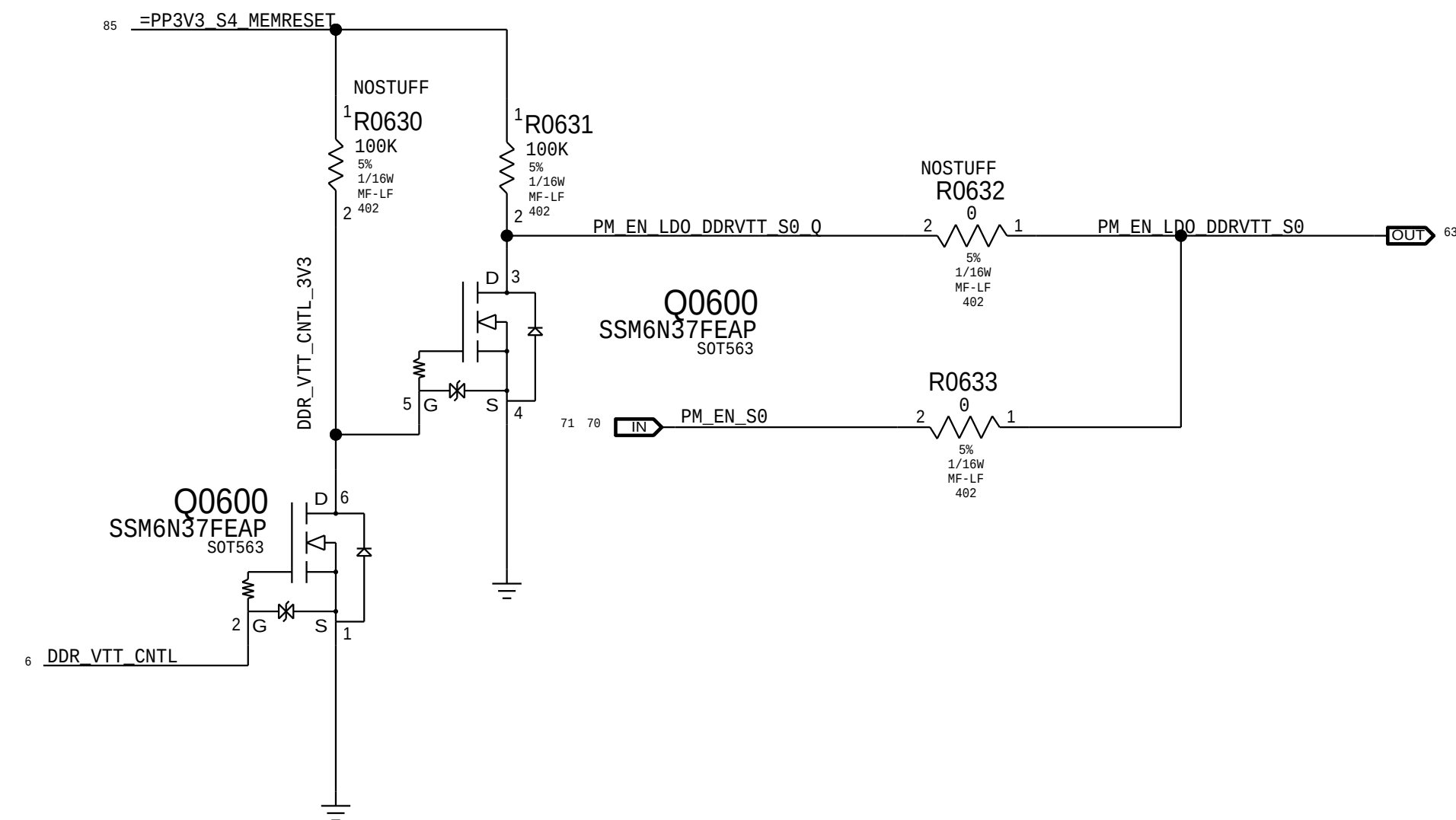
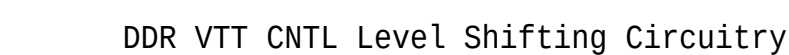
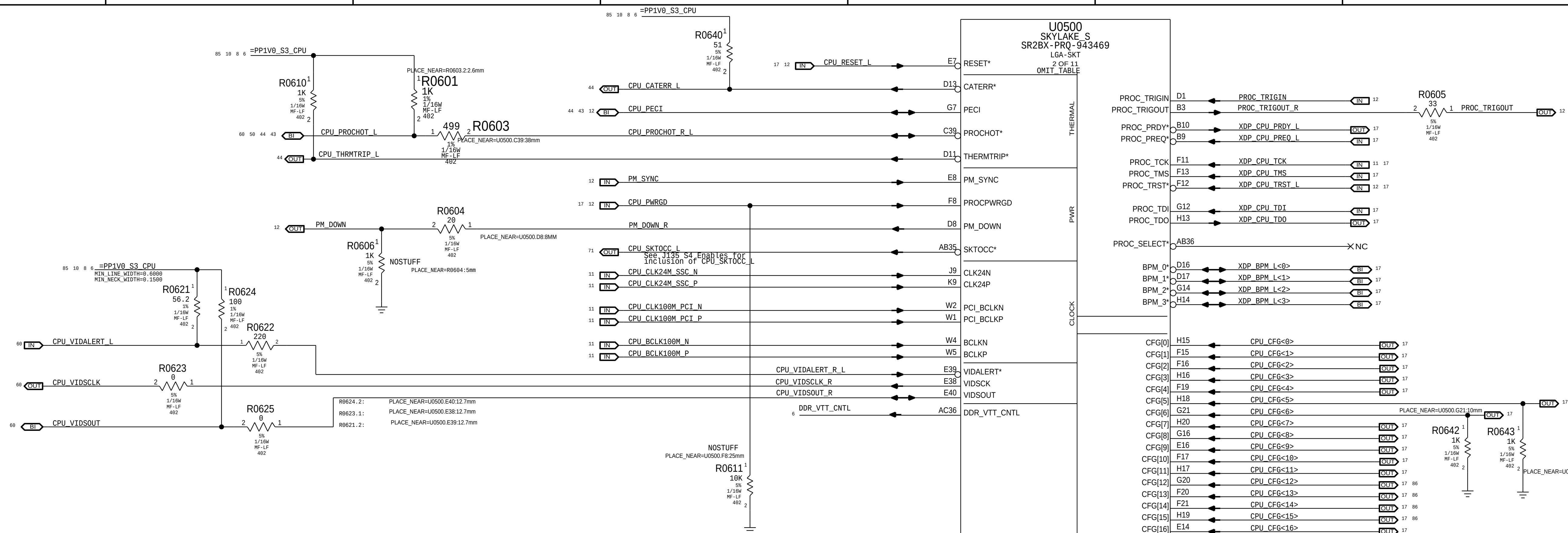
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APN: 860-00314



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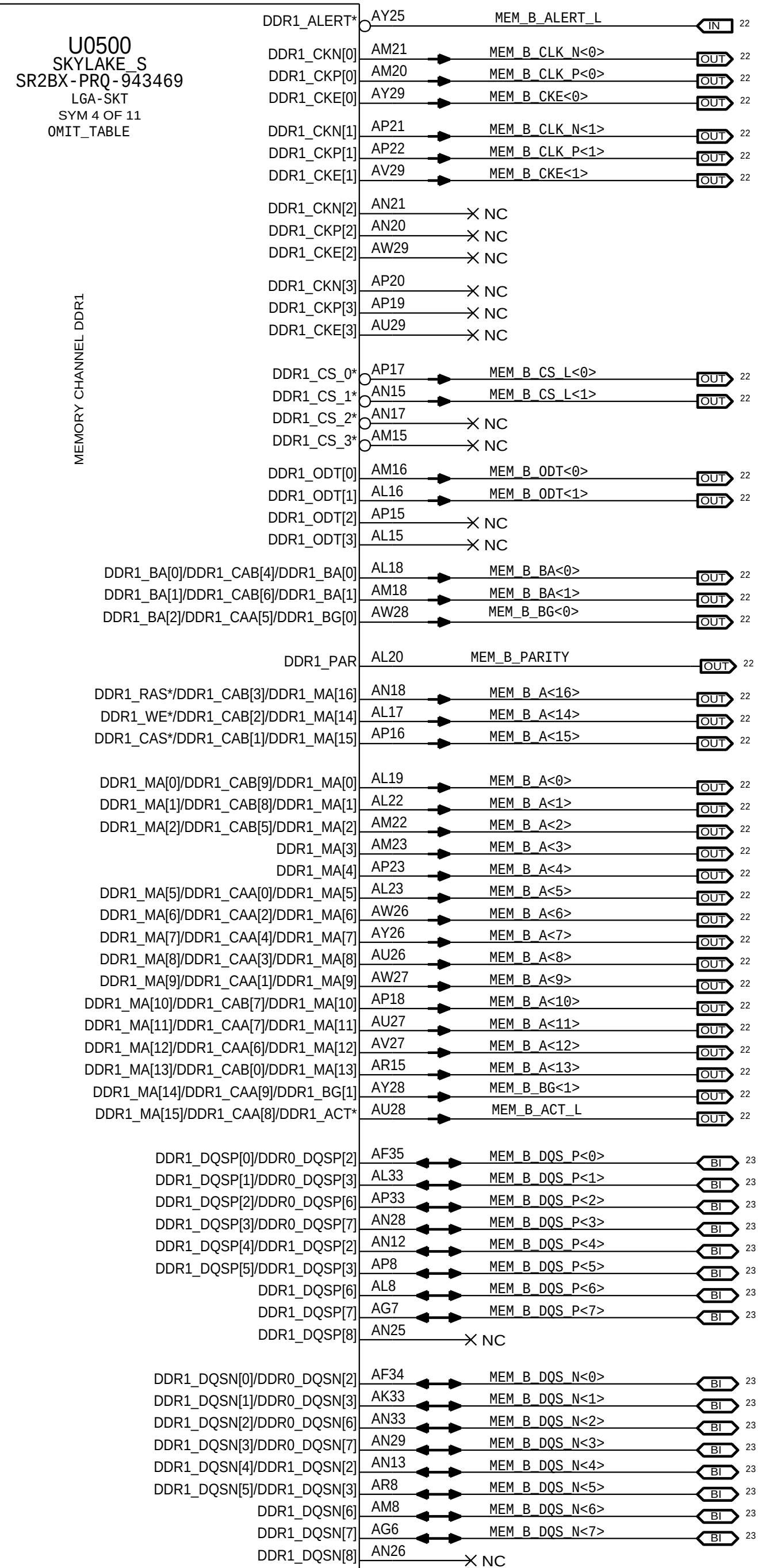
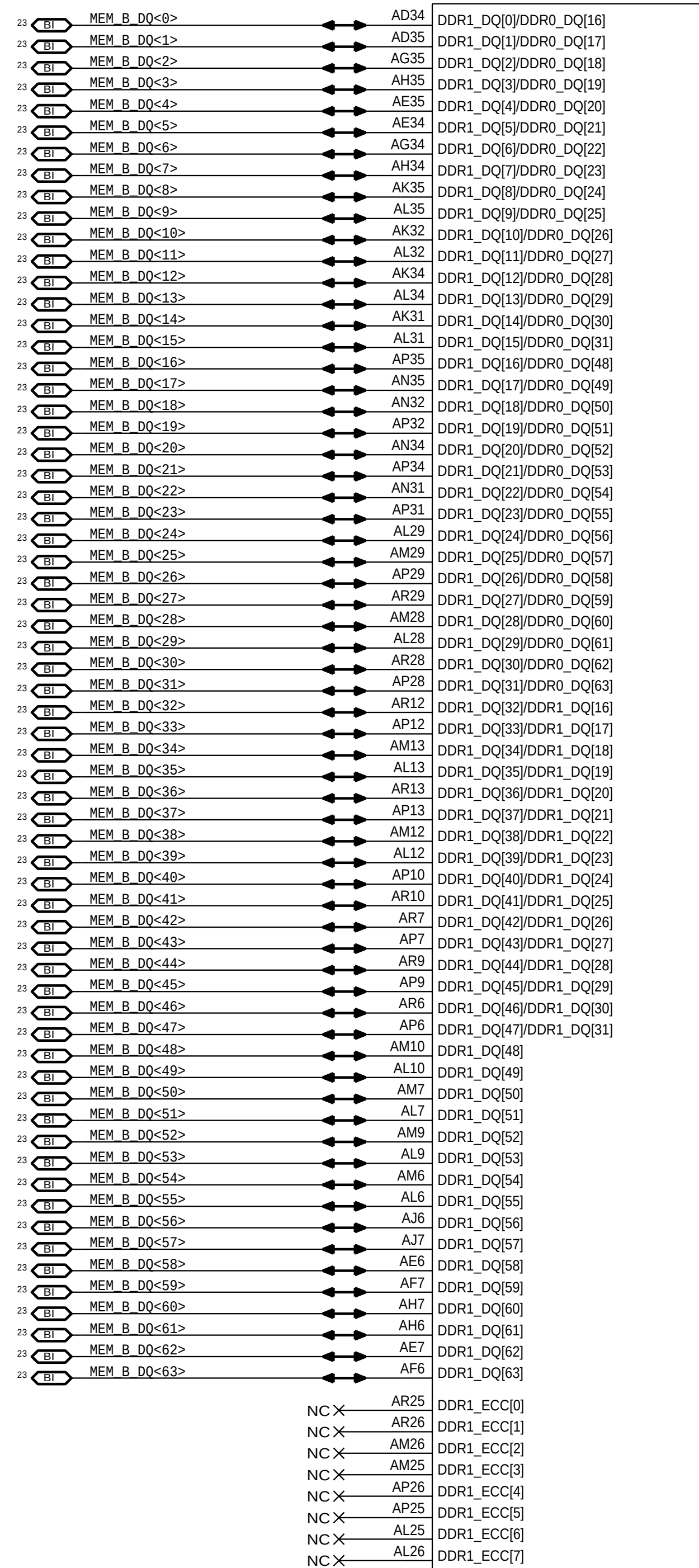
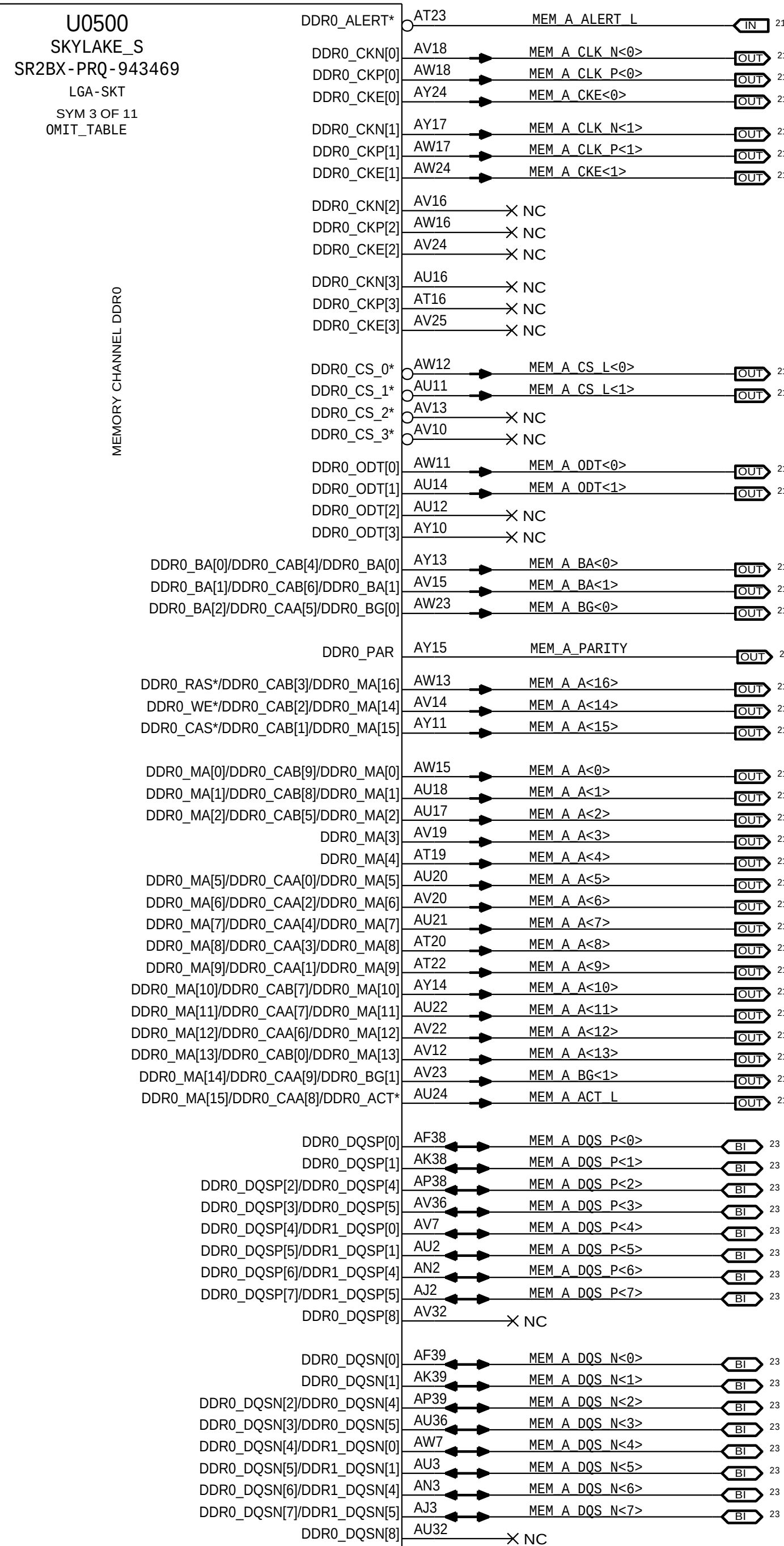
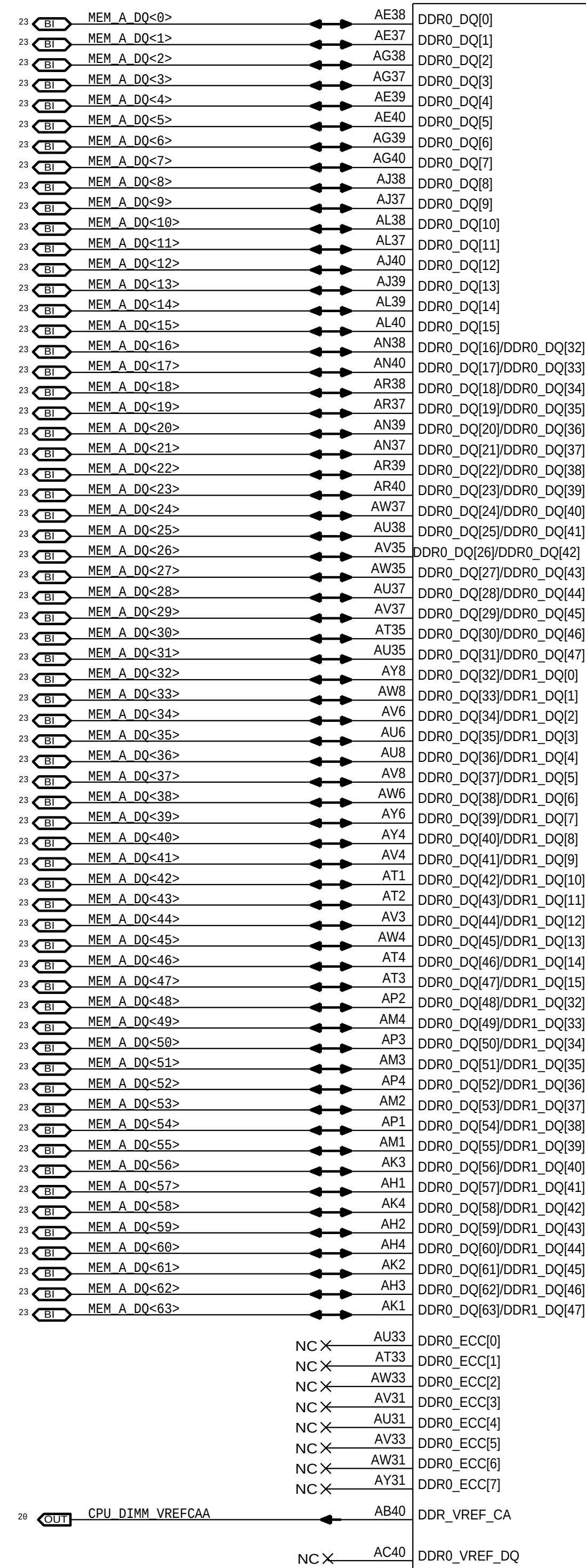
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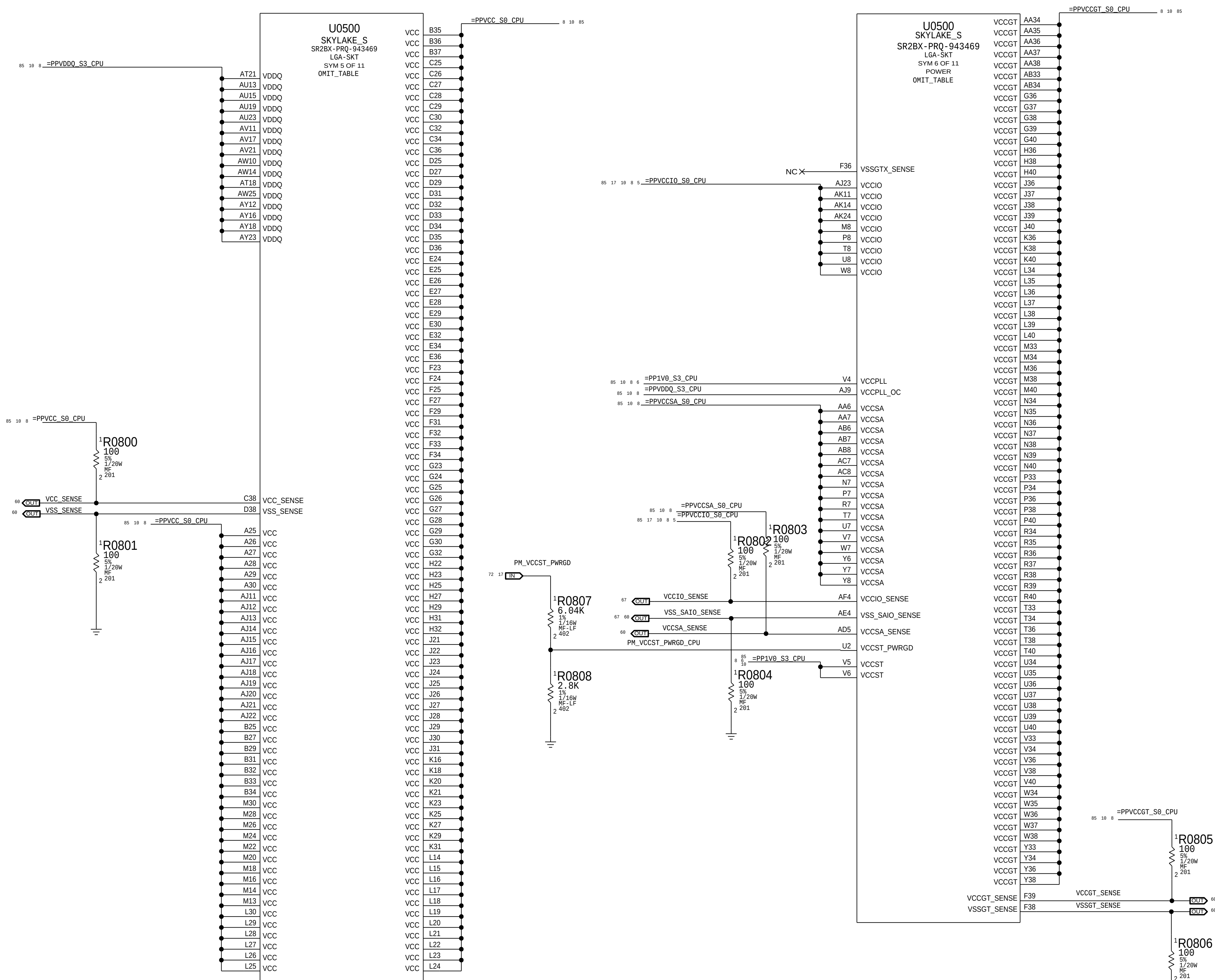


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CFG [7] :PEG DEFER TRAINING          1 = (default) IMMEDIATELY AFTER xxRESETB  0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION          11 = 1 X16 (default) 10 = 2 X8  01 = RSVD  00 = X8, X4, X4
CFG [4] :x0P ENABLE/DISABLE          1 = DISABLED(default)  0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL        1 = NORMAL OPERATION(default)  0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL       1 = NORMAL OPERATION(default)  0 = LANES REVERSED
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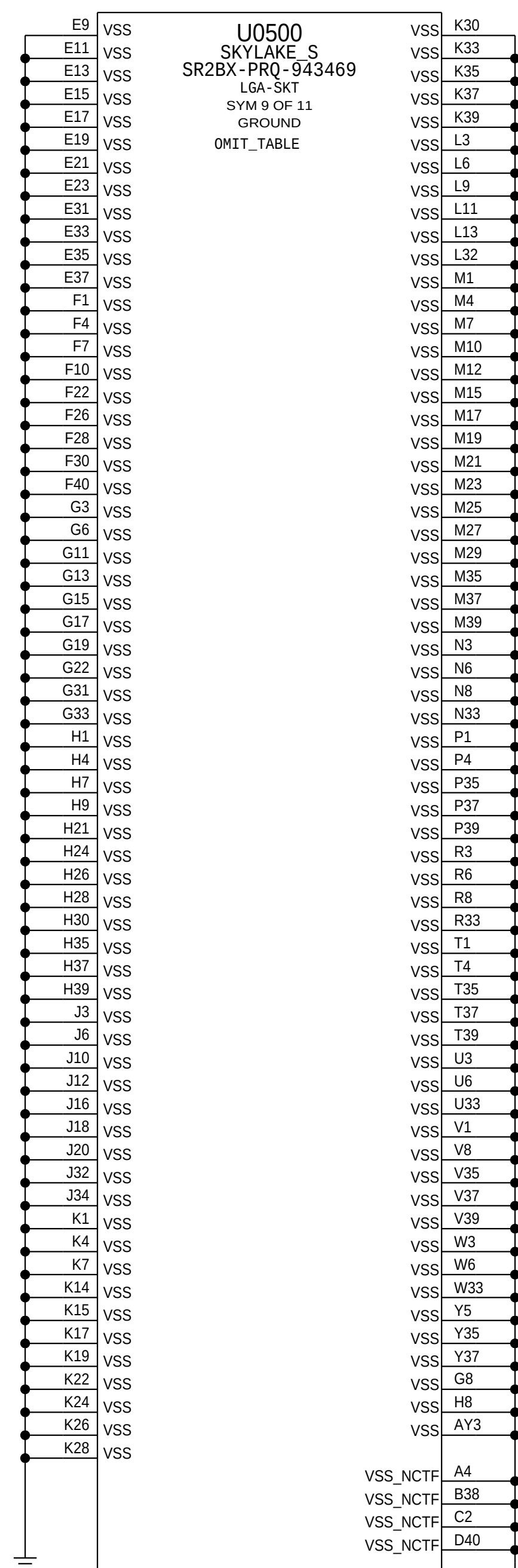
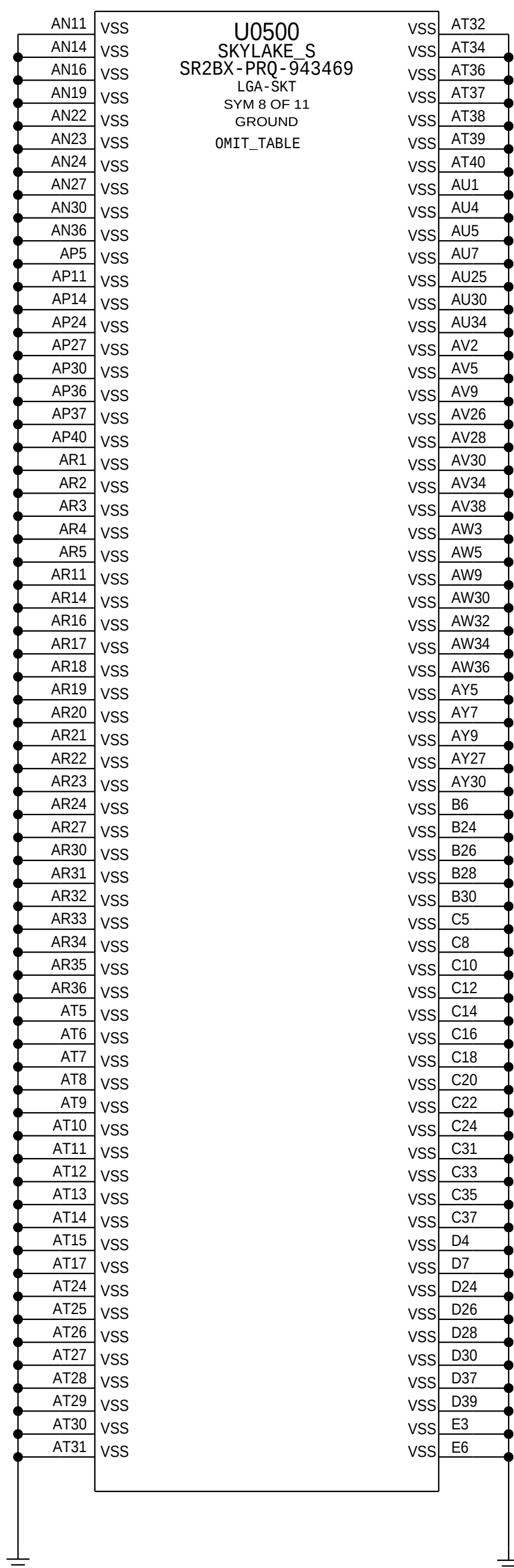
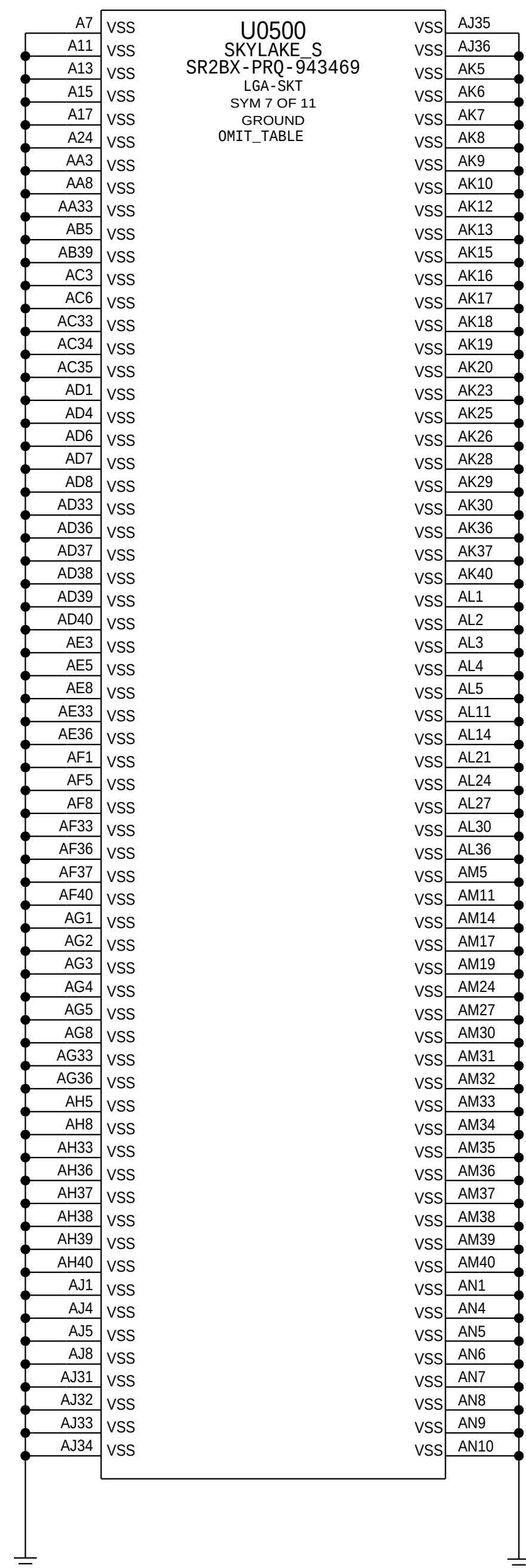
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BOM_COST_GROUP=CPU & CHIPSET





BOM_COST_GROUP=CPU & CHIPSET

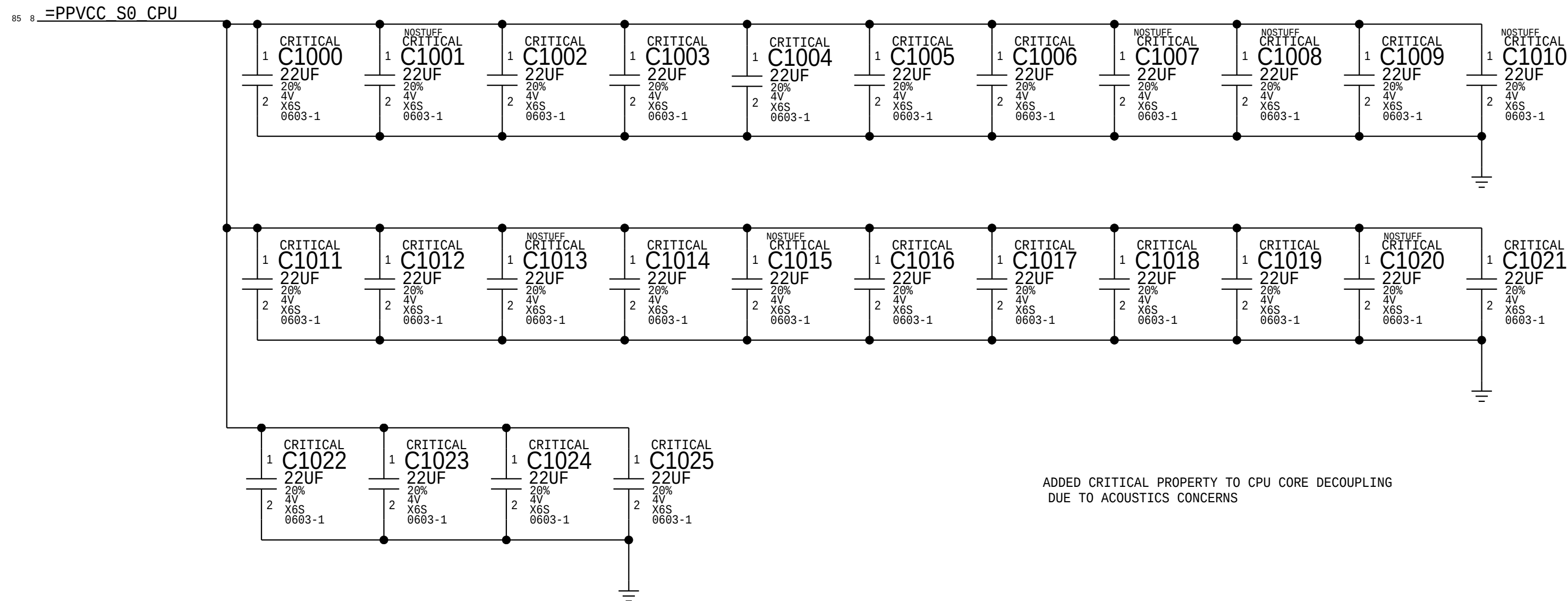


CPU VCORE DECOUPLING

Intel Recommendation: 12x 22UF 0805 (top side cavity)
6x 22UF 0603 (top side cavity)
5x 22UF 0805 (top side outside cavity)

Apple Implementation: 26x 22UF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



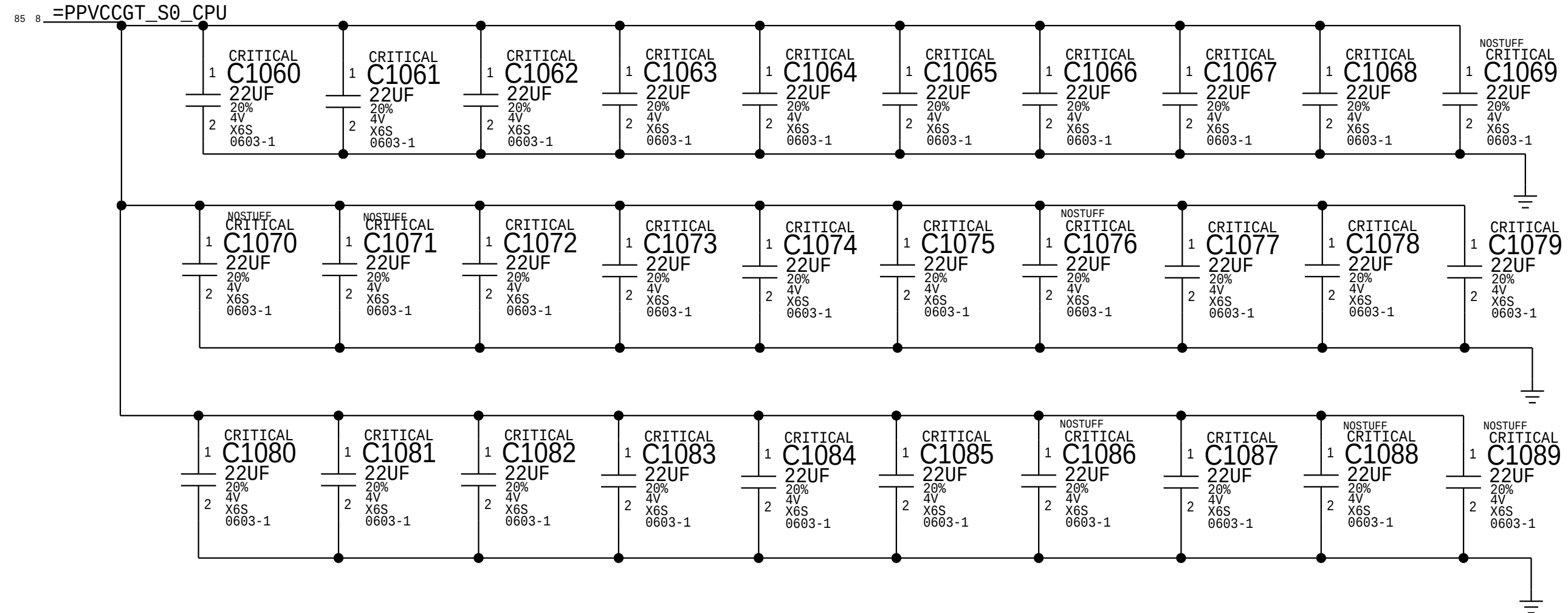
ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

CPU GT DECOUPLING

Intel Recommendation: 9x 47UF 0805 (top side cavity)
4x 47UF 0805 (top side outside cavity)

Apple Implementation: 30x 22uF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

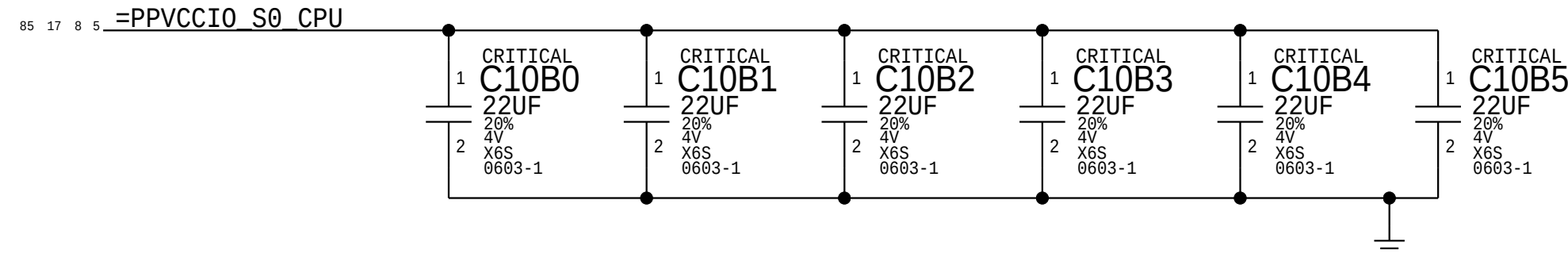


CPU VCCIO DECOUPLING

```
Intel Recommendation:5x 22UF 0603 (top side cavity)
                    1x 22UF 0805 (top side cavity)
```

Apple Implementation:(following Intel recommendation w/ 0603)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

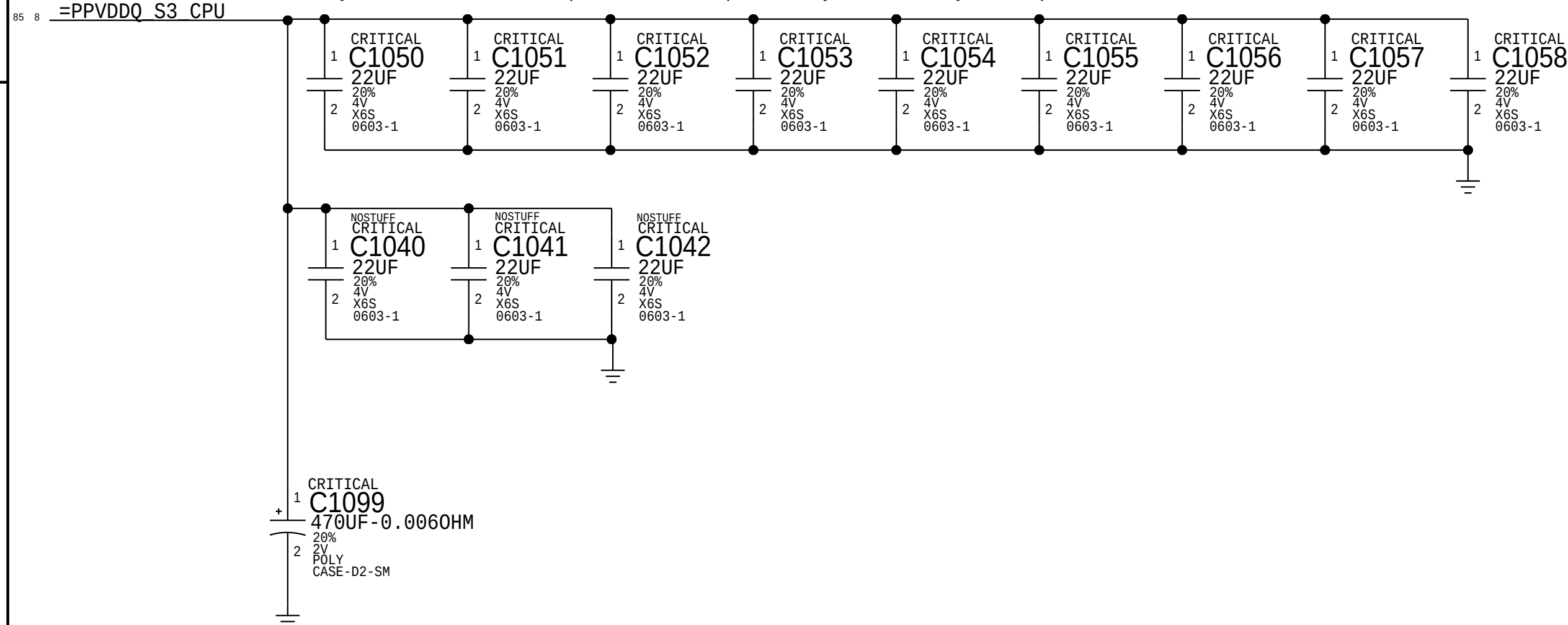


Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:4x 22UF 0603 (top side outside cavity)

Apple Implementation: 9x 22UF 0603 (J78 carry over)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

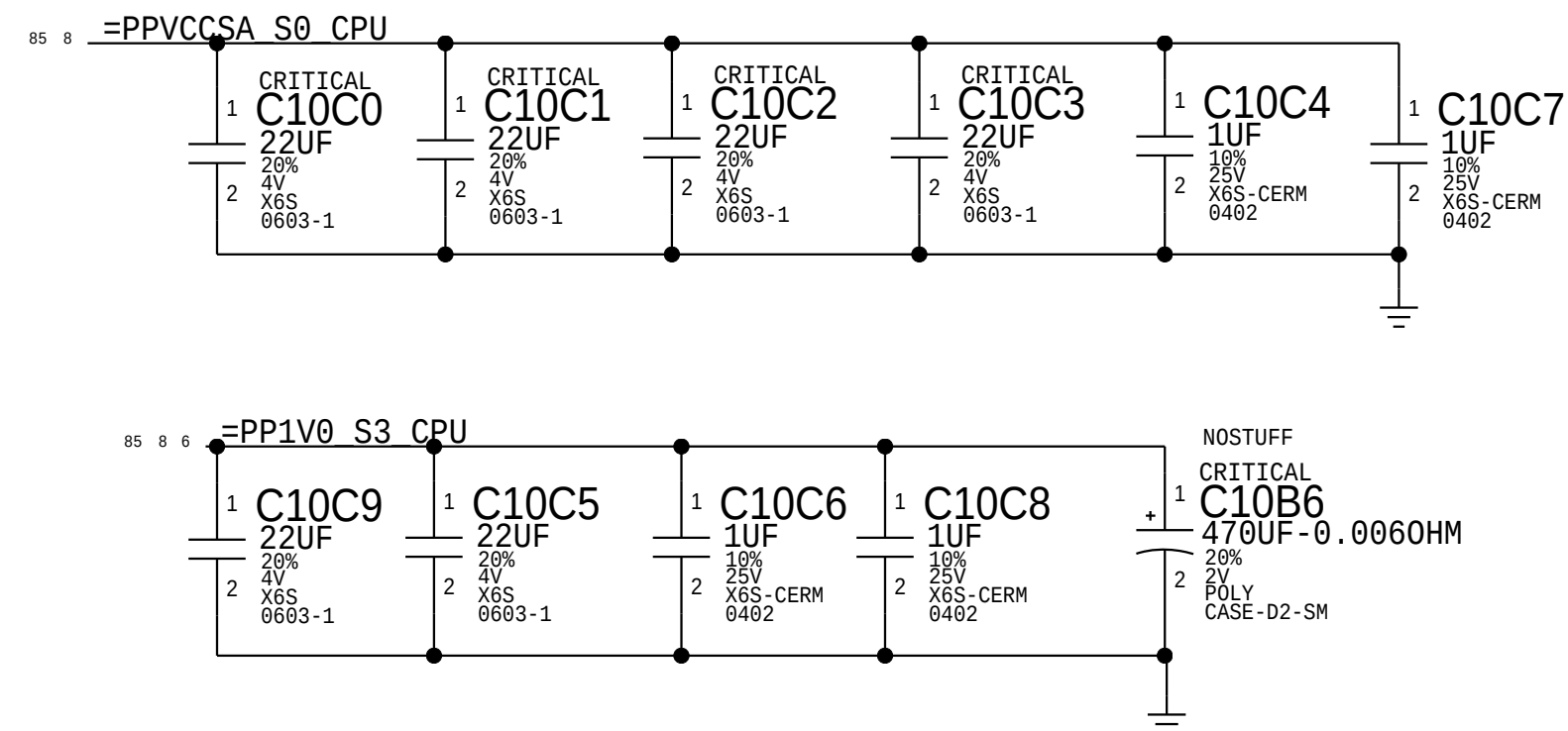


CPU VCCSA / VCCST+VCCPLL DECOUPLING

Intel Recommendation: 2x 22UF 0603 near top side cavity

Apple Implementation: VCCST/VCCPLL: 1X 22UF 0603/2X 1UF 0402

Layout Note: These caps should be placed on top side cavity.



BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=harper		SYNC_DATE=06/08/2016	
PAGE TITLE			
CPU & CHIPSET: CPU Decoupling			
	Apple Inc.		DRAWING NUMBER 051-01477
			SIZE D
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		BRANCH	
		PAGE 10 OF 119	
		SHEET 10 OF 87	

D

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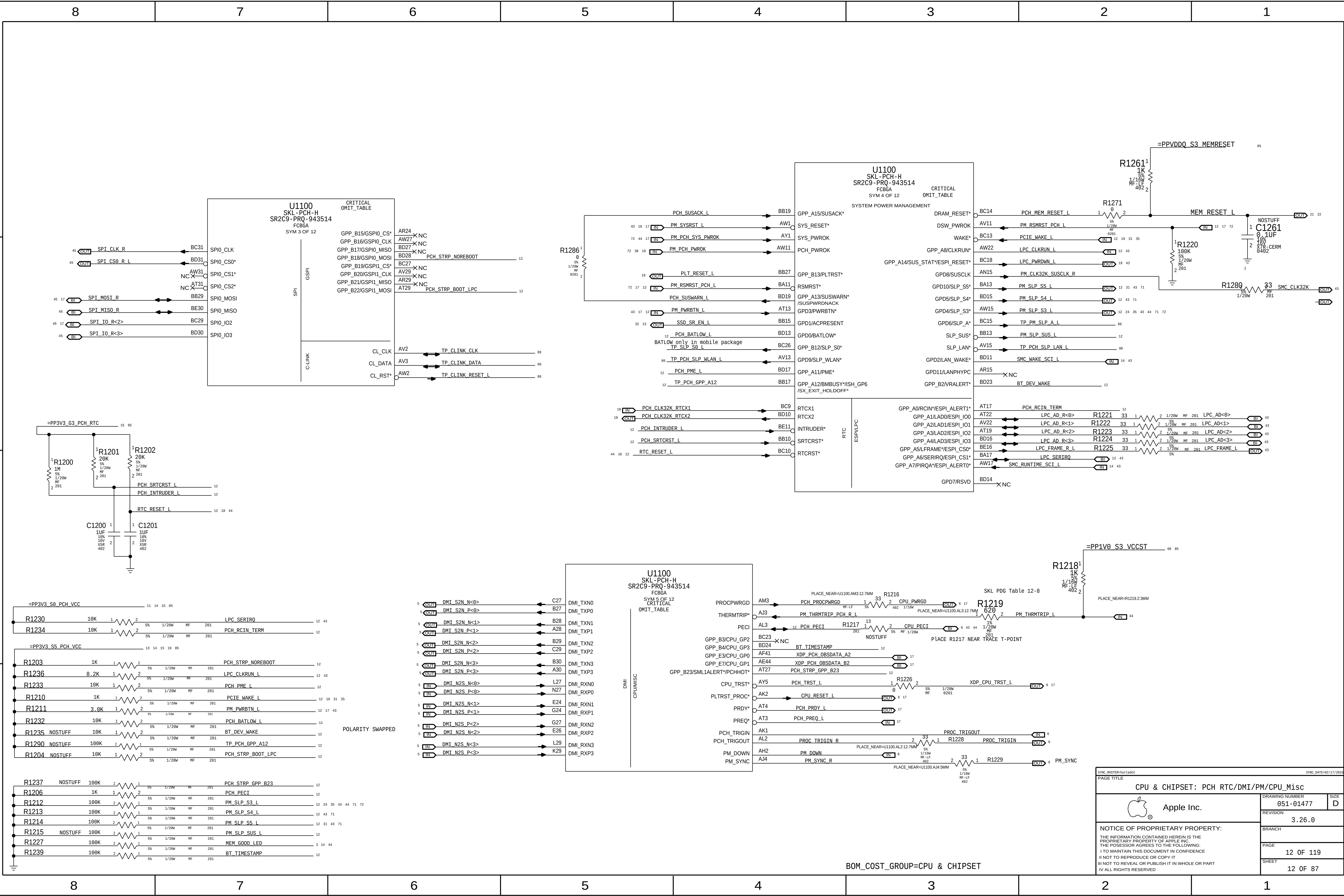
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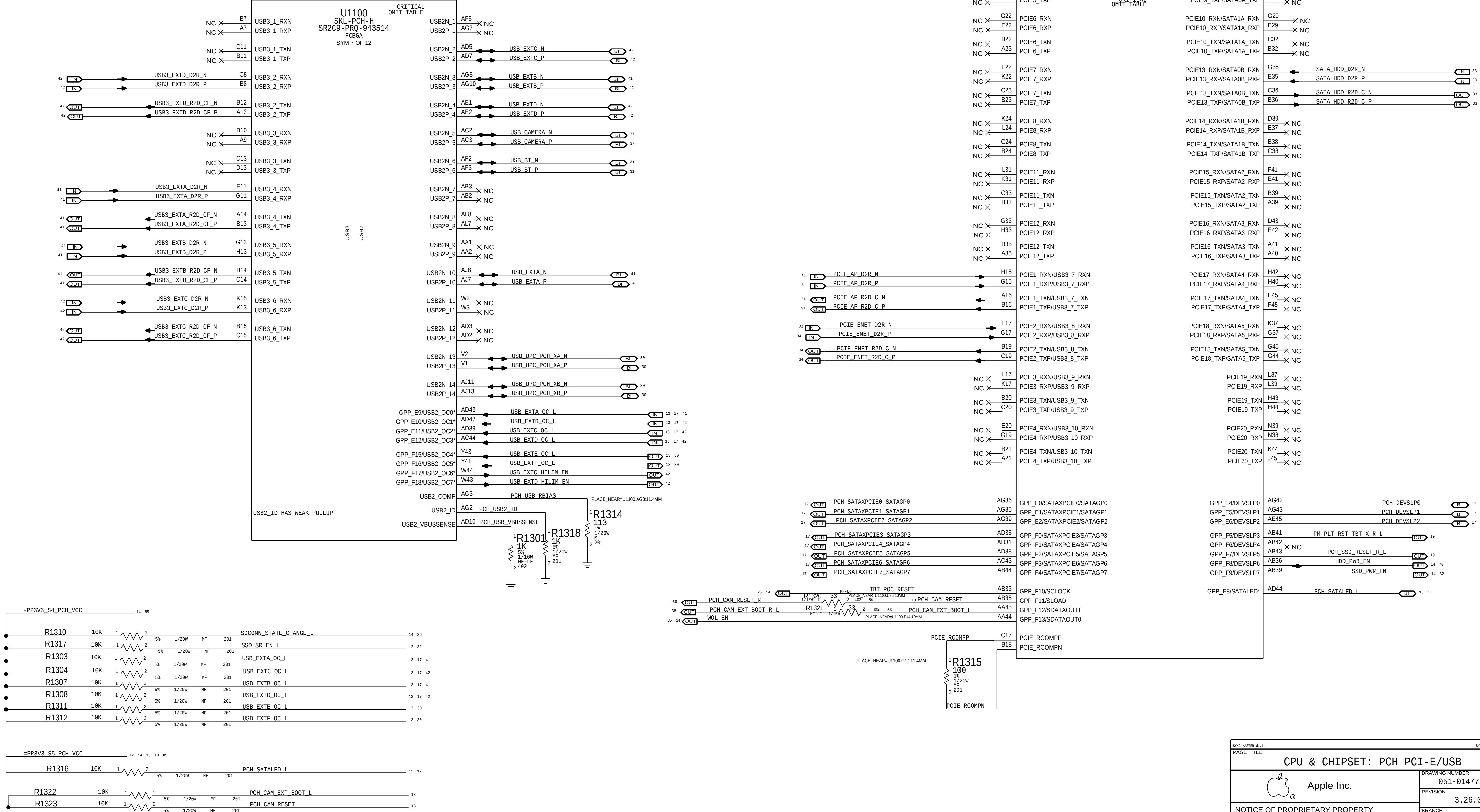
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SYNC_MASTER=HurLadot		SYNC_DATE=82/17/2016	
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CPU & CHIPSET: PCH RTC/DMI/PM/CPU_Misc			
	DRAWING NUMBER		SIZE
	051-01477		D
Apple Inc.	REVISION		3.26.0
	BRANCH		
	PAGE		12 OF 119
	SHEET		12 OF 87
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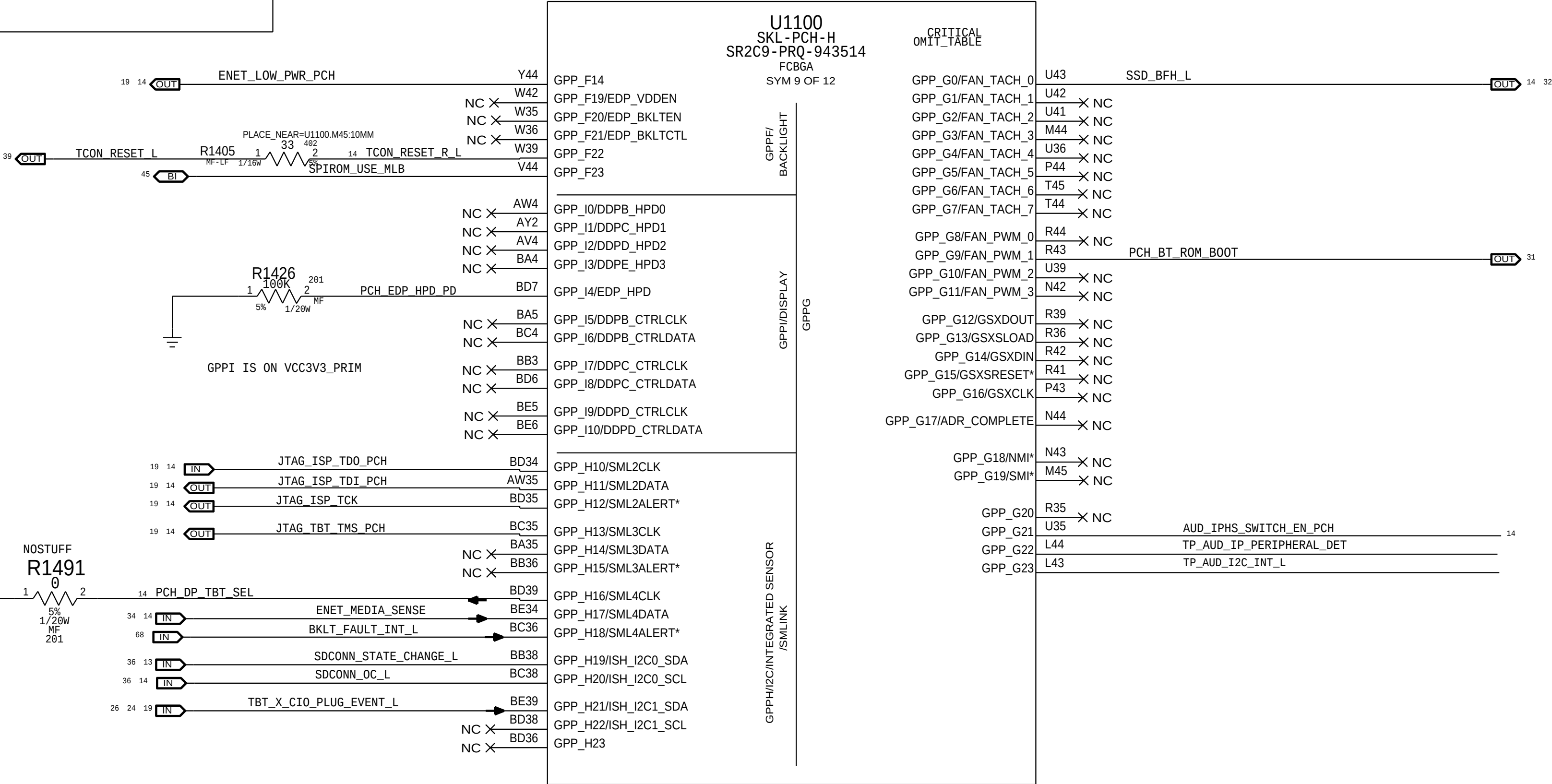
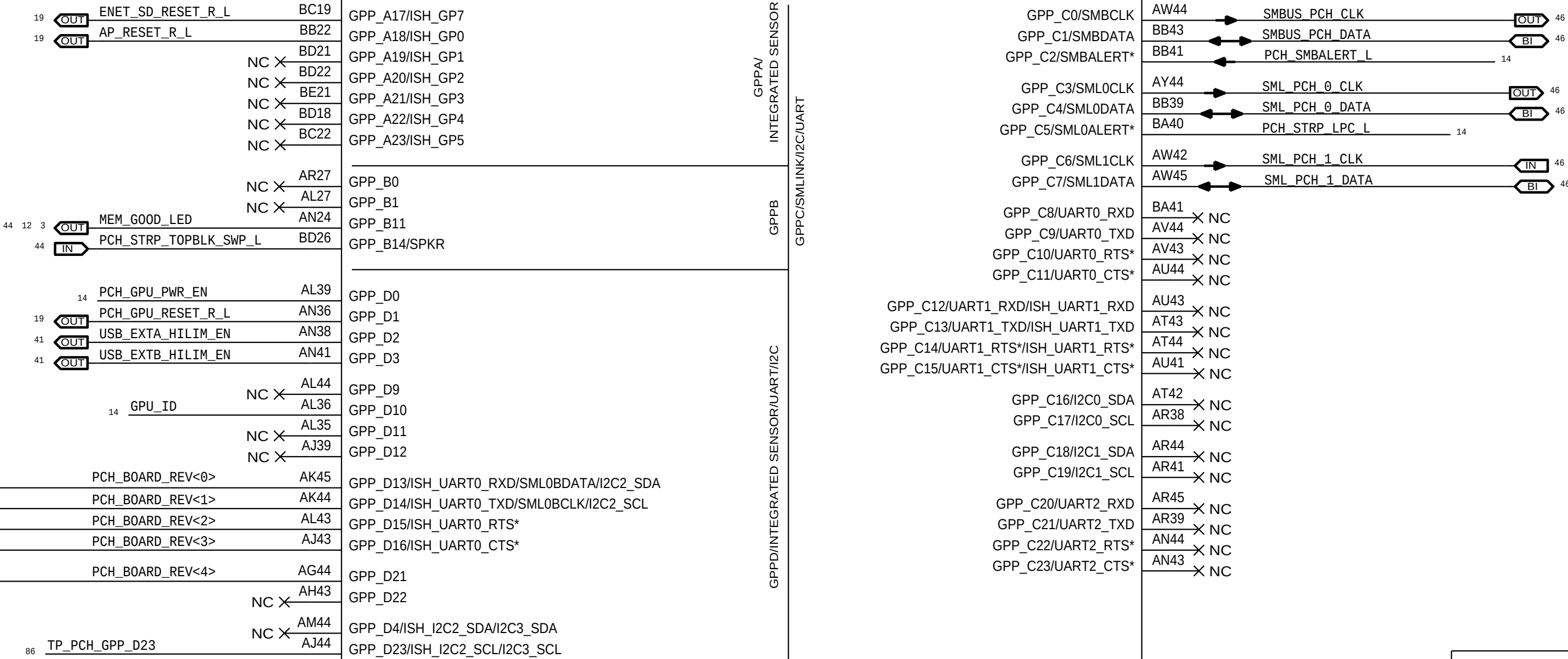


Board Rev[4:0]

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Proto0: 11110
Proto1: 11101
EVT: 11100 <<

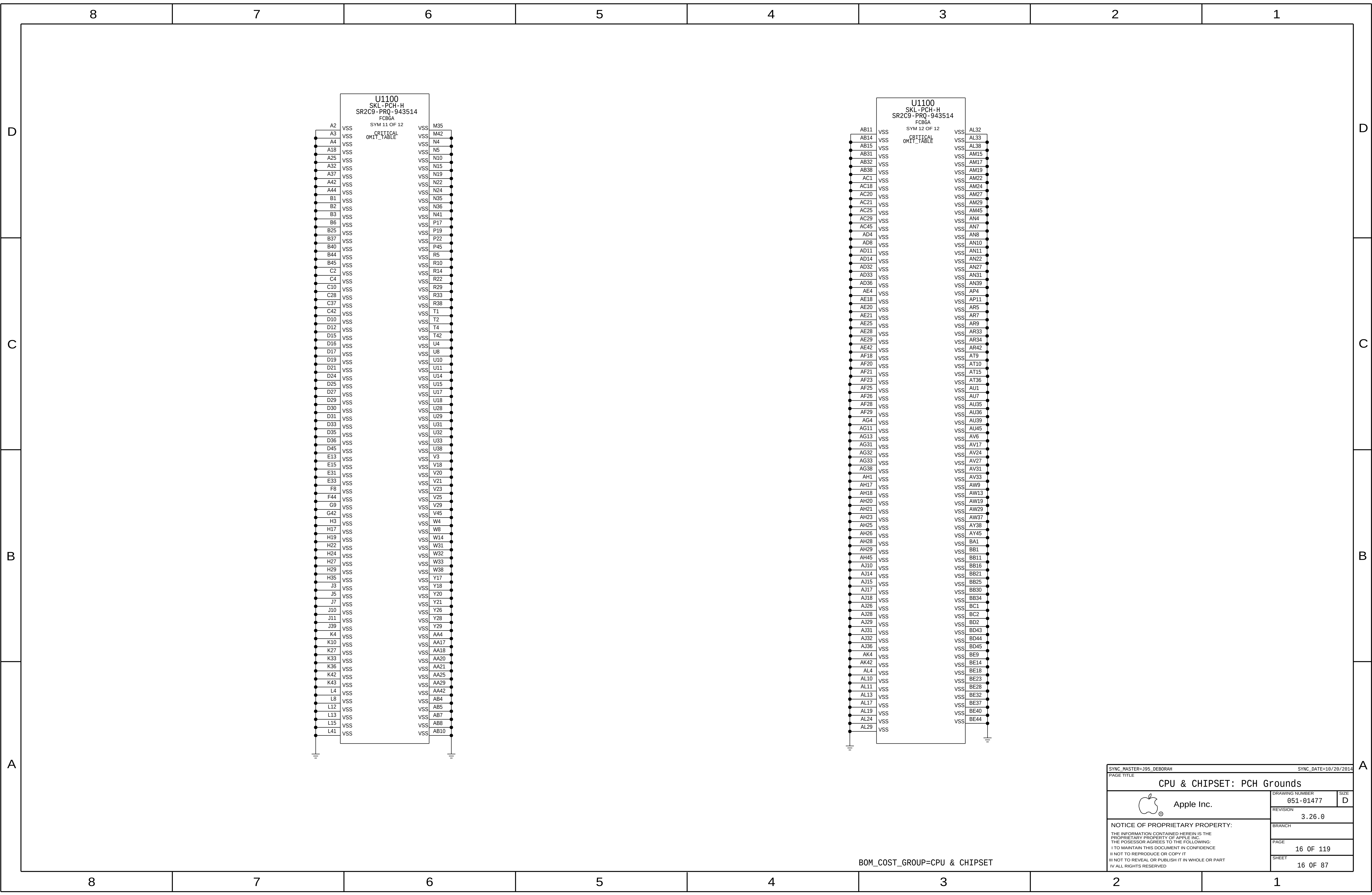
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SYM 6 OF 12

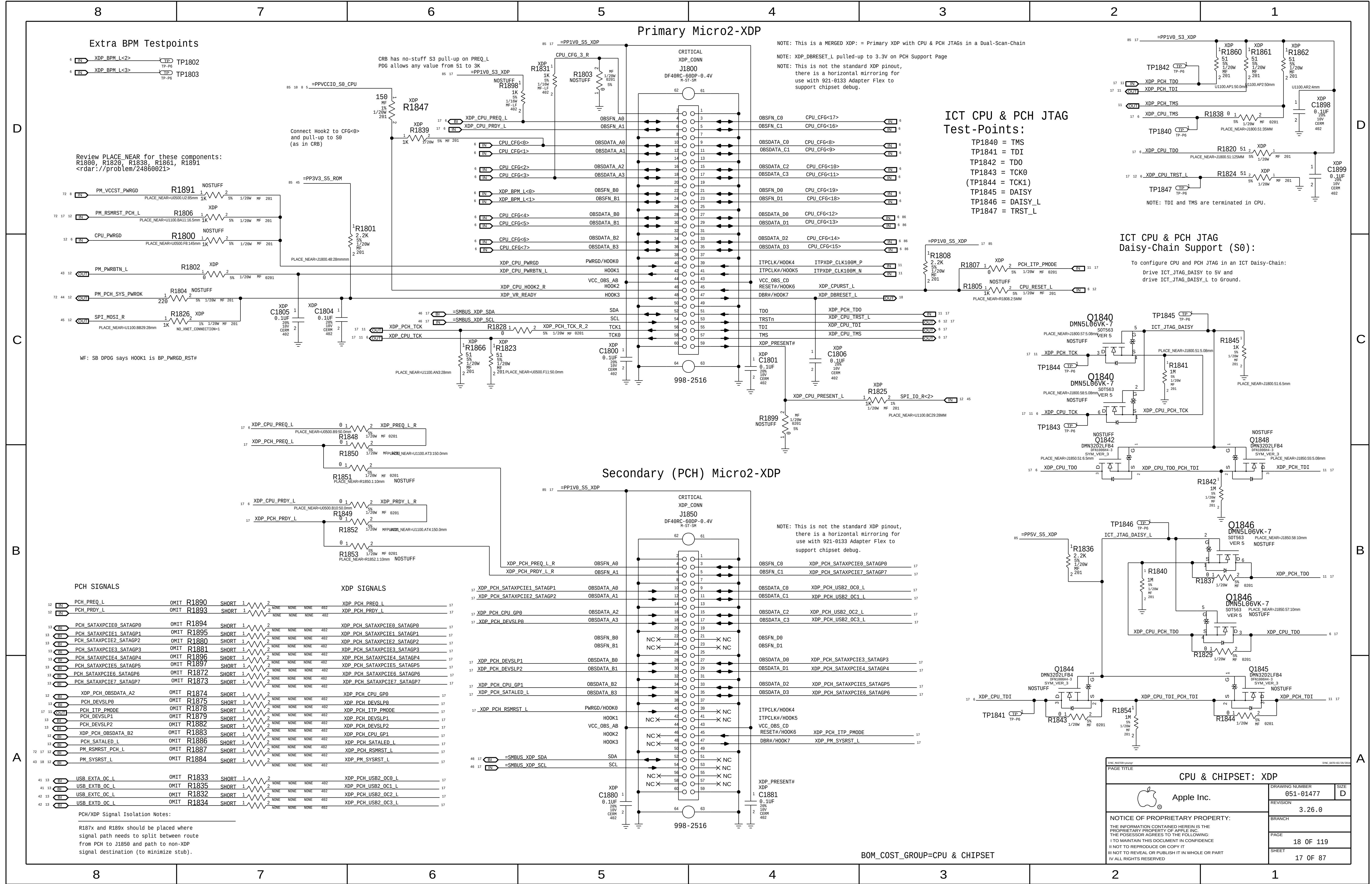
CRITICAL
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CPU & CHIPSET: PCH GPIO/Misc		
Apple Inc.	DRAWING NUMBER	051-01477
	REVISION	3.26.0
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BOM_COST_GROUP=CPU & CHIPSET





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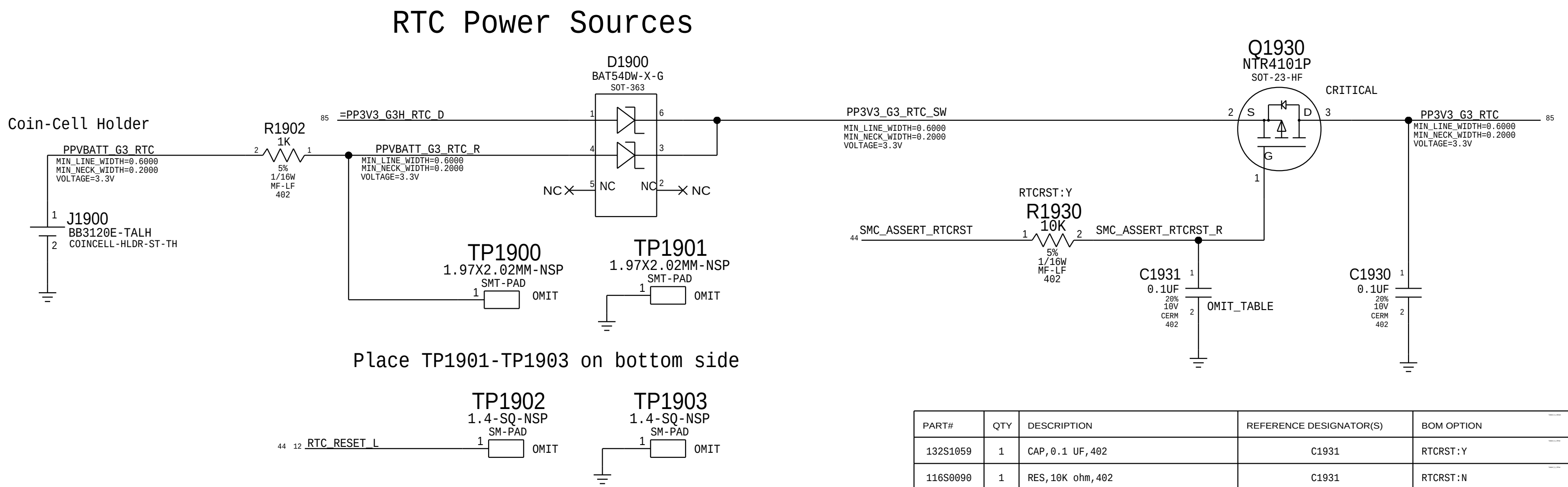
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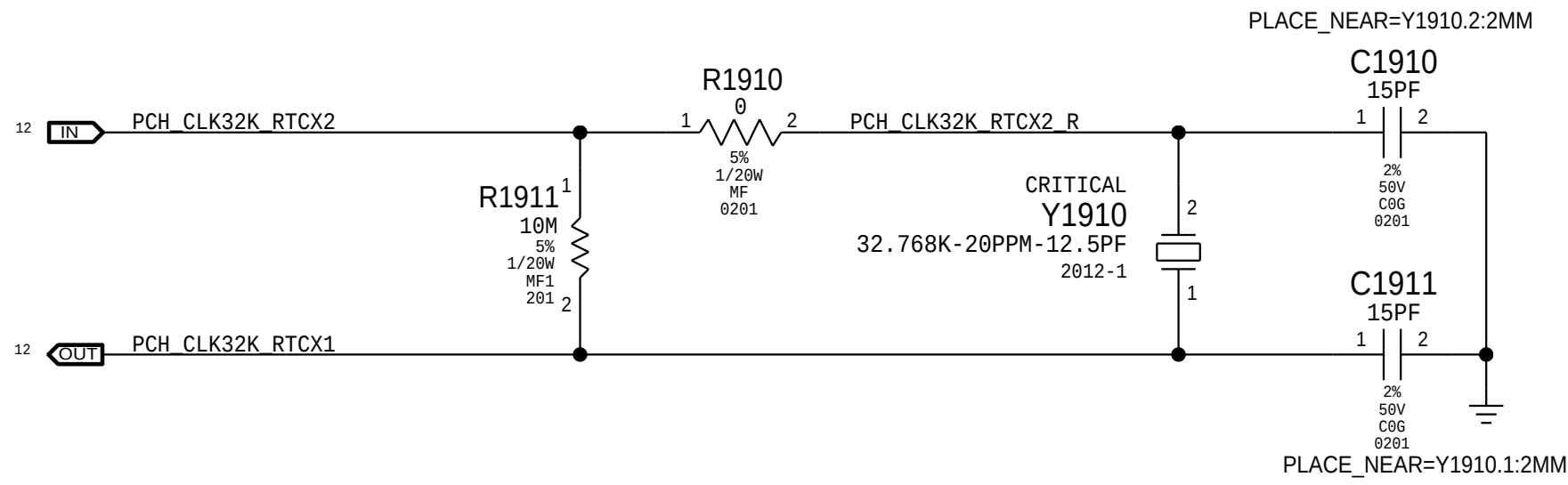
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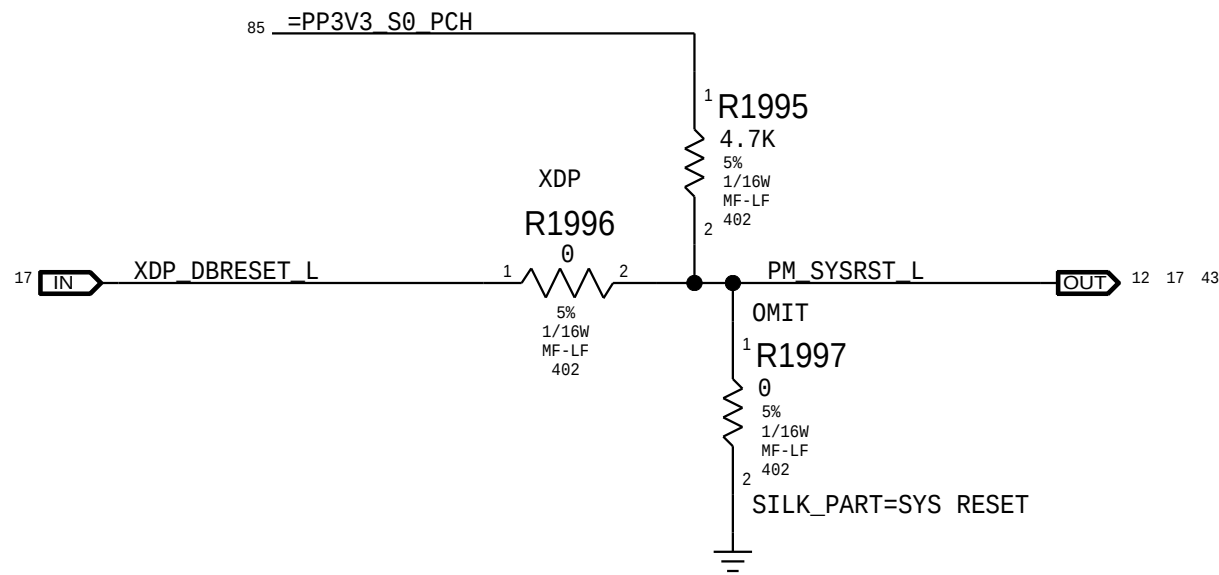
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PCH RTC Crystal

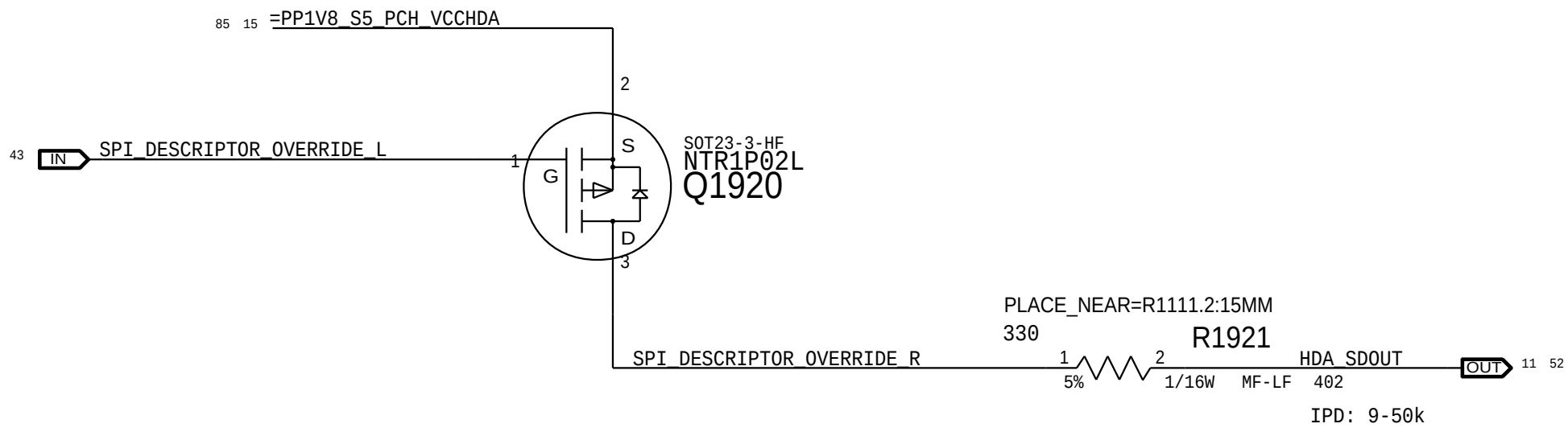


PCH Reset Button



PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.

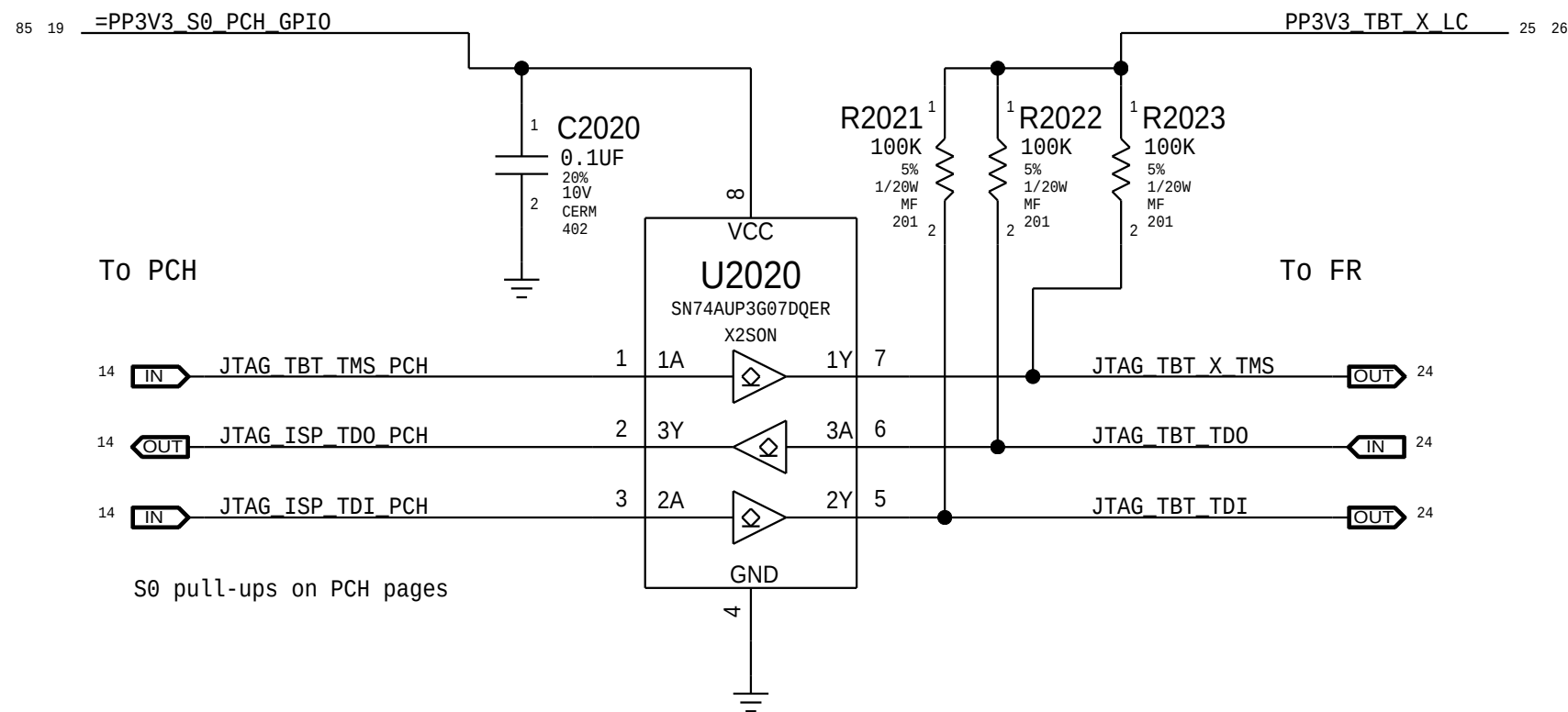


BOM_COST_GROUP=CPU & CHIPSET

SYNC_MASTER=hurtadot		SYNC_DATE=12/09/2015			
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Chipset Support					
 Apple Inc.	DRAWING NUMBER		051-01477	SIZE	D
	REVISION		3.26.0		
	BRANCH				
	PAGE		19 OF 119		
	SHEET		18 OF 87		
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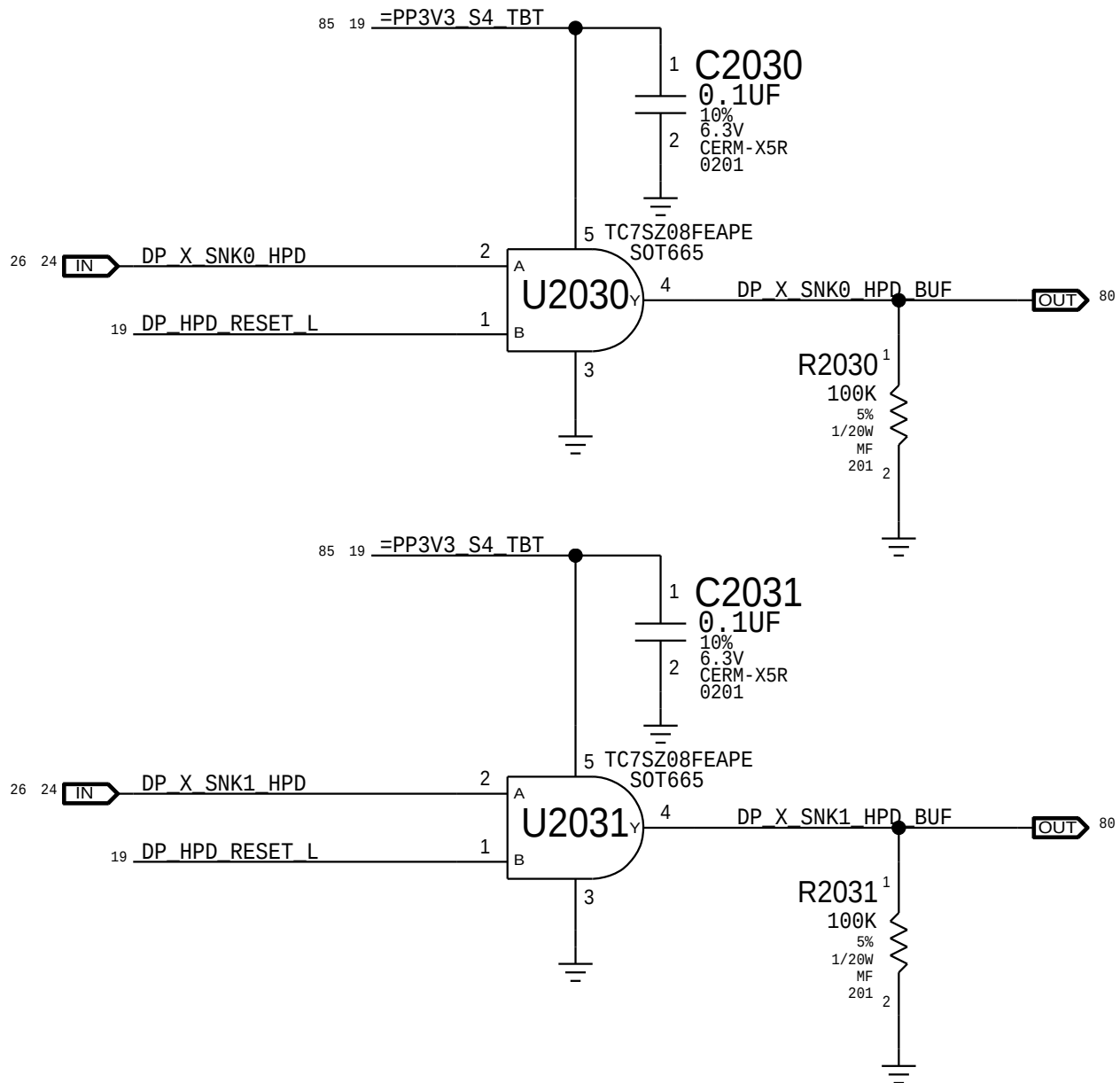
Alpine Ridge JTAG Isolation

TBTLC can be on when S0 is off, and vice-versa
Isolation ensures no leakage to RR or PCH



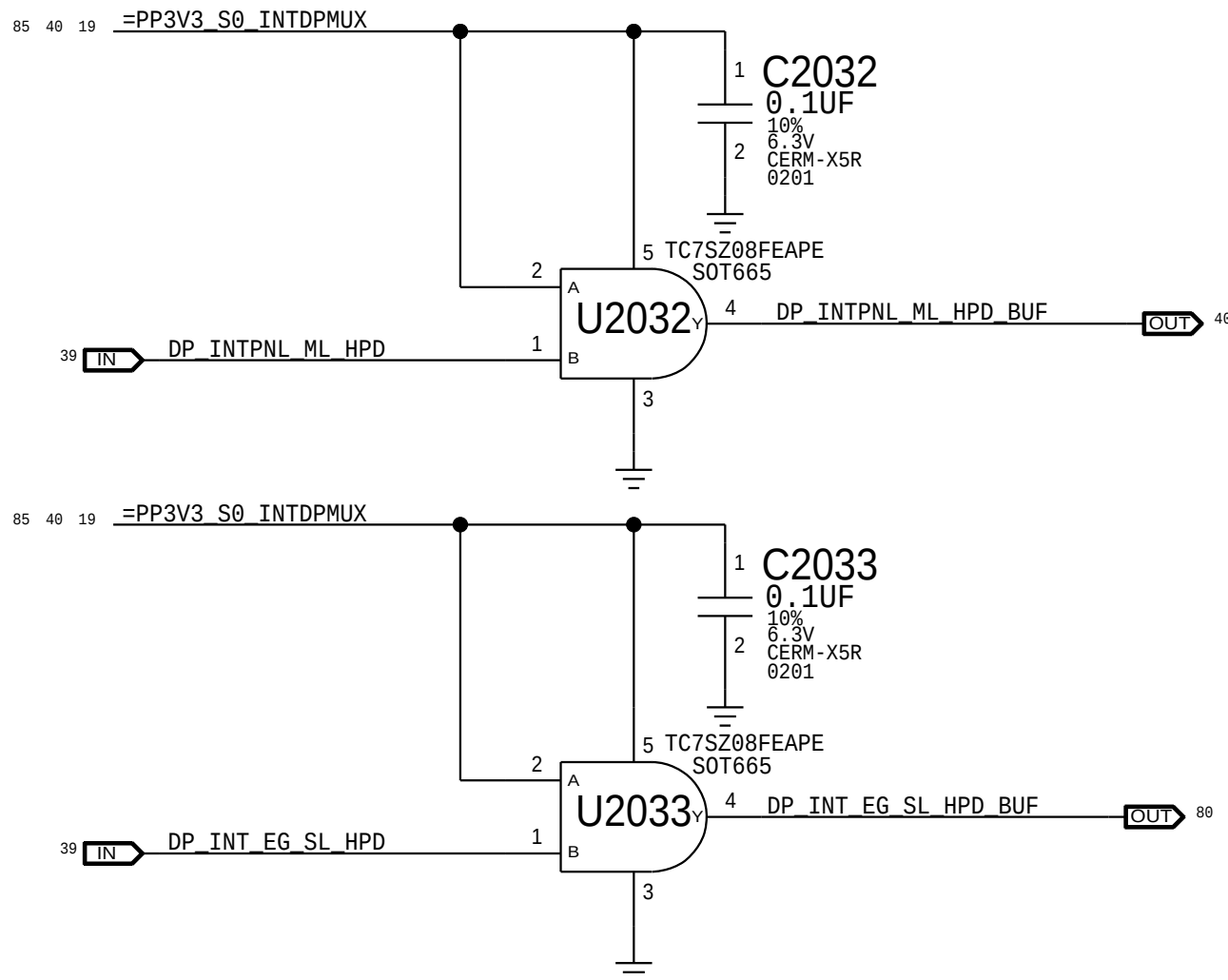
Alpine Ridge HPD Isolation

NOTE: PLT_RESET_L used as other input to the AND gate so that HPD is only driven high to the PCH in S0.

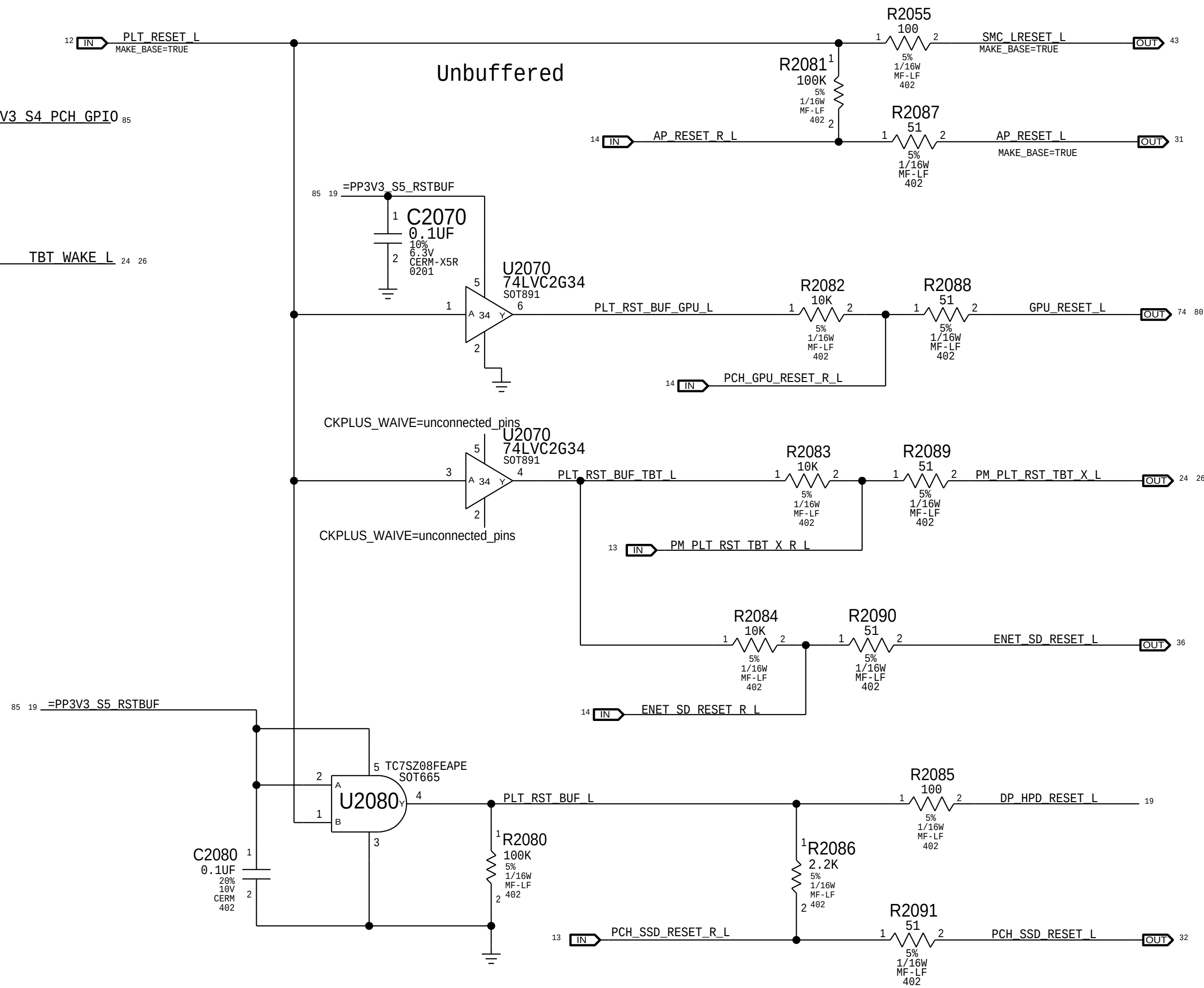


Internal Display HPD Isolation

NOTE: PP3V3_S0 used as other input to the AND gate so that HPD is only driven high when PCH 3V3_S0 is up.

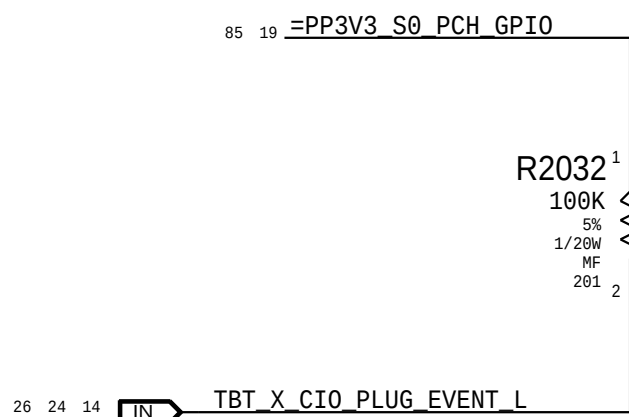


Platform Reset Connections

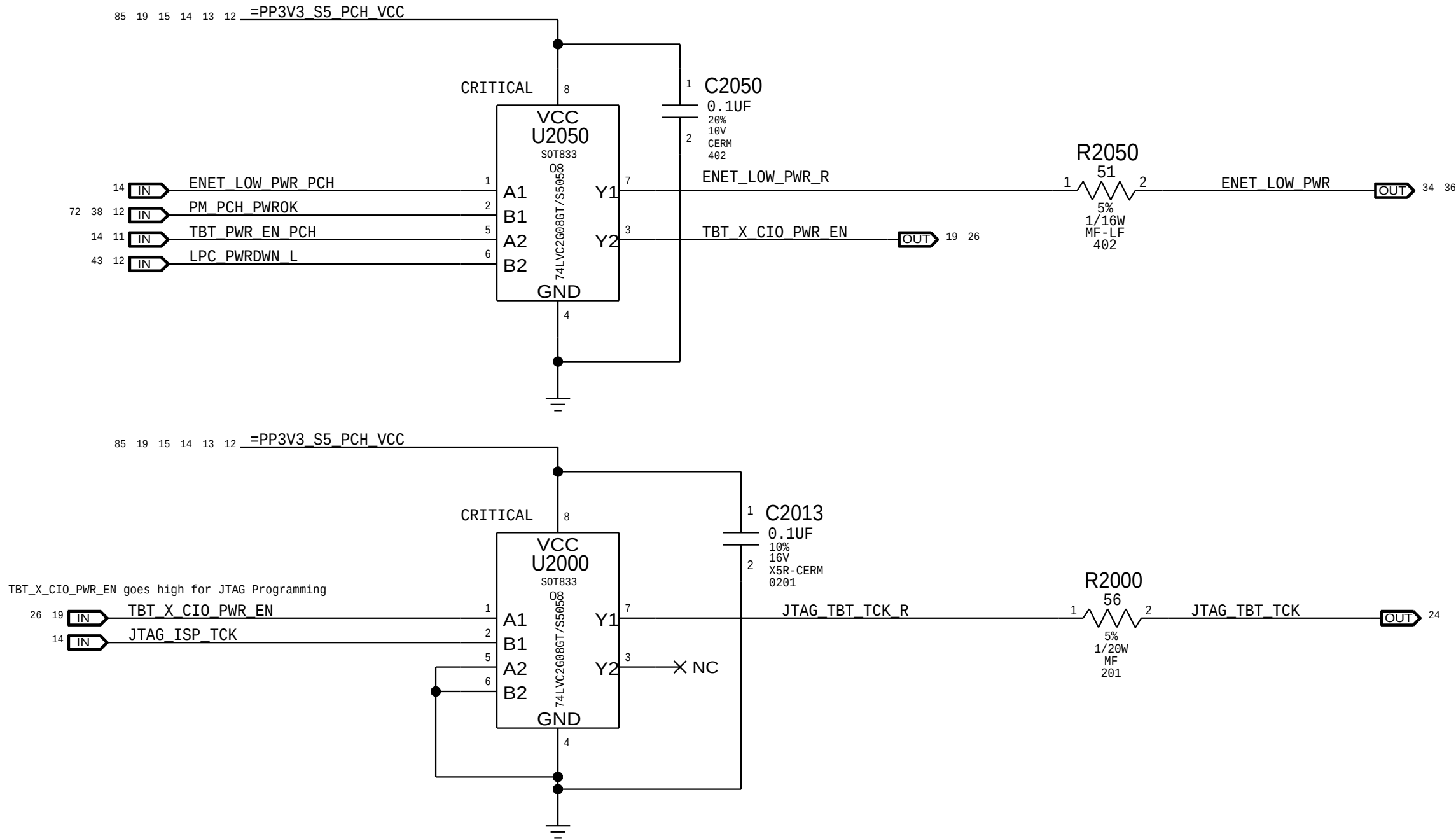


Alpine Ridge Support

FR output is open-drain, no isolation necessary



GPIO Glitch Prevention

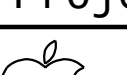


SYNCHMASTER:david1

SYNCHDATE:06/22/2016

PAGE TITLE

Project-Specific Chipset Support



Apple Inc.

DRAWING NUMBER

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REVISION

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PAGE

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BOM_COST_GROUP=CPU & CHIPSET

Page Notes

Power aliases required by this page:

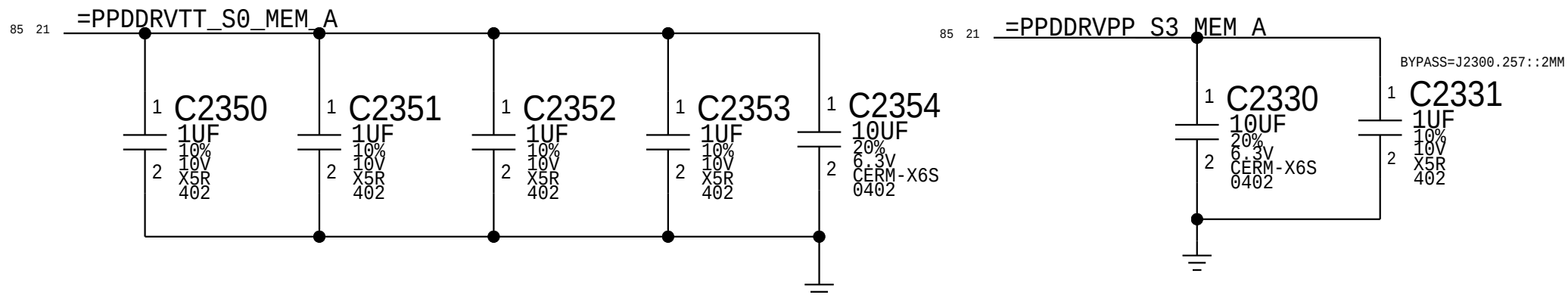
- =PPDDRVP_S3_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

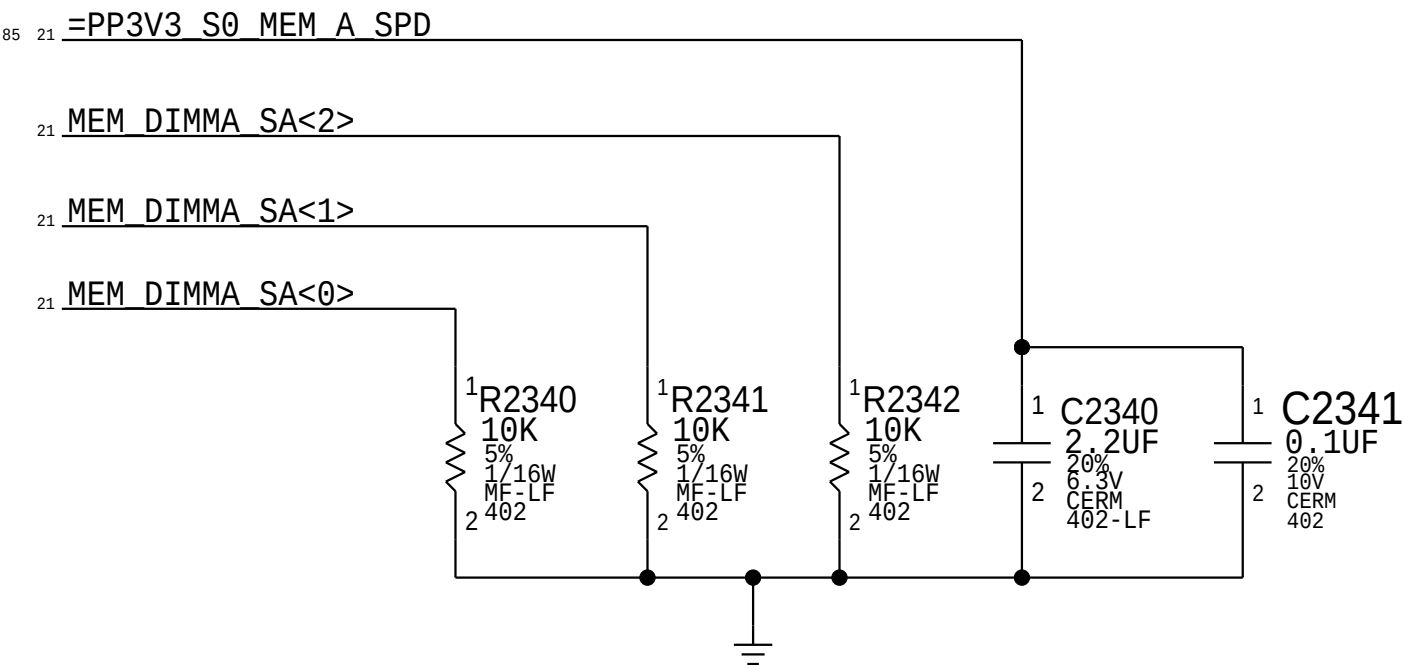
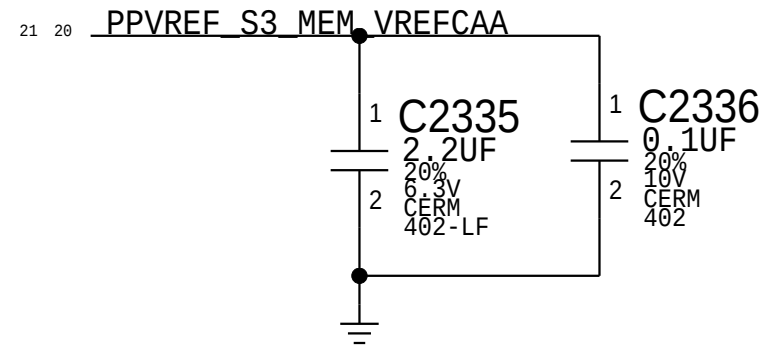
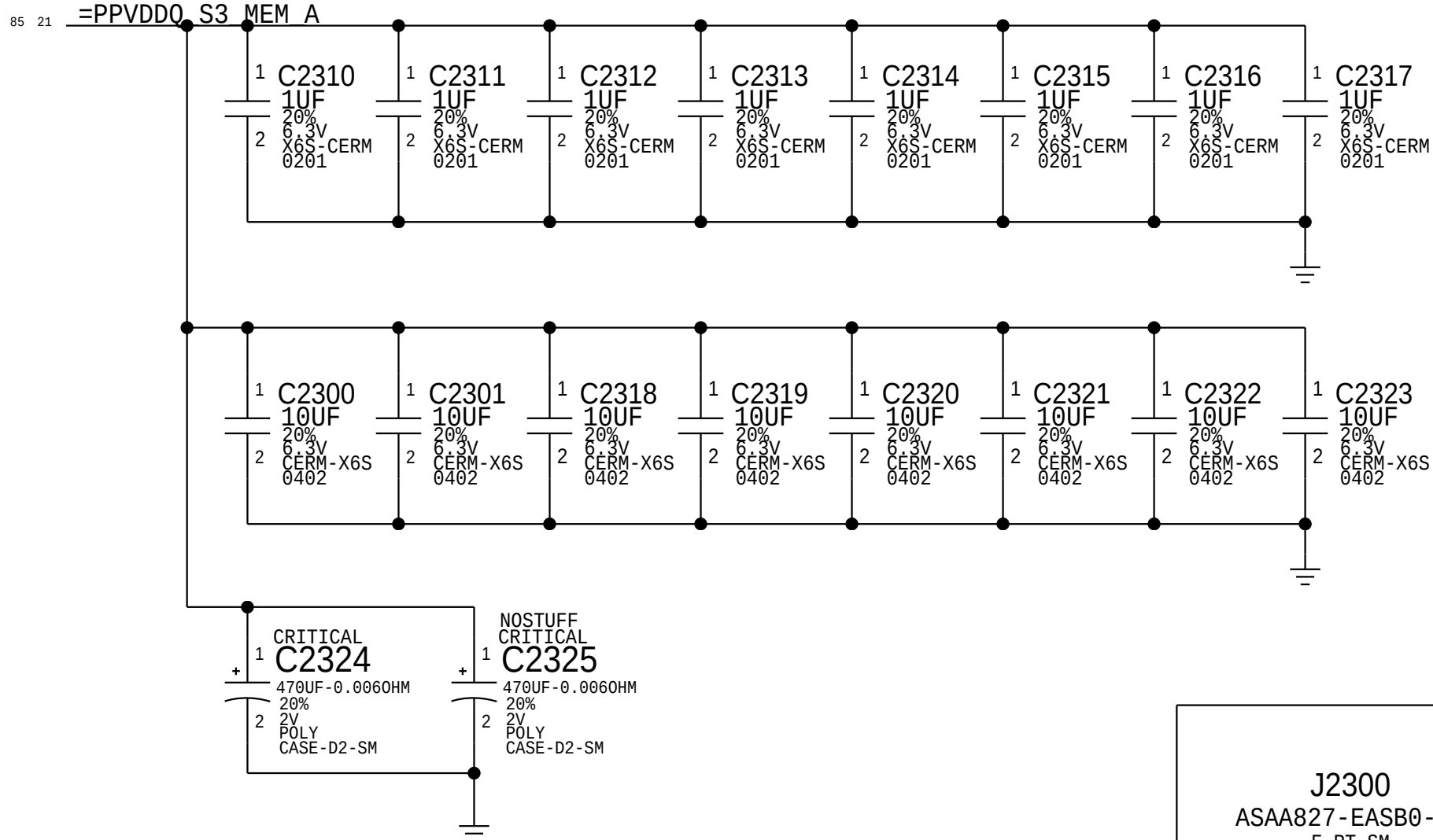
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- =I2C_S0DIMMA_SDA

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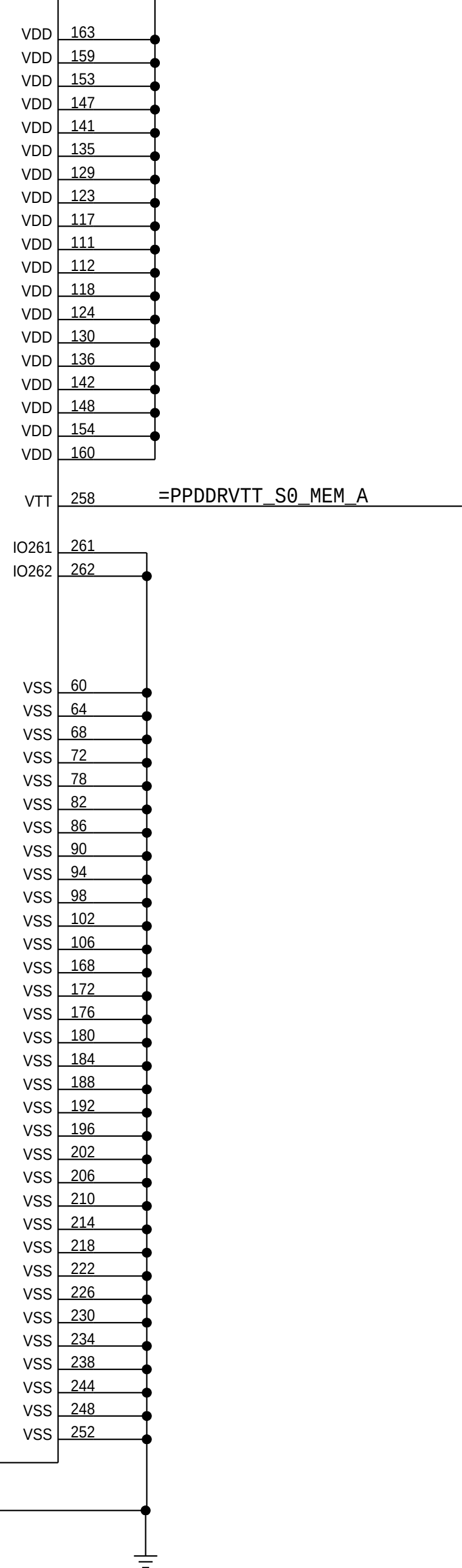
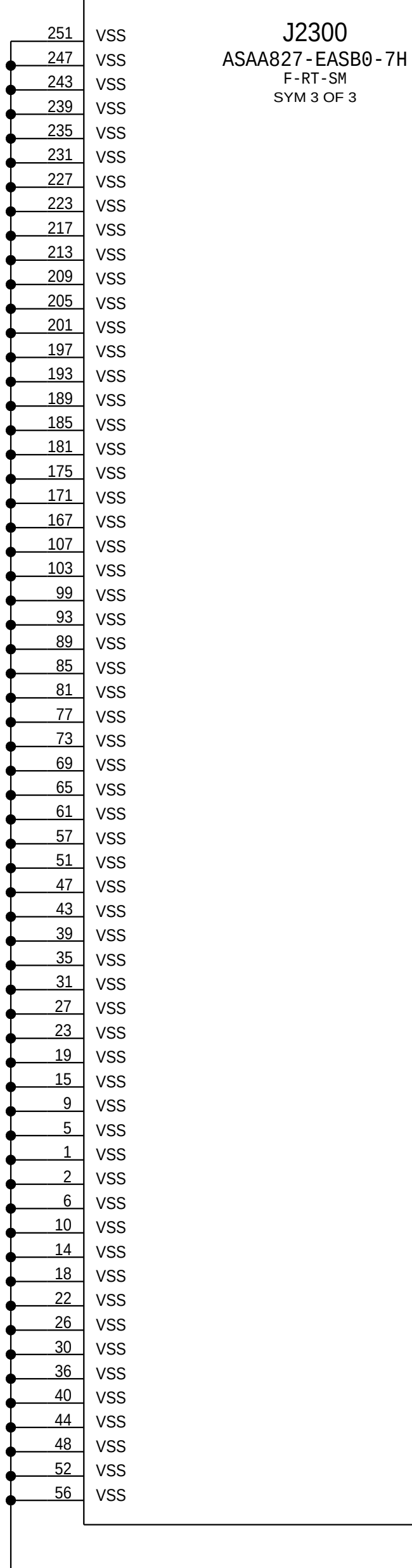
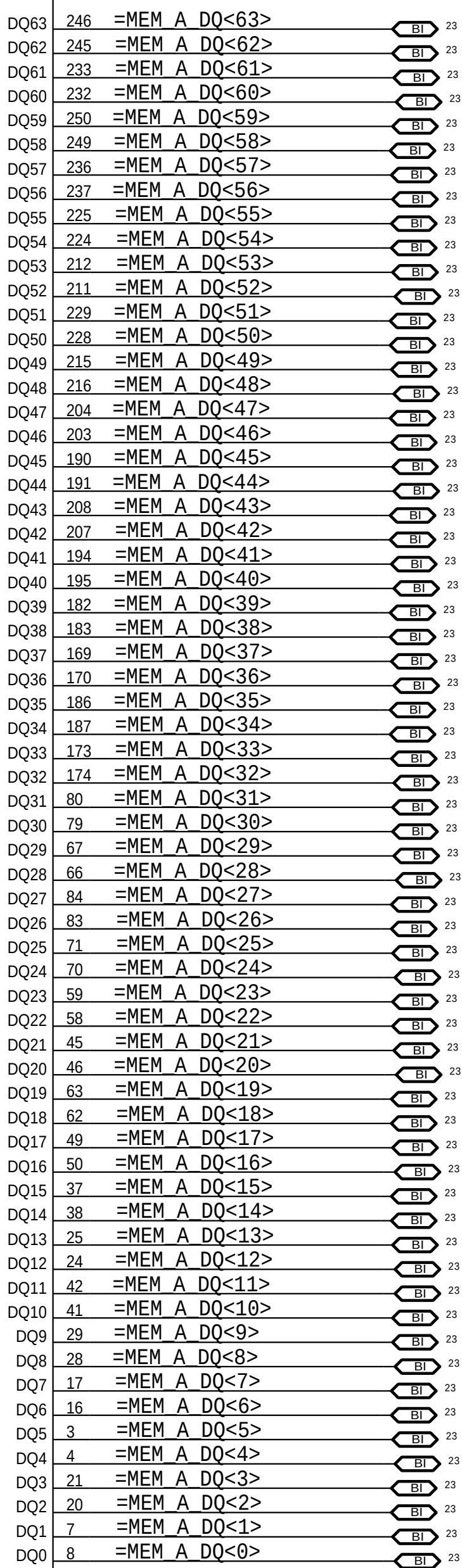
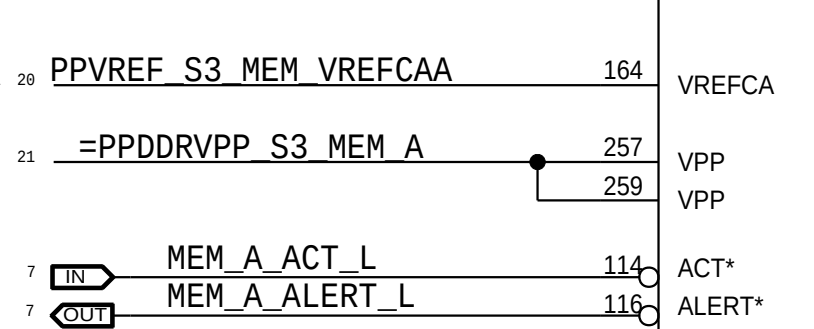
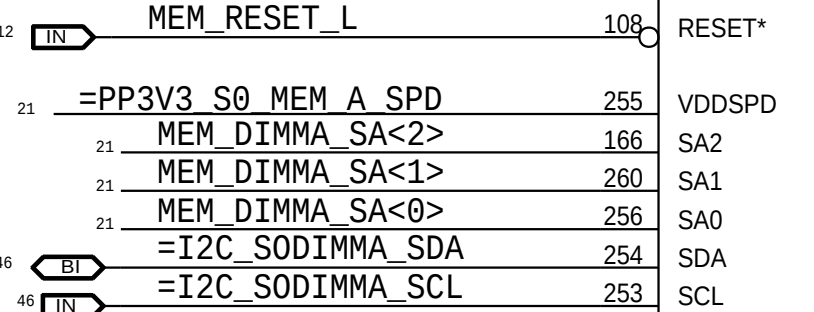
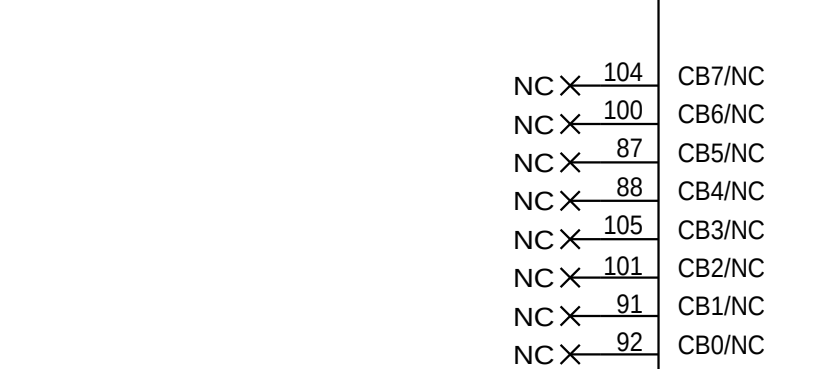
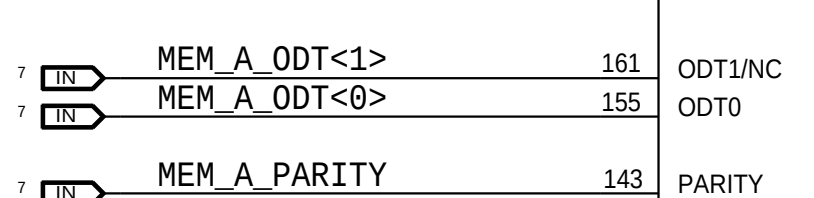
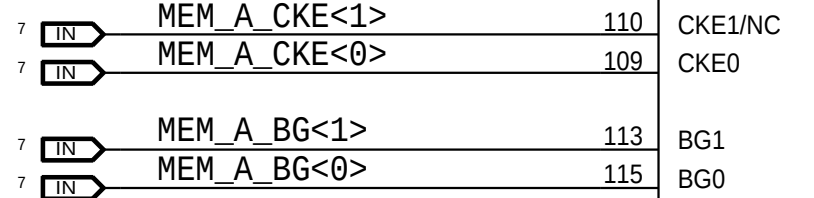
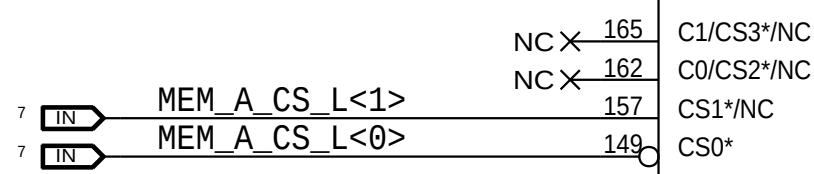
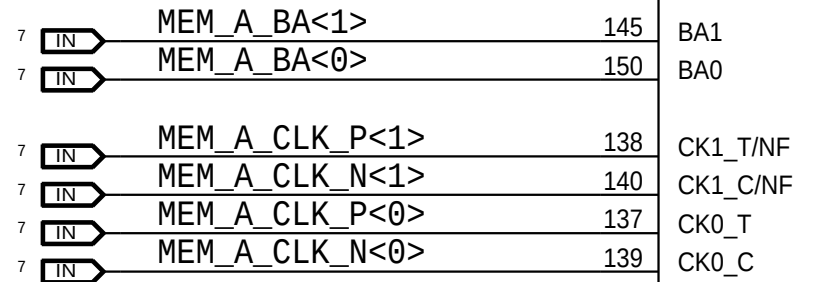
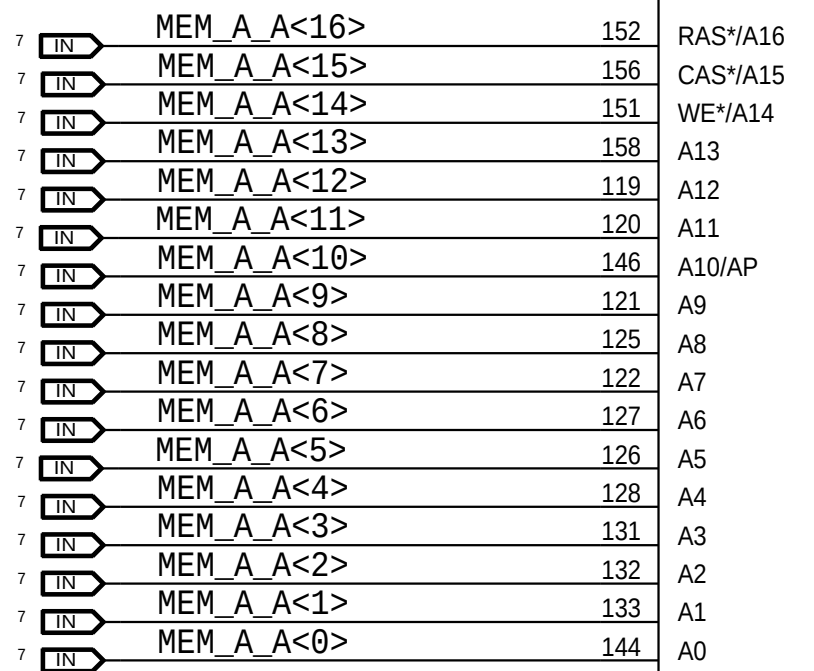


DDR4 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

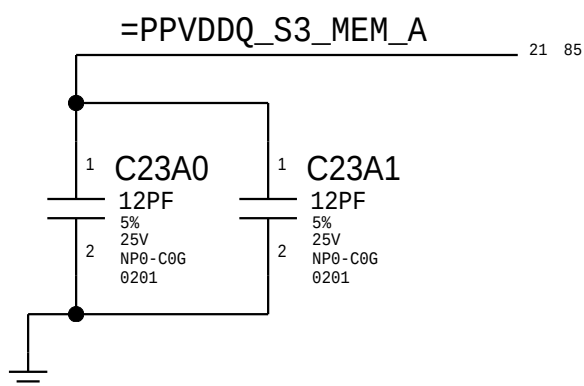


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F-RT-SM
SYM 1 OF 3

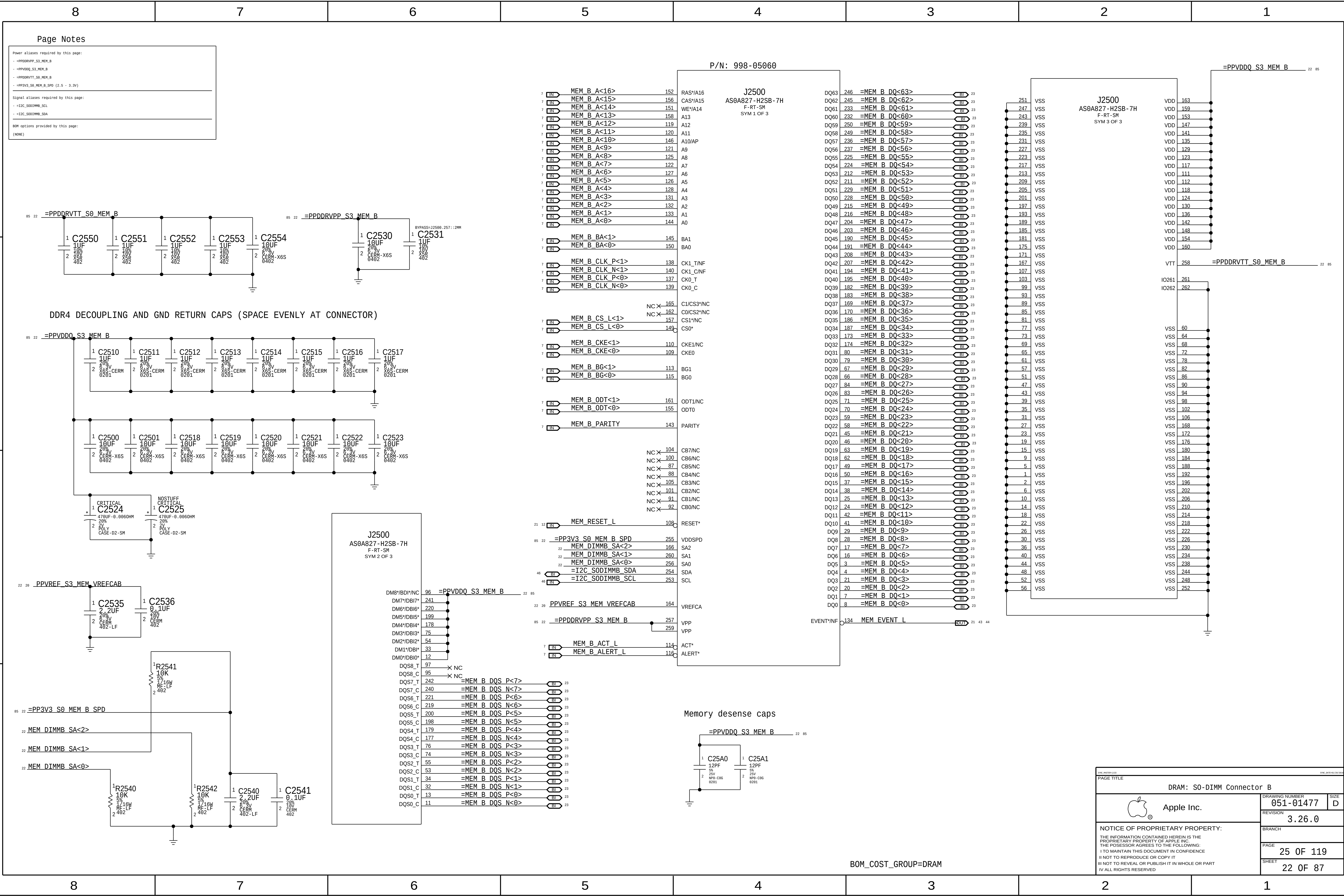


Memory desense caps



BOM_COST_GROUP=DRAM

DRAWING NUMBER 051-01477		SIZE D
REVISION 3.26.0		BRANCH
PAGE 23 OF 119		SHEET 21 OF 87



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DRAM: ALIASES AND BITSWAPS															
Apple Inc.															
051-01477															
3.26.0															
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SYNC_MASTER=ddressler

SYNC_DATE=11/09/2015

DRAM: ALIASES AND BITSWAPS

 Apple Inc.

DRAWING NUMBER051-01477

REVISION3.26.0

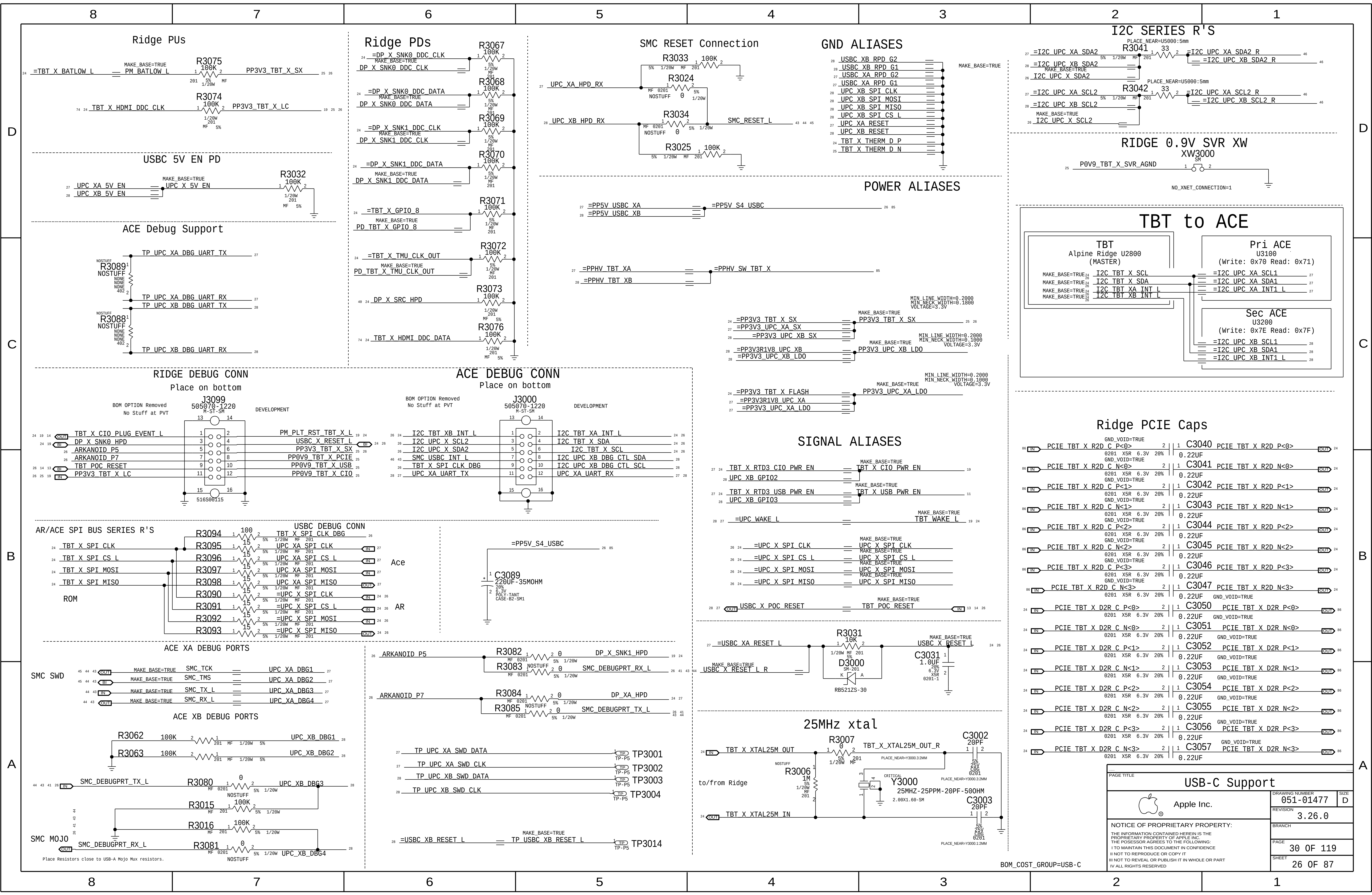
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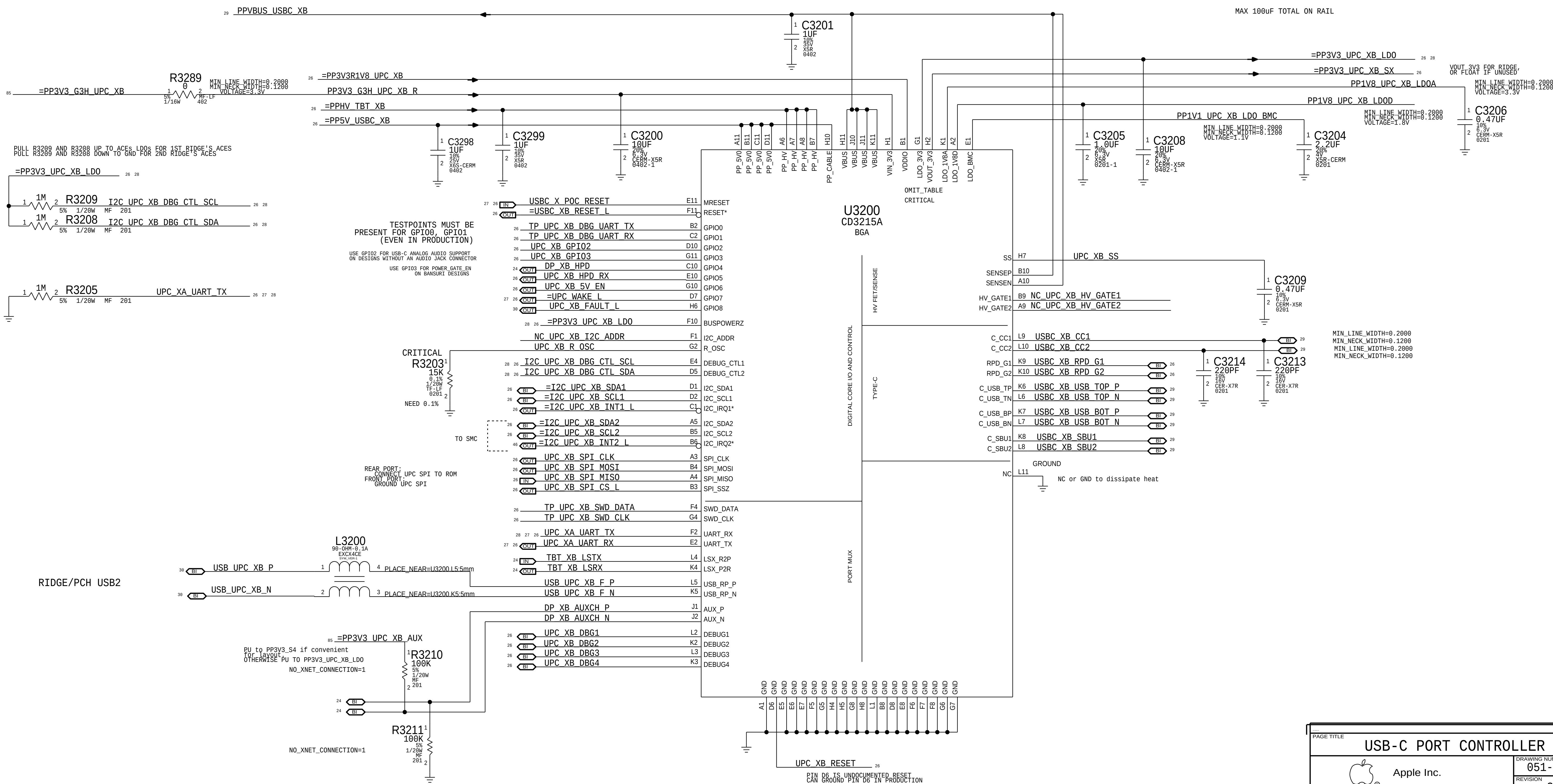
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SECONDARY ACE USB-C PORT CONTROLLER (UPC)



PAGE TITLE			
USB-C PORT CONTROLLER B			
 Apple Inc.	DRAWING NUMBER		SIZE
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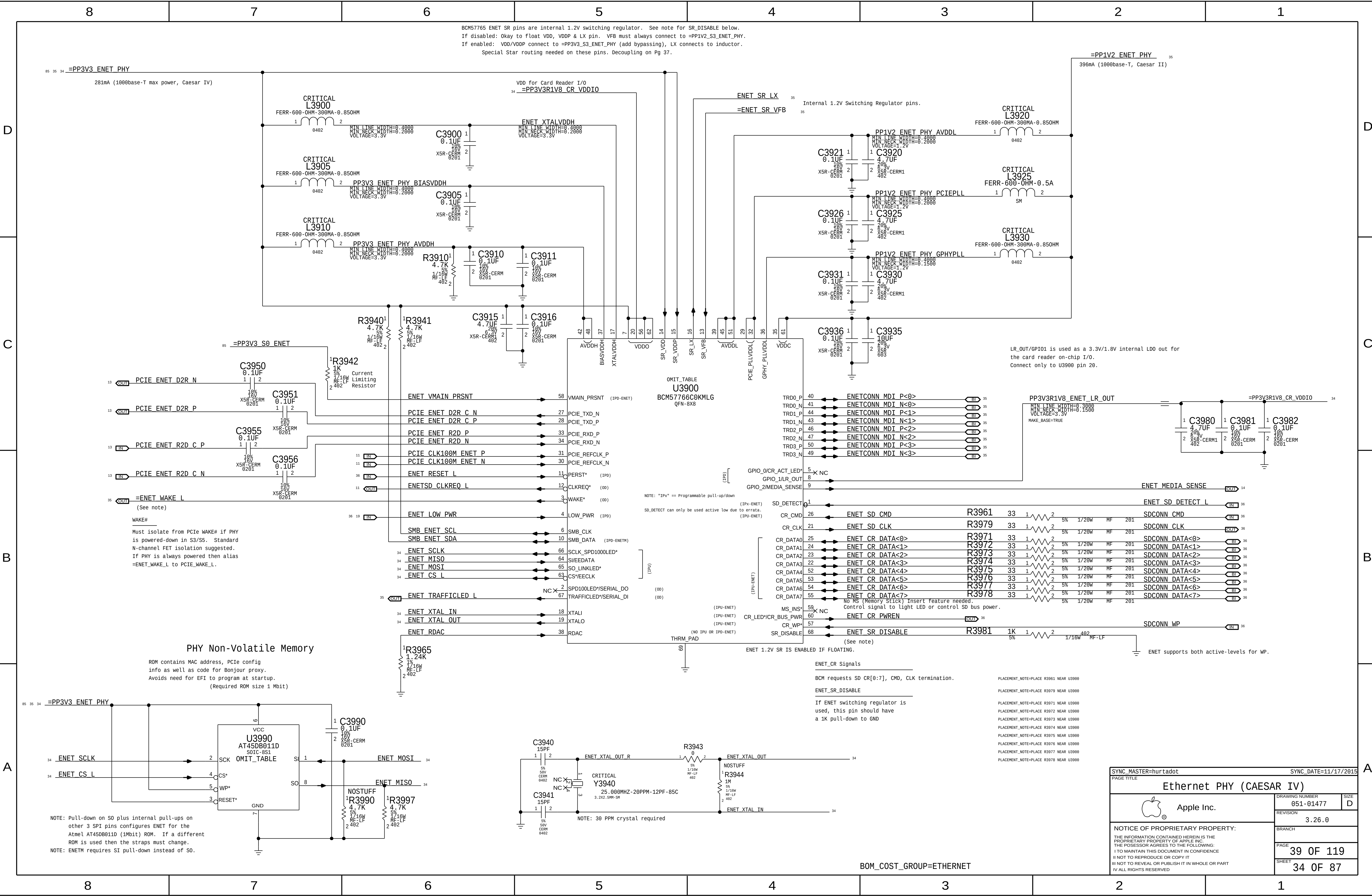
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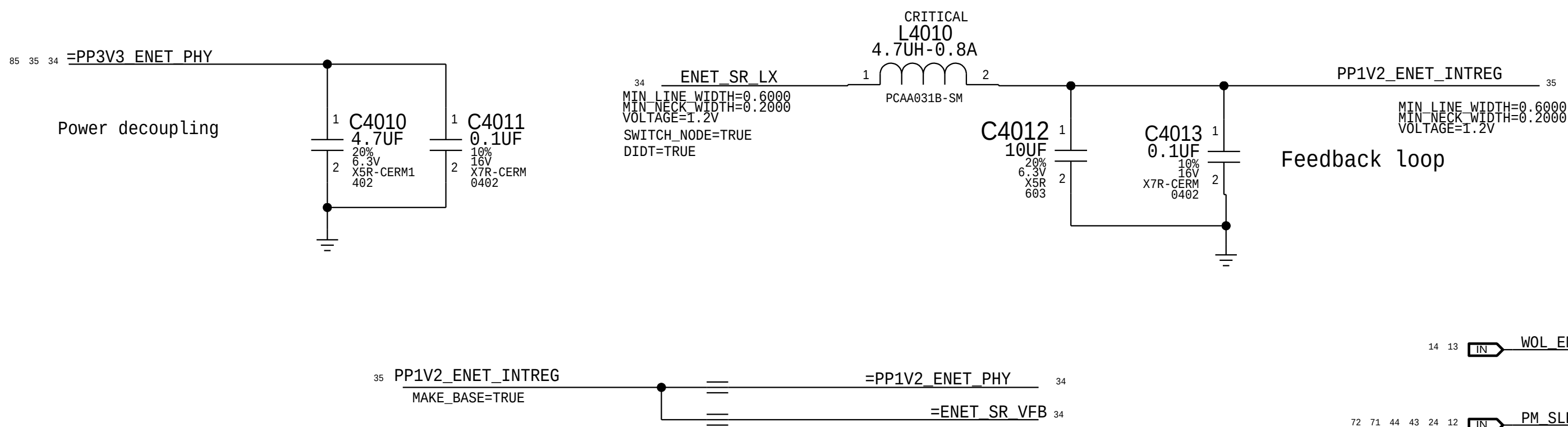


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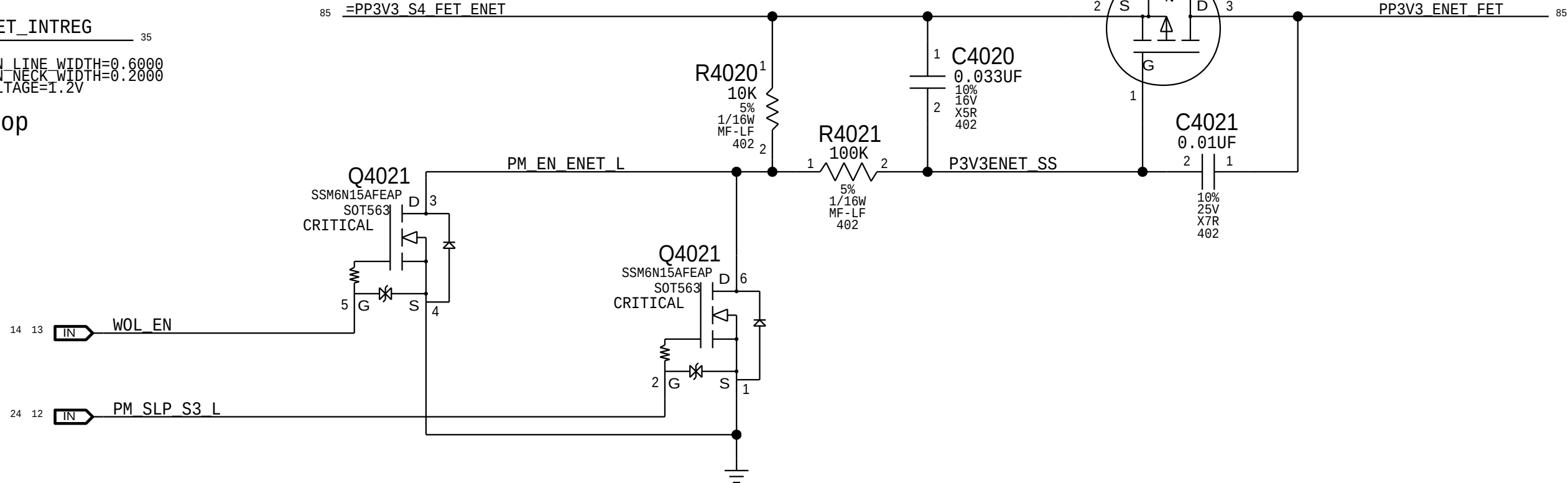


CAESAR IV 1.2V INT.VR CMPTS

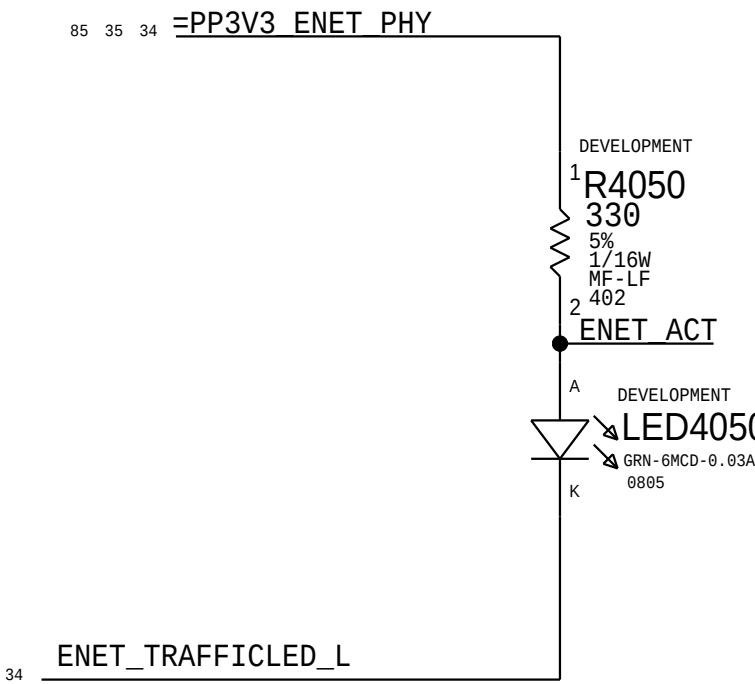


ENET Enable Generation

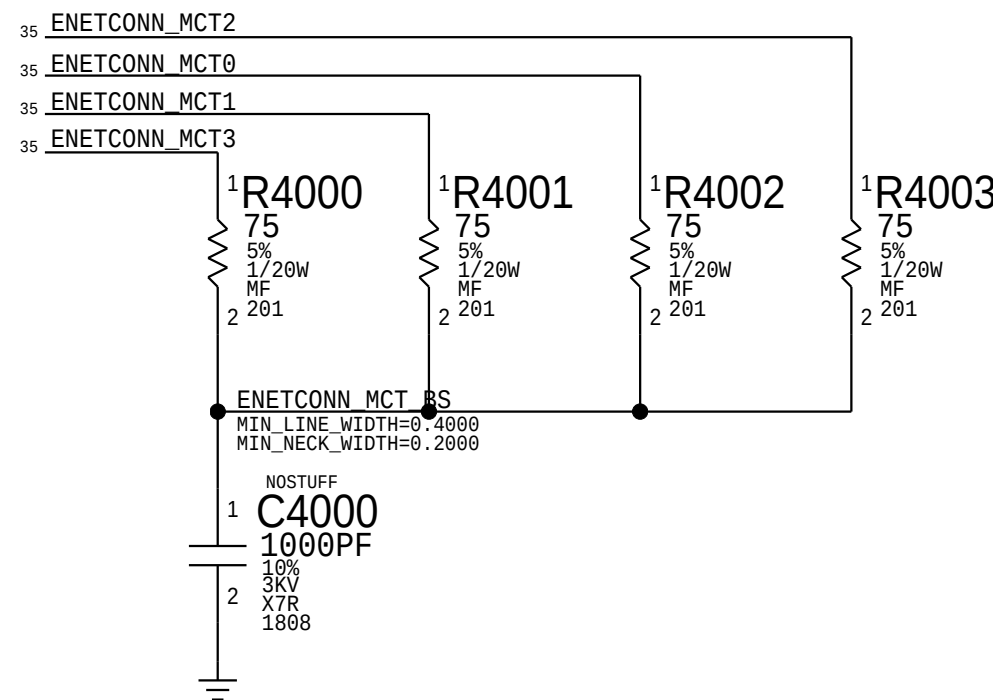
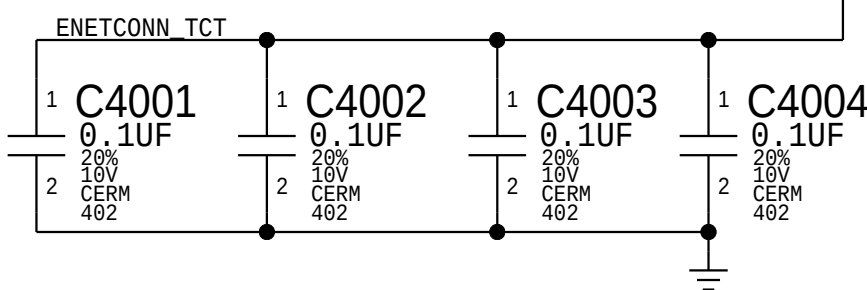
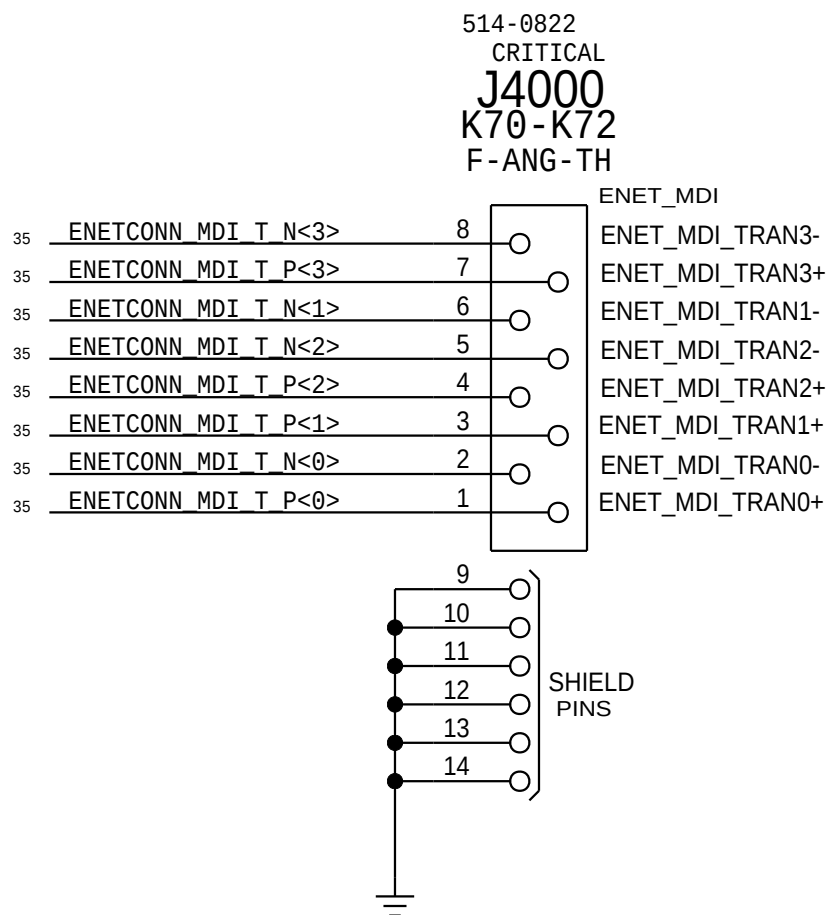
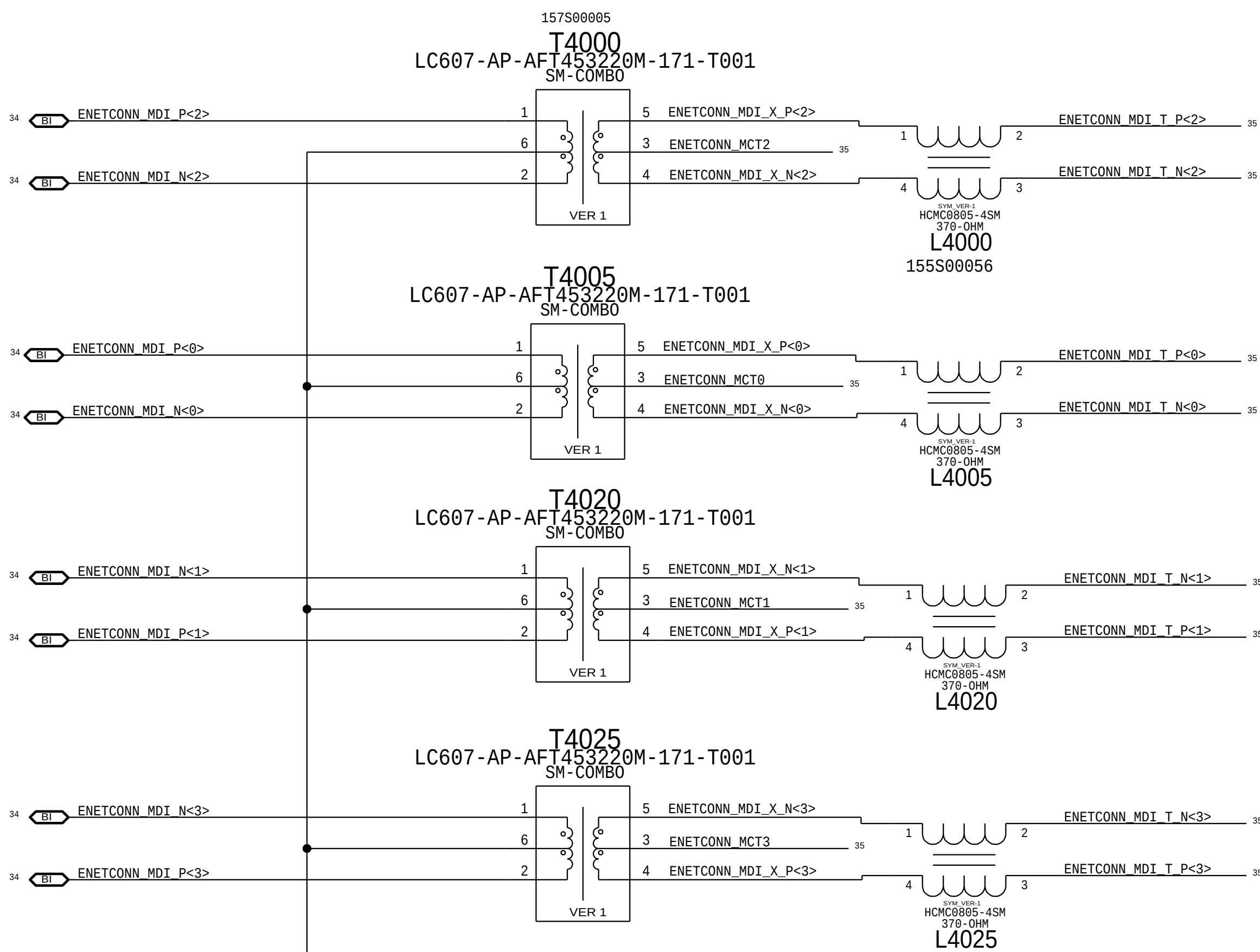
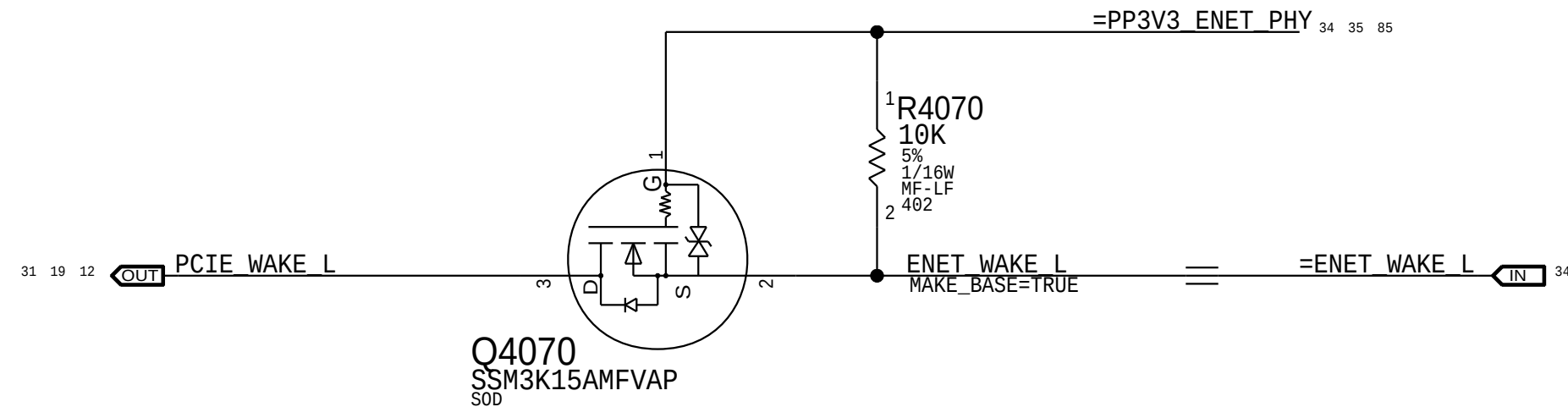
ENET is enabled when in S0 or when (S4 & WOL_EN) is present



CAESAR IV ACTIVITY LED



CAESAR IV WAKE# ISOLATION



SYNC_MASTER=yeungr		SYNC_DATE=03/30/2016	
PAGE TITLE			
Ethernet Support & Connector			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-01477	D
		REVISION	3.26.0
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BOM_COST_GROUP=ETHERNET

D

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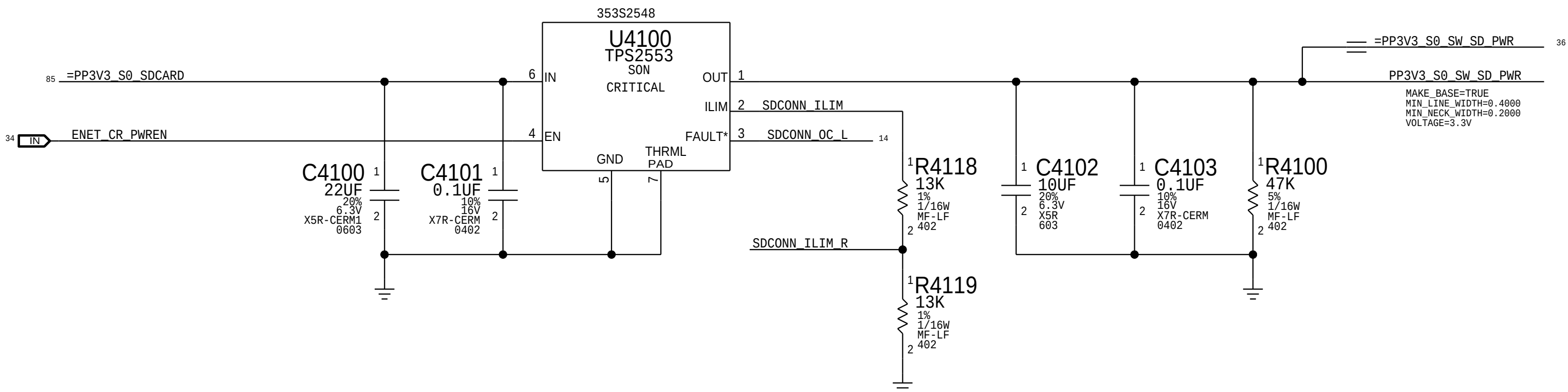
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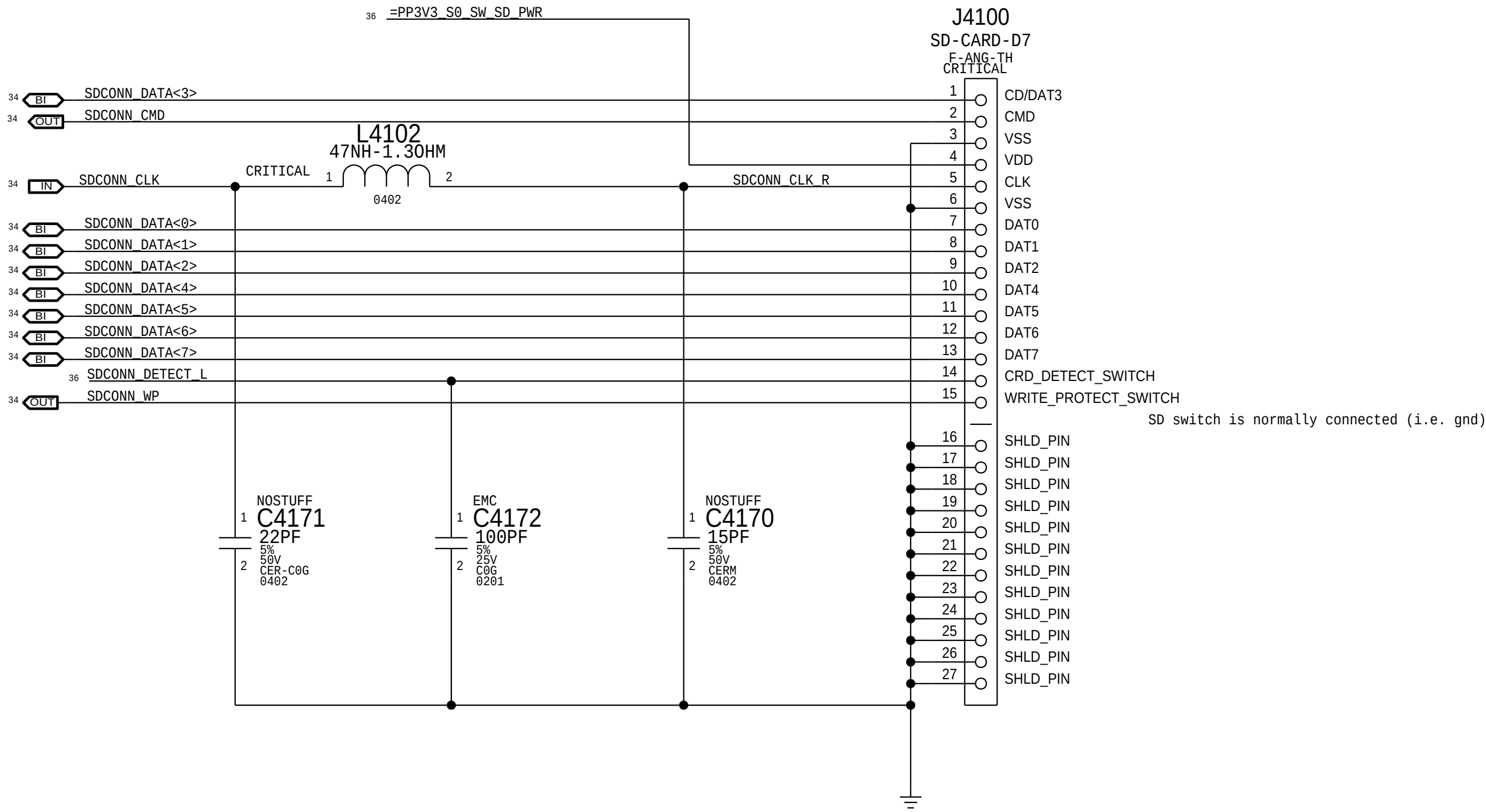
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SD CARD 3.3V OVERCURRENT PROTECTION CHIP

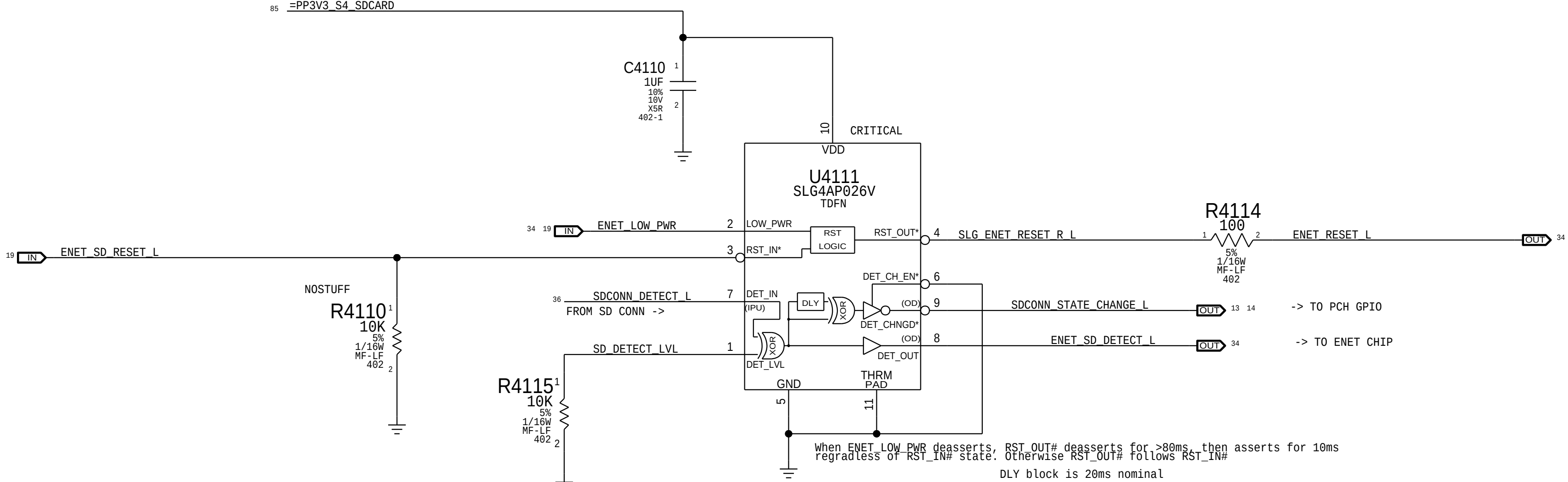


J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



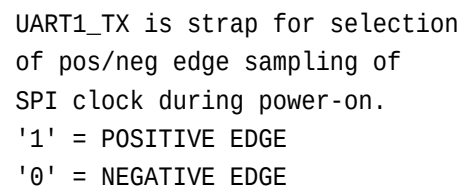
SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



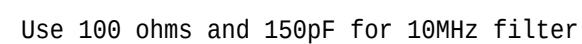
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SYNC_MASTER=J117_MLB		SYNC_DATE=08/18/2014	
PAGE TITLE			
SD Reader Connector			
	Apple Inc.	DRAWING NUMBER	SIZE
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A



SYNC_MASTER=J117_MLB		SYNC_DATE=11/06/2015	
PAGE TITLE			
Camera Controller			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-01477		D
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			3.26.0
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BRANCH			
PAGE		42 OF 119	
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8

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6

5

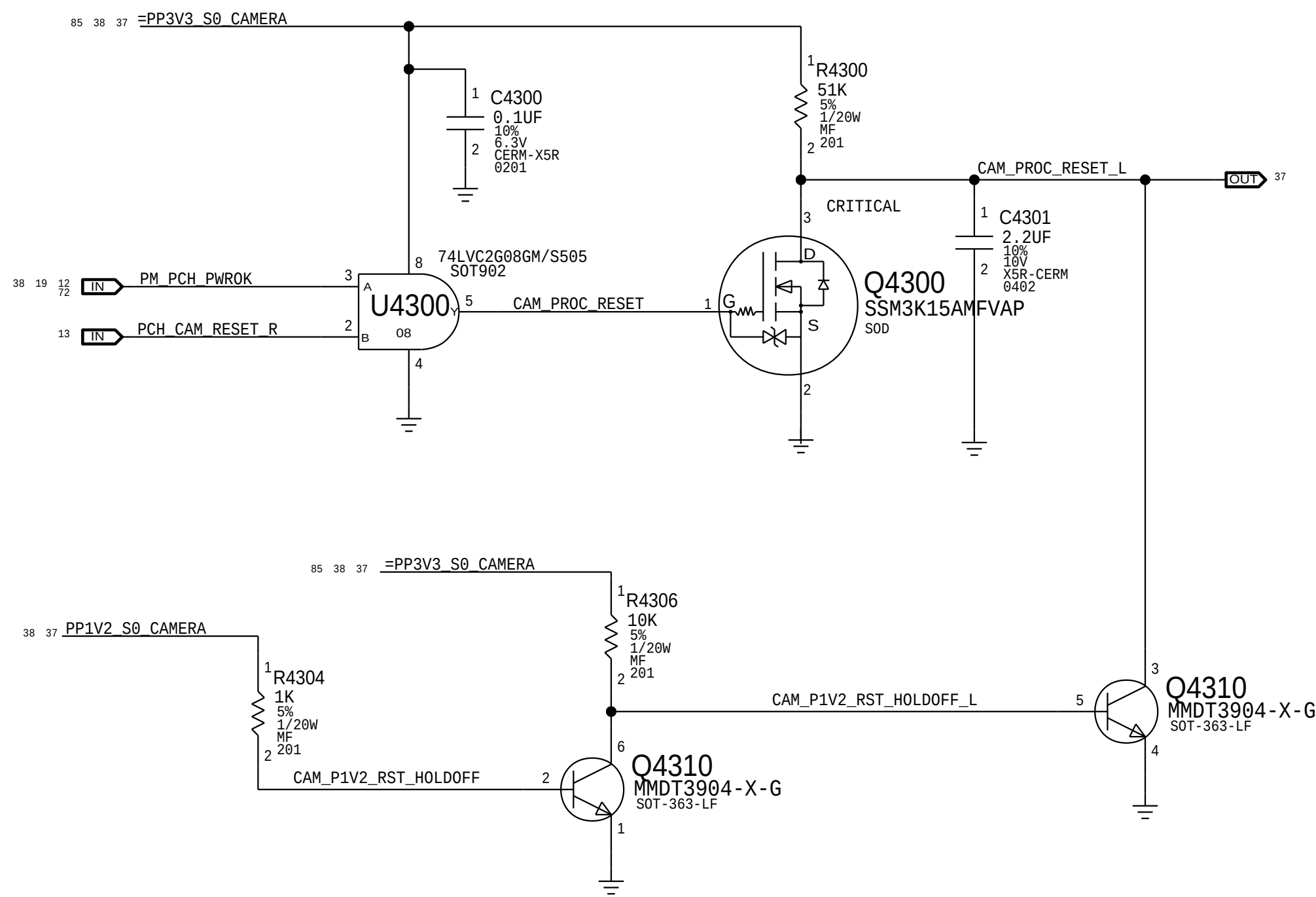
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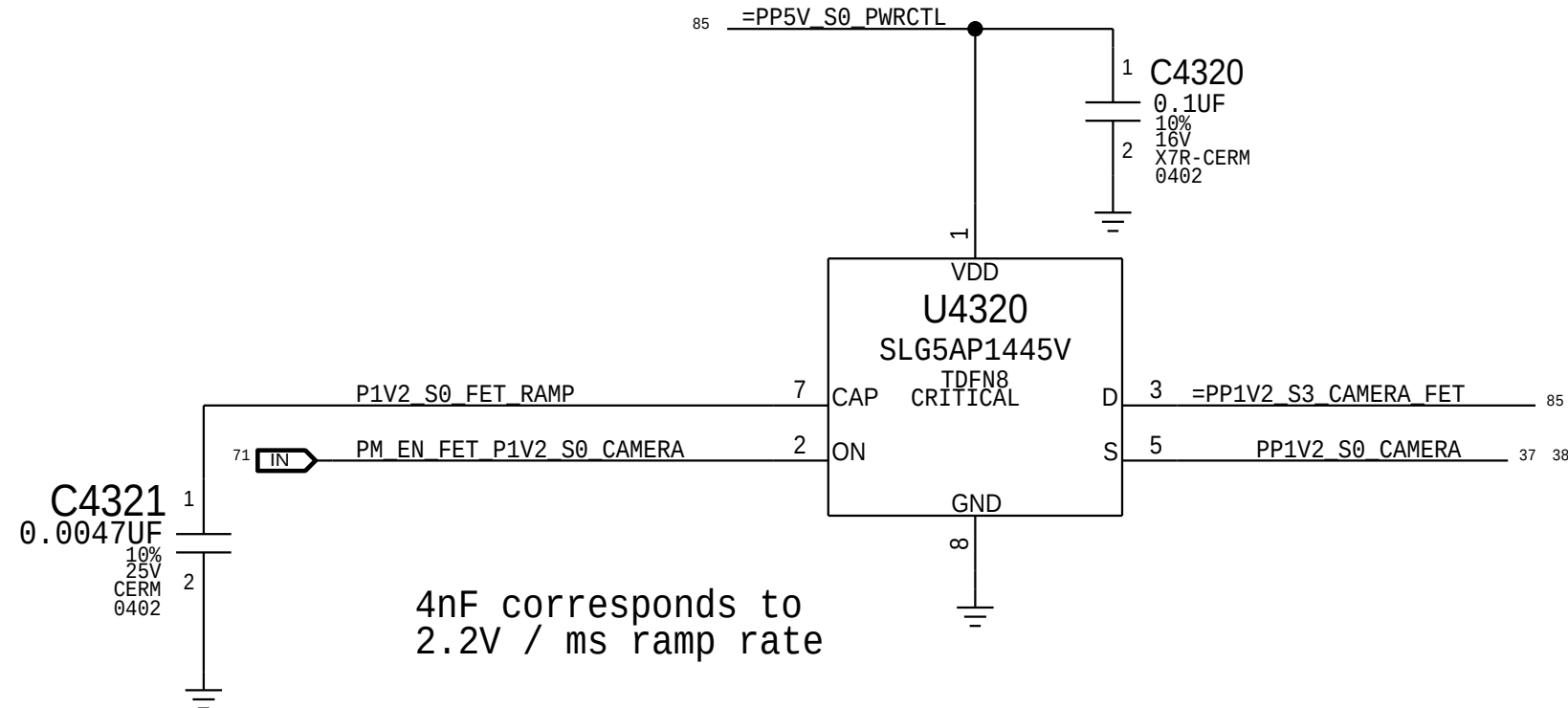
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1

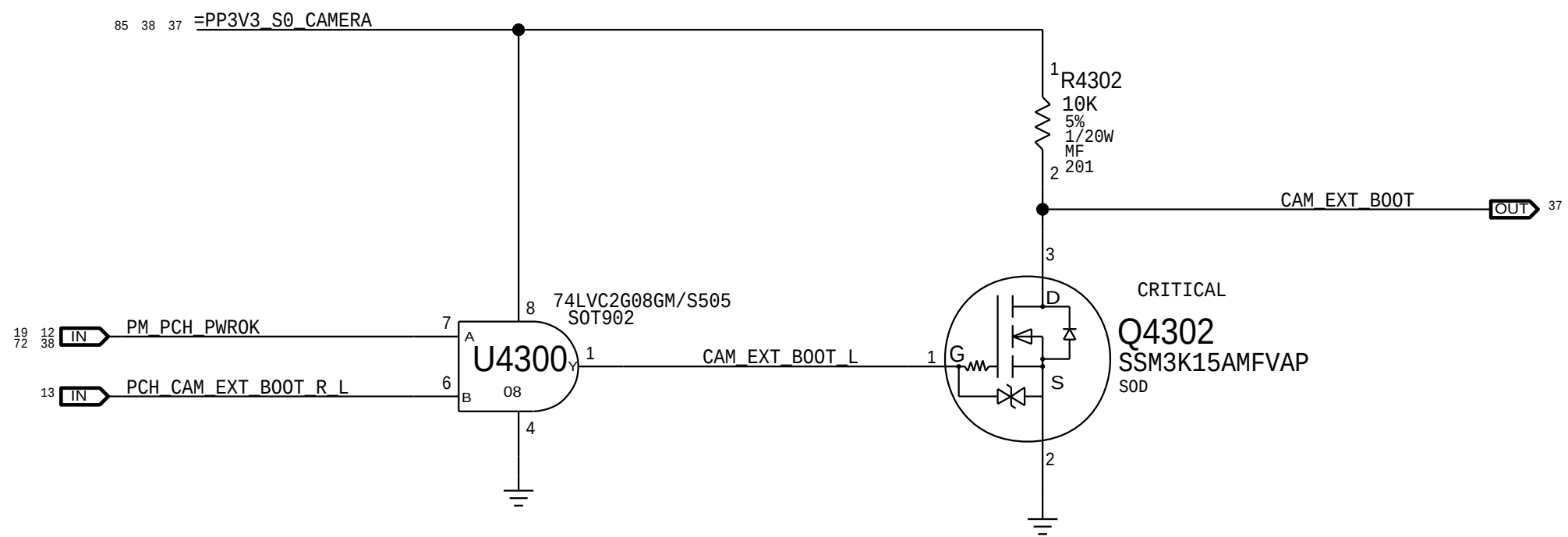
Camera Processor Reset



1.2V S0 Camera Pass FET



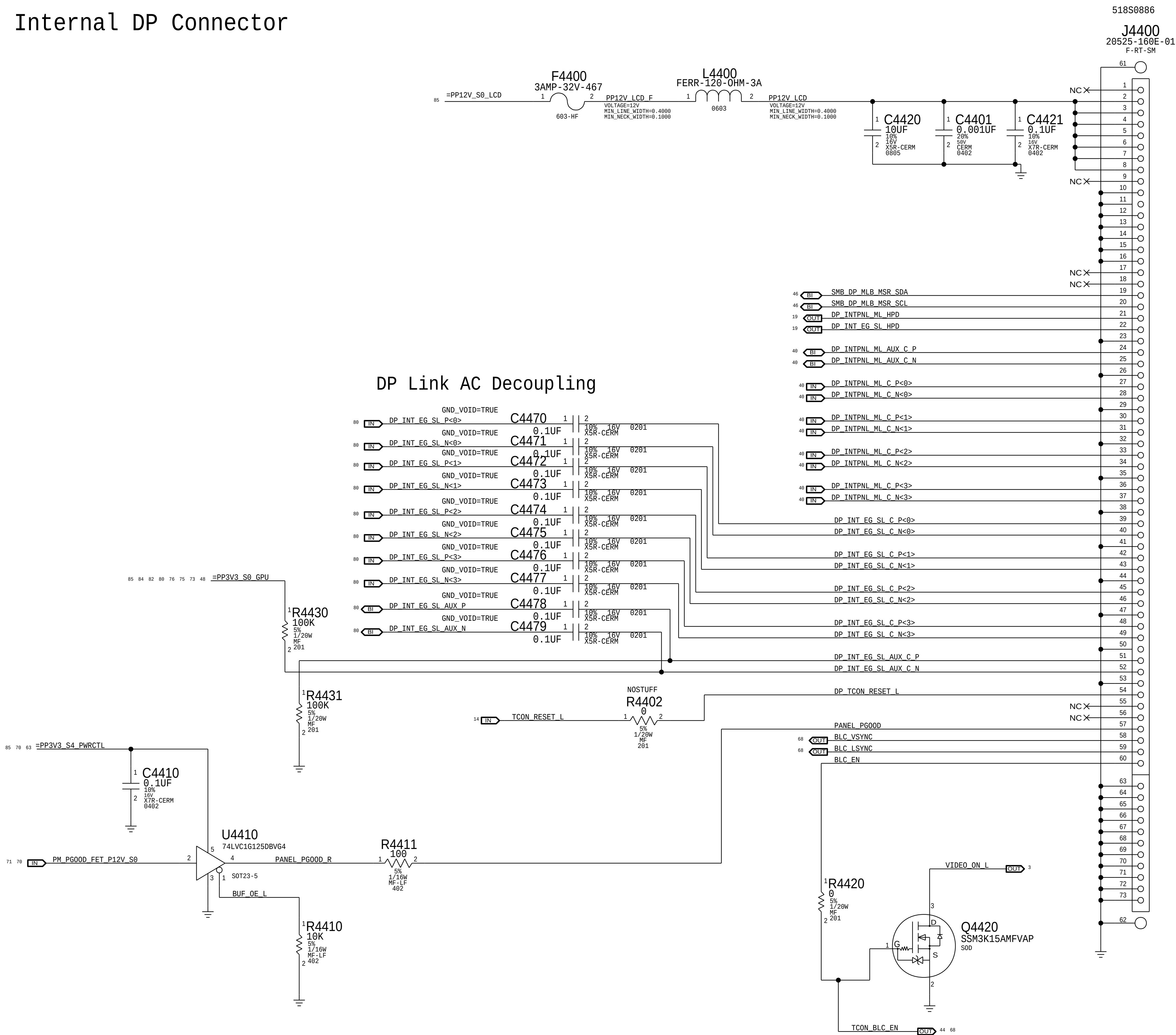
Camera Processor ExtBoot Cntl



BOM_COST_GROUP=CAMERA

SYNC_MASTER=J117_MLB		SYNC_DATE=08/18/2014	
PAGE TITLE			
Camera Controller Support			
 Apple Inc.	DRAWING NUMBER 051-01477		SIZE D
	REVISION 3.26.0		BRANCH
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Internal DP Connector



SYNC_MASTER=andres		SYNC_DATE=08/16/2016	
PAGE TITLE			
Internal DP Support			
	Apple Inc.	DRAWING NUMBER	051-01477
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		PAGE	44 OF 119
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BOM_COST_GROUP=DISPLAY

D

C

B

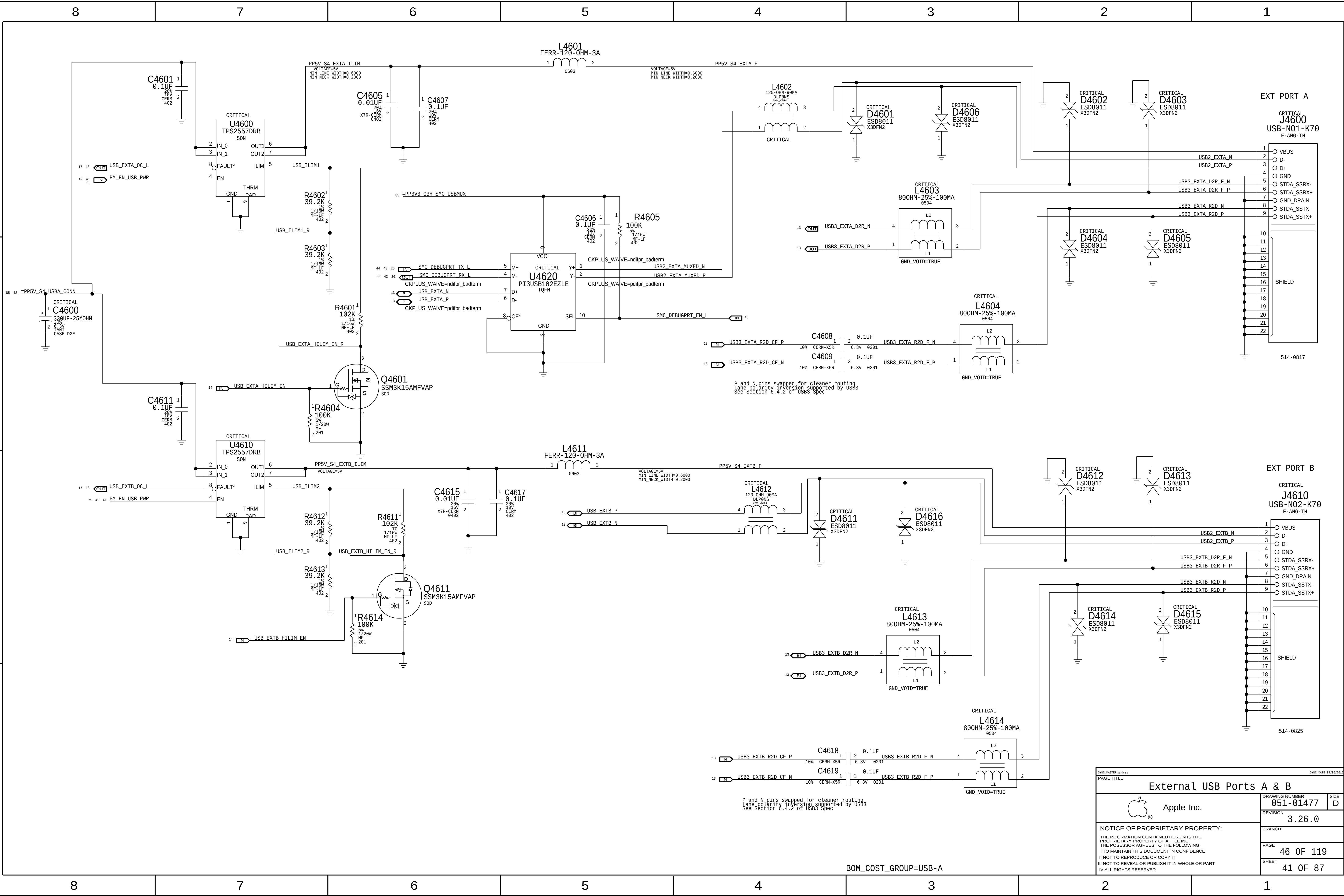
A

D

C

B

A



P and N pins swapped for cleaner routing.
Lane polarity inversion supported by USB3
See Section 6.4.2 of USB3 Spec

BOM_COST_GROUP=USB-A

EVAL_MASTER=andres		SYNC_DATE=09/06/2016	
PAGE TITLE			
External USB Ports A & B			
	Apple Inc.	DRAWING NUMBER	051-01477
		SIZE	D
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D

C

B

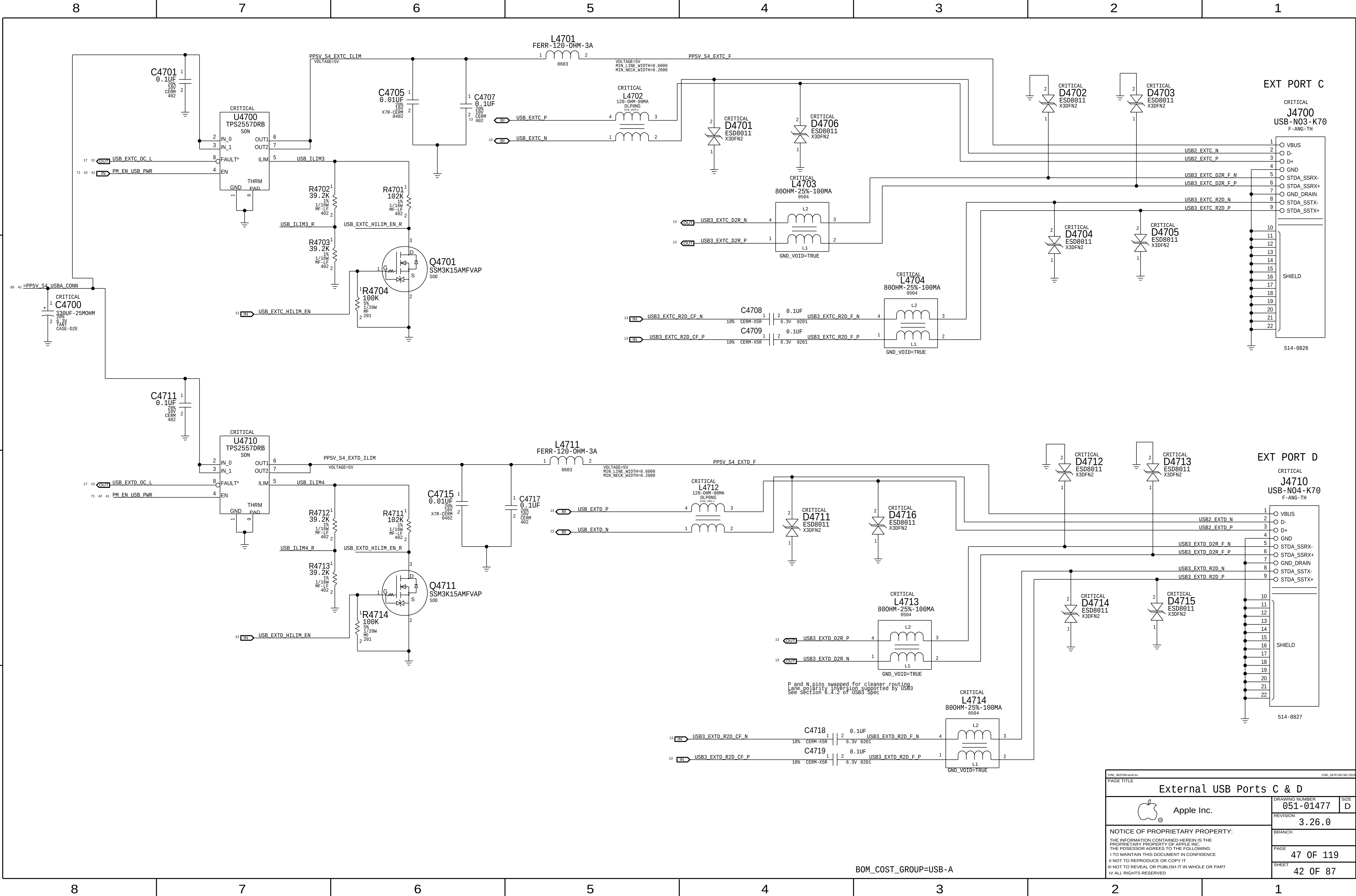
A

D

C

B

A



BOM_COST_GROUP=USB-A

SYMC_MASTER=andres

SYMC_DATE=89/06/2819

PAGE TITLE

External USB Ports C & D



Apple Inc.

DRAWING NUMBER

051-01477

SIZE

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3.26.0

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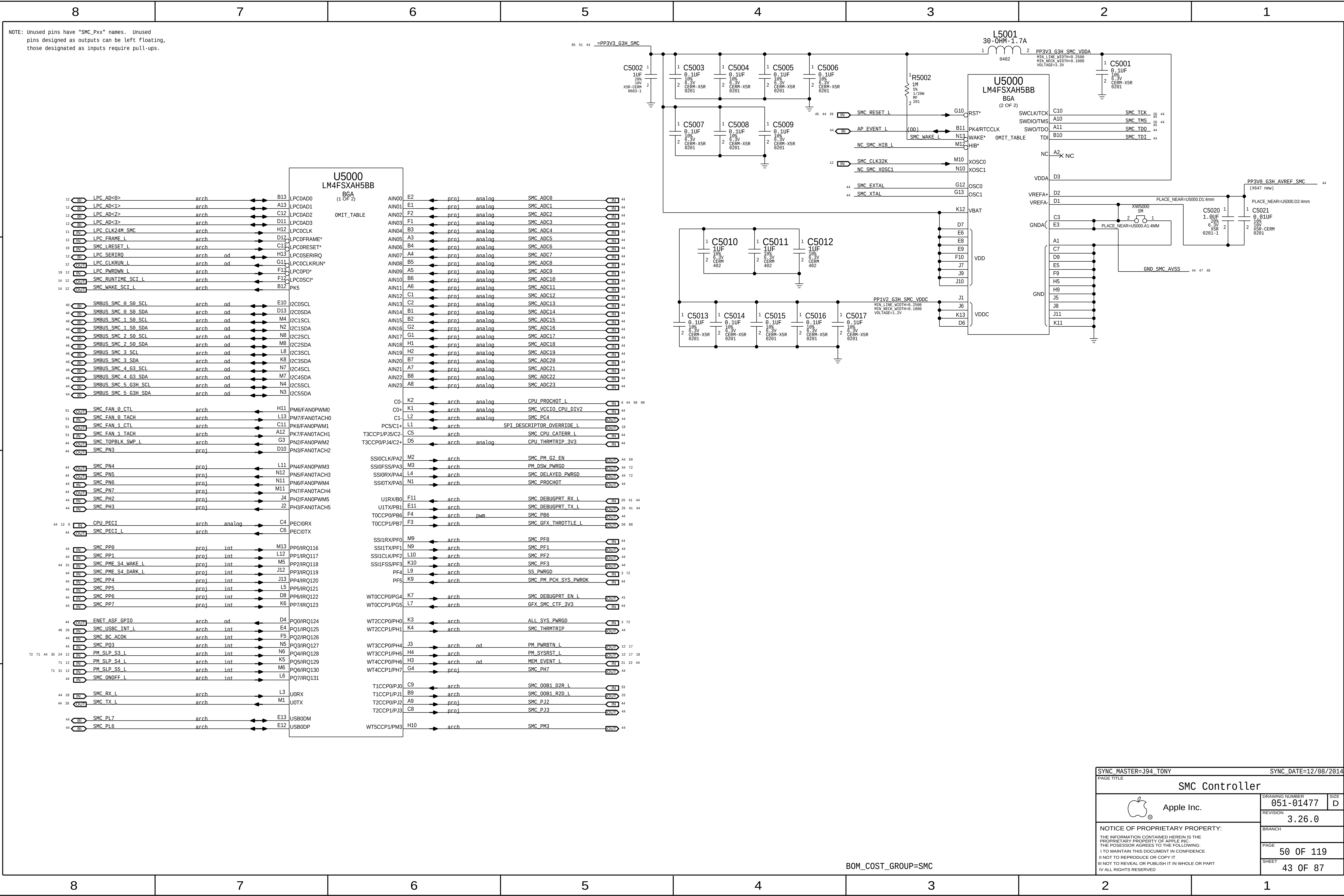
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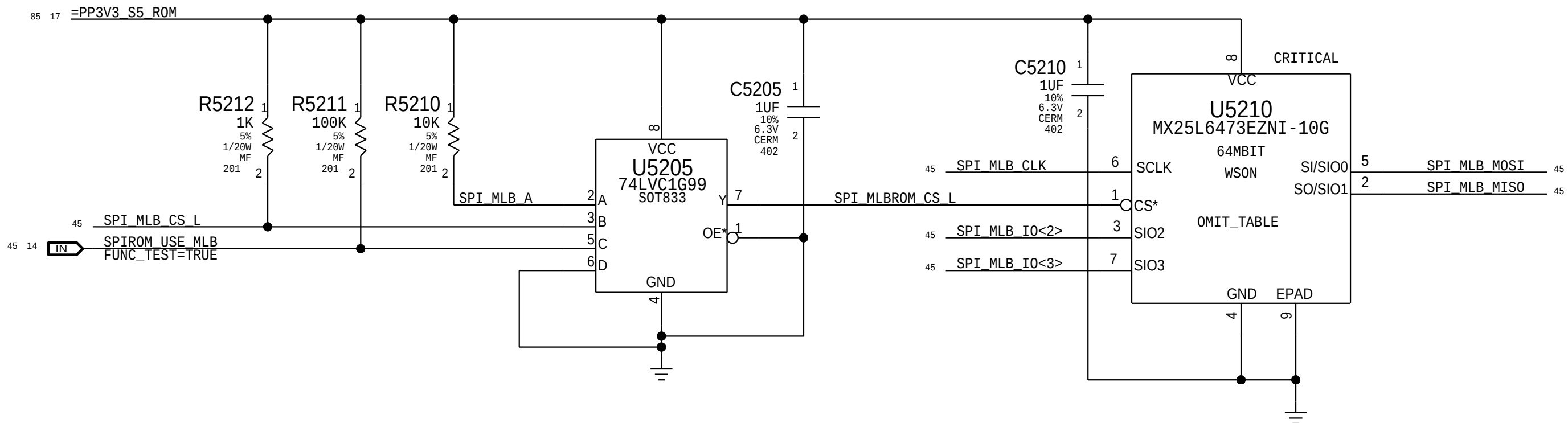
SHEET

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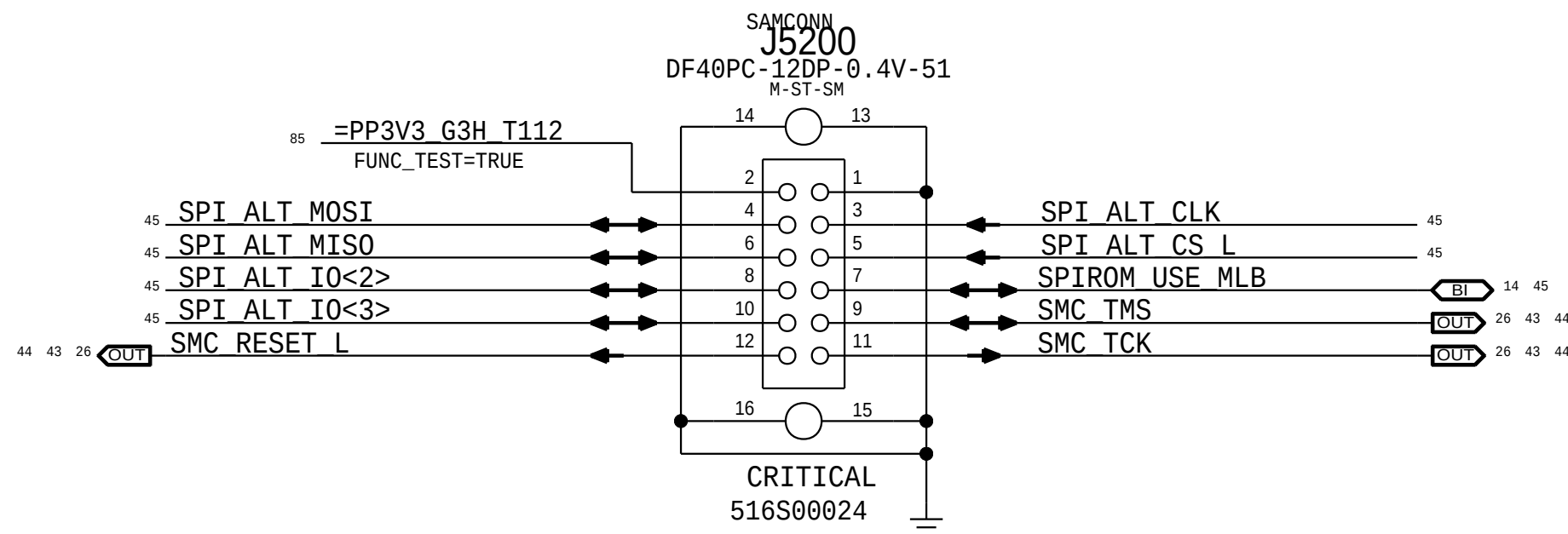


SPI BootROM

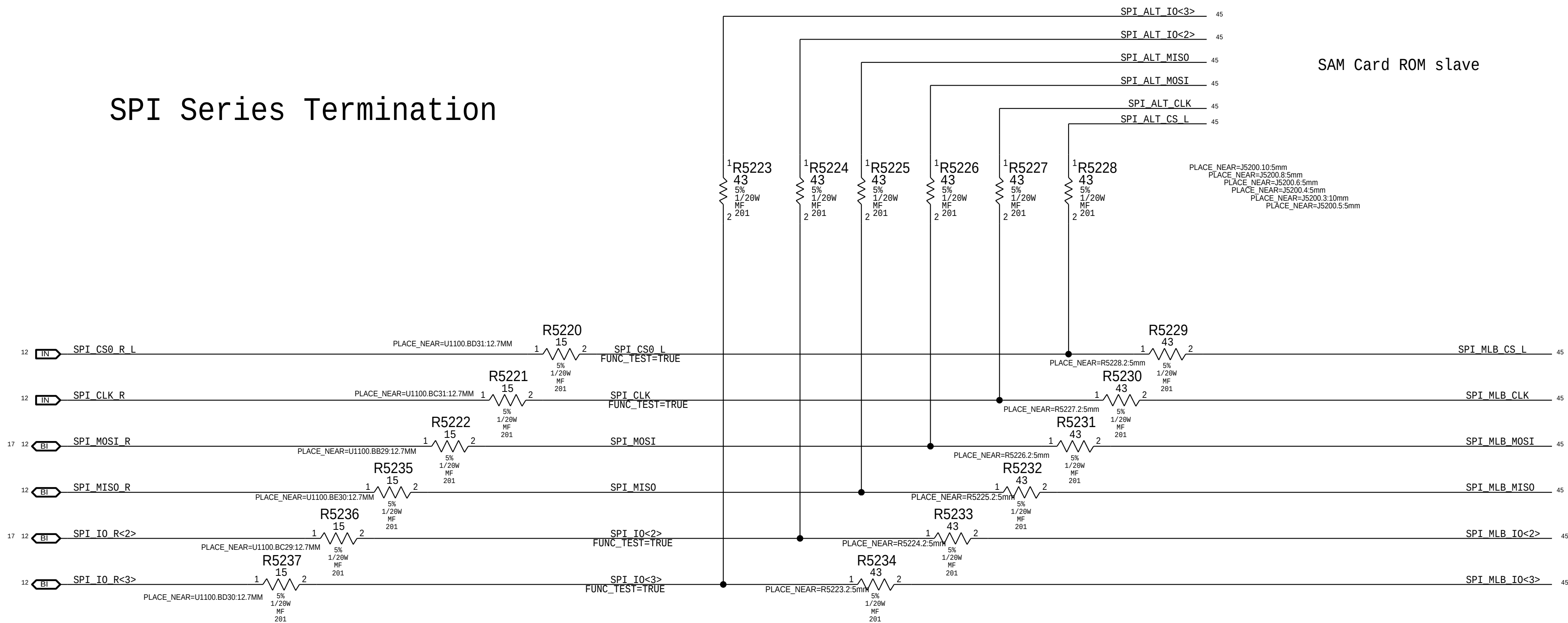
Quad_IO Mode (Mode 0 & 3) supported
SPI Freq: 50MHz for PCH, 20 MHz for SMC



SPI+SWD SAM Connector

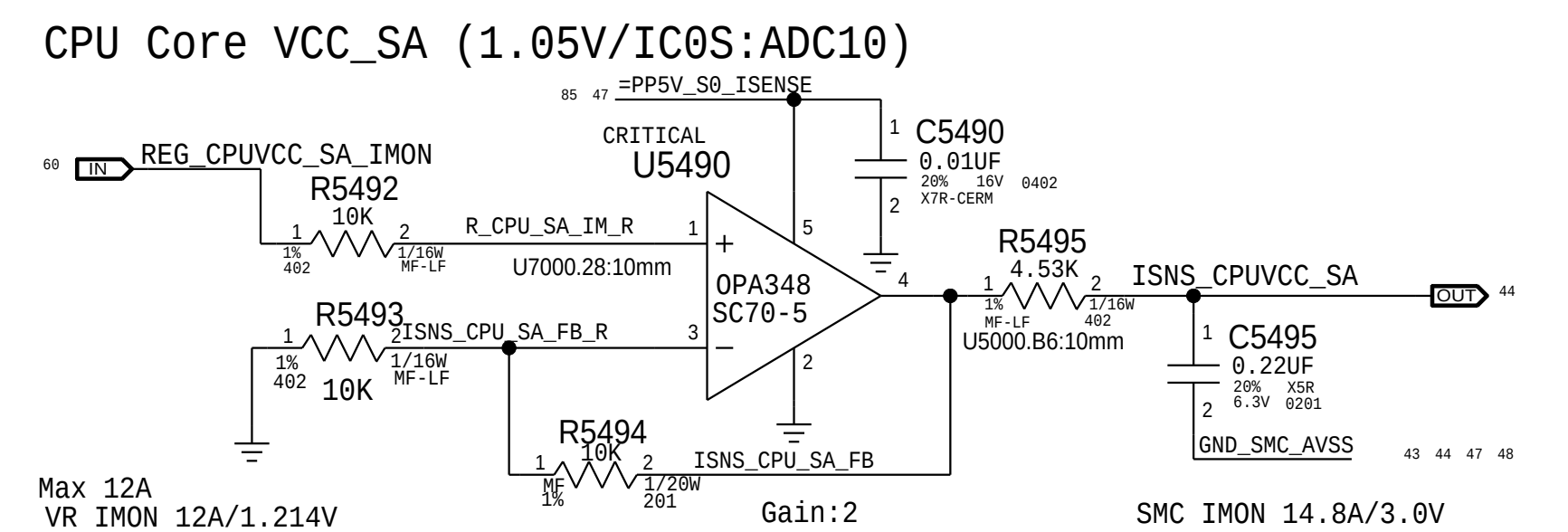
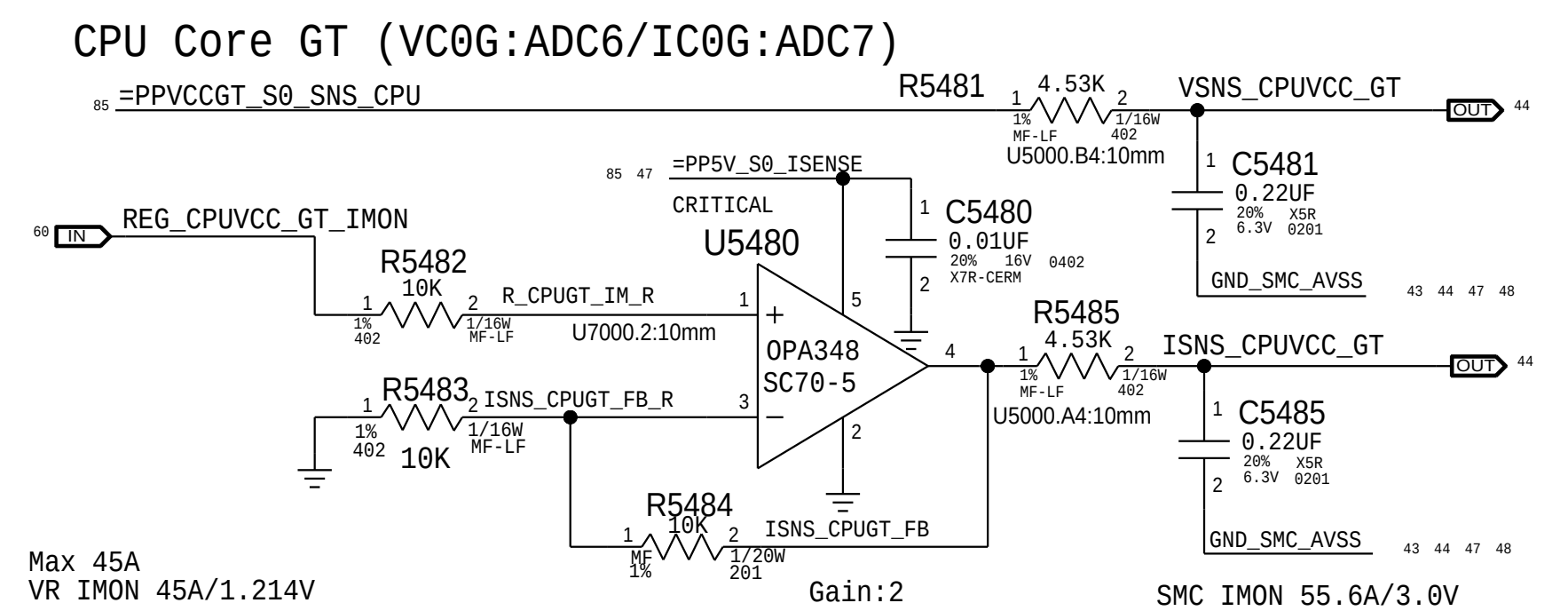
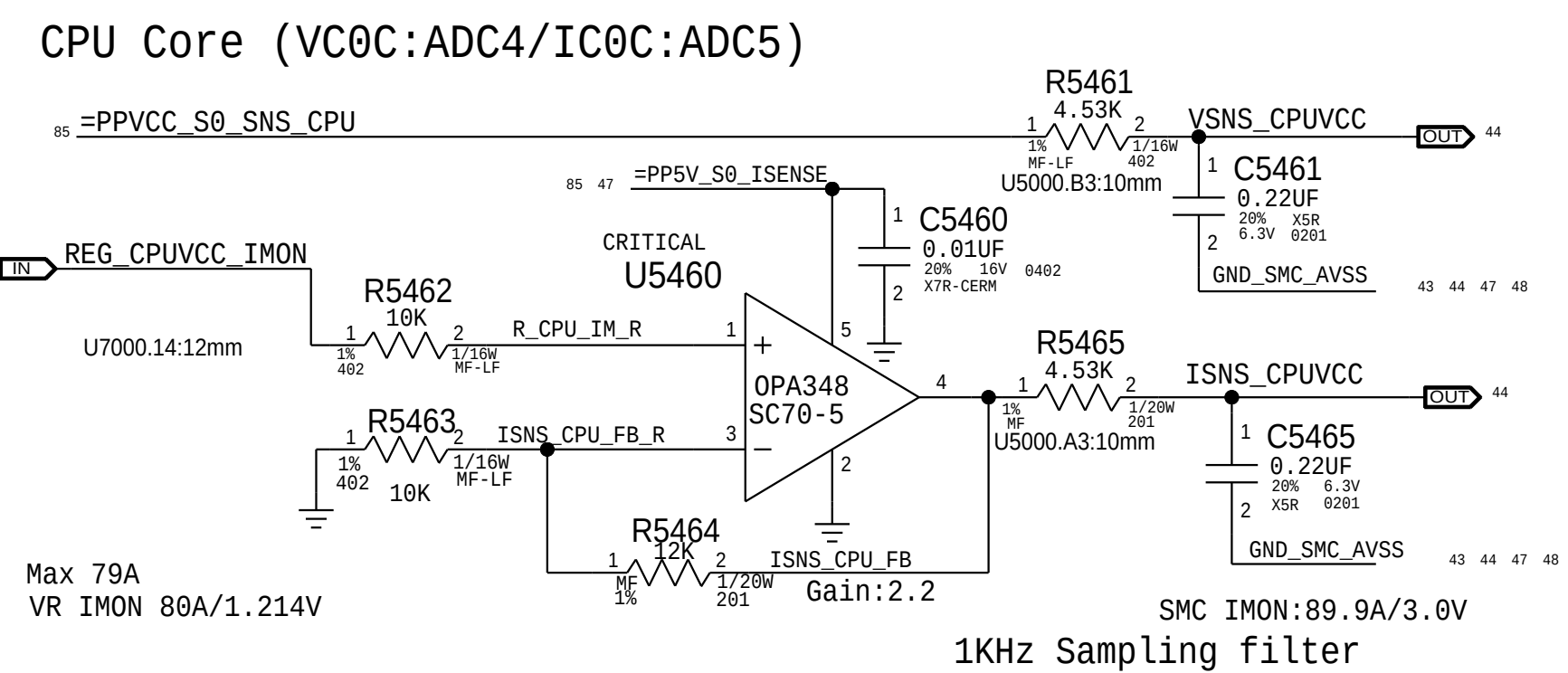
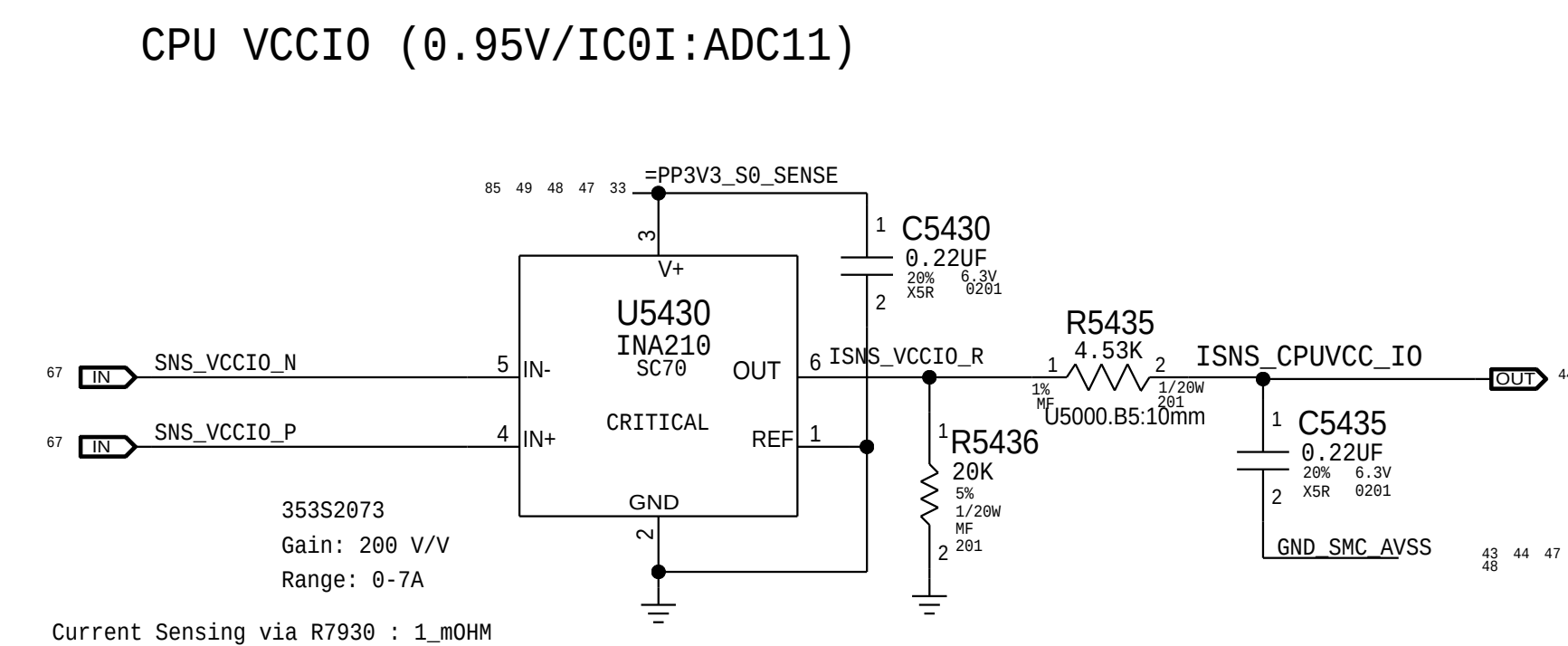
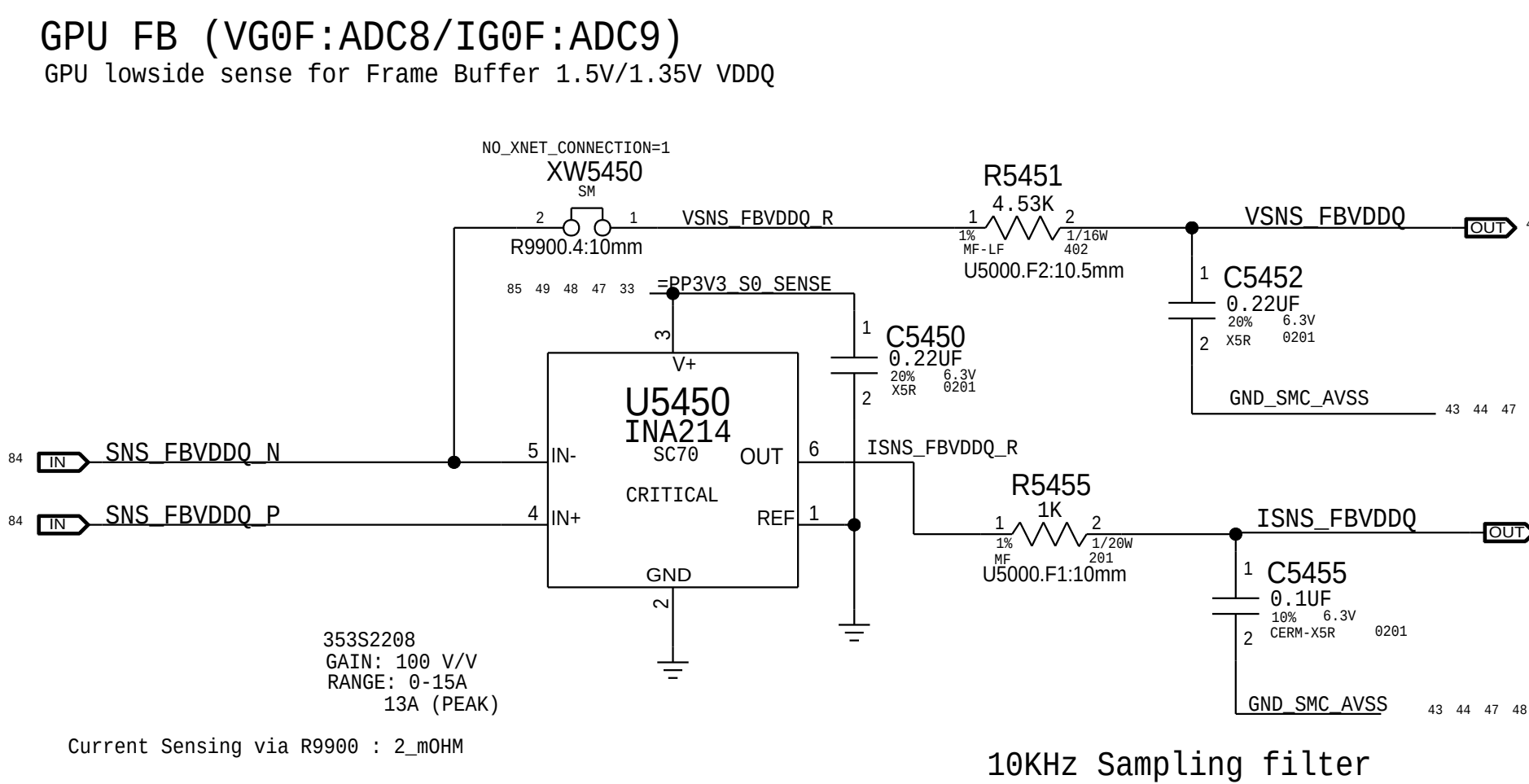
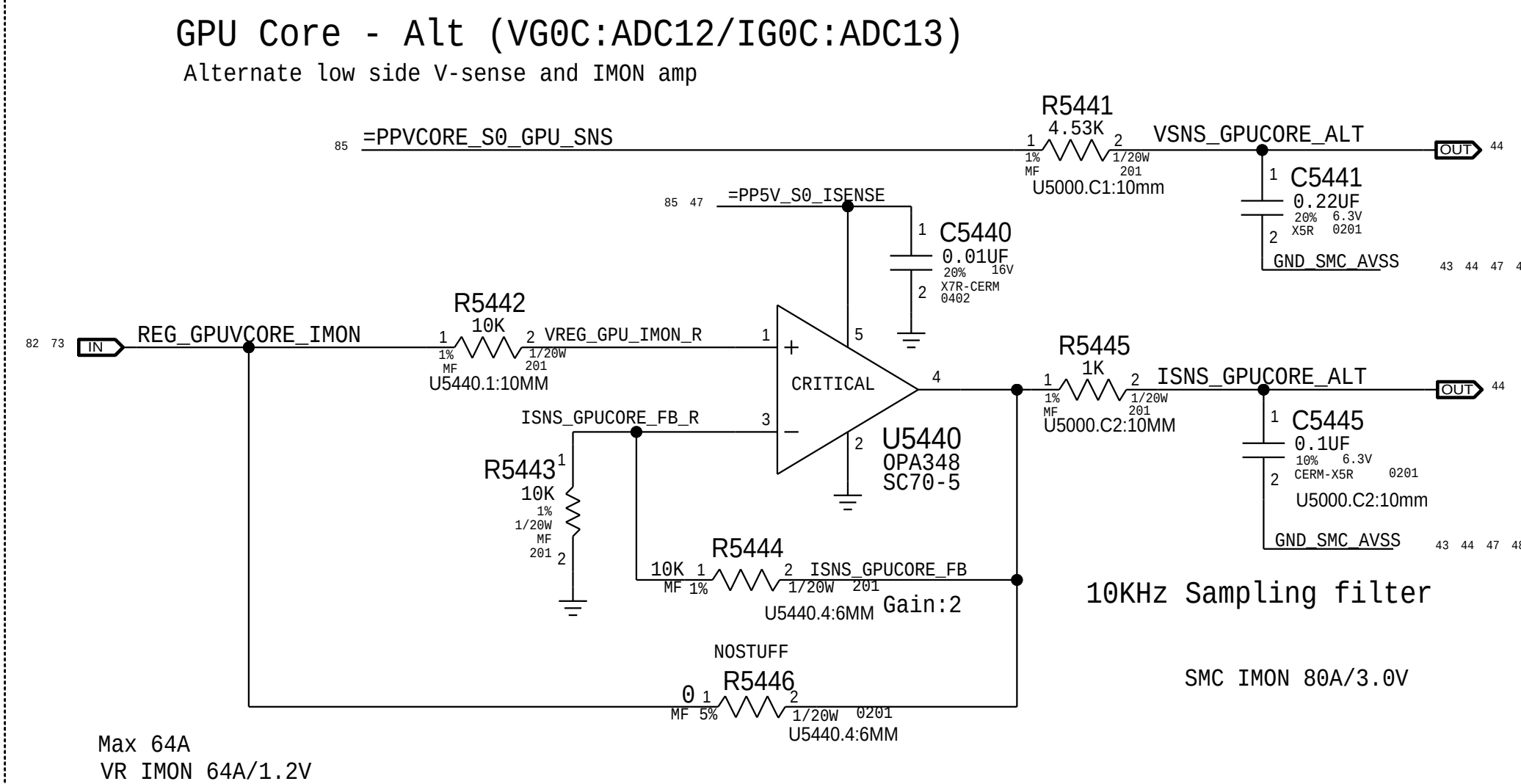
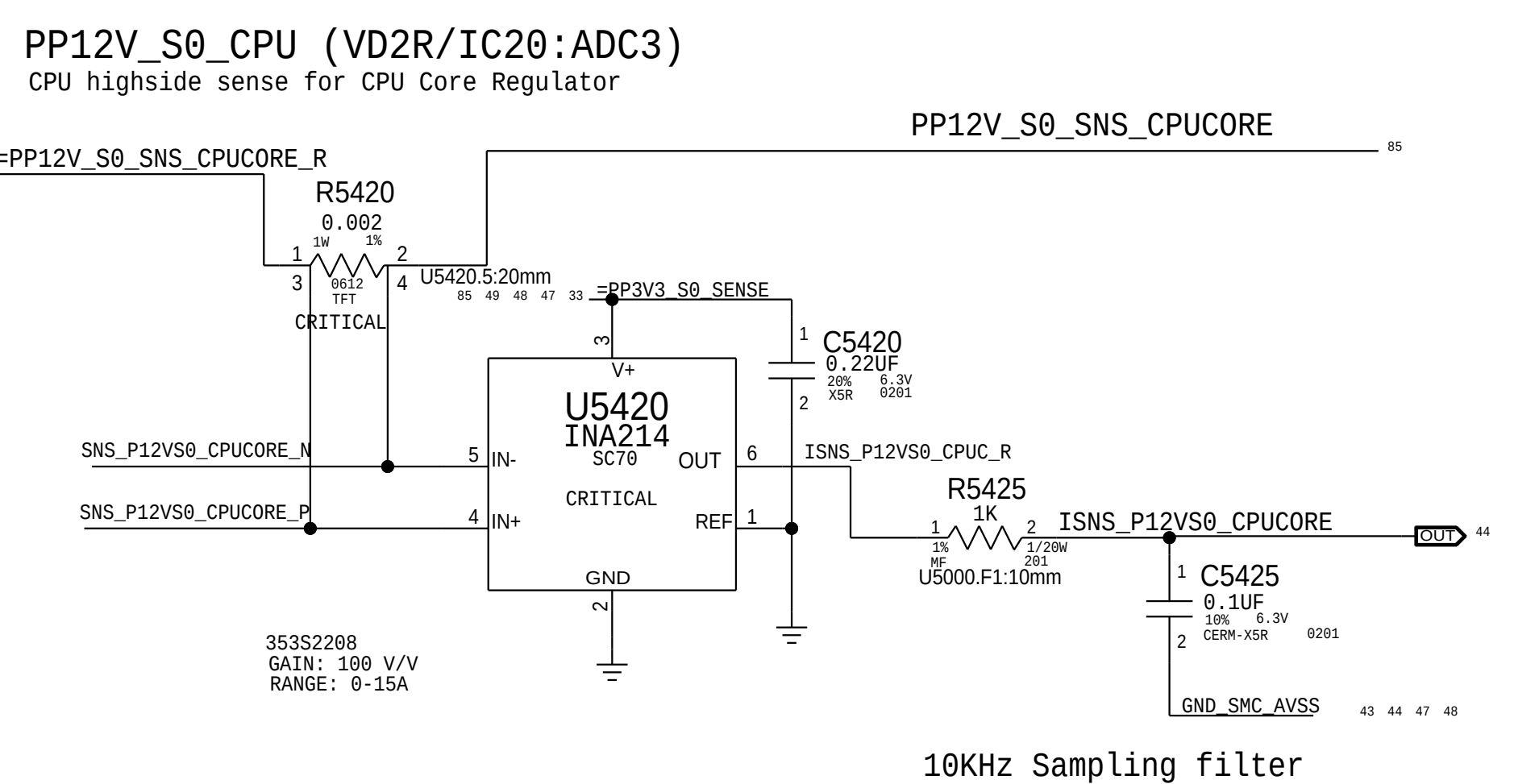
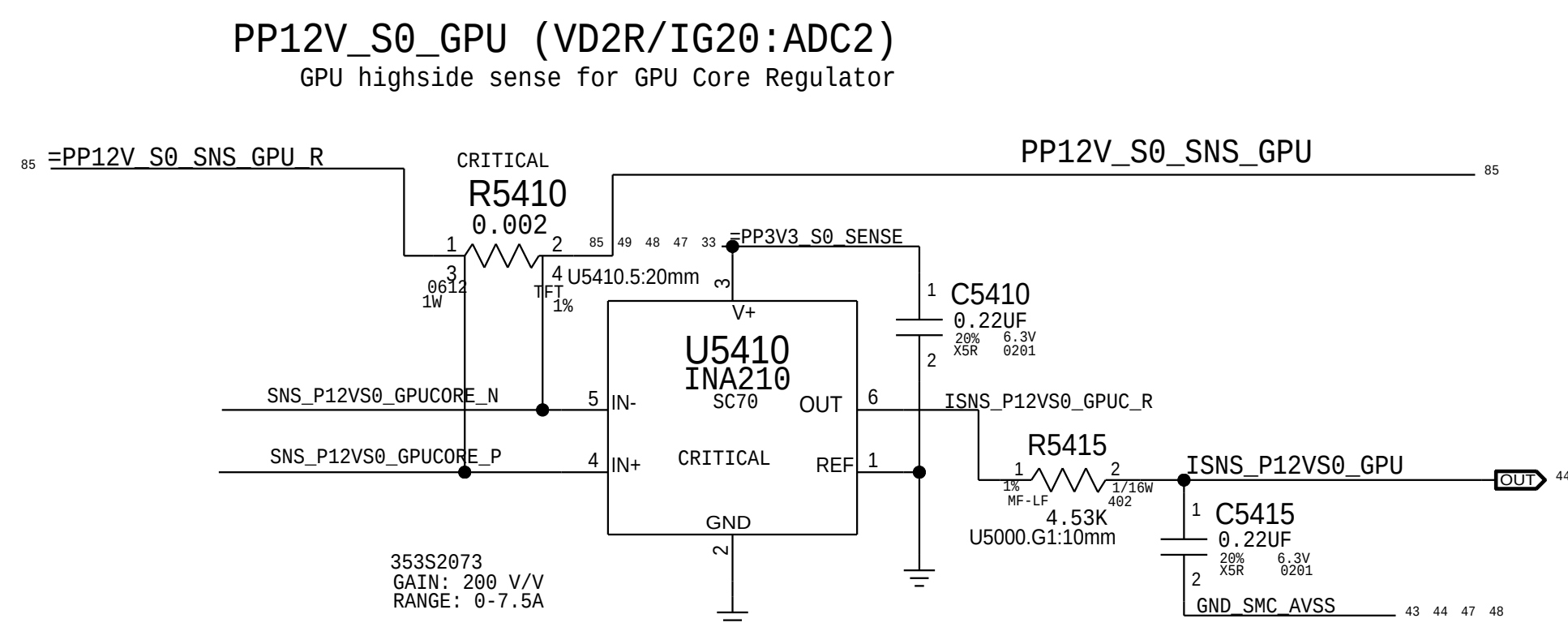
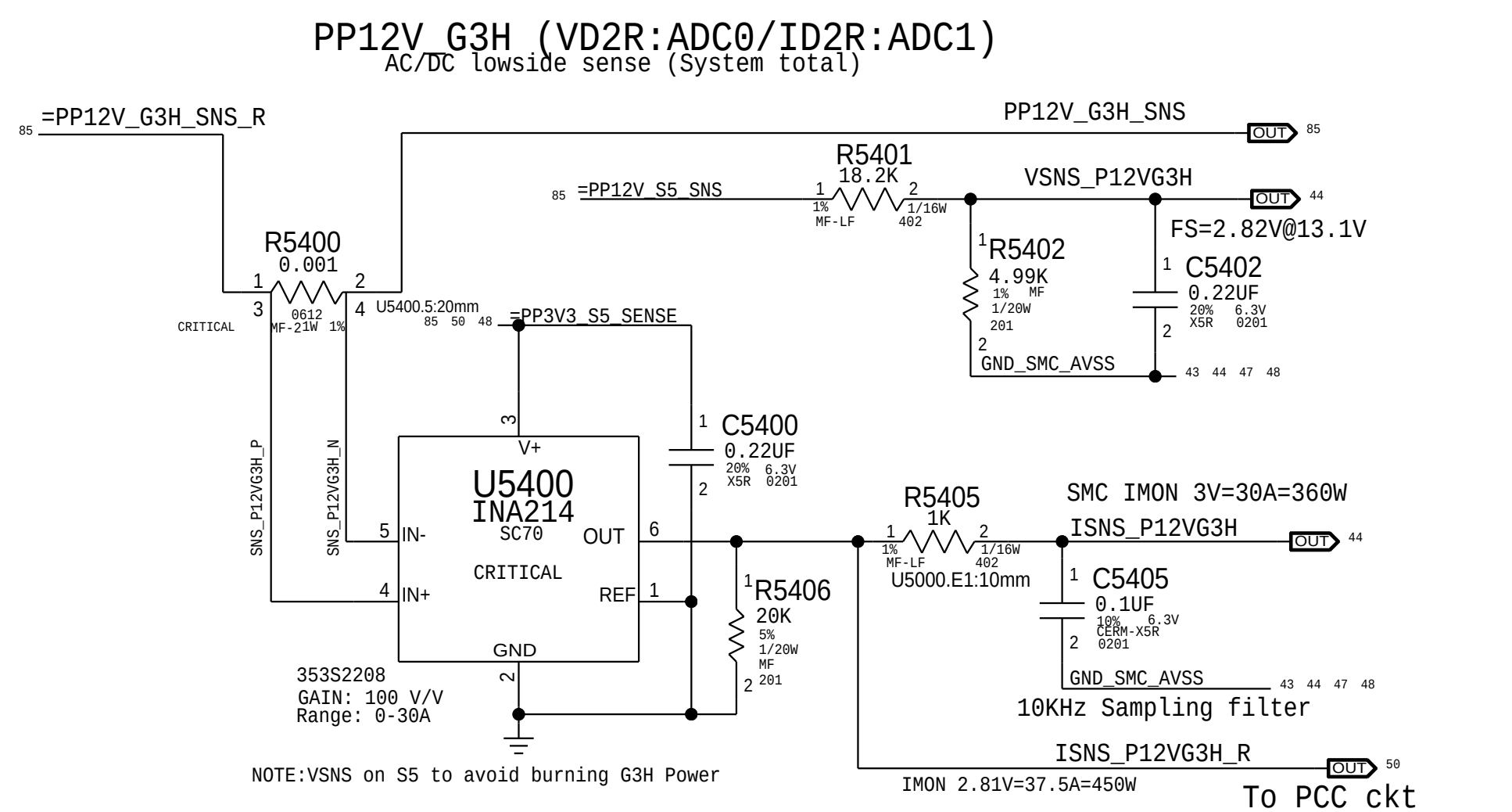


SPI Series Termination



SYNC_MASTER=J94_TONY		SYNC_DATE=10/02/2014	
PAGE TITLE			
SPI and Debug Connector			
 Apple Inc.	DRAWING NUMBER		SIZE
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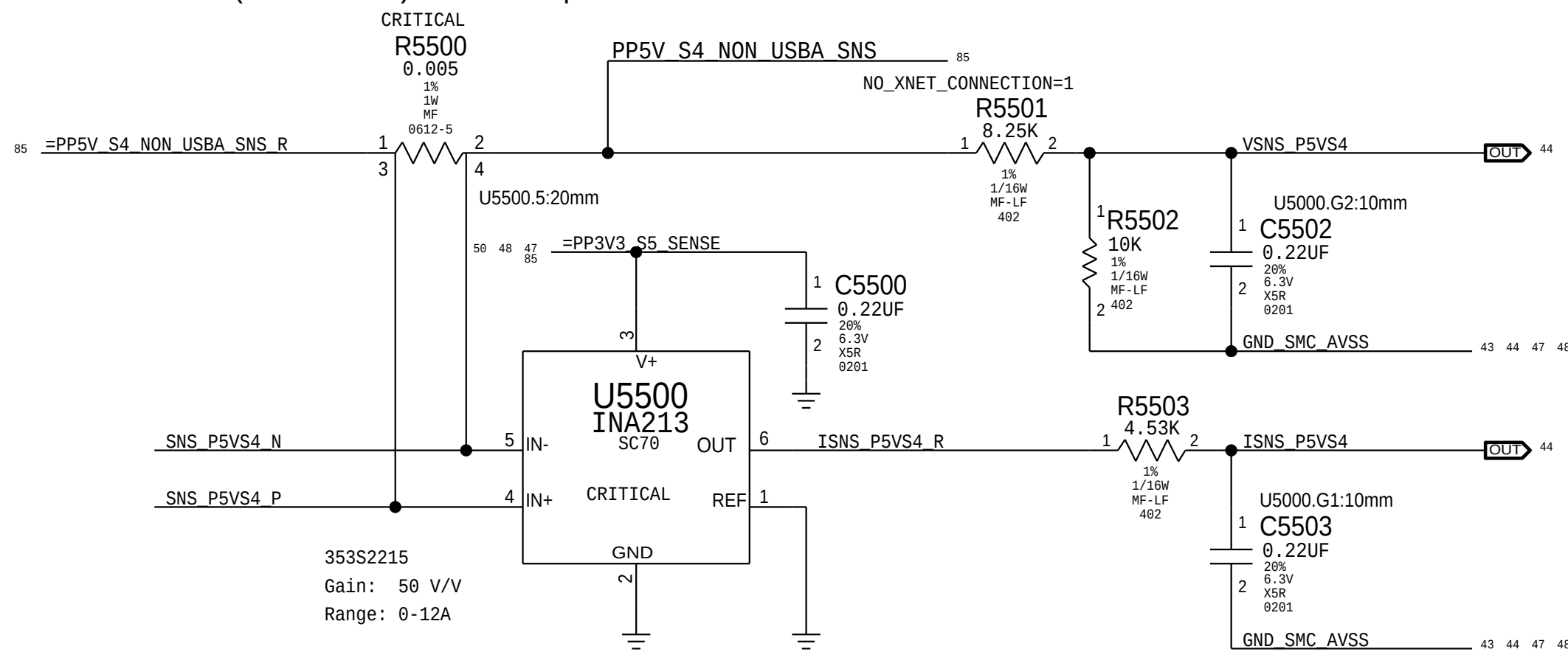
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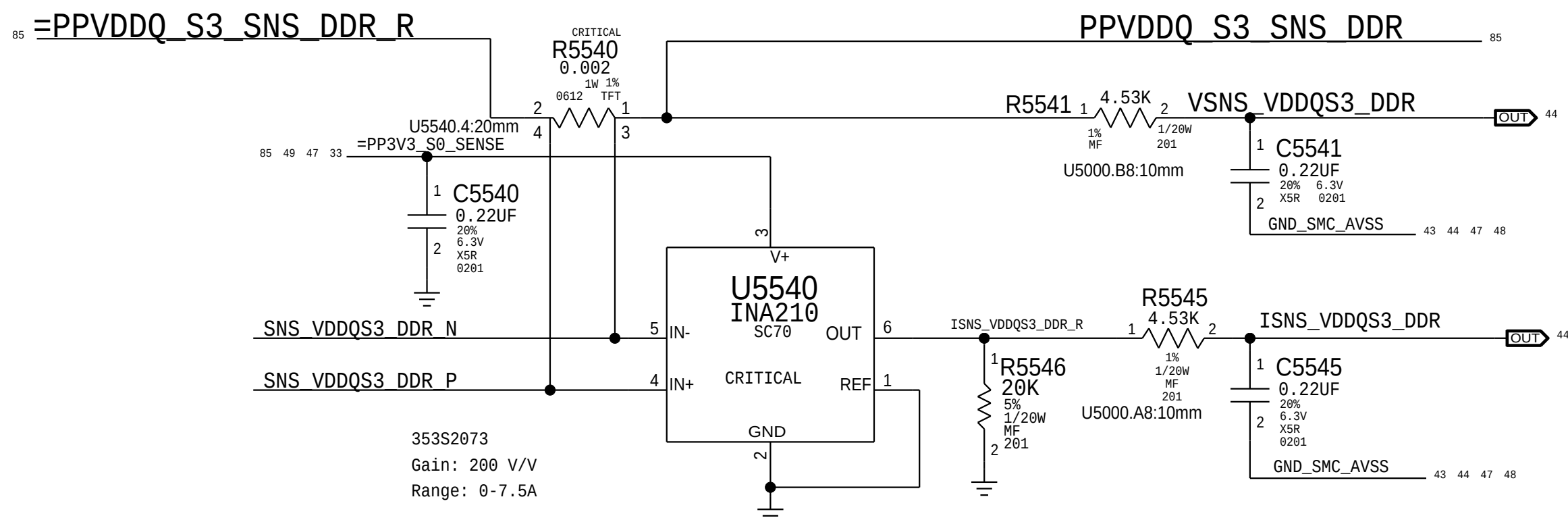
PP5V_S4 (VR54:ADC16/IR54:ADC17)

PP5V S4 (Non-USB-A) low side power distribution



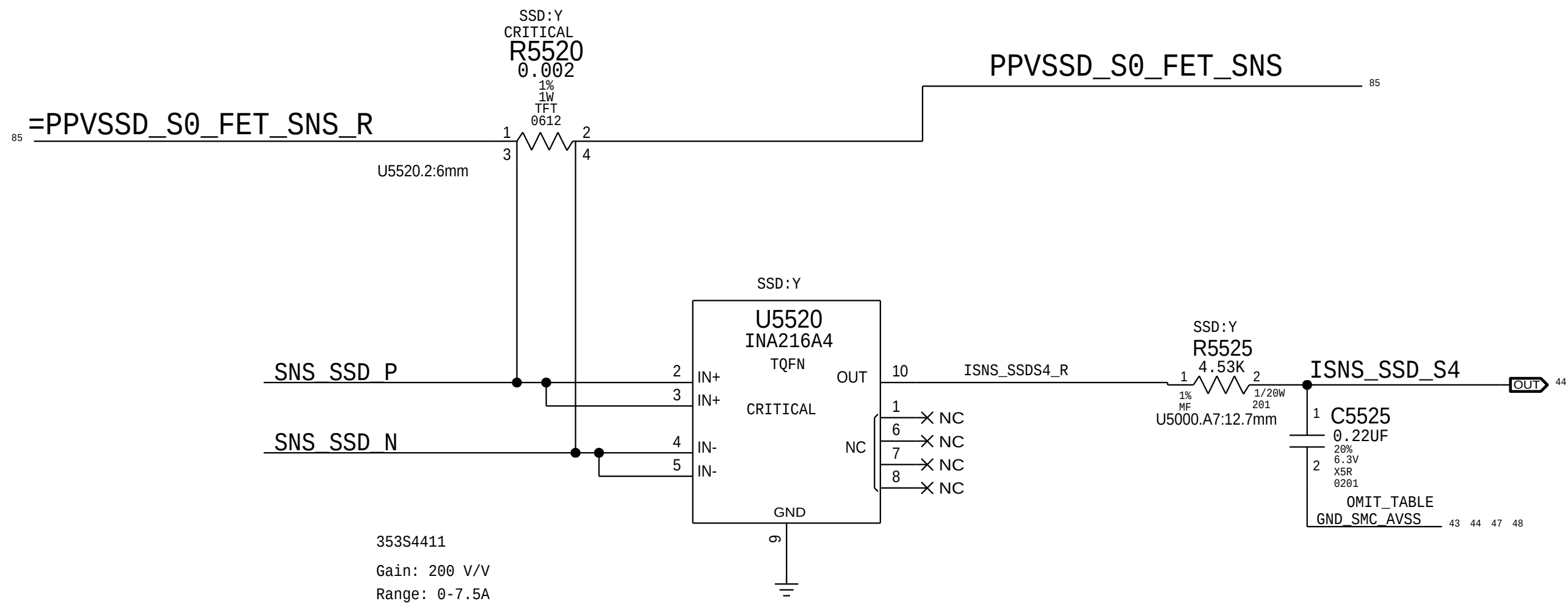
VDDQ S3 (VM0R:ADC22/IR13:ADC23)

VDDQ lowside sense for CPU Mem Controller (IC0M) and S0-DIMM modules



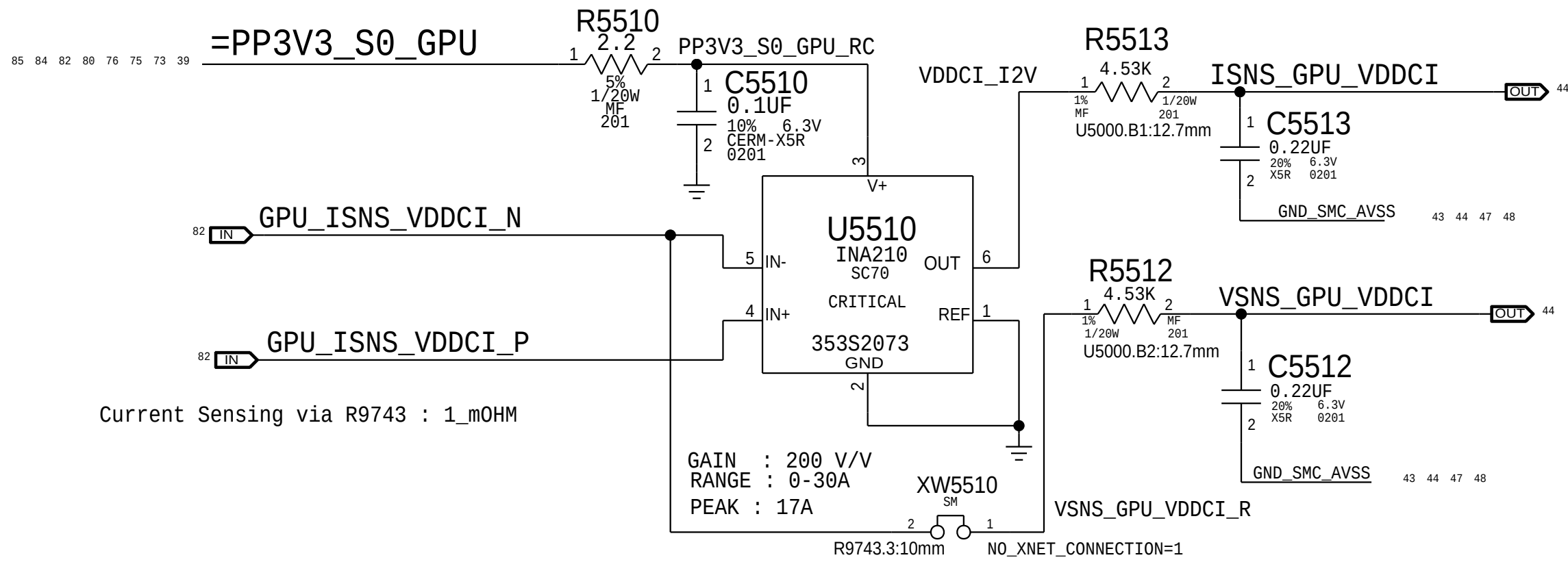
SSD S0 (VR35/IH1R:ADC21)

I-SENSE FOR SSD / V-SENSE FOR PPSSD_S0)



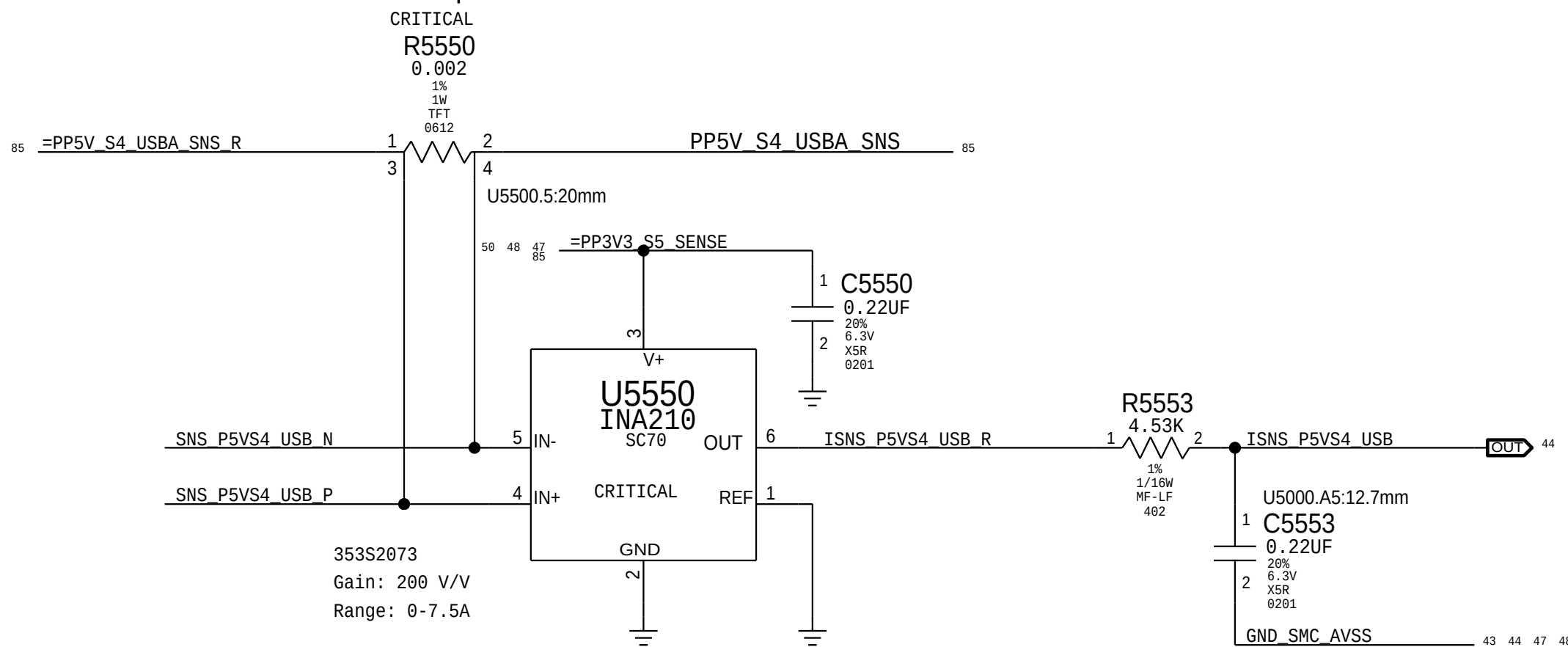
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S9394	1	CAP, 0.22UF, 201	C5525	SSD:Y
117S9201	1	RES, 0 OHM, 201	C5525	SSD:N

GPU_VDDCI S0 (VG0I:ADC14/IG0I:ADC15)



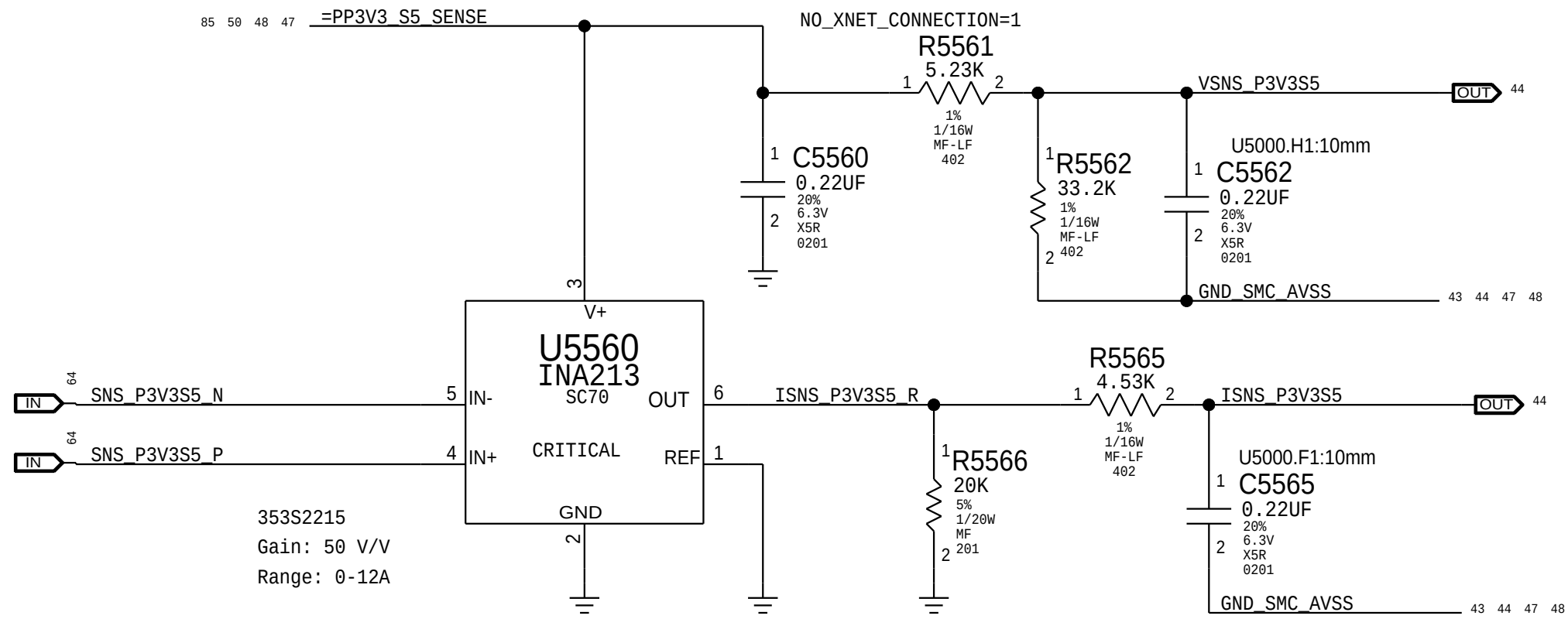
PP5V_S4_USB (VR54/IU54:ADC20)

PP5V S4 USB low side power distribution



PP3V3_S5 (VR35:ADC18/IR35:ADC19)

PP3V3 S5 low side power distribution



C

B

A

D

C

B

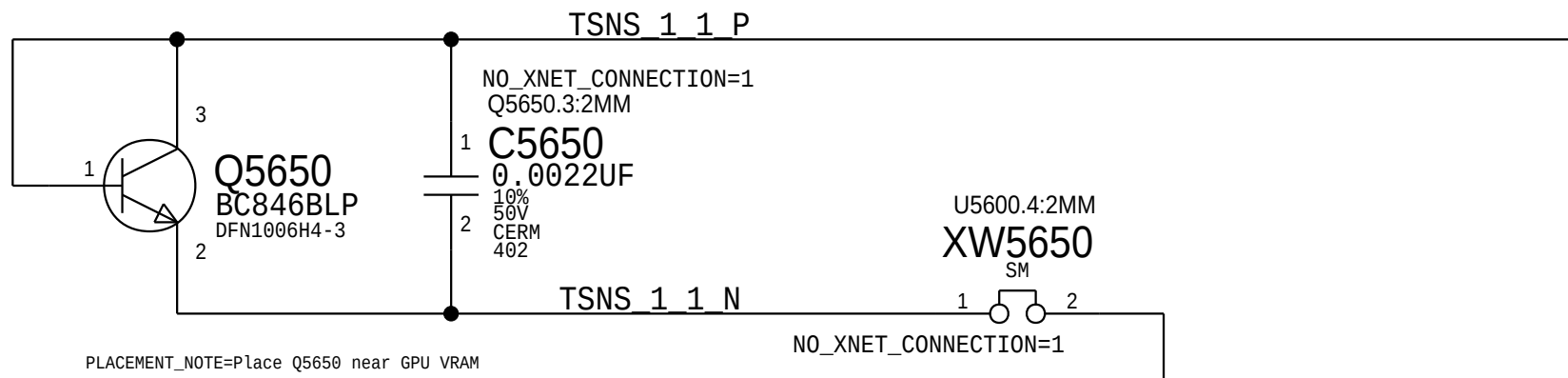
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BOM_COST_GROUP=SENSORS

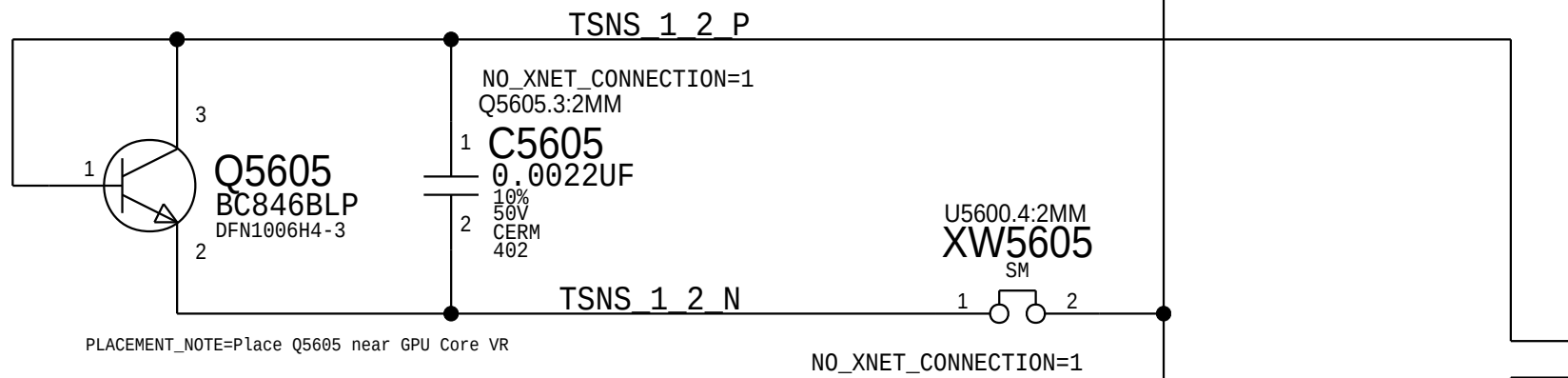
DRAWING NUMBER 051-01477		SIZE D
REVISION 3.26.0		BRANCH
PAGE 55 OF 119		SHEET
48 OF 87		

TEMPERATURE SENSOR T1

GPU VRAM Prox (TG1p)

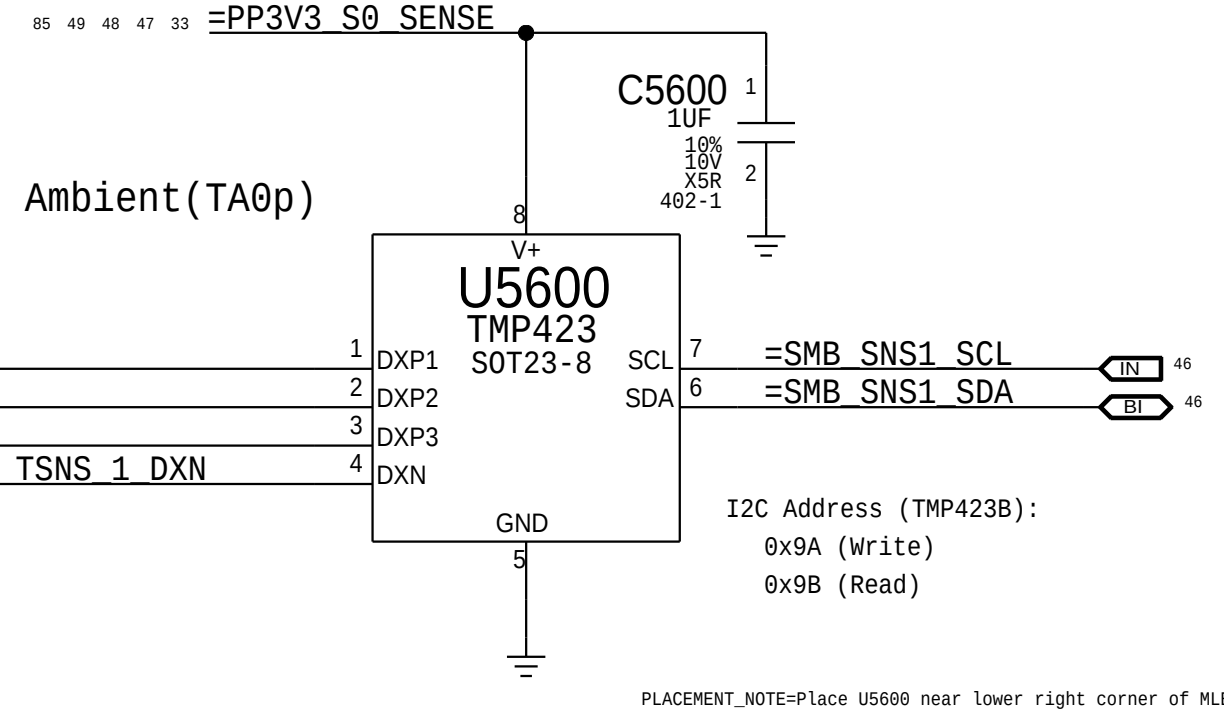
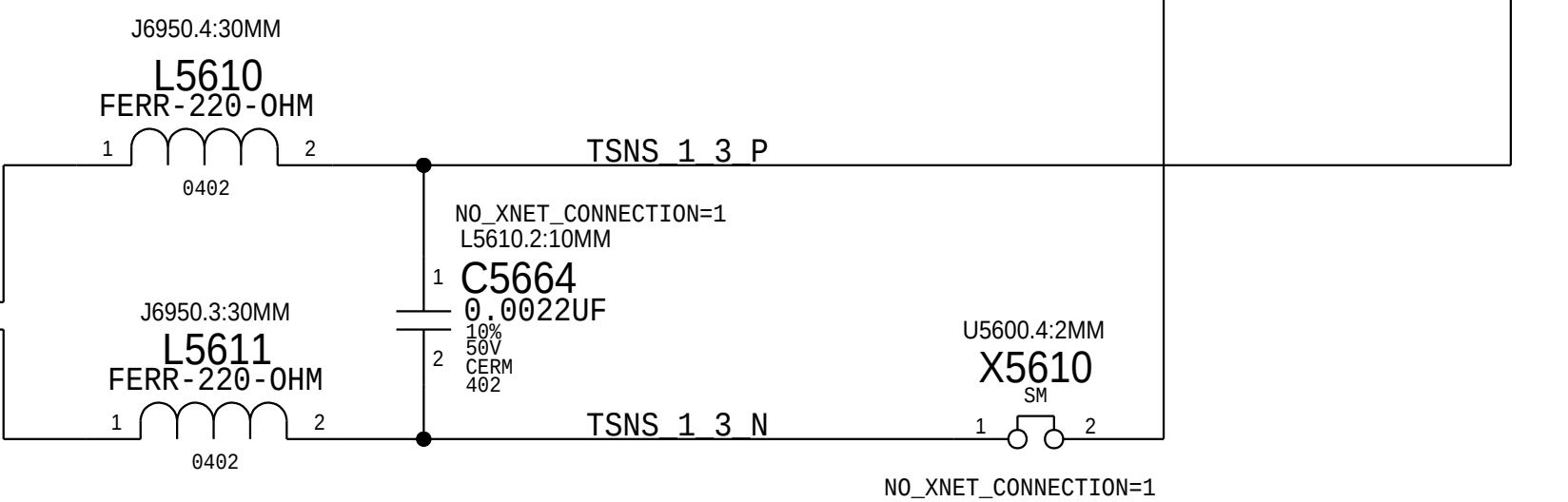


MLB Prox (Tm2p)



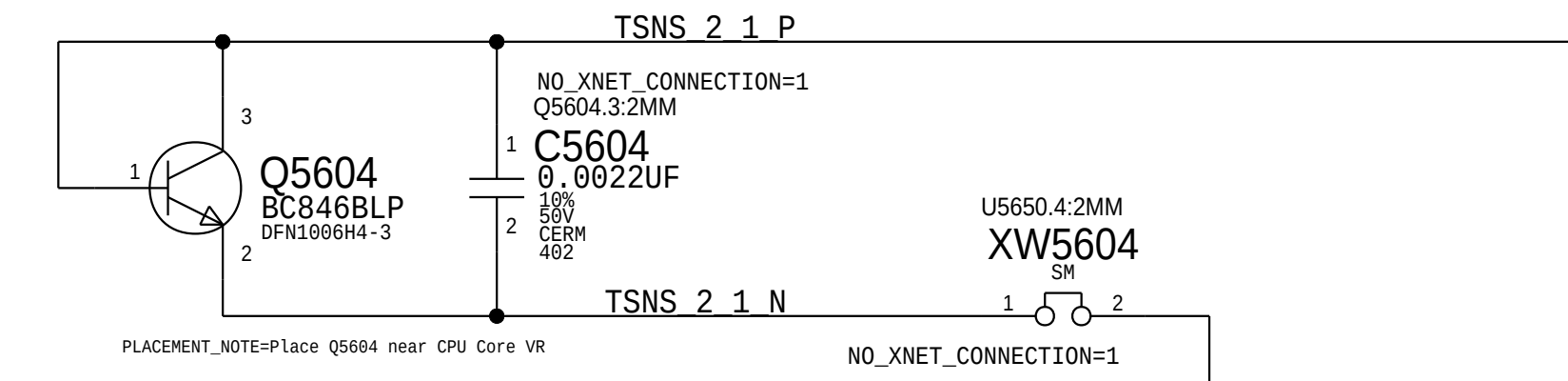
AC/DC(Tp2h)

SNS ACDC P
SNS ACDC N

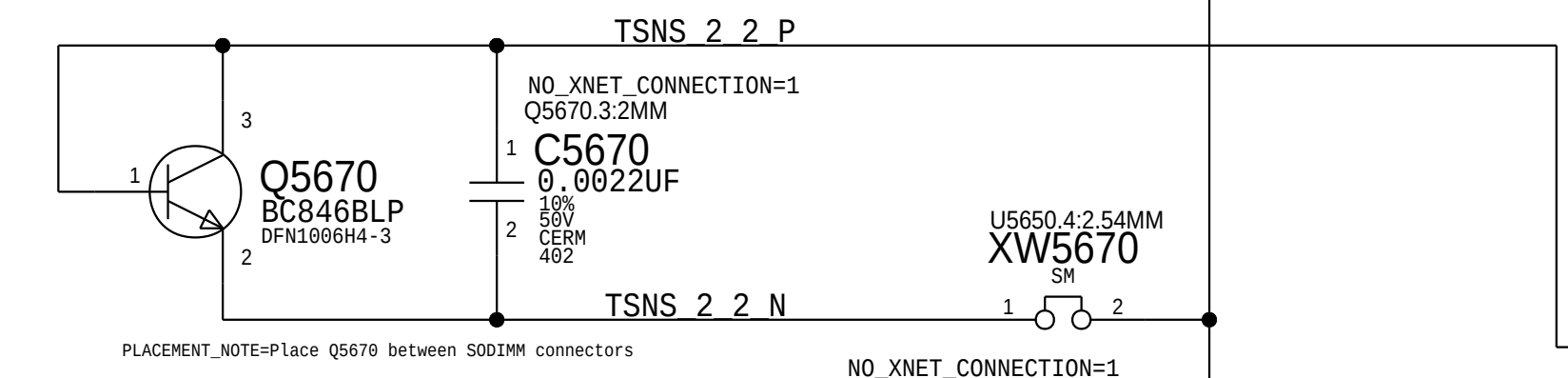


TEMPERATURE SENSOR T2

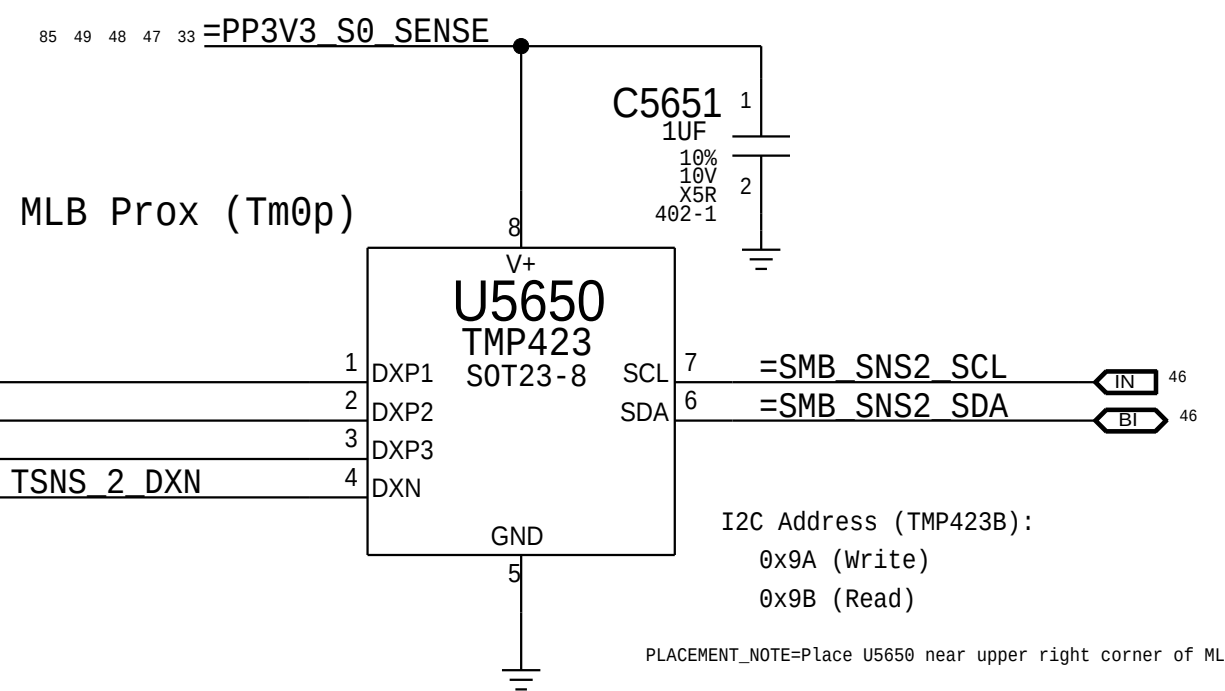
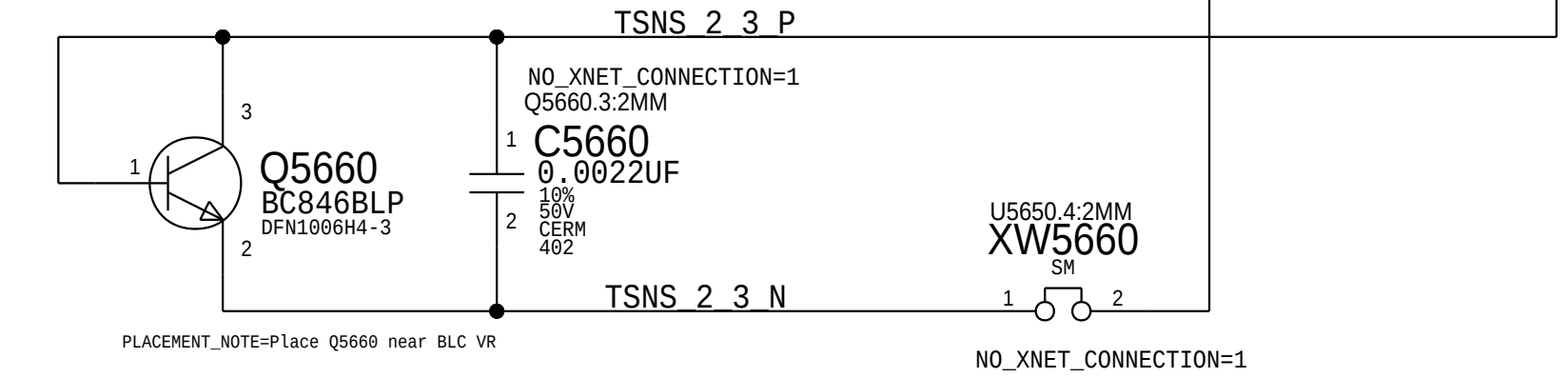
CPU Prox (TC0p)



DIMM Prox (TM0p)



BLC Prox (Tb0p)



BOM_COST_GROUP=SENSORS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

SYNC_MASTER=YEUNGR		SYNC_DATE=03/16/2016	
PAGE TITLE			
SENSORS: Temperature Sensors			
	Apple Inc.	DRAWING NUMBER	051-01477
		SIZE	D
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		BRANCH	
		PAGE	56 OF 119
		SHEET	49 OF 87

D

C

B

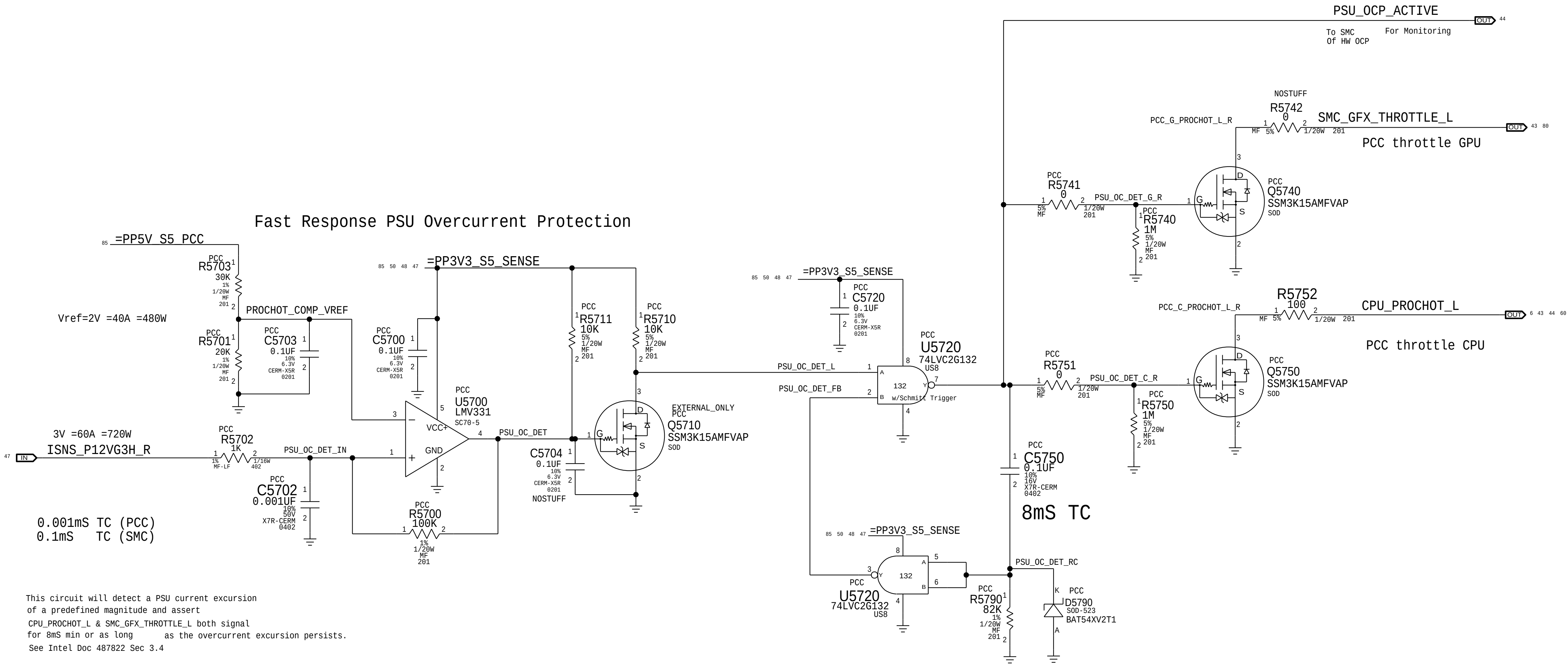
A

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A



BOM_COST_GROUP=SENSORS

SYNC_MASTER=jerry

SYNC_DATE=11/28/2015

PAGE TITLE

SENSORS: PSU/PCC



Apple Inc.

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051-01477

SIZE

D

REVISION

3.26.0

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D

C

B

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C

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A

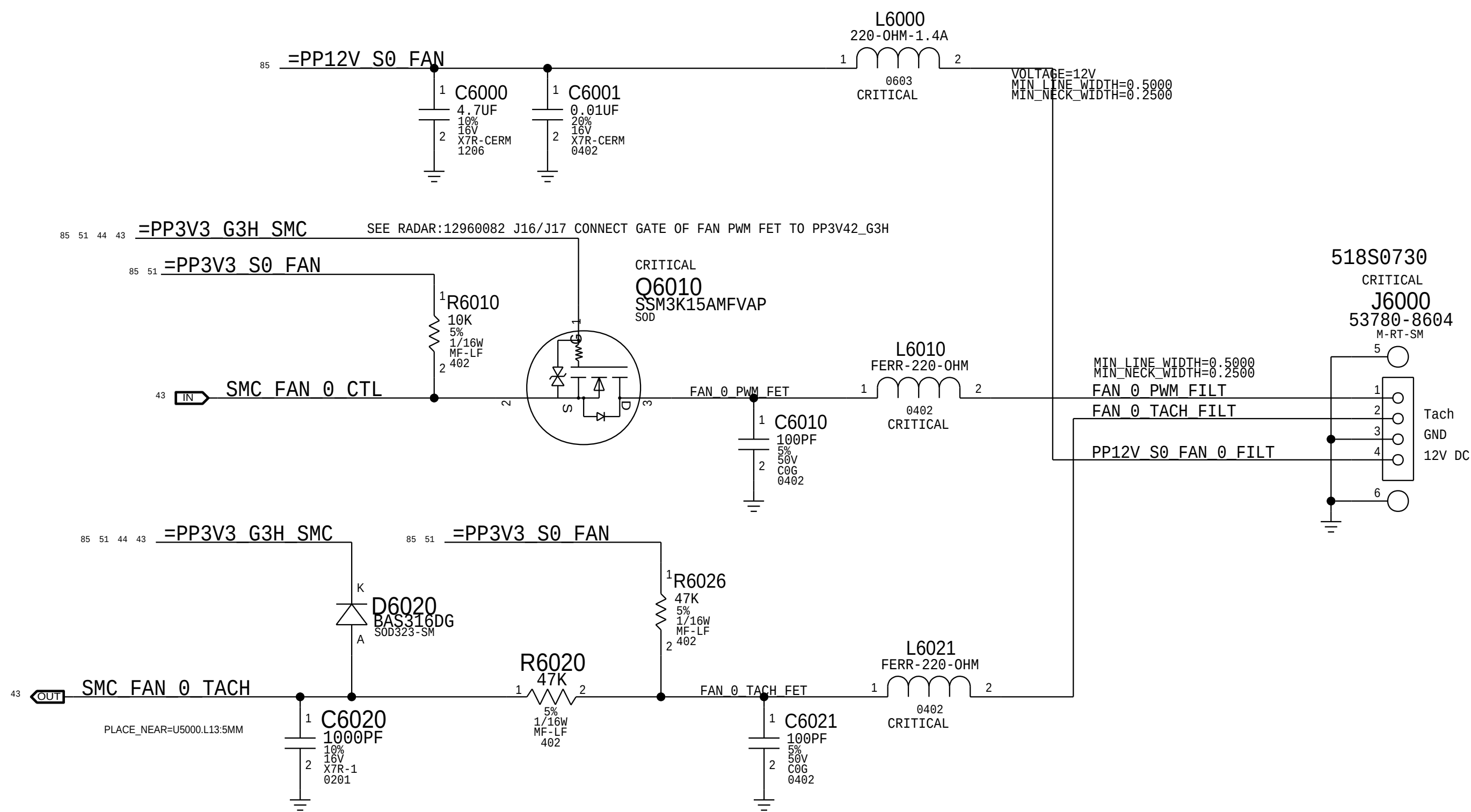
SMC Fan 0 (System)

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

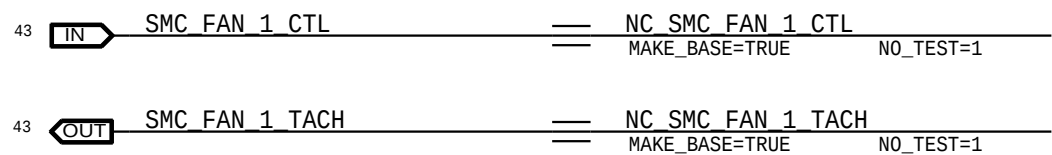
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

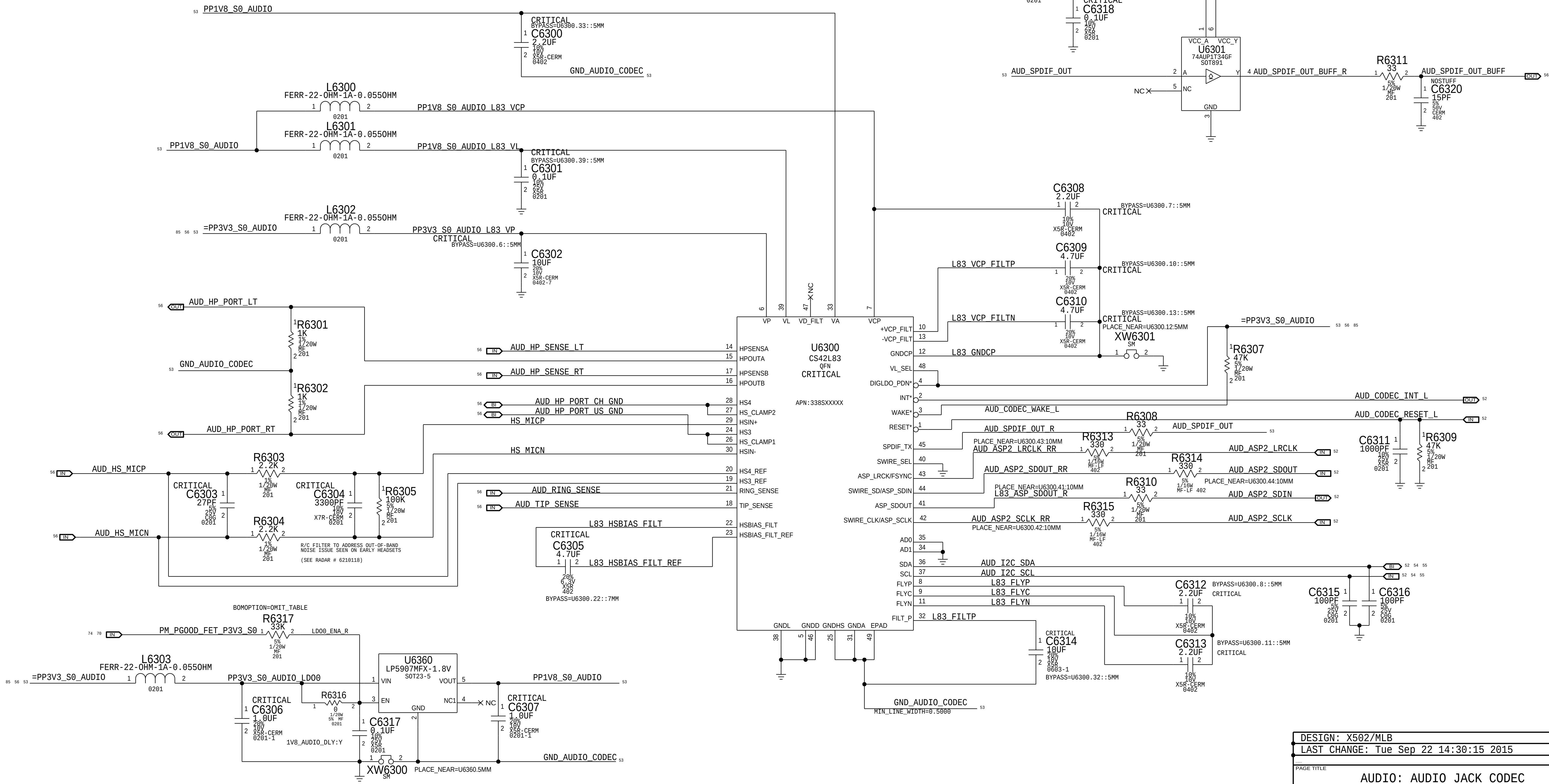
SMC Fan 1 (Unused)



BOM_COST_GROUP=FAN

SYNC_MASTER=J117_MLB		SYNC_DATE=08/01/2014	
PAGE TITLE			
System Fan Connector			
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		BRANCH	
		PAGE 60 OF 119	
		SHEET 51 OF 87	

AD1 AD0 I2C_ADDR
GND GND 0X48 <--
GND 1.8V 0X49
1.8V GND 0X4A
1.8V 1.8V 0X4B



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11750083	1	RES, 33K, 0HM, 201	R6317	1V8_AUDIO_DLY:Y
11750201	1	RES, 0 OHM, 201	R6317	1V8_AUDIO_DLY:N

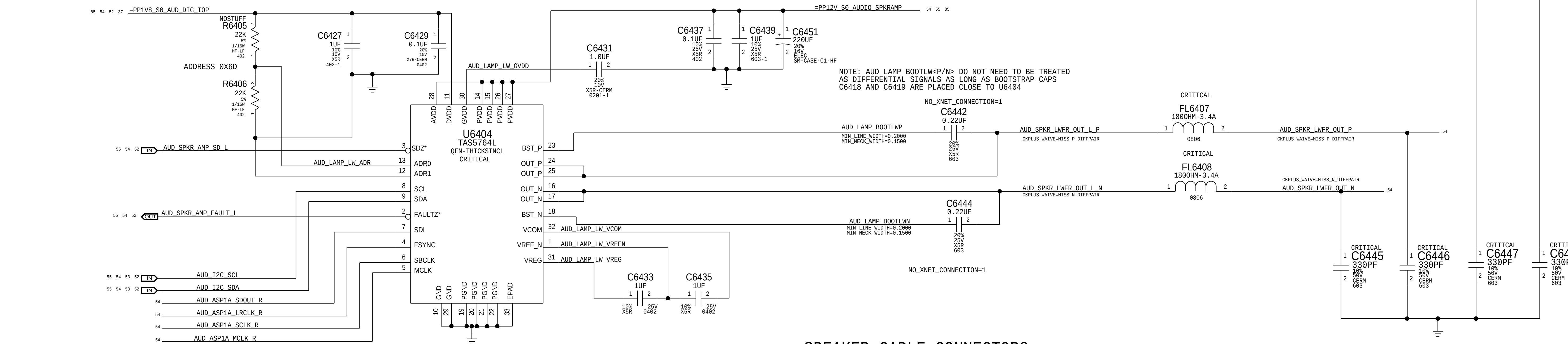
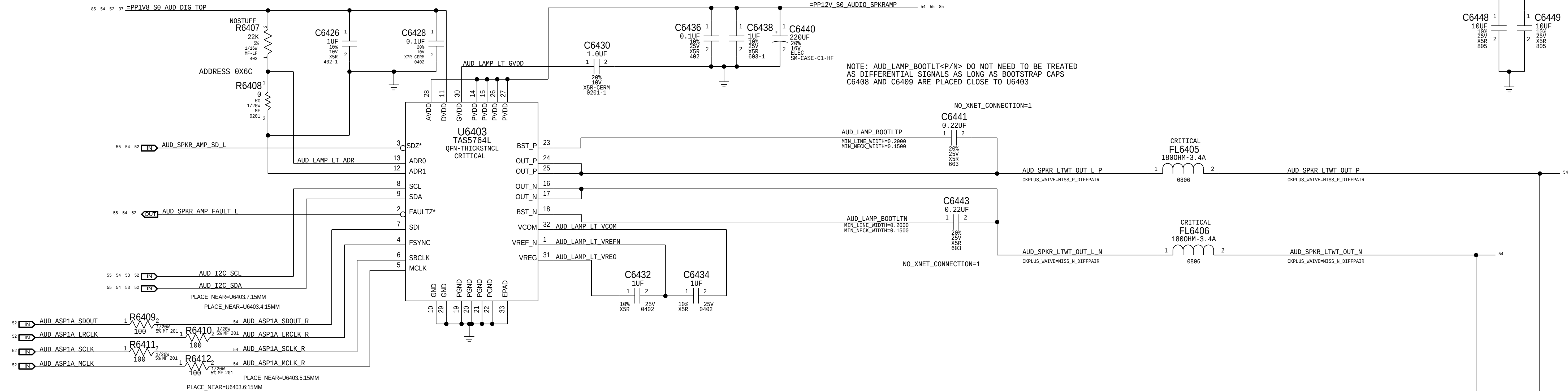
----- J133/134
----- J135

DESIGN: X502/MLB		
LAST CHANGE: Tue Sep 22 14:30:15 2015		
PAGE TITLE		
AUDIO: AUDIO JACK CODEC		
 Apple Inc.	DRAWING NUMBER	051-01477
	REVISION	3.26.0
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		SHEET 53 OF 87

BOM_COST_GROUP=AUDIO0

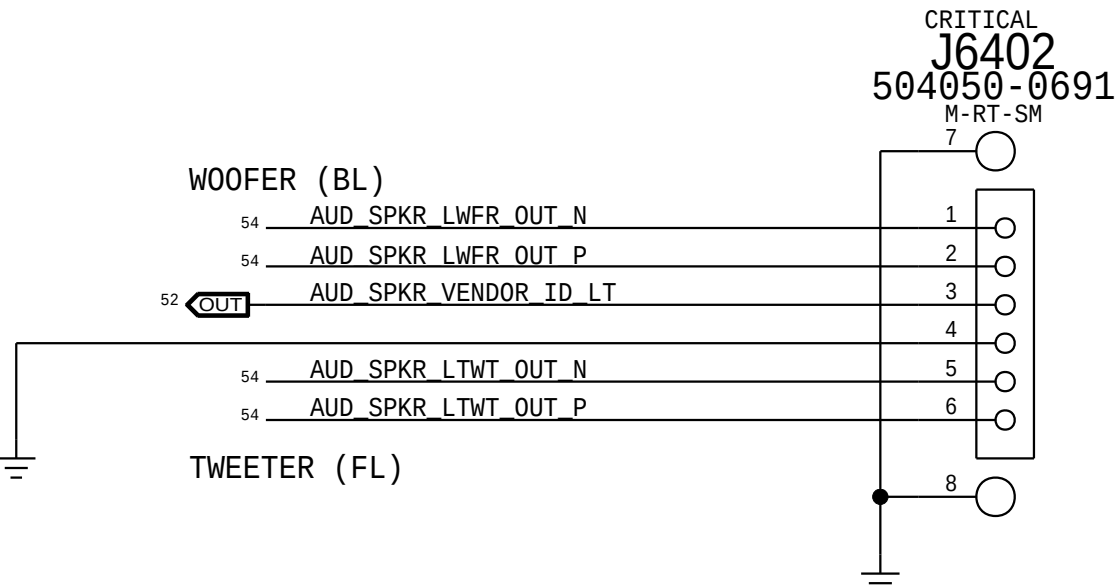
2X MONO SPEAKER AMPLIFIERS (TAS5764L)

SPEAKER AMPLIFIERS - LEFT CHANNEL



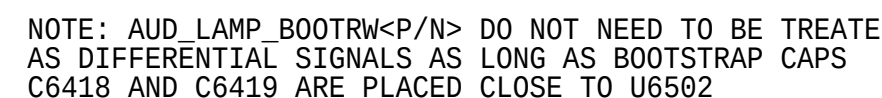
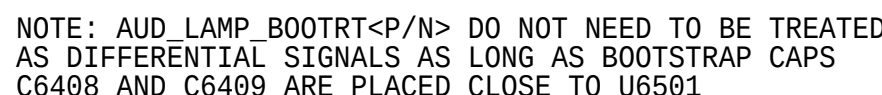
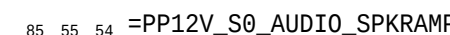
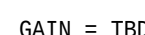
SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862



DESIGN: X502/DEV_MLB_U	
LAST CHANGE: Wed Feb 18 17:12:24 2015	
PAGE TITLE	
AUDIO: LEFT SPKR AMP	
 Apple Inc.	DRAWING NUMBER 051-01477
	REVISION 3.26.0
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	PAGE 64 OF 119
	SHEET 54 OF 87

BOM_COST_GROUP=AUDIO0



DESIGN: X502/DEV_MLB_U			
LAST CHANGE: Wed Feb 18 17:12:24 2015			
PAGE TITLE			
AUDIO: RIGHT SPKR AMP			
 Apple Inc.	DRAWING NUMBER 051-01477		SIZE D
	REVISION 3.26.0		
	BRANCH		
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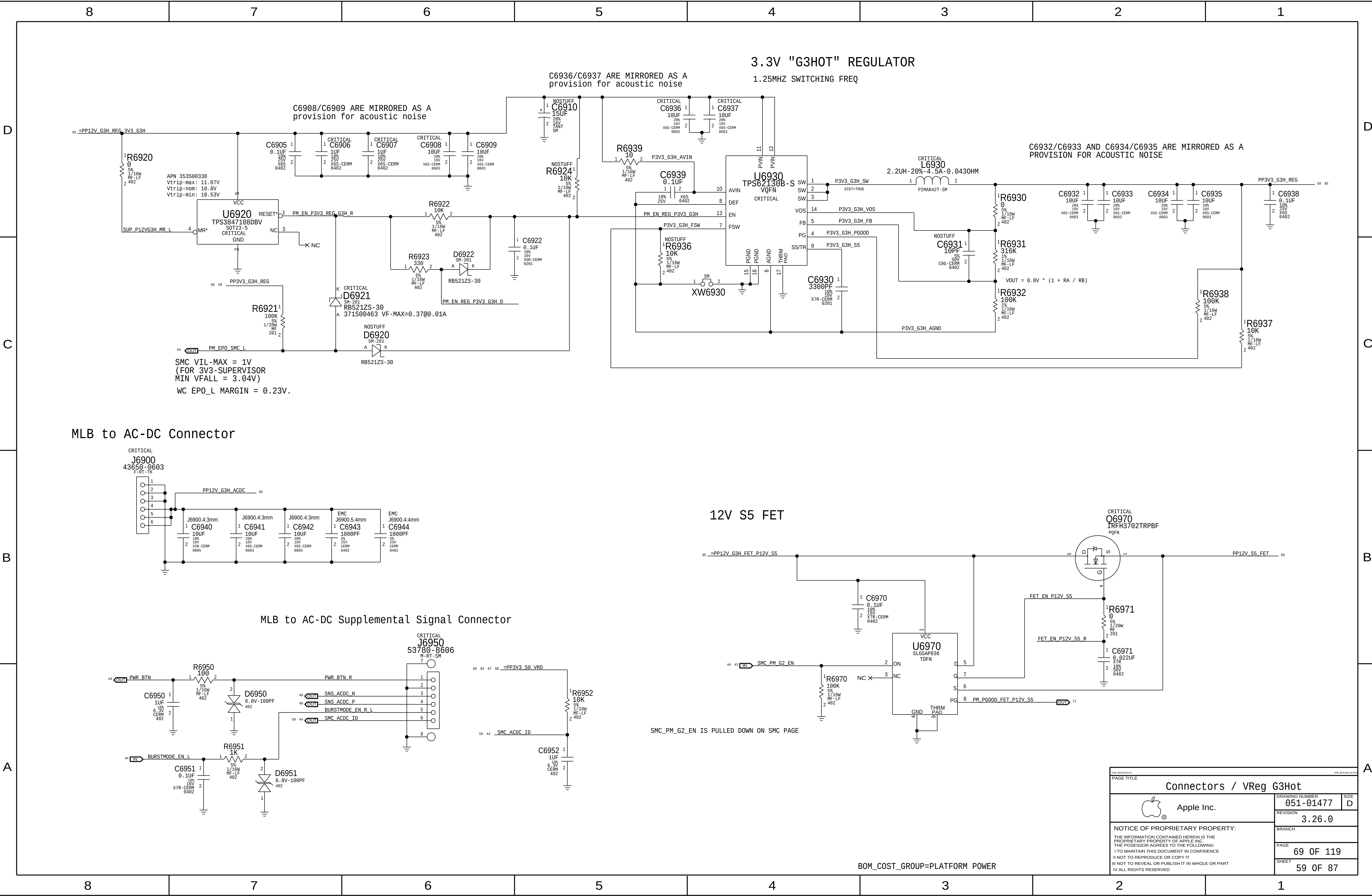
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																D
																C
																B
																A
8		7		6		5		4		3		2		1		

DATE: 06/08/2015		
PAGE TITLE		
AUDIO: EMPTY		
 Apple Inc.		DRAWING NUMBER 051-01477
		REVISION 3.26.0
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		SHEET 57 OF 87

SYMC_MASTER-1118

SYMC_DATE-04/14/2018

PAGE TITLE			AUDIO: EMPTY		
 Apple Inc.		DRAWING NUMBER	051-01477	SIZE	D
		REVISION	3.26.0		
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Connectors / VReg G3Hot		
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	REVISION	3.26.0
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BOM_COST_GROUP=PLATFORM POWER



CPU VCC Regulator
EDC = 79A
TDC = 61A

CPU VCORE Phase 1

CPU VCORE Phase 2

CPU Phase 3

CPU VCORE OUTPUT DECOUPLING

PPVCC_S0_CPU_REG

BOM_COST_GROUP=CPU & CHIPSET

DRAWING
LAST_MODIFIED=Wed Sep 14 23:01:32 2016
PAGE TITLE
CPU & CHIPSET: CPU CORE VR (VCC)

Apple Inc.
DRAWING NUMBER
051-01477
REVISION
3.26.0
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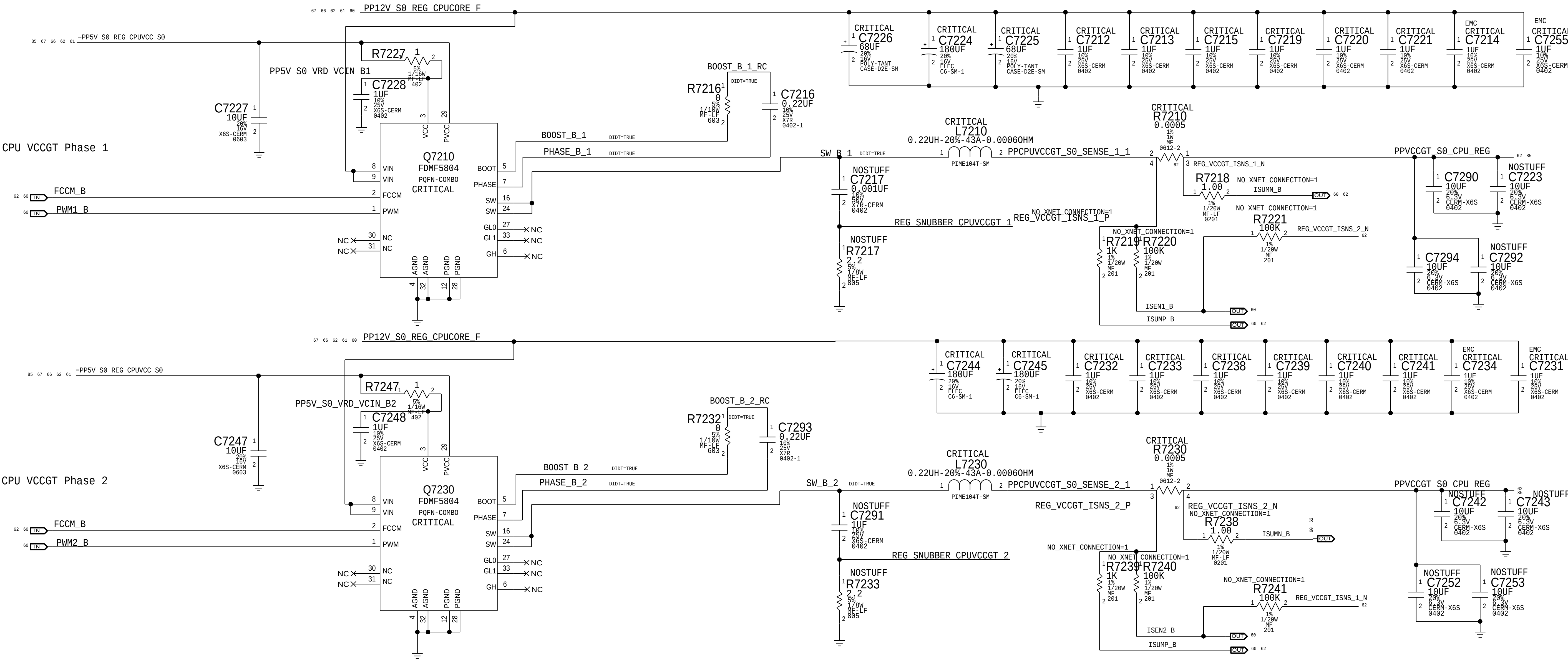
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1

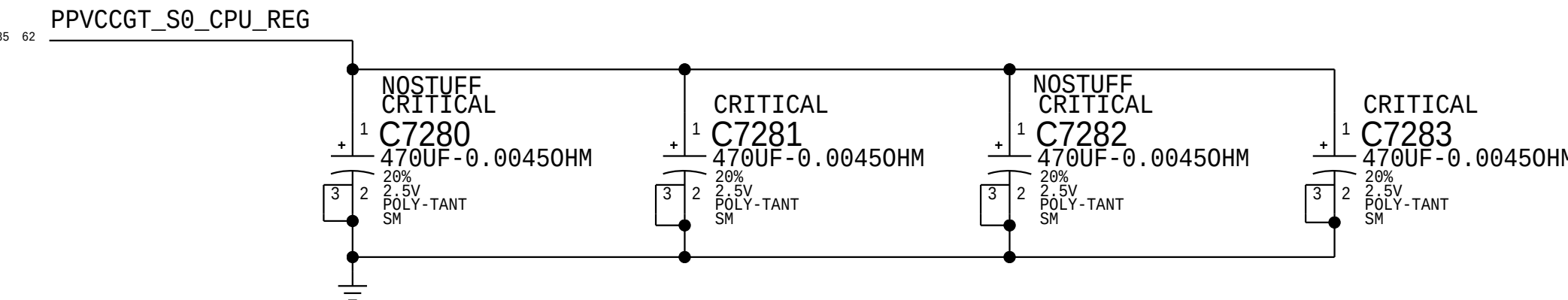
CPU VCCGT Regulator

EDC = 45A

TDC = 30A



CPU VCCGT OUTPUT DECOUPLING



BOM_COST_GROUP=CPU & CHIPSET

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PAGE TITLE			CPU & CHIPSET: CPU CORE VR (VCCGT)	
 Apple Inc.			DRAWING NUMBER	051-01477
			REVISION	3.26.0
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			PAGE	72 OF 119
			SHEET	62 OF 87

D



C

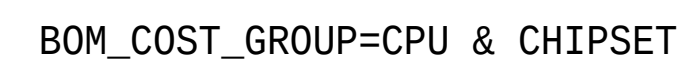
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A

PAGE TITLE		DRAWING NUMBER		SIZE	
CPU & CHIPSET: CPU VDDQ VR		051-01477		D	
 Apple Inc.		REVISION		3.26.0	
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		SHEET		63 OF 87	

EDC = 11.1A
TDC = 10A



DRAWING		LAST_MODIFIED=Wed Sep 14 23:01:33 2016	
SINC: WATER-Apple2		SINC: DATE=06/09/2016	
PAGE TITLE			
CPU & CHIPSET: CPU CORE VR (VCCSA)			
 Apple Inc.		DRAWING NUMBER 051-01477	
		REVISION 3.26.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
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		SHEET 66 OF 87	

D



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5

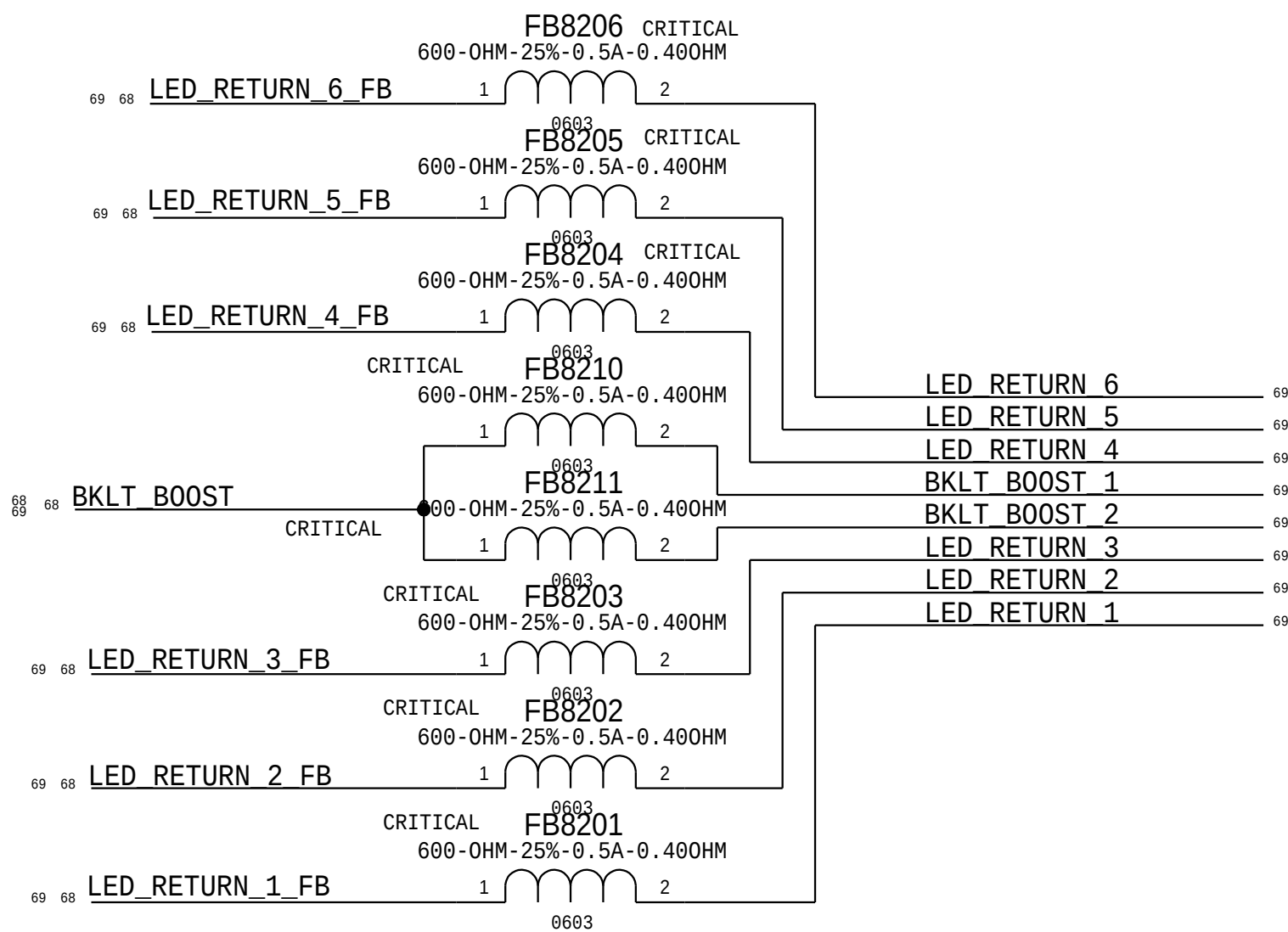
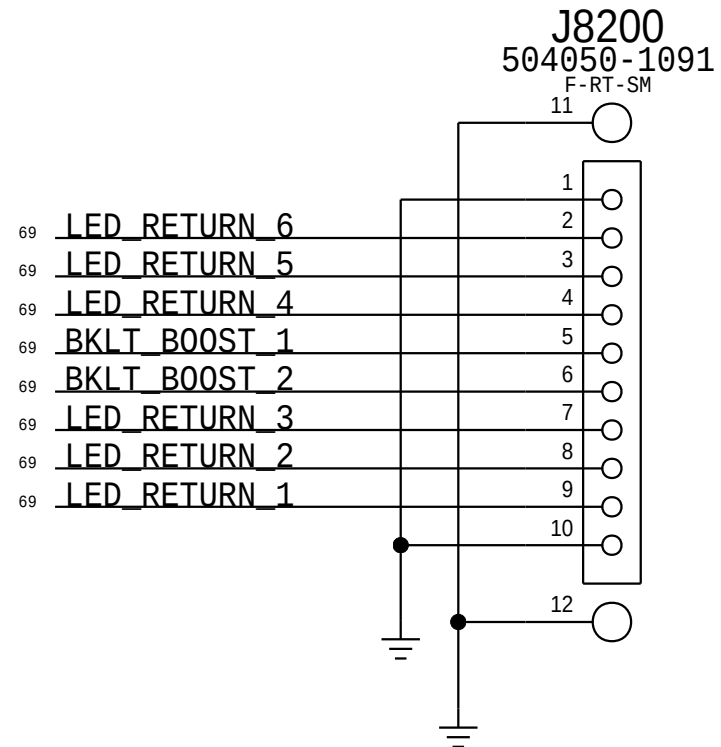
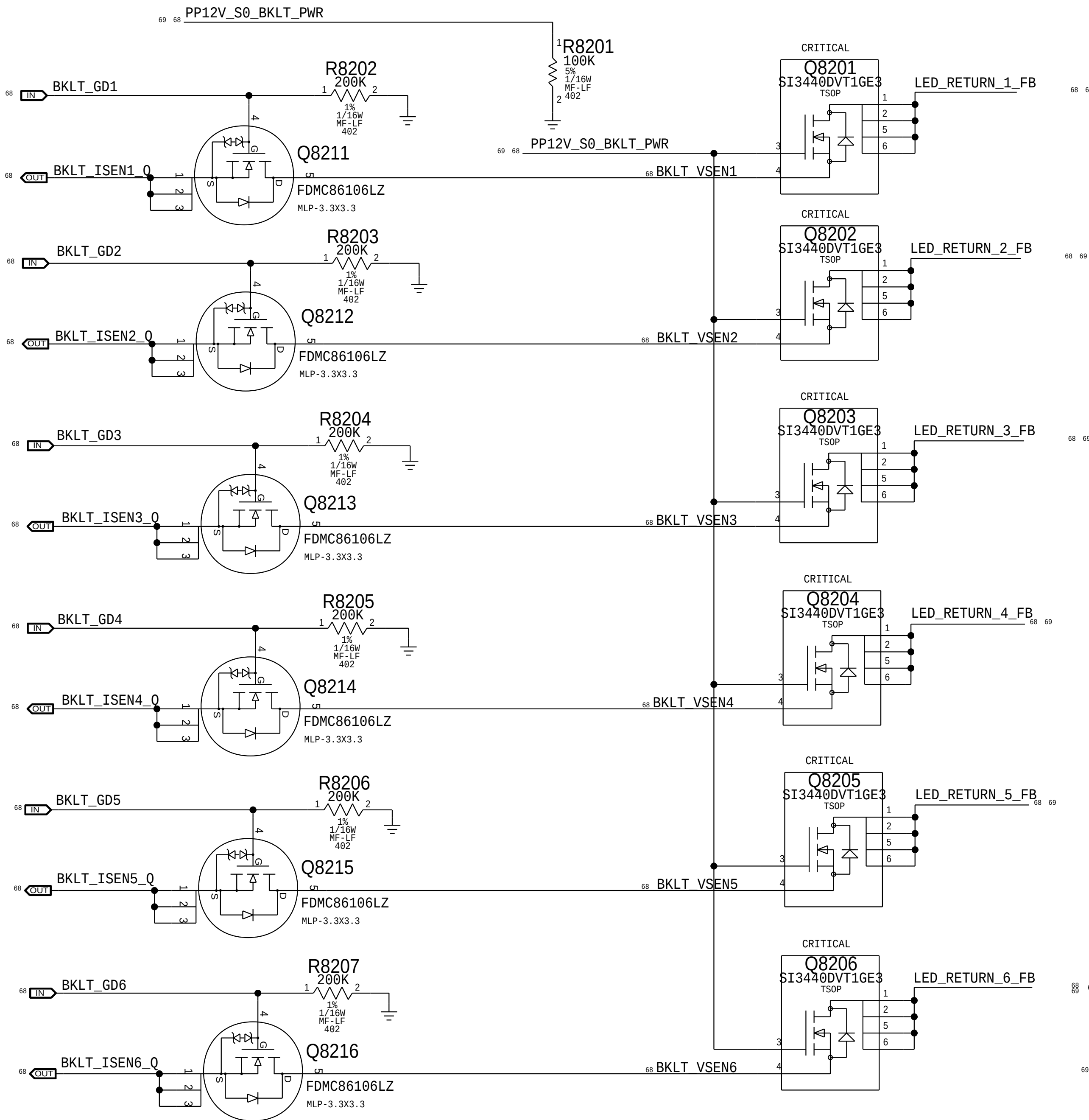
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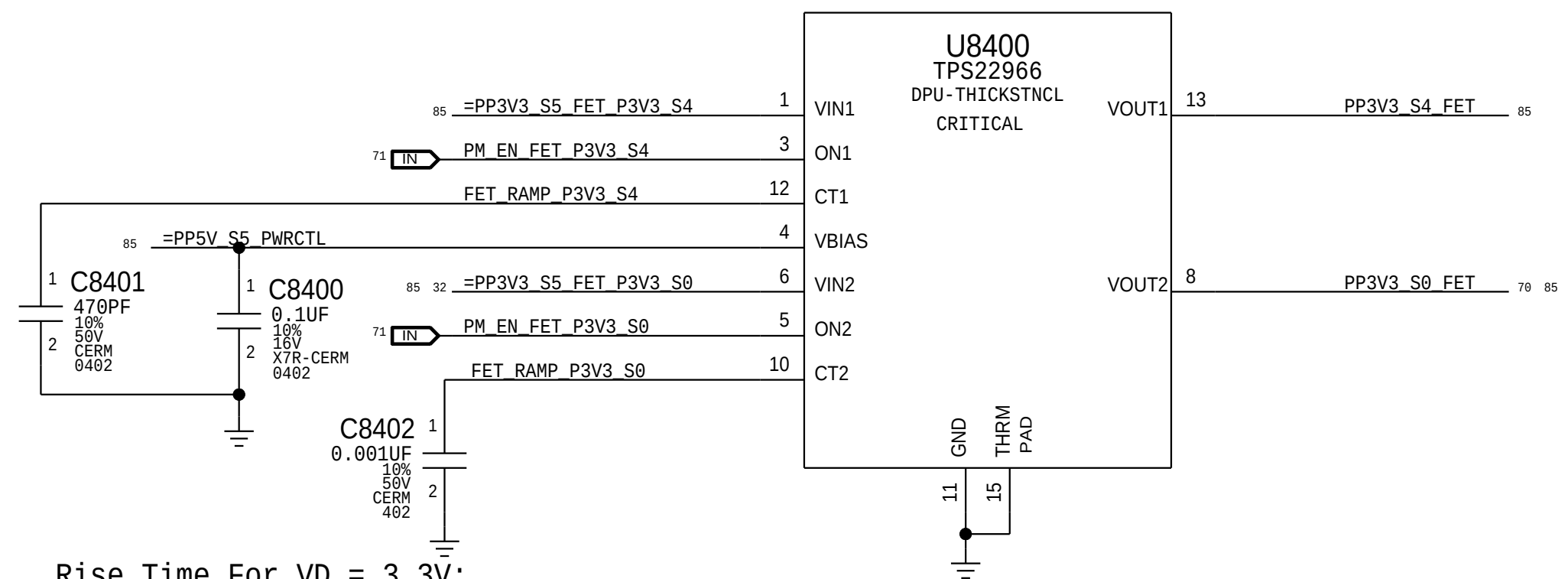
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1256	376S1973		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8201 TO FB8211



SYNC_MASTER=J94_JINGDONG		SYNC_DATE=08/21/2014	
PAGE TITLE			
LCD Backlight Driver 2			
 Apple Inc.	DRAWING NUMBER		SIZE
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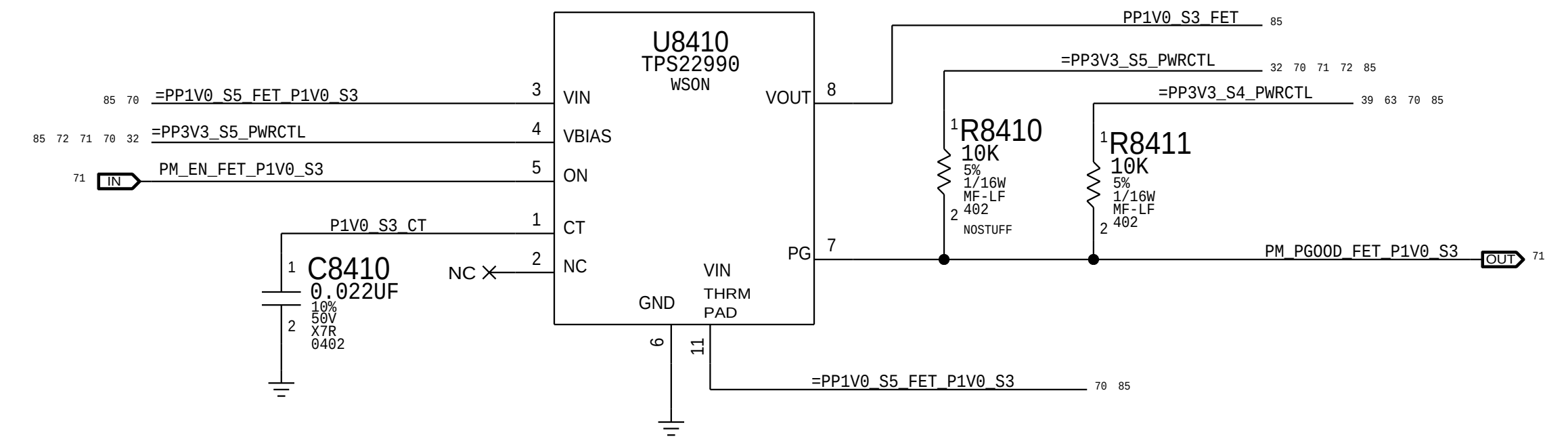
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3.3V S4/S0 FET

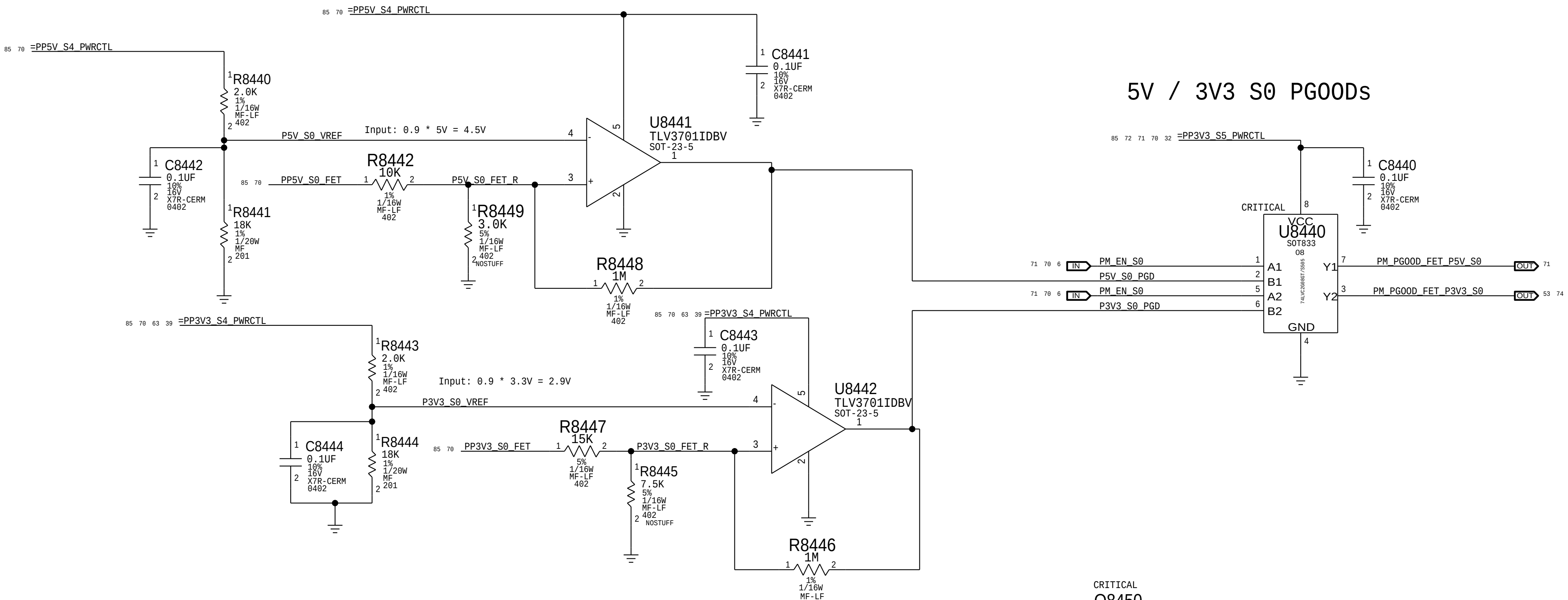


Rise Time For VD = 3.3V:
470 pF -- 603 us
1000 pF -- 1185 us

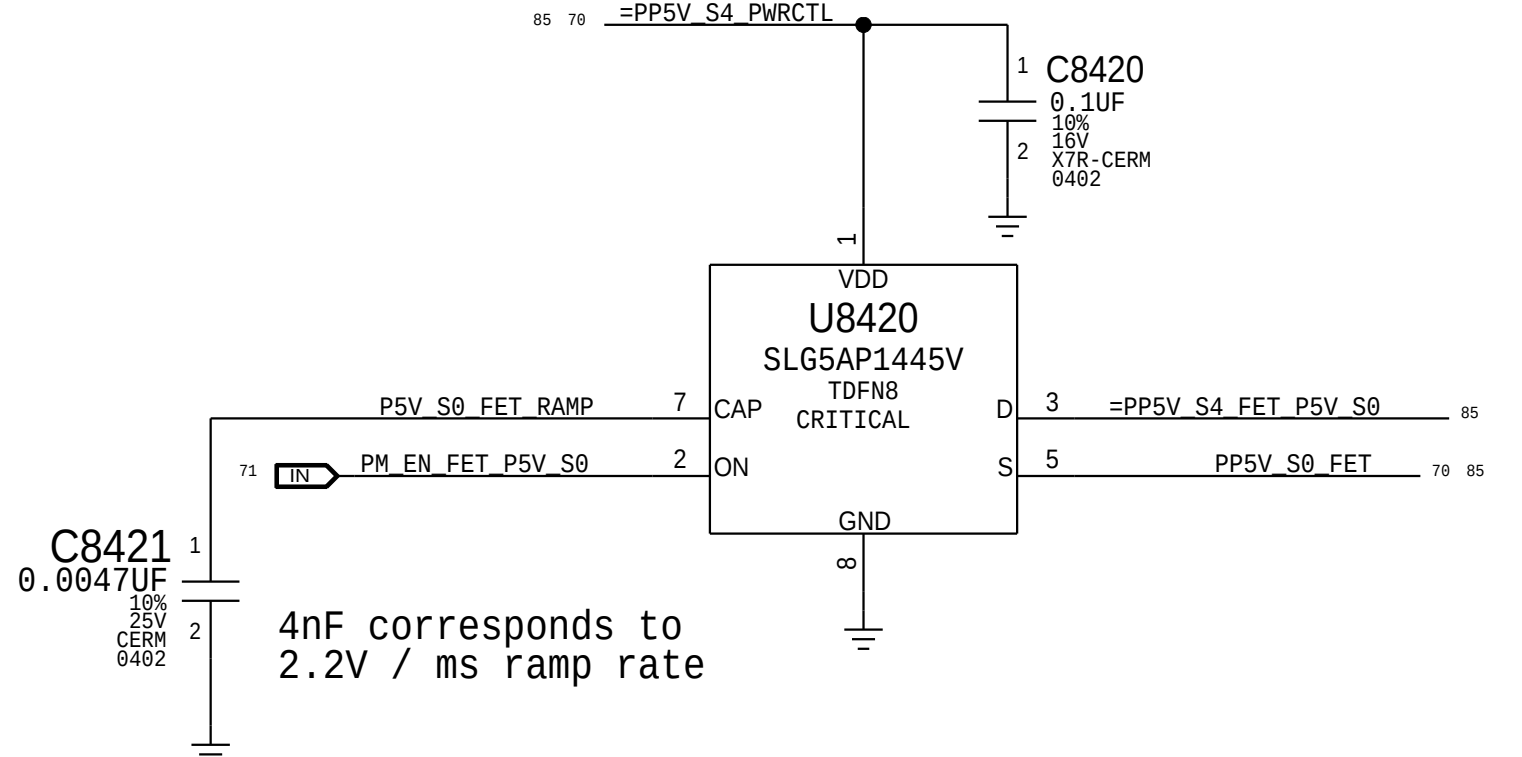
1.0V S3 FET



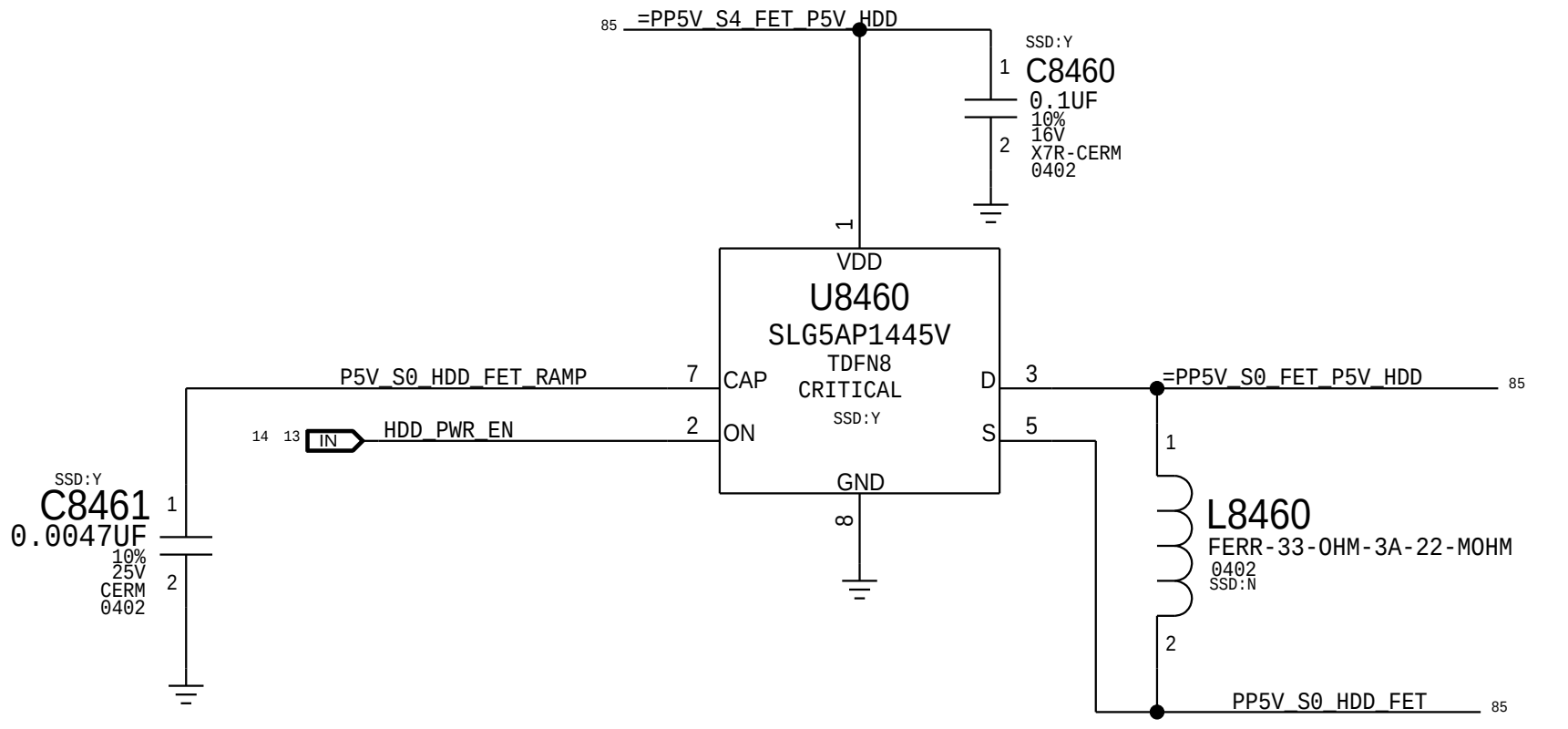
5V / 3V3 S0 PG00Ds



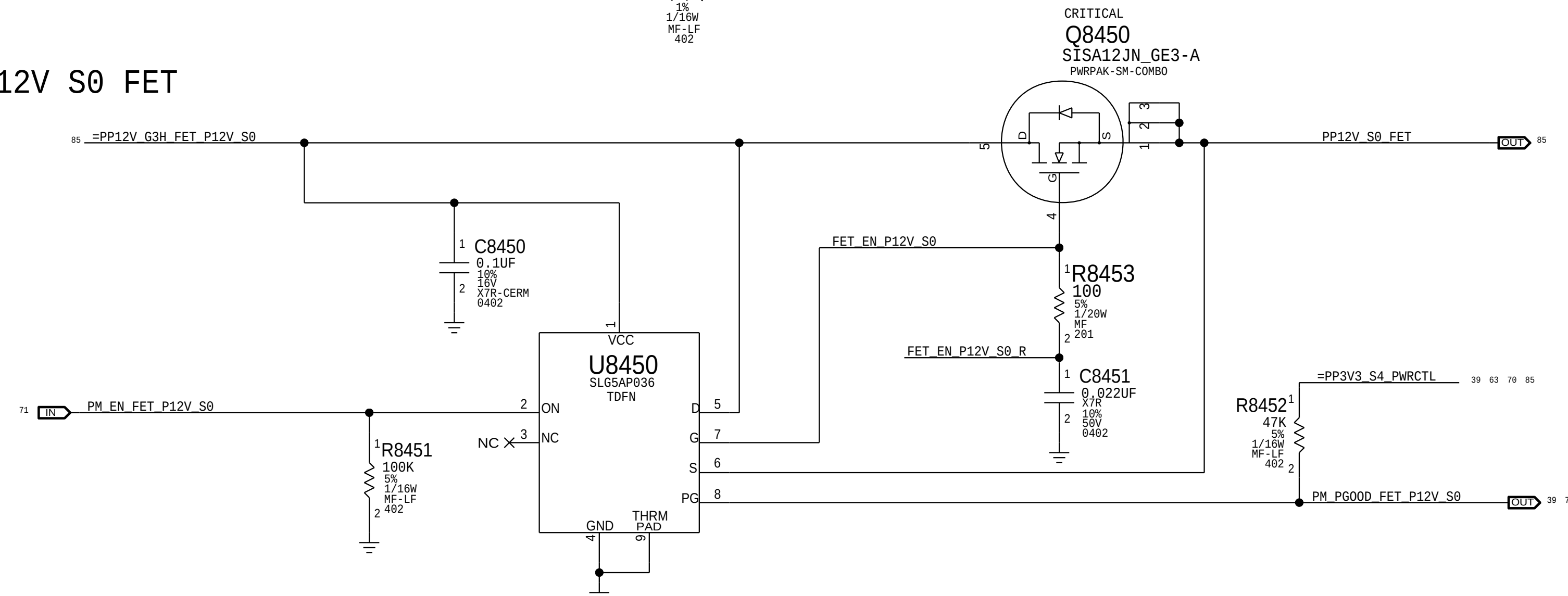
5V S0 FET



5V HDD FET



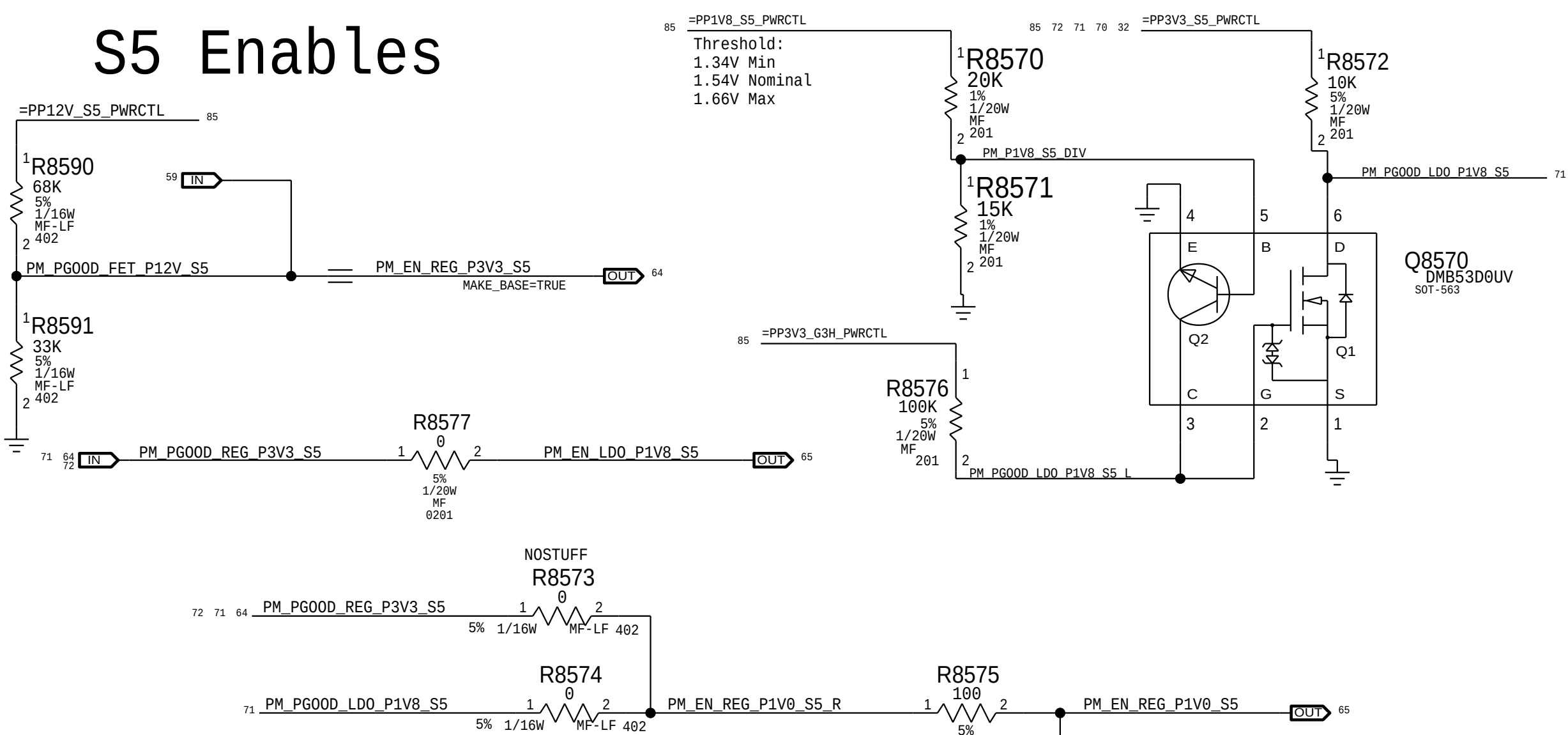
12V S0 FET



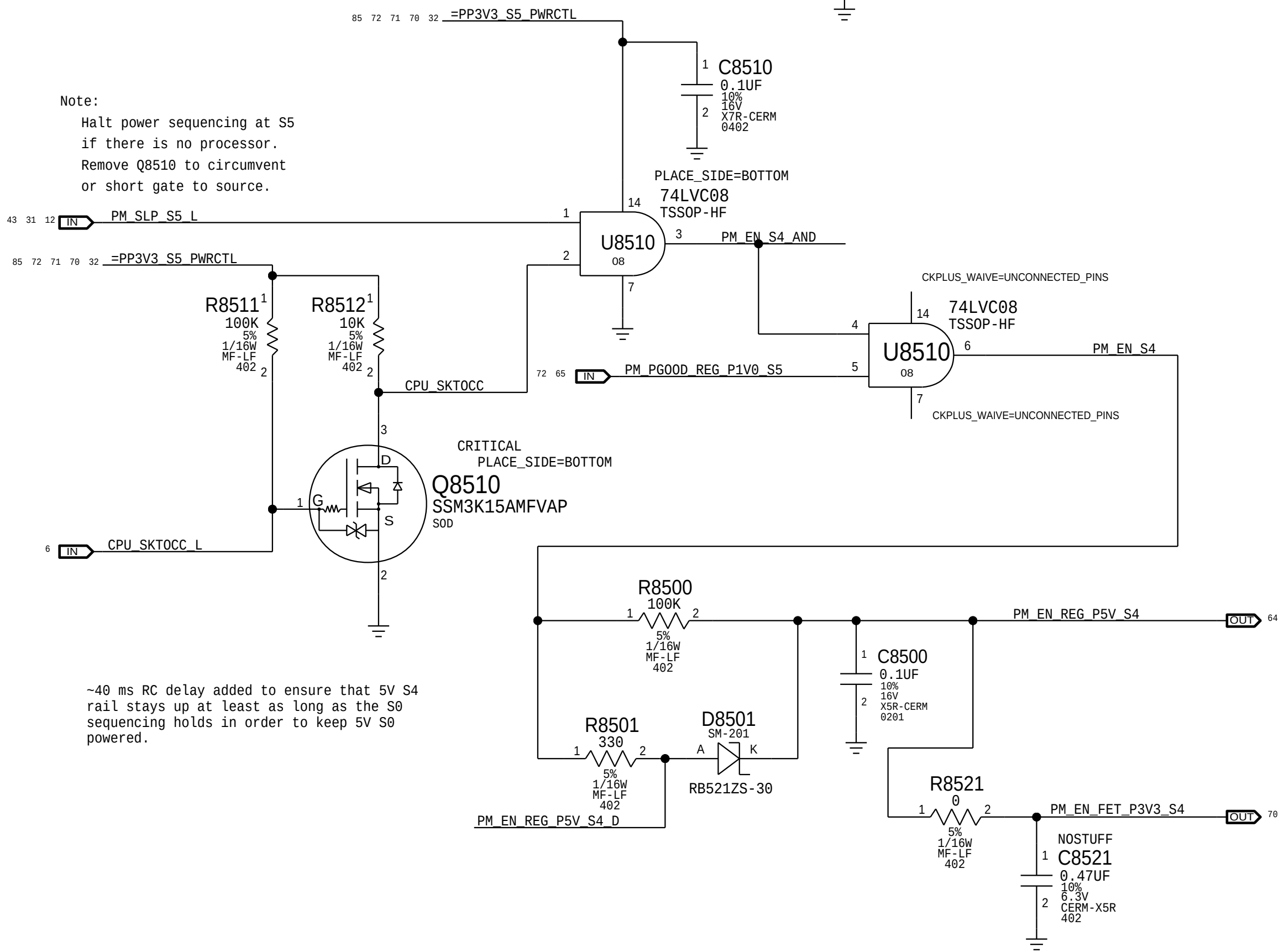
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PAGE TITLE			
PM FETS/LD0s			
 Apple Inc.	DRAWING NUMBER 051-01477		SIZE D
	REVISION 3.26.0		
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BOM_COST_GROUP=PLATFORM POWER

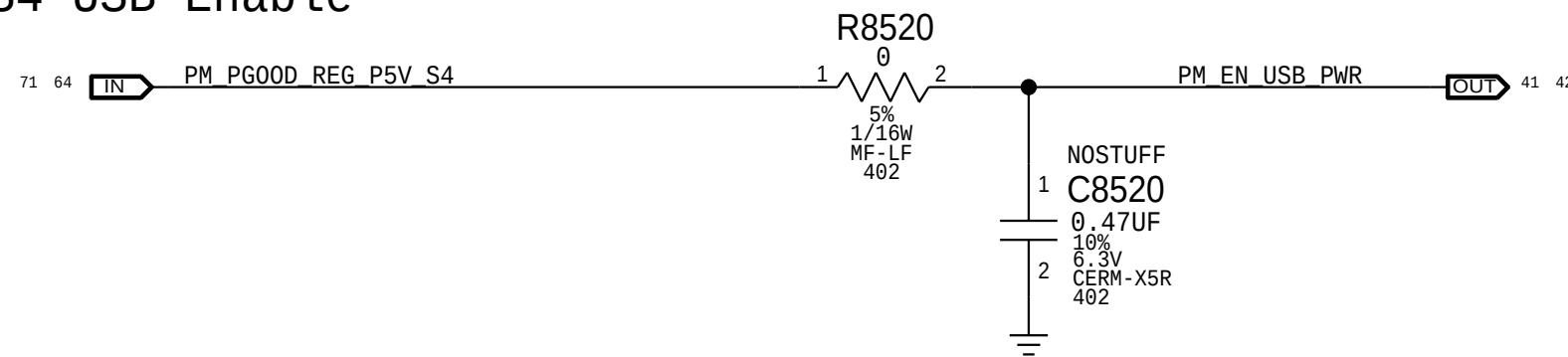
S5 Enables



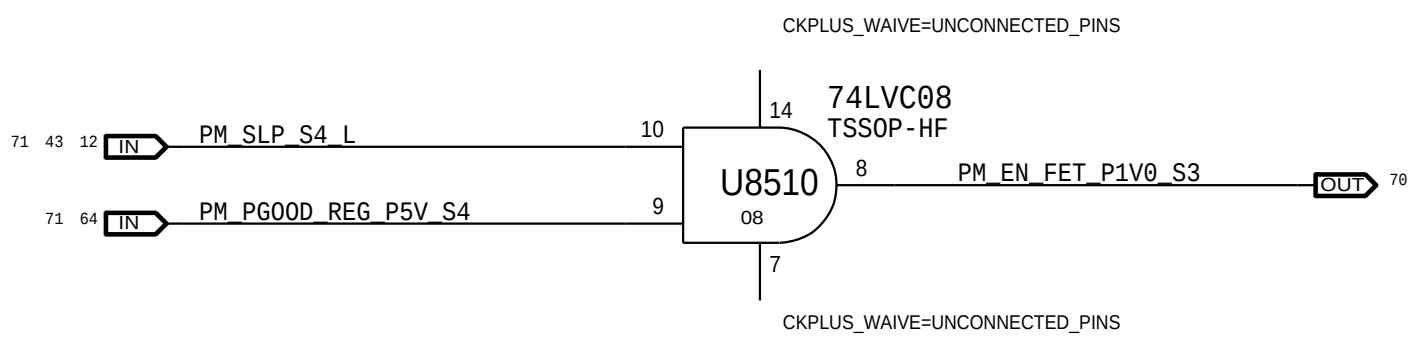
S4 Enables



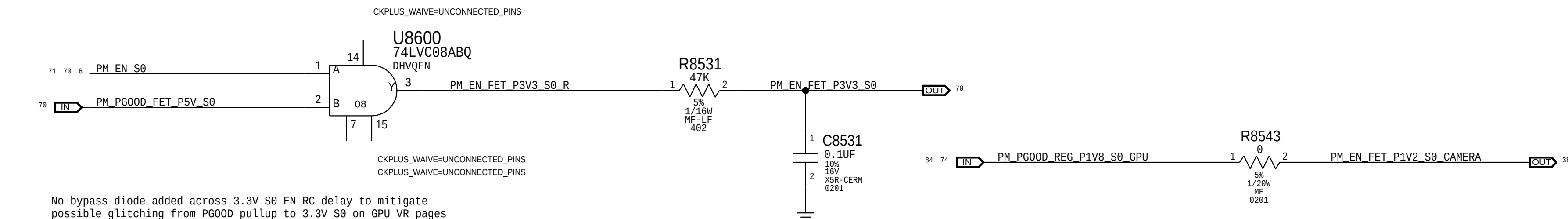
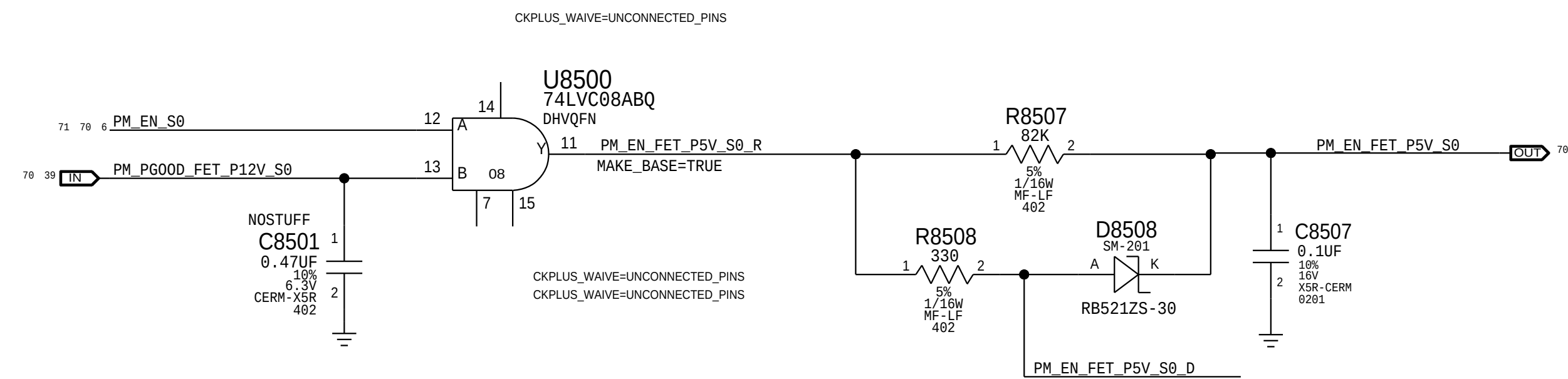
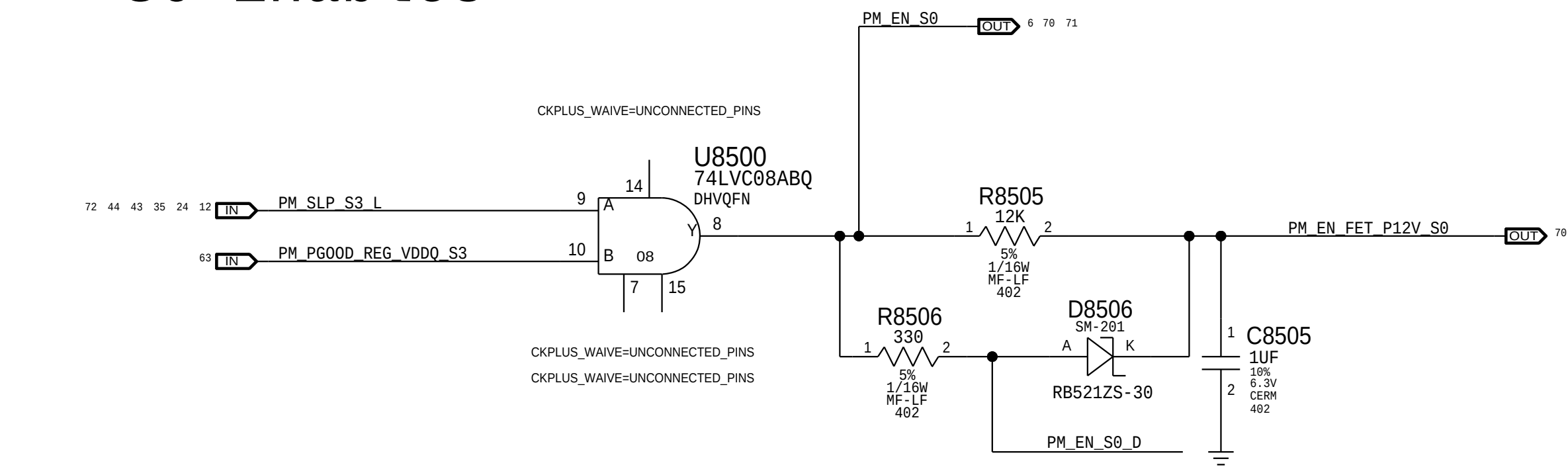
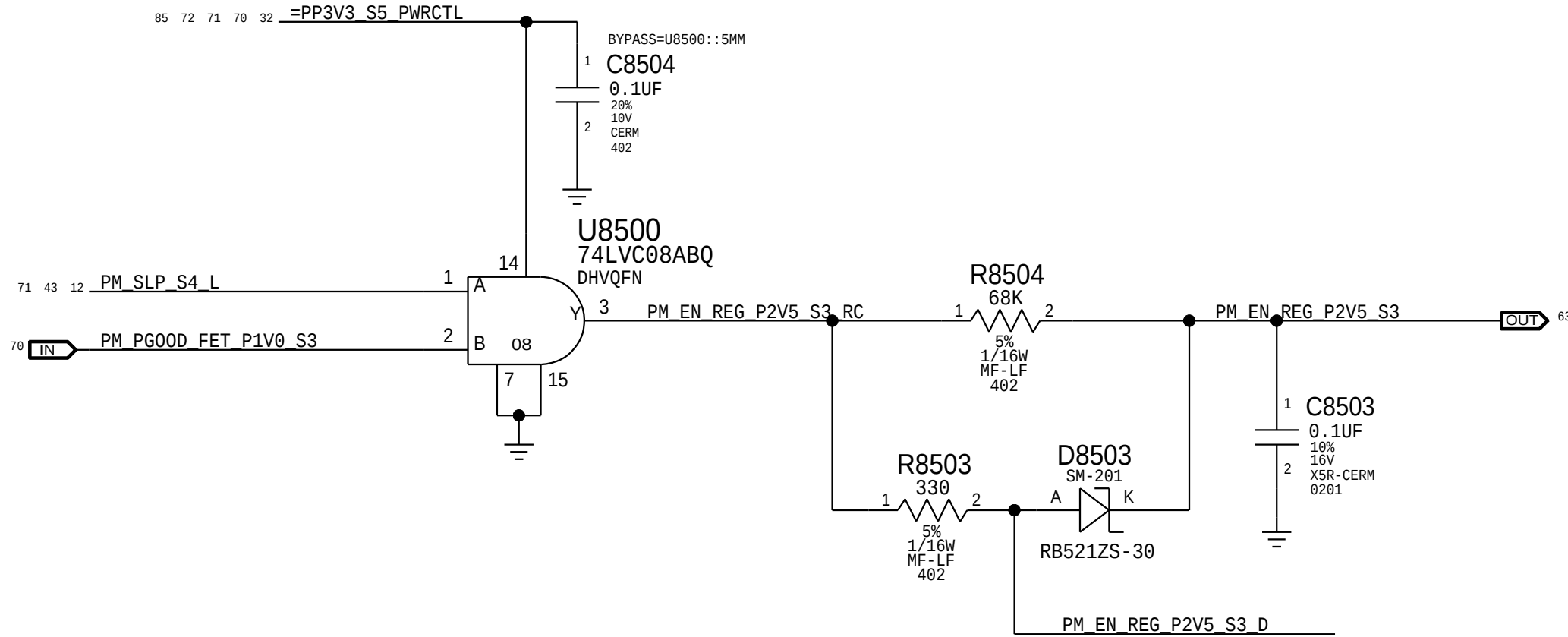
S4 USB Enable



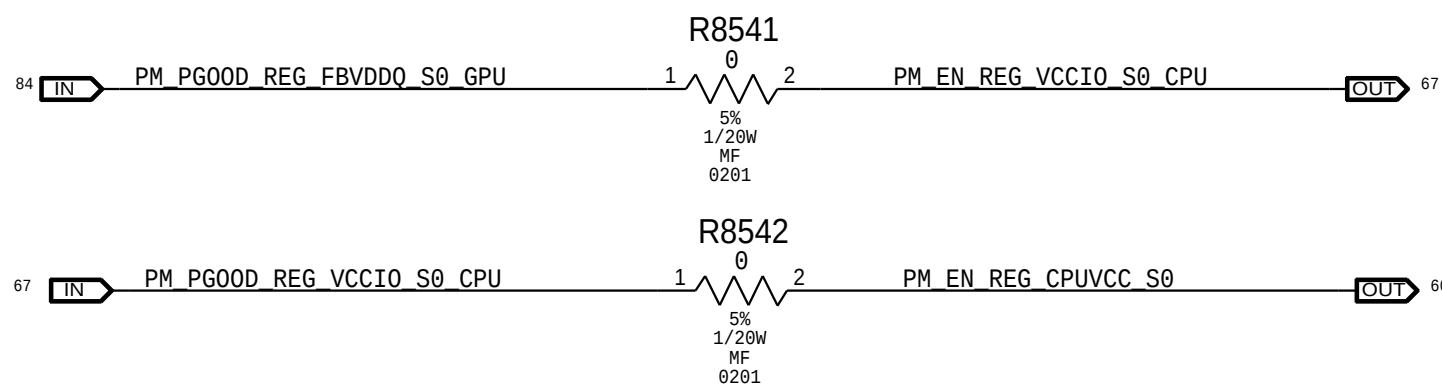
S3 Enables



S0 Enables



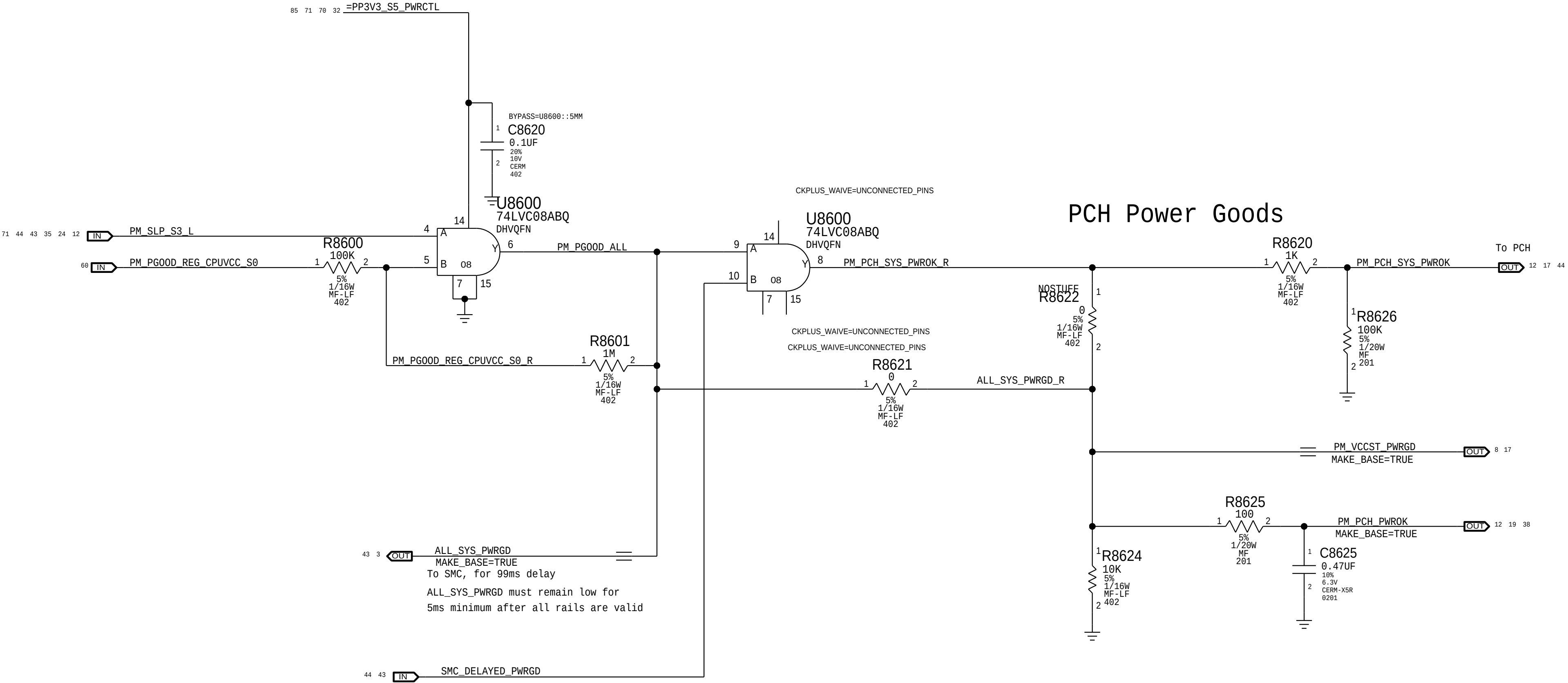
S0 CPU Sequencing



SYNC_MASTER=dtuzman		SYNC_DATE=04/01/2016	
PAGE TITLE			
PM Regulator Enables			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-01477	D
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		BRANCH	
		PAGE	
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BOM_COST_GROUP=PLATFORM POWER

ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation



Rail definitions

Platform: All processor non-Core and non-Graphics (5V, 3.3V, 1.5V, 1.05V for PCH/TBT/GPU)
Uncore: 2.5V and 1.2V for DDR4

Notes on sequencing requirements

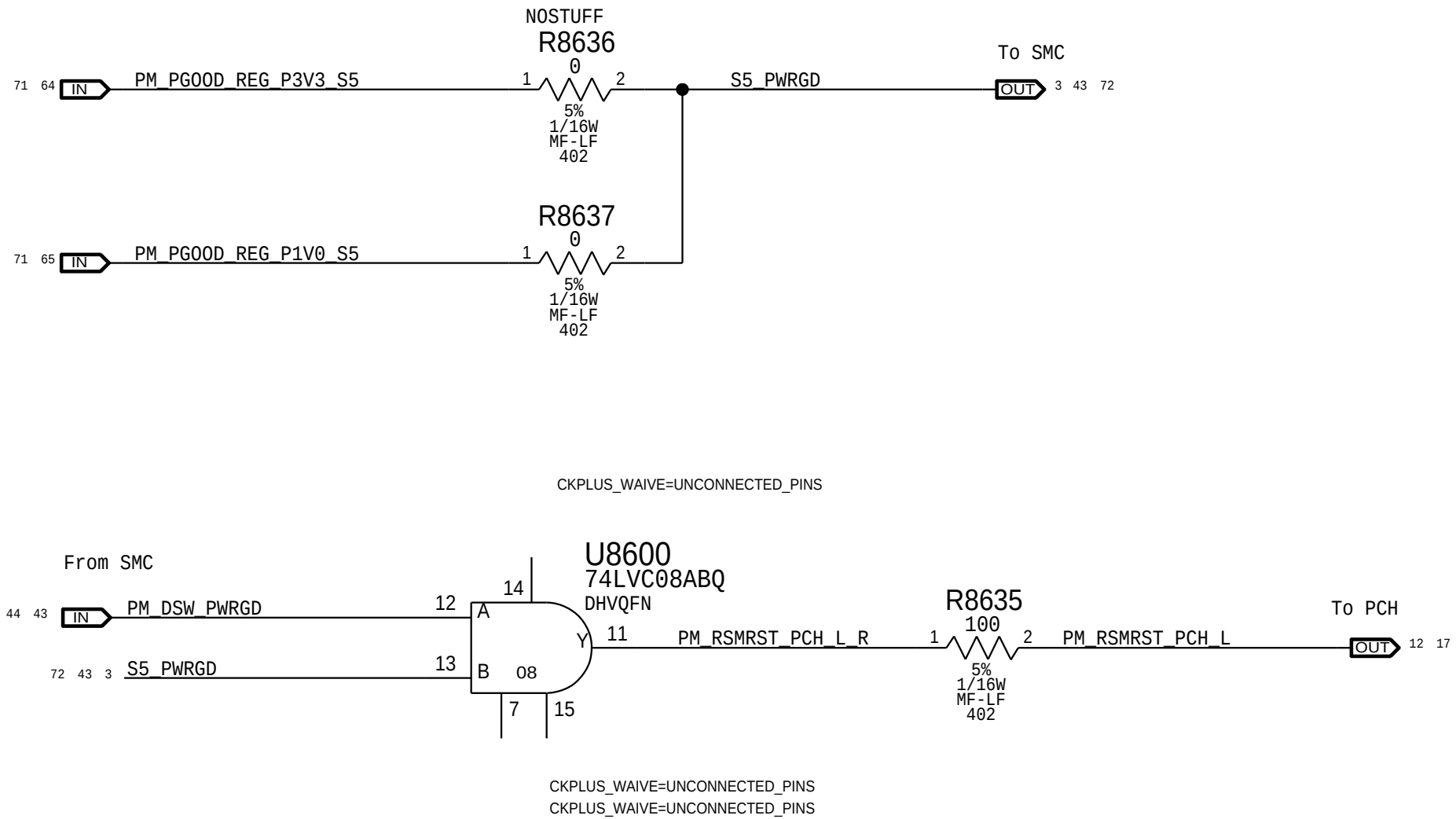
- Intel:
1. No hard specification on platform rails
 2. SMC guarantees timing on PCH DPWROK and PWROK
 3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
 4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
 5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
 6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

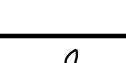
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

- Note:
- The iMac J70 design does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together
- Requirements:
- Power on:
- Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
- Transition to 0.0V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

- Method:
- The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
- RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=hurtadot		SYNC_DATE=06/01/2016	
PAGE TITLE			
PM Power Good			
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BOM_COST_GROUP=PLATFORM POWER

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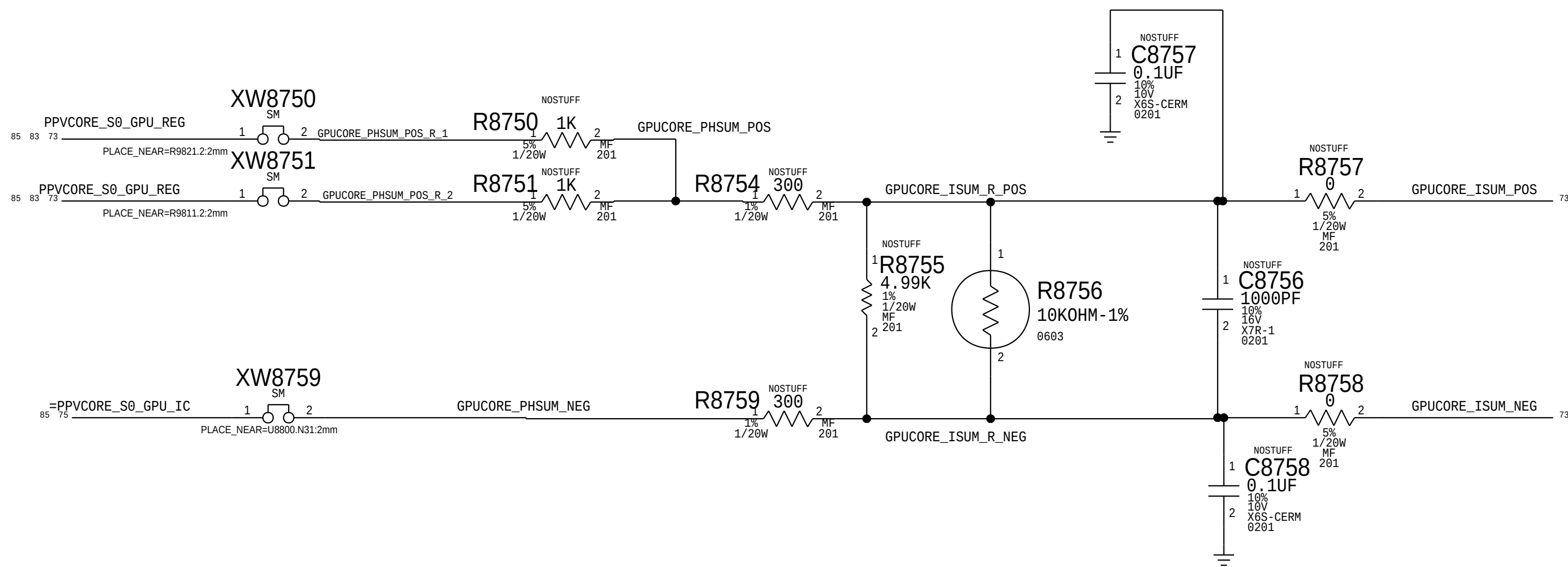
D

C

B

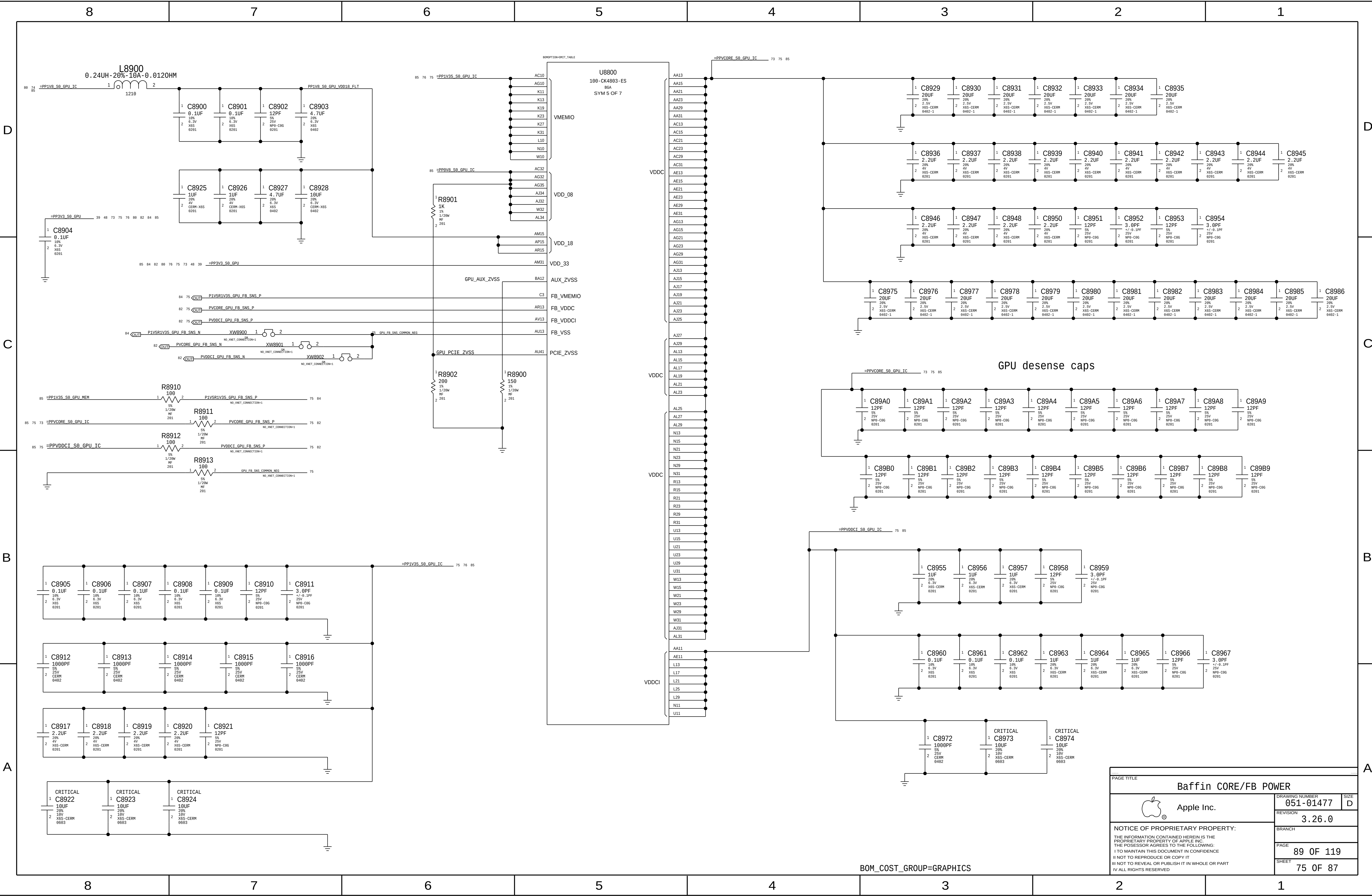
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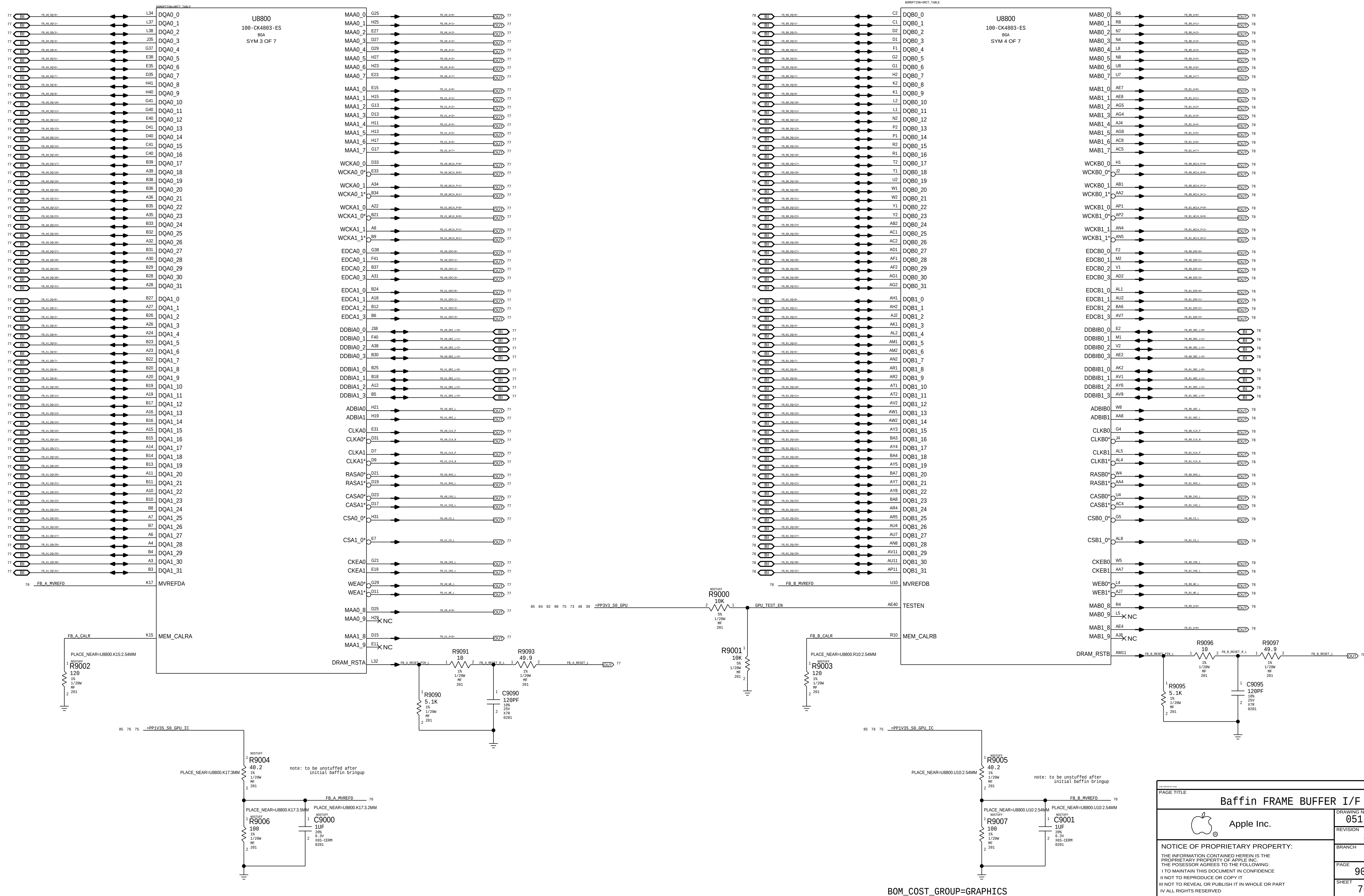
Copper sense apparatus

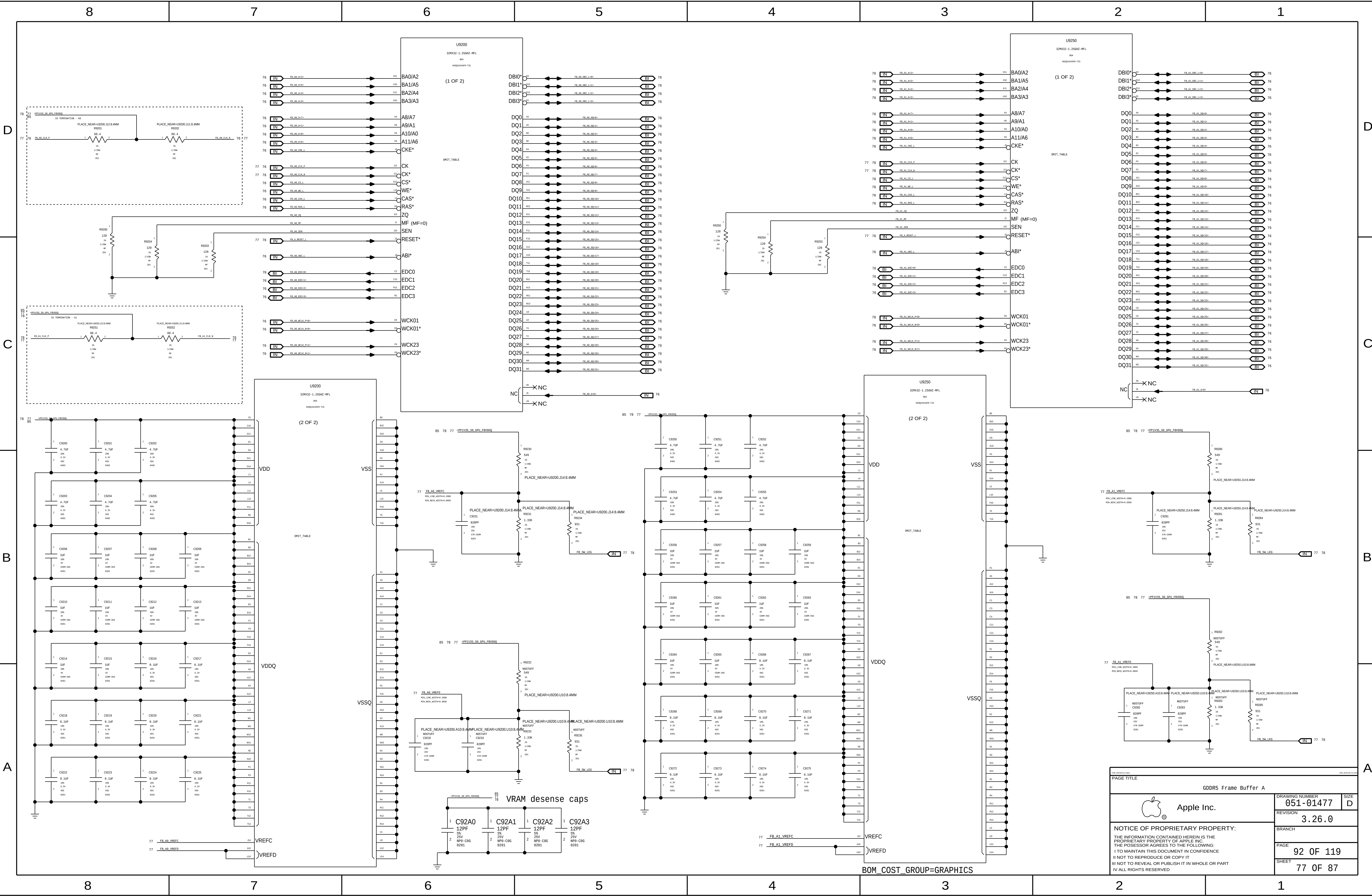


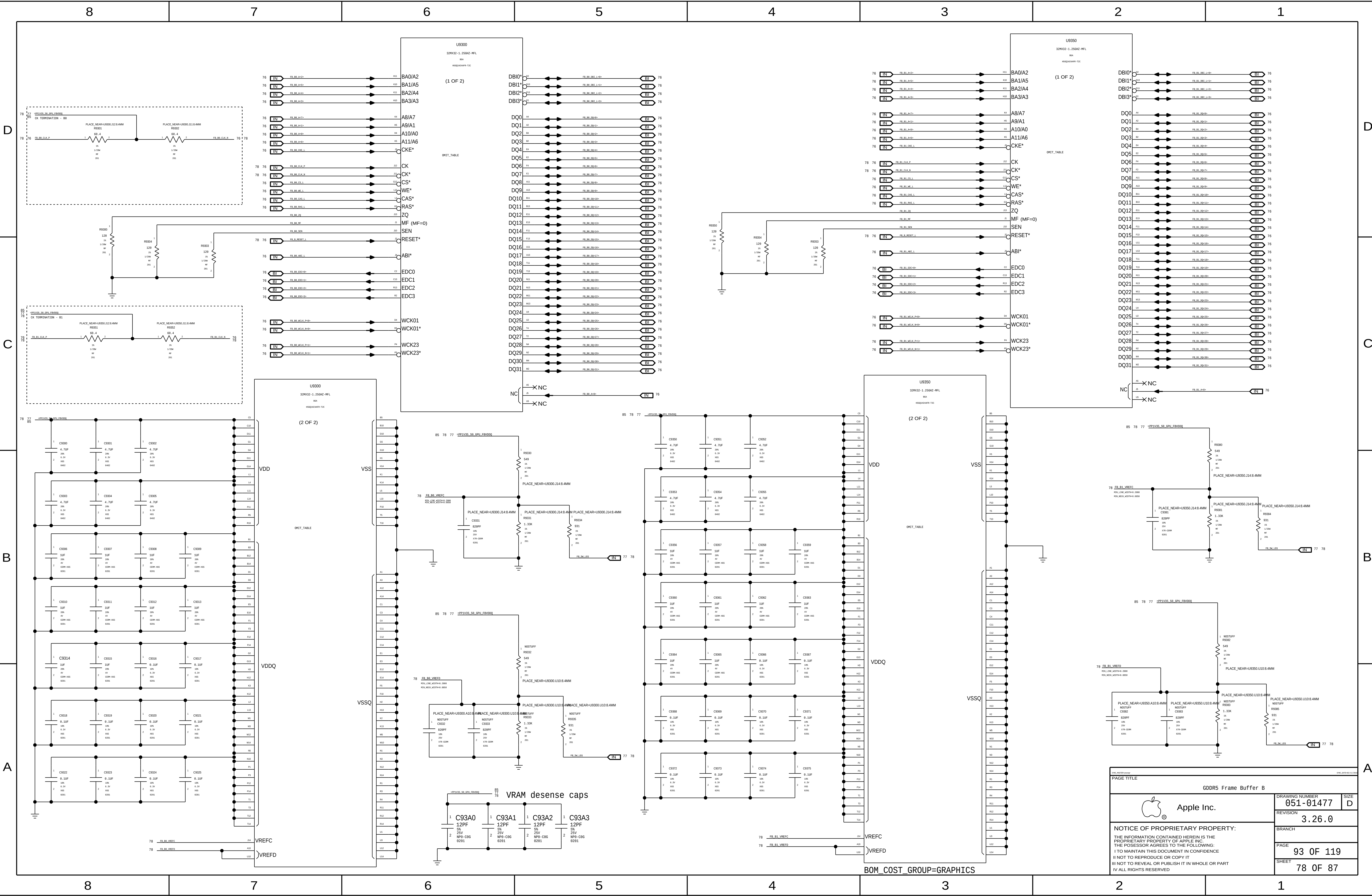
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GPU PCC		
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SPEED MLPS STRAPS

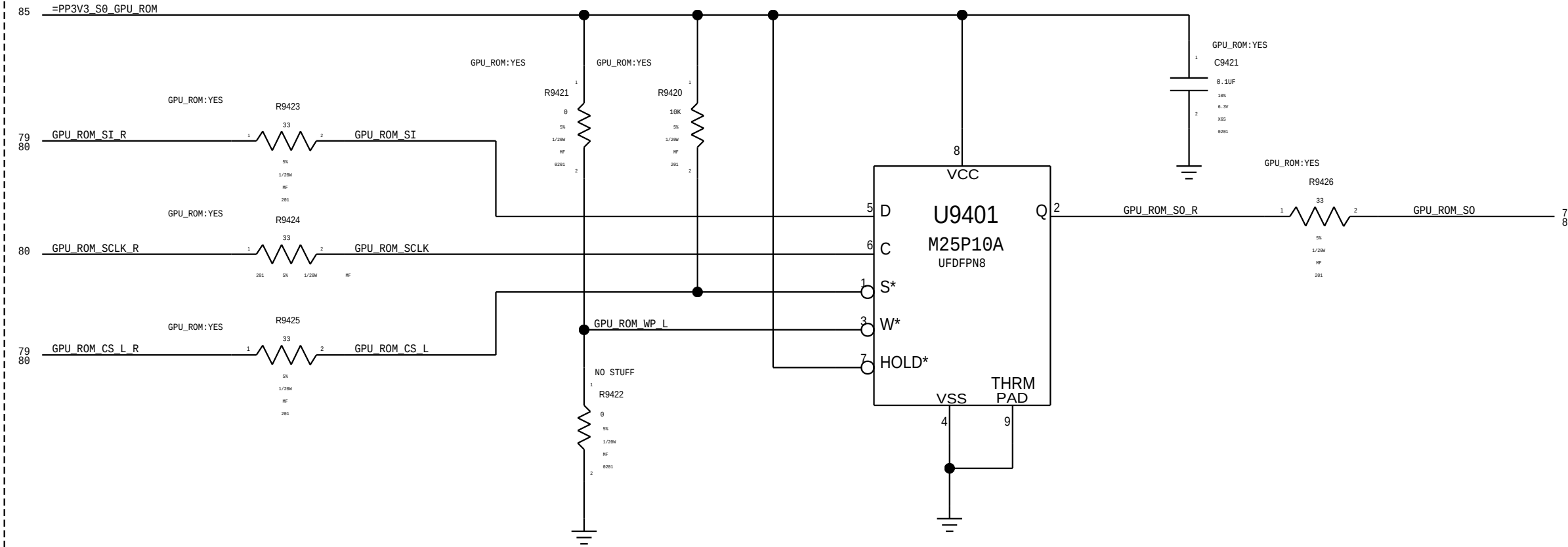
PCIe Gen3 enabled, Half-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled
All power audio-capable display output
256MB FB Aperture Size
PS_0: 01001 82nF 8.45k 2k
PS_1: 10001 10nF 8.45k 2k
PS_2: 10000 10nF NC 4.75k
PS_3: 00000 680nF NC 4.75k

NOT REQUIRED FOR BAFFIN

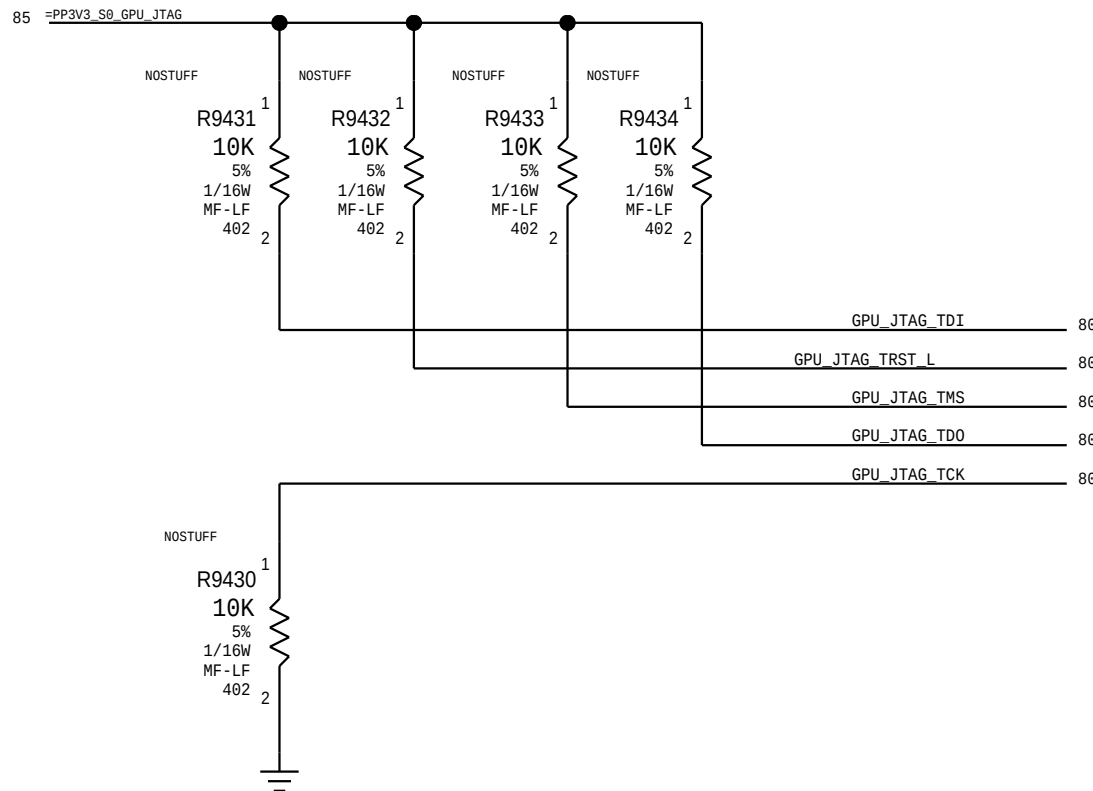
GPU SVI2 VOLTAGE TRANSLATION
SPEED ONLY

NOT REQUIRED FOR BAFFIN

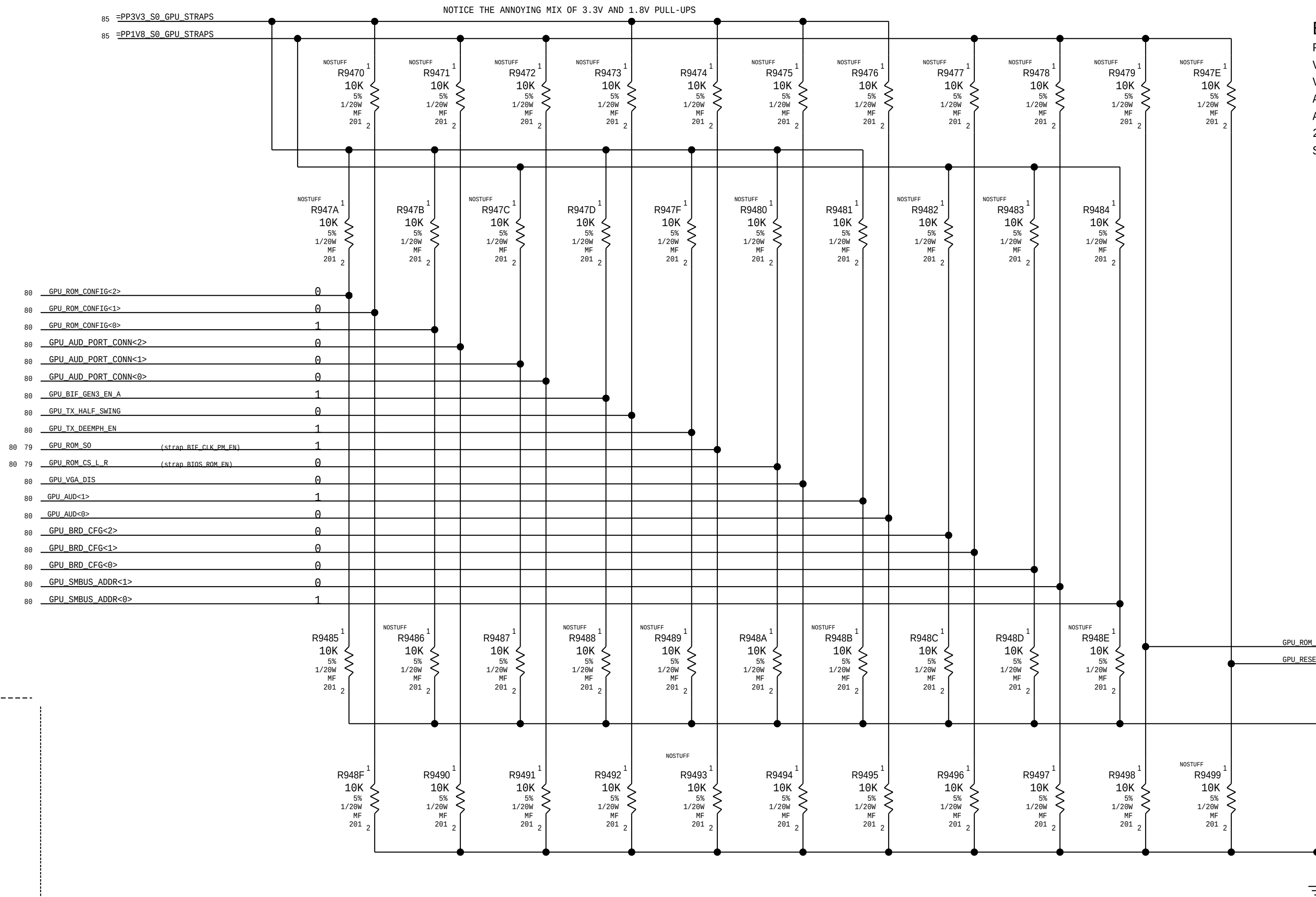
GPU ROM



GPU JTAG



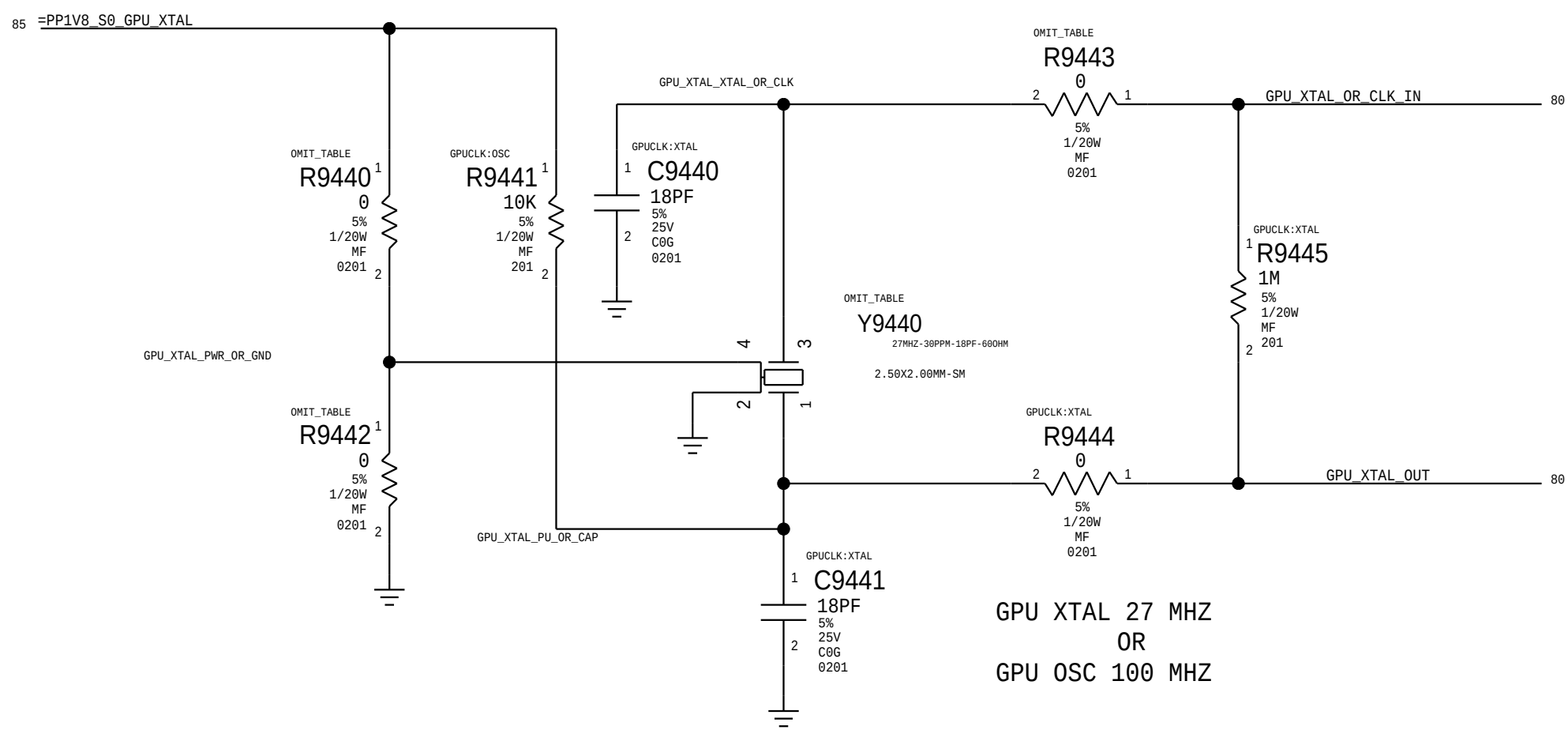
NOTICE THE ANNOYING MIX OF 3.3V AND 1.8V PULL-UPS



BAFFIN STRAPS

PCIe Gen3 enabled, Half-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled, CLKREQ_L enabled
All power audio-capable display output
Audio for DP or HDMI dongle
256MB FB Aperture Size
SMBUS address 41

GPU CLKREQ GATING



GPU XTAL 27 MHZ
OR
GPU OSC 100 MHZ

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0499	1	XTAL, 27.000MHZ, 38PPM, 12PF, 2.5x2.0MM	Y9440	CRITICAL	GPUCLK:XTAL
197S00056	1	OSC, MEMS, 100MHZ, +/-20PPM, 1.8V, 2520	Y9440	CRITICAL	GPUCLK:OSC
155S0387	1	FERRITE BEAD, 4700HM, 6-1A, 1.5MOHM DCR, 668	R9440	CRITICAL	GPUCLK:OSC
117S0201	1	RES, 8 OHM, 5%, 0201	R9442	CRITICAL	GPUCLK:XTAL
132S0444	1	CAP, CER, X5R, 8.1UF, 10V, 6.3V, 2501	R9442	CRITICAL	GPUCLK:OSC
117S0201	1	RES, 8 OHM, 5%, 0201	R9443	CRITICAL	GPUCLK:XTAL
117S00080	1	RES, 33 OHM, 5%, 0201	R9443	CRITICAL	GPUCLK:OSC

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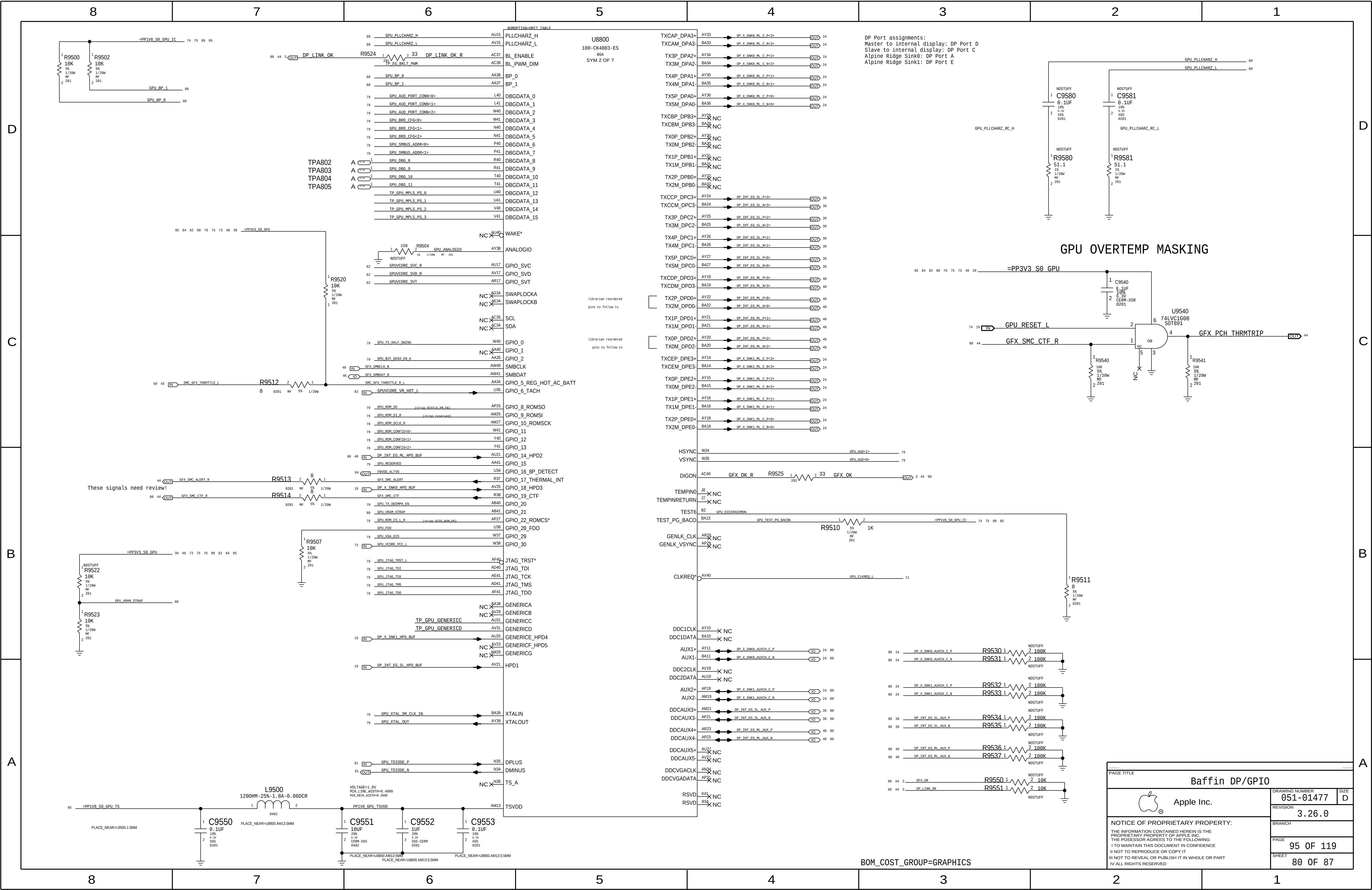
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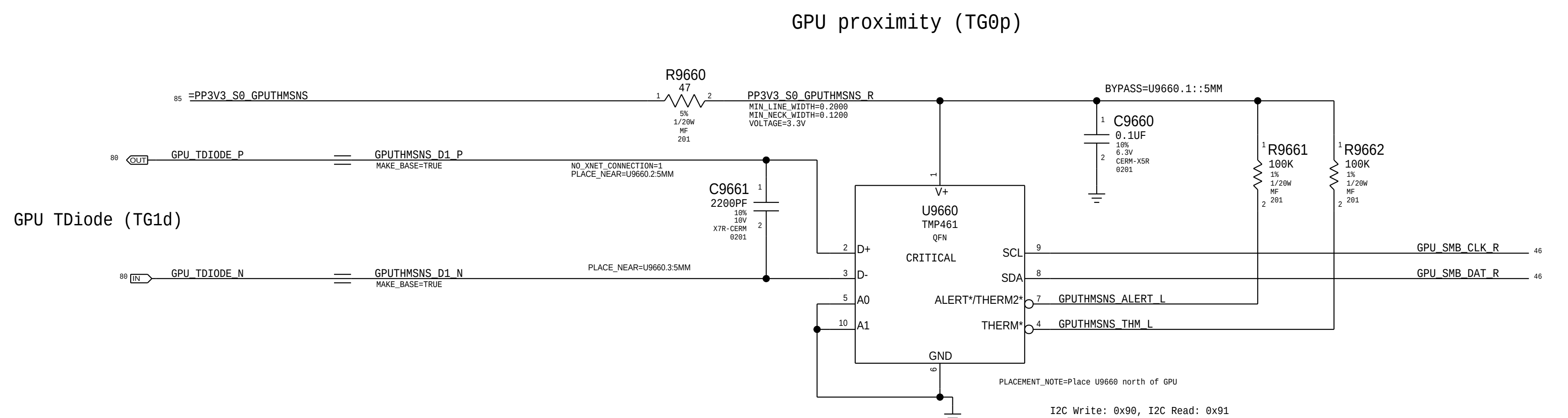
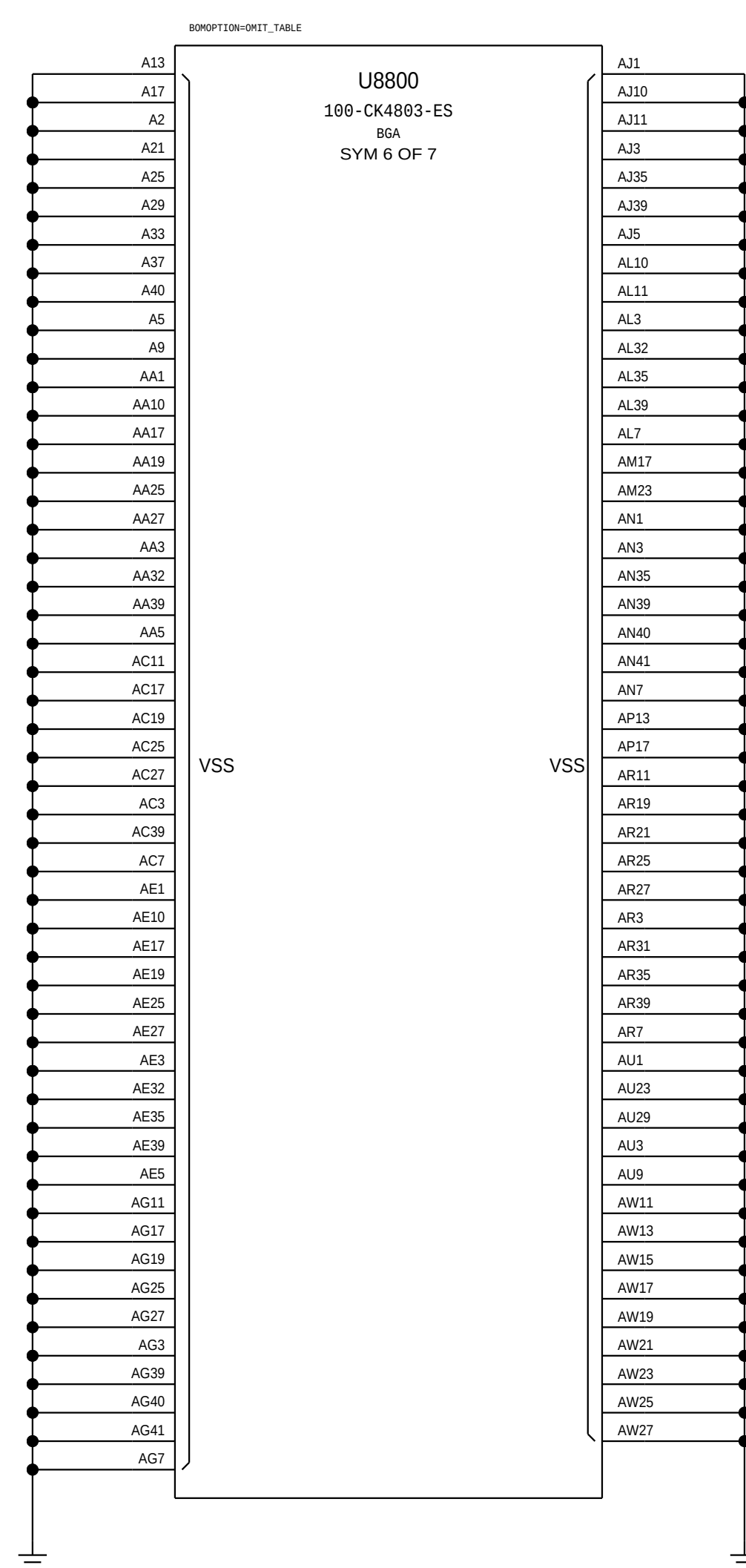
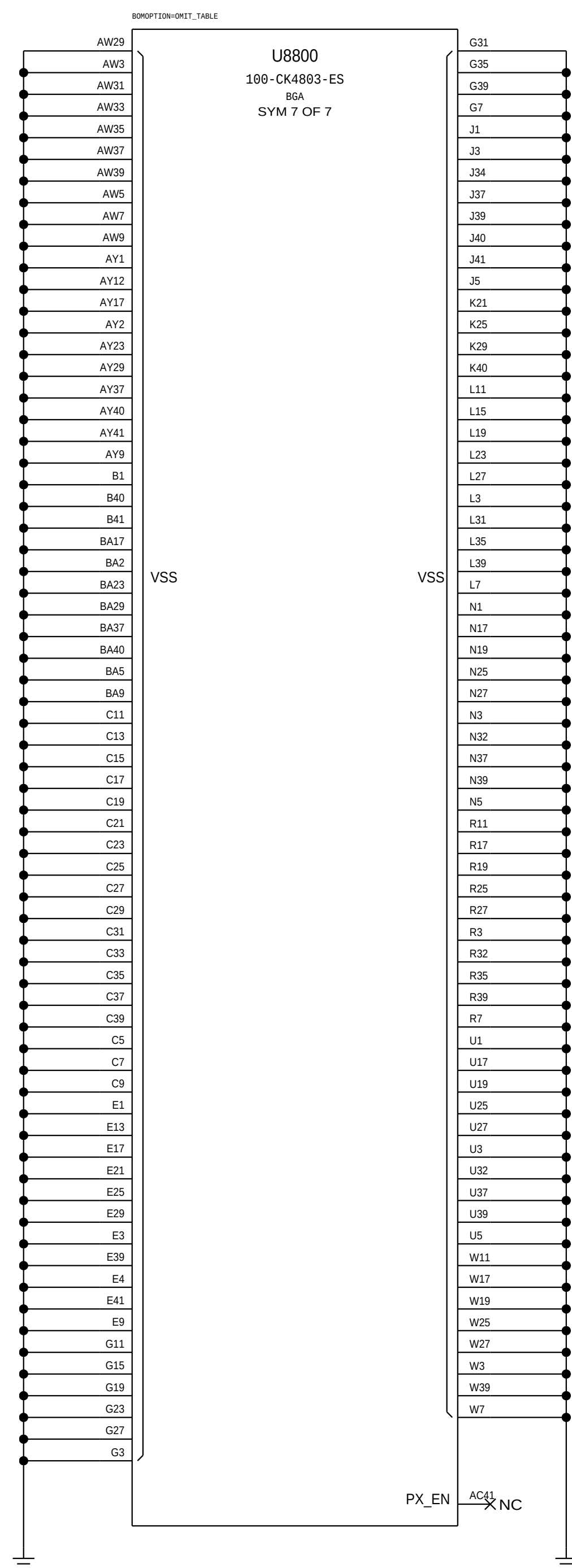
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Baffin GPIOs,CLK & Straps

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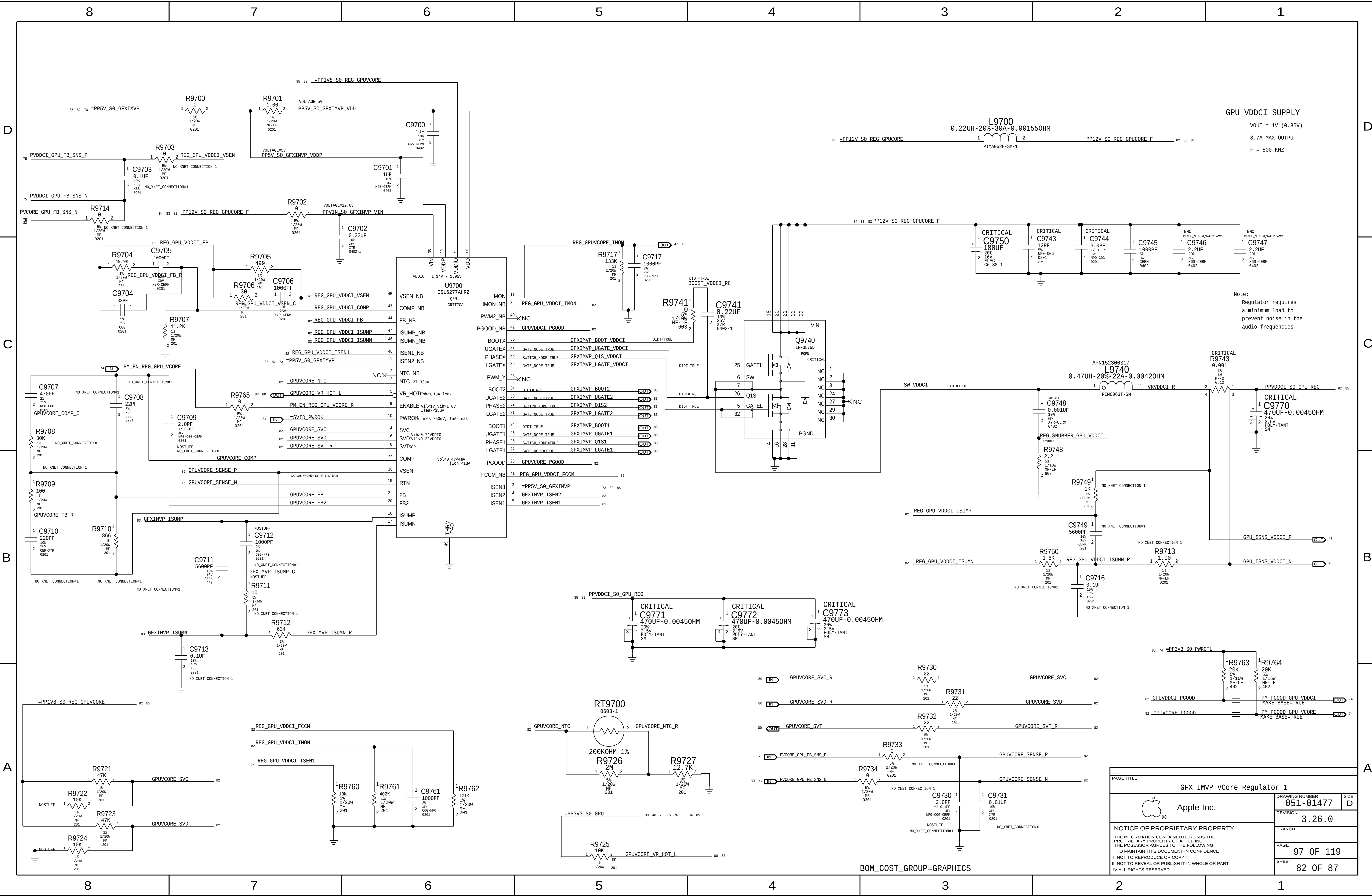
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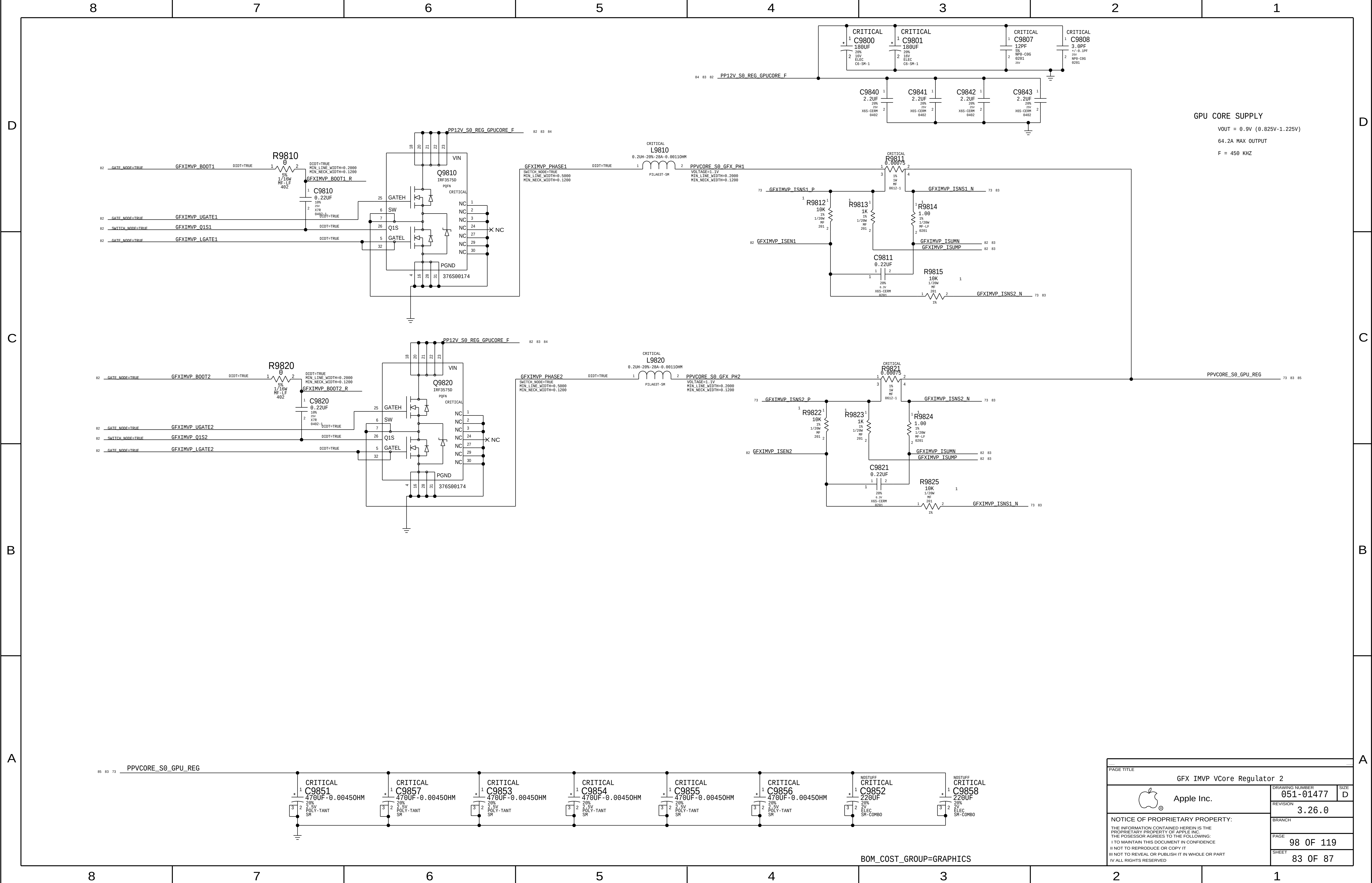




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BOM_COST_GROUP=GRAPHICS





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GFX IMVP VCore Regulator 2		
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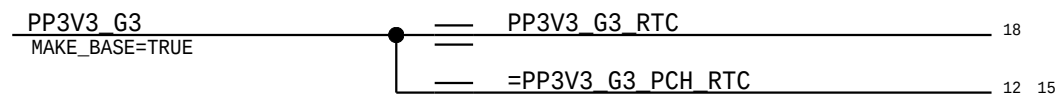
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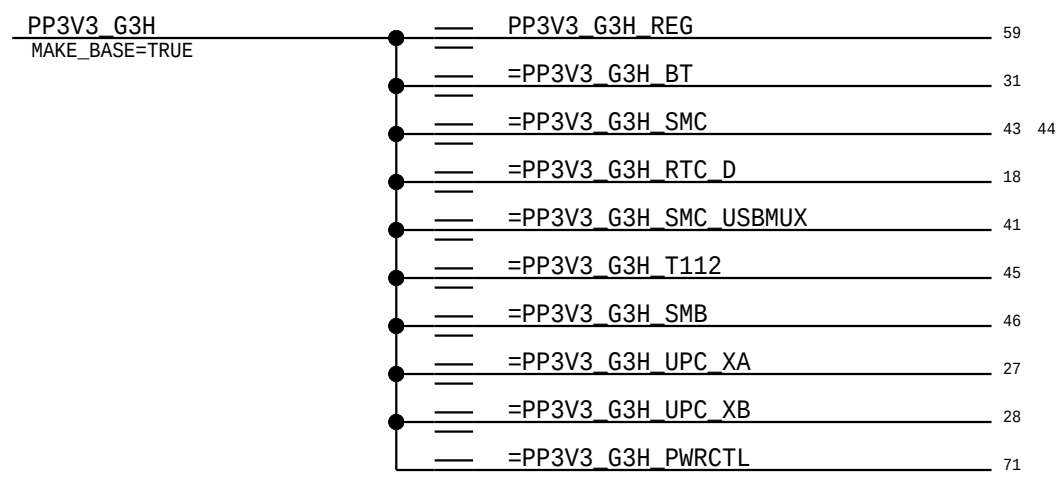
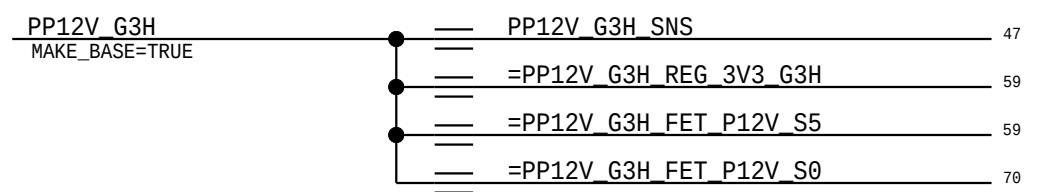
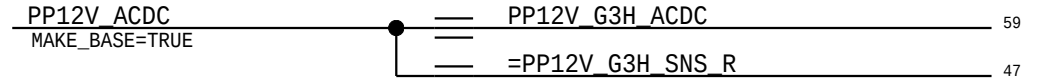
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G3 Rails

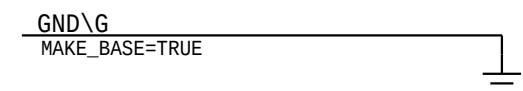
Always on: Keeps the PCH RTC alive



G3H Rails

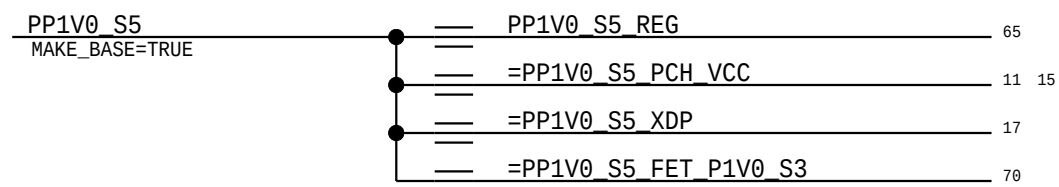
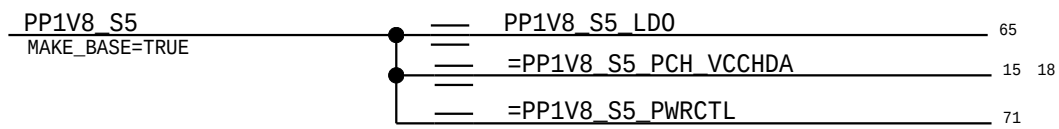
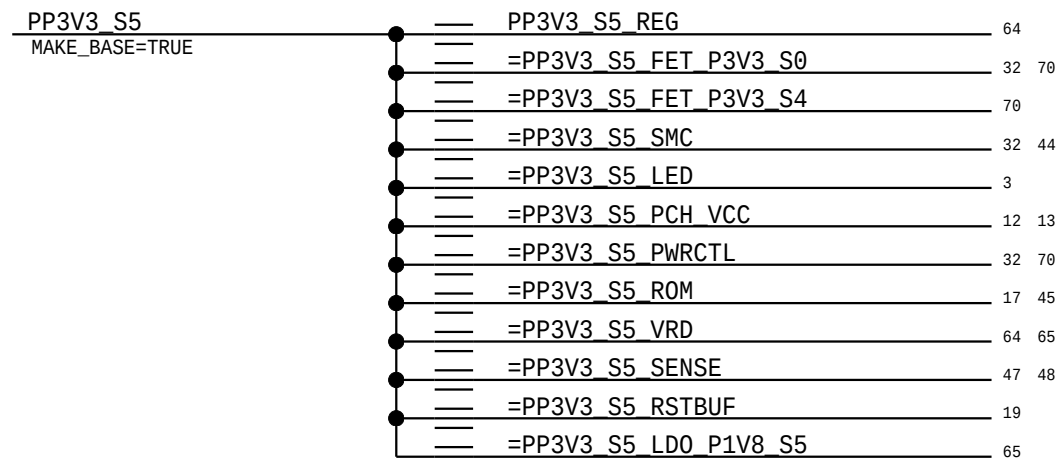
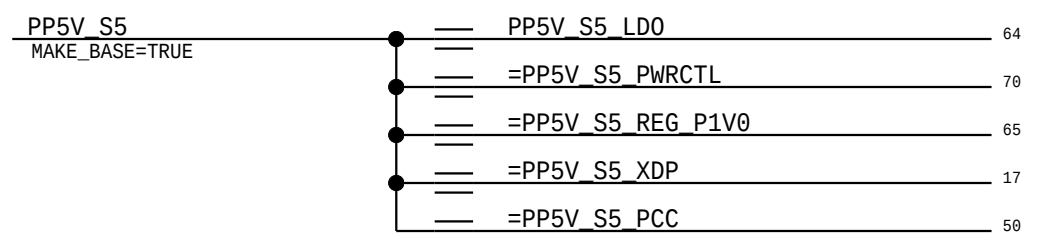
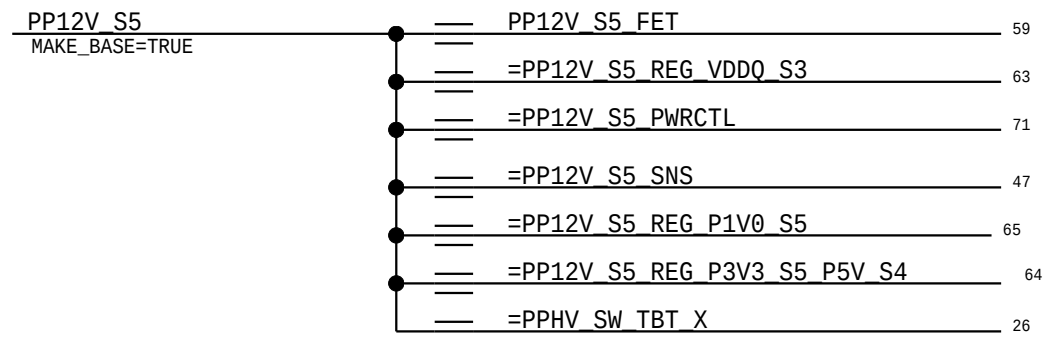


Ground/Common



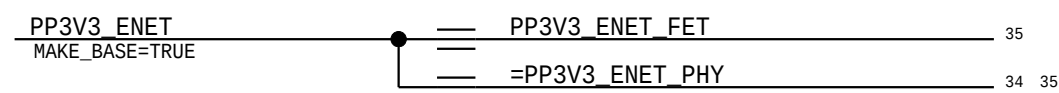
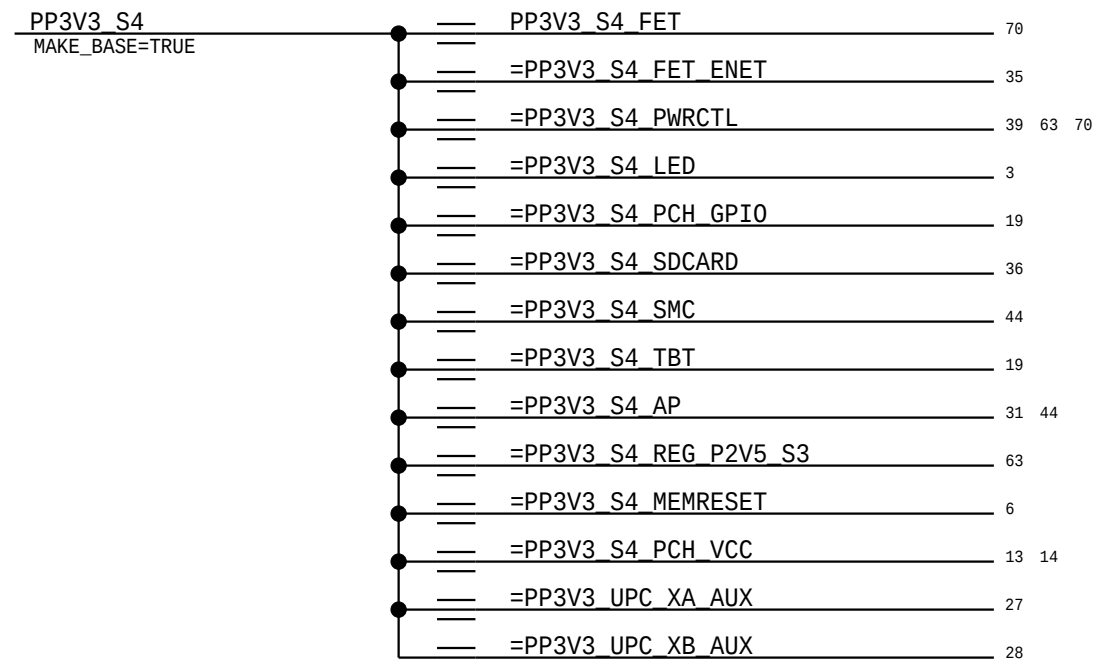
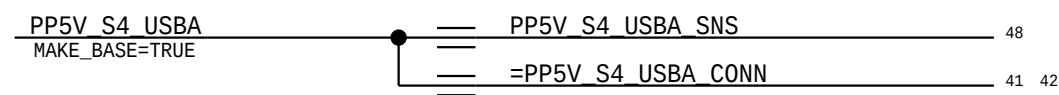
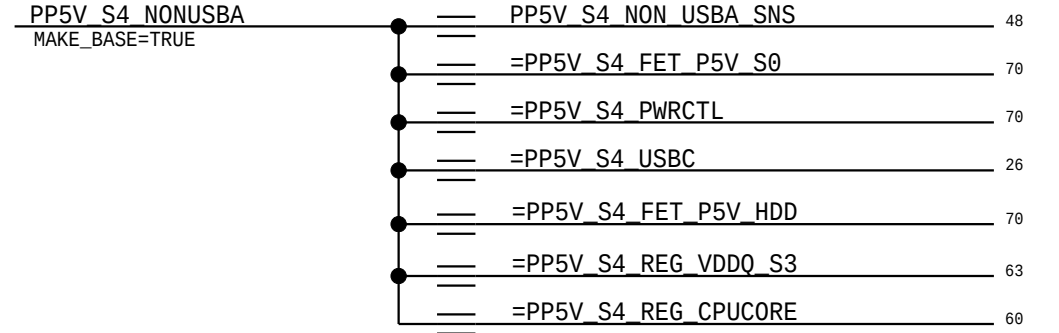
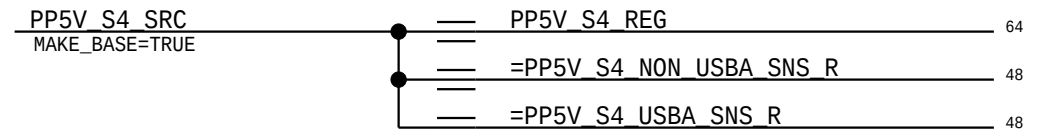
S5 Rails

Enabled when system has AC and is in S5



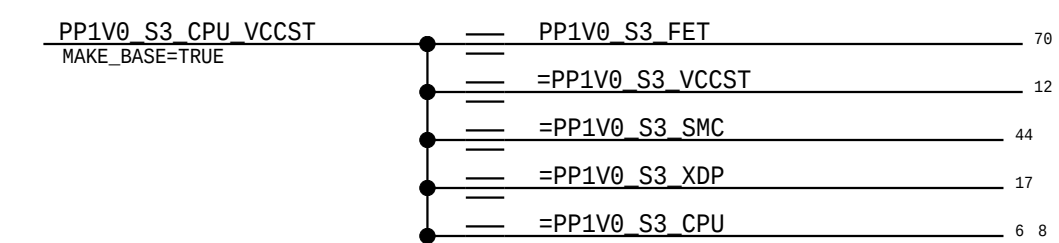
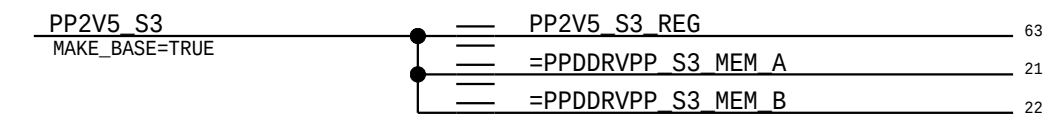
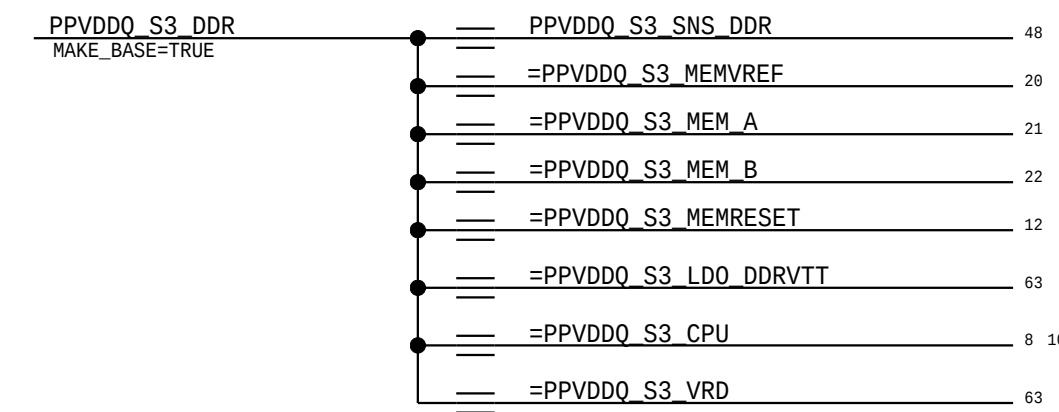
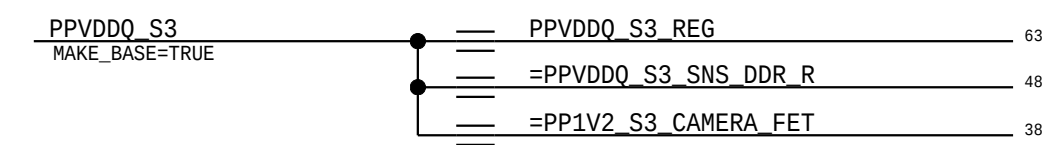
S4 Rails

Enabled when system has AC and is in run or sleep



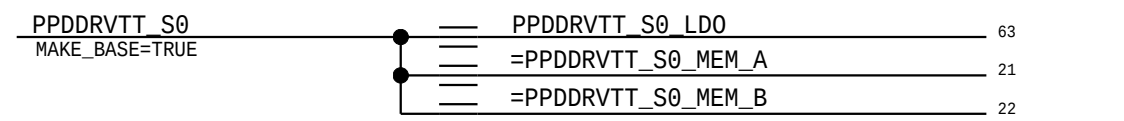
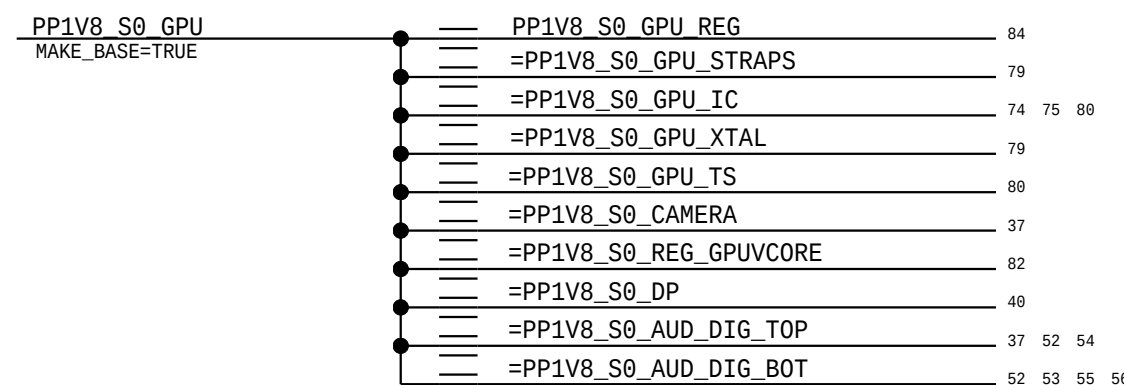
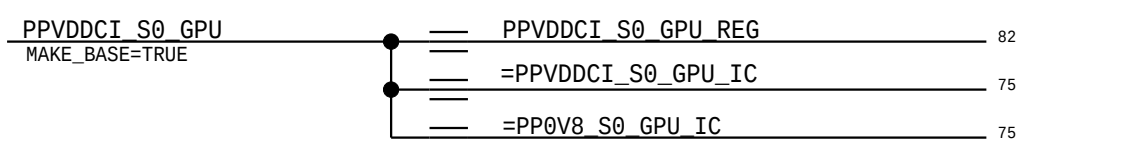
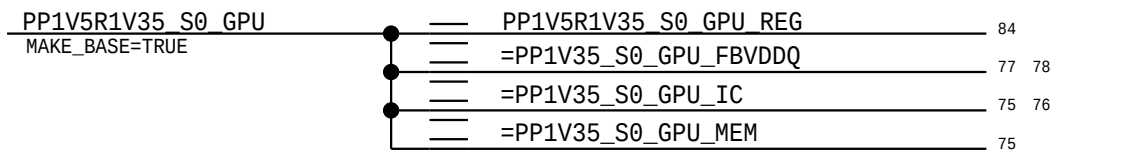
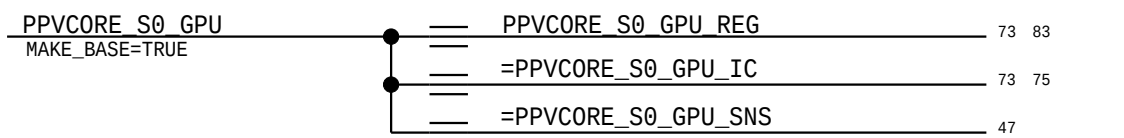
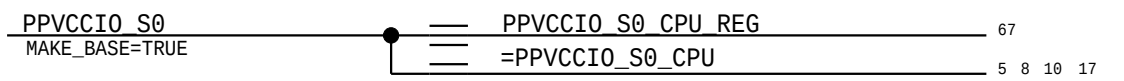
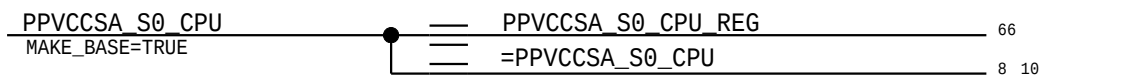
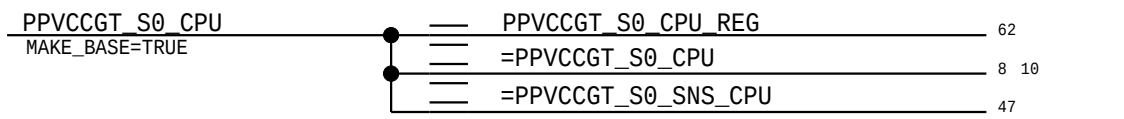
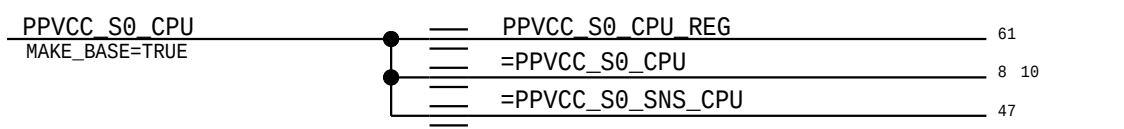
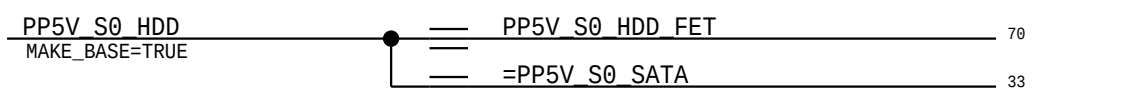
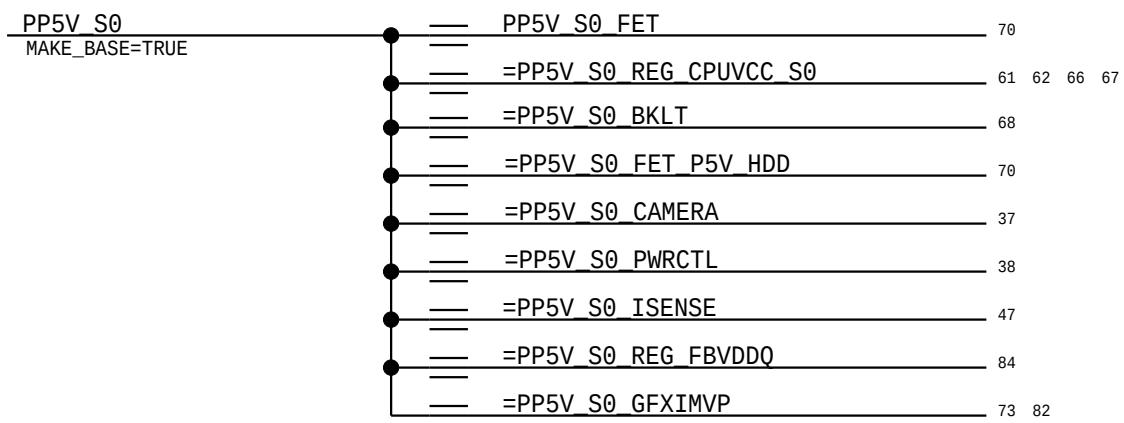
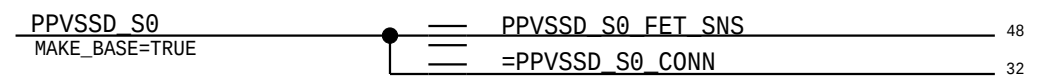
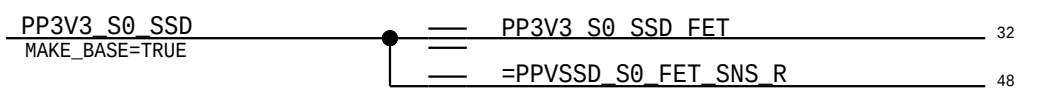
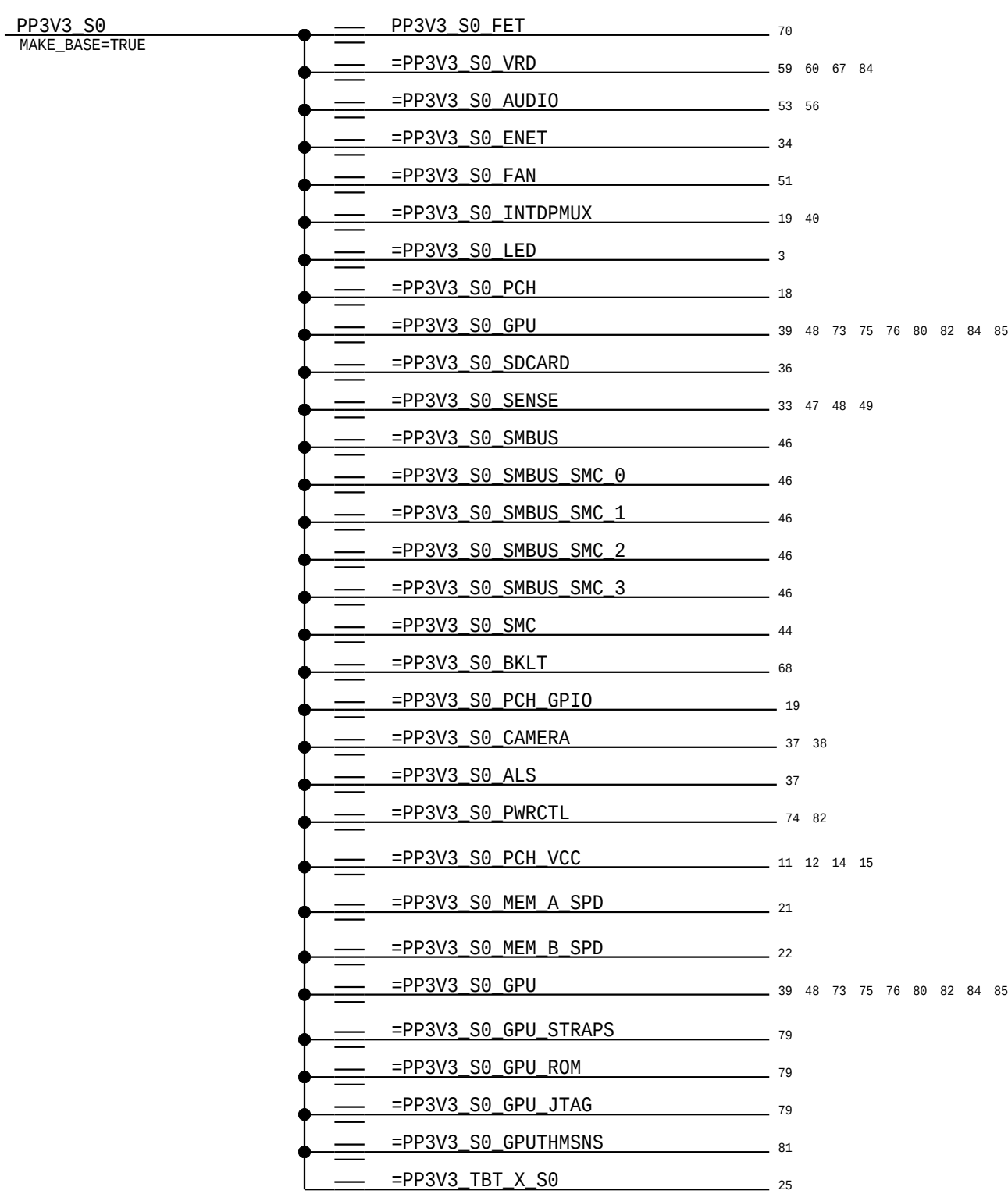
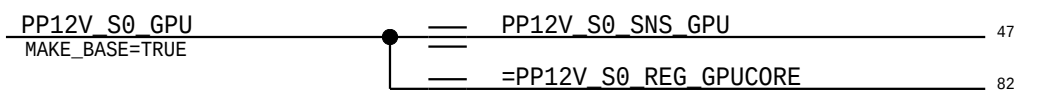
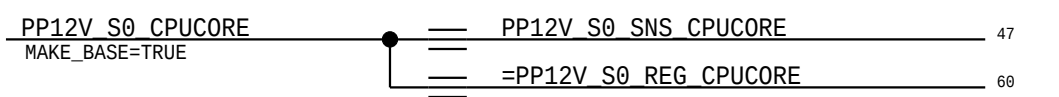
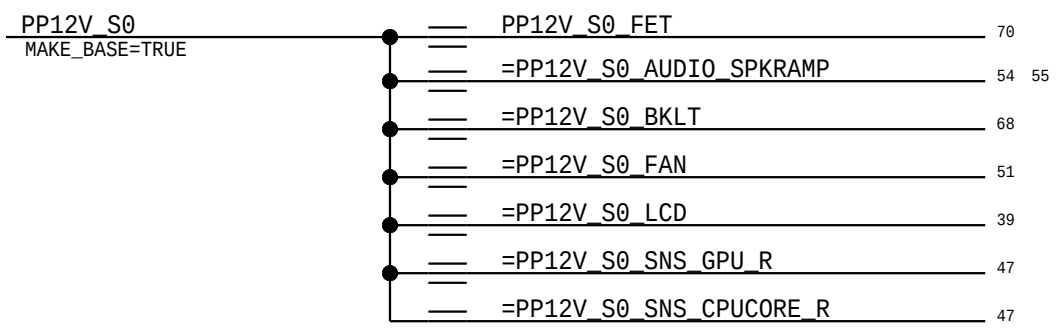
S3 Rails

Enabled when system is in run or sleep



S0 Rails

Enabled when system is in run



SYNCHMASTERandres

SYNCHDATE=06/22/2016

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Power Aliases



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